

XXV International Symposium on Solid State Dosimetry

Local Venue: Centro de Desenvolvimento da Tecnologia Nuclear (CDTN)
September 22-26, 2025 — Belo Horizonte, Brazil

Book of Proceedings: Volume 3

Editors

Telma C. F. Fonseca

Bruno M. Mendes

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Ester M. R. Andrade



BOOK OF PROCEEDINGS

Volume 3

XXV International Symposium on Solid State Dosimetry

Edited by

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CHAPTER 1

Preface

It is with great pleasure that we welcome all participants to the XXV International Symposium on Solid State Dosimetry (ISSSD) in September of 2025, for the first time, at Centro de Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN) and Universidade Federal de Minas Gerais (UFMG) in Belo Horizonte - Minas Gerais - Brazil. This milestone edition marks over two decades of continuous commitment to the advancement of radiation dosimetry through solid-state materials, techniques, and applications.

Since its inception, the ISSSD has served as a vital platform for scientists, engineers, and professionals to share knowledge, discuss new developments, and foster collaborations in the field of radiation detection and measurement. This 25th edition continues this proud tradition, bringing together leading experts and emerging researchers from across the globe to present their latest findings and insights.

The program of this symposium reflects the diversity and depth of the field, covering topics ranging from fundamental studies of dosimetric materials to applied research in medical physics, environmental monitoring, and radiation protection. Special attention has been given to emerging technologies and interdisciplinary approaches that are shaping the future of dosimetry.

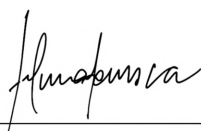
We are especially grateful to all contributors, authors, reviewers, session chairs, and members of the organizing, scientific, and support committees, whose dedication and effort have made this event possible. We also extend our sincere appreciation to our institutional partners and sponsors for their continued support.

We hope that this symposium will not only be an opportunity for scientific enrichment but also a space for meaningful exchanges, new collaborations, and the strengthening of our global community.

Welcome to the XXV ISSSD, and may this be a rewarding and inspiring experience for all.

Organizing Committee

XXV International Symposium on Solid State Dosimetry



Telma C. F. Fonseca
Chair at XXV ISSSD



Bruno M. Mendes
Chair at XXV ISSSD



Juan Azorin Nieto
President of the Sociedad Mexicana de
Irradiación y Dosimetría (SMID)



CHAPTER 2

Summary and Perspectives

The XXV International Symposium on Solid State Dosimetry (ISSSD-2025) brought together researchers, professionals, and students from across the globe, reinforcing the vitality and relevance of solid state dosimetry in advancing radiation science and applications.

The 250 submitted works for the event were distributed across eight thematic areas, reflecting a wide range of research interests. Dosimetry (G2) had the highest representation with 90 works (36% of total), followed by Medical Physics (G4) with 36 contributions (14.4%) and Applications of Thermoluminescence (G1) with 35 works (14%). Other areas included Luminescent Materials (G8, 24 works, 9.6%), Radiation Protection (G5, 21 works, 8.4%), Radiobiology (G7, 19 works, 7.6%), Radiation Sources (G6, 15 works, 6%), and Ionizing and Non-Ionizing Radiation (G3, 10 works, 4%).

Regarding the presentation format, a total of 99 oral communications were delivered, with 54 in person and 45 online, while 151 poster presentations were held, including 96 in person and 55 online. Dosimetry (G2) again showed strong engagement with 23 in-person oral presentations, 10 online oral presentations, 38 posters in person, and 19 online posters. Applications of Thermoluminescence (G1) accounted for 8 oral in person, 7 online, 12 posters in person, and 8 online, while Medical Physics (G4) registered 7 oral in person, 9 online, 11 posters in person, and 9 online. Other areas contributed more modest numbers, such as Luminescent Materials (G8) with 19 works, mainly presented as posters, and Radiobiology (G7) and Radiation Protection (G5) with 19 and 21 contributions, respectively, shared between oral and poster sessions. Radiation Sources (G6) and Ionizing and Non-Ionizing Radiation (G3) had smaller representations but maintained coverage of fundamental aspects of the field. Overall, the data demonstrate diverse scientific participation, with a healthy balance between oral and poster formats, and a substantial proportion of online presentations, highlighting the hybrid and inclusive nature of the event.

Reflecting the truly international character of the symposium, participants represented 18 countries, totaling 248 presentations, ensuring broad participation and accessibility despite geographical distances. An analysis of the data reveals a notable predominance from Brazil, with 175 presentations. Mexico followed with a significant participation of 37 presentations. Other Latin American countries also had a notable presence, including Colombia (7), Chile (4), Peru (4), and Argentina (3). The international representation was broadened by Saudi Arabia and Kazakhstan, both with 3 presentations, and Indonesia with 2. Finally, contributing to the event's diversity, Bolivia, Canada, Ecuador, Spain, Guatemala, Malaysia, the United Kingdom, Switzerland, and Uruguay each participated with a single presentation.



CHAPTER 3

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CHAPTER 4

Event Organization, Supporting Institutions, and Sponsors

4.1 ORGANIZATION



4.2 SUPPORTING INSTITUTIONS



4.3 SPONSORS





CHAPTER 5

Schedule

General Schedule

Hour

Registration			Development of PET Radiopharmaceuticals - Dr. Marina Bicalho Silveira Dr. Isabel Dregely Dr. Emerson Bernardes Soares (Spectrum)		Introduction to Radiation Detectors - Dr. Marco A Polo Labarrios (Lattice)		8h			
							9h			
							9h30			
							10h			
							10h30			
COFFE BREAK							11h			
							11h30			
LUNCH								12h		
			Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico - Dr. Silvia S Hidalgo Tobón (Spectrum)		Solid State Dosimetry: Principles and Applications & OSL Advanced Materials - Dr. Ivonne Berenice Lozano Rojas Dr. Alida Tamar-Gabor (Lattice)		13h			
							13h30			
							14h			
							14h30			
							15h			
COFFE BREAK							15h30			
							16h			
							16h30			
							17h			
Tuesday, 23 Sep			Wednesday, 24 Sep		Thursday, 25 Sep		Friday, 26 Sep	Hour		
Registration			Oral Sessions (Spectrum and Lattice)	Poster Session: 10h00 to 12h30 - Referees for evaluation (Hall, Virtual Room)	Invited Speaker: Luciana Carvalho (Lumina)		Invited Speaker: Rolf Behrens (Lumina)	8h		
Opening Ceremony (Lumina)					Invited Speaker: Bernardo Maranhão Dantas Maria Jose Neves (Lumina)		Invited Speaker: Susana Lalic (Lumina)	8h40		
COFFE BREAK					COFFE BREAK		Invited Speaker: Thiago Viana Miranda Lima (Lumina)	9h20		
Invited Speaker: Wilson Calvo (Lumina)			Oral Sessions (Spectrum and Lattice)	Poster Session: 10h00 to 12h30 - Referees for evaluation (Hall, Virtual Room)	Invited Speaker: Ademar Lugão (Lumina)		Oral Sessions (Lumina and Spectrum)	10h		
Invited Speaker: Yvone Maria Mascarenhas (Lumina)					Invited Speaker: Richard Hugtenburg (Lumina)		COFFE BREAK	10h30		
LUNCH					LUNCH		Oral Sessions (Lumina, Spectrum and Lattice)	11h		
Oral Sessions (Lumina and Spectrum)	Poster Session: 14h30 to 16h30 - Referees for evaluation (Hall)	Roundtable: Radon in Environment: Tracer Applications and Public Health Risks (Lattice)	Technical Visit	Oral Sessions (Virtual Room)	Invited Speaker: Helio Yoriyaz (Lumina)	Poster Session: 14h30 to 17h - Referees for evaluation - (Hall)	Oral Sessions - (Spectrum and Lattice)	Invited Speaker: Tarcisio Campos and Alberto Campos (Lumina)	11h30	
		Roundtable: AI's challenges and Opportunities: Connecting research, industry and society (Lumina)			Invited Speaker: Georgia Joana (Lumina)			Invited Speaker: Carmen Bueno (Lumina)	12h	
COFFE BREAK					COFFE BREAK			COFFE BREAK	12h30	
	Poster Session: 14h30 to 16h30 - Referees for evaluation (Hall)	Invited Speaker: Silvia Tobón (Lumina)			Helen Khoury Mentorship WIN-Brasil (Spectrum)		Poster Session: 14h30 to 17h - Referees for evaluation - (Hall)	Roundtable: Leadership in Scientific and Technological Research (Lumina)	Closing Session: Honor award & Helen Khoury award for best works (Lumina)	13h30
Welcome & Honors Night (Lumina)					Oral Sessions (Lumina and Lattice)					14h
								14h30		
								15h		
								15h30		
								16h		
								16h30		
								17h		
								17h30		
								18h00		
								20h00		

Monday, 09/22/2025

08:00 Registration

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Marina Bicalho	Chair: Álvaro M. L. Gómez
10:30 COFFE BREAK	08:00 Dr. Marina Bicalho (CDTN/CNEN-Brazil), Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria) and Dr. Emerson Bernardes (IPEN/CNEN-Brazil) <i>Development of PET Radiopharmaceuticals</i>	08:00 Dr. Marco Antonio Polo Labarrios (UAM-Mexico) and Dr. Álvaro M. L. Gómez (DEN/UFMG-Brazil) <i>Introduction to Radiation Detectors</i>

12:00 LUNCH

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Angel Ramirez	Chair: Carlos E. Velasquez
15:00 COFFE BREAK	13:00 Dr. Silvia S. Hidalgo Tobón (UAM/Mexico) <i>Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico</i>	13:00 Dr. Ivonne Berenice Lozano Rojas (CICATA-Mexico) <i>Solid State Dosimetry: Principles and Applications & OSL Advanced Materials</i>

Tuesday, 09/23/2025

08:00 Registration

Opening Cerimony - Lumina Auditorium

09:00 Opening Ceremony – **Amenônia Maria Ferreira Pinto** – Centro do Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN), **Bruno Melo Mendes (ISSSD 2025 – Chair)**, **Henrique Resende Martins** – Vice-director of Escola de Engenharia of Universidade Federal de Minas Gerais (UFMG), **Silvia S. Hidalgo Tobón (ISSSD 2025 – Chair)** – Universidad Autónoma Metropolitana (UAM), **Telma Cristina Ferreira Fonseca (ISSSD 2025 – Chair)** – Universidade Federal de Minas Gerais (UFMG), **Wilson Aparecido Parejo Calvo** – Comissão Nacional de Energia Nuclear (CNEN).

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Silvia Hidalgo

10:30 *Invited Speaker* – Wilson Aparecido Parejo Calvo (CNEN-Brazil),
Current Status of Strategic Projects at CNEN.

11:20 *Invited Speaker* – Yvone Maria Mascarenhas (SAPRA Landauer-Brazil),
Personal Dosimetry: The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry Services in Latin America and the Caribbean.

12:30 LUNCH

Roundtable Session (Lattice Auditorium)

13:30 Radon in Environment: Tracer Applications and Public Health Risks
Chair: Talita Santos (CDTN/CNEN-Brazil),
Panelists: Elydio Soares, Laura Takahashi, and Nathan da Silva

Radiobiology	Luminescent Materials/ Applications of Thermoluminescence
Spectrum Auditorium	Lumina Auditorium
Chair: Tarcísio Campos	Chair: Susana Lalic

13:30 Jhon Brandon Felix Ferreira
G07-002 - *Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress*

13:45 Lucas Siqueira de Lima Neto
G07-013 - *Radiometric Analysis by gamma spectrometry of copaiba oil using a Hyper*

14:00 Iris Bomfim Martins Ferreira
G07-015 - *Radiometric analysis of the bivalve mollusk Perna*

13:30 João Pedro
VanellusRad - *The First Brazilian Real-Time IoT Electronic Dosimeter for Radiation Monitoring*

14:00 Ángel Ramírez Luna
G01-002 - *Afinando la Cronología Relativa de Cerámicas de la Zona de Playa Vicente, Veracruz, México Aplicando el Método de Datación Absoluta por Termoluminiscencia*

Roundtable Session (Lumina Auditorium)

14:30 AI's Challenges and Opportunities: Connecting Research, Industry, and Society
Chair: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ana Carolina Costa, Andre Henrique Alves Carneiro and Rita Sebastião

13:30 to 16:30 - Poster Session (Main Hall)

**Note: 14:30 to 16:30 - Referees for evaluation
In-Person**

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-001** - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
- G01-011** - Samira Batista da Silva, *Enzymatic biocatalysis of alginate spheres and iodinated amino acids for applications in embolization guided by SPECT*
- G01-026** - Roseli Künzel, *A low-cost integrated system for thermoluminescence and optically stimulated luminescence measurements*
- G01-029** - Yuri Avelino Costa, *Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using X-ray micro-computed tomography and linear regression*
- G01-032** - Danilo O. Junot, *Crab shell biowaste composites as luminescent radiation detectors*
- G01-035** - Luiz Matheus de Souza Aguiar Leite, *Development of a radiochromic film using a polyacetylene compound*
- G01-036** - Thalles Oliveira Campagnani, *Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes*
- G01-038** - Ketrin Regina Souza, *Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership*
- G01-040** - Raissa Alexia Camargo Guassu, *Estimation of absorbed dose in critical organs during interventional procedures: experimental and computational approach*
- G01-041** - Danilo O. Junot, *Influence of precursor purity on the synthesis and TL/OSL response of CaSO₄-based radiation detectors*
- G01-044** - Vinicius Costa Silva, *Radiological characterization of the waters feeding the Camorim Reservoir by alpha-beta – Pedra Branca State Park*
- G01-045** - Margarete Cristina Guimarães, Thessa Cristina Alonso, *Standardization of gamma irradiation procedures for the preservation of bibliographic heritage*

G5 - Radiological Protection

- G05-003** - Adriana de Souza Medeiros Batista, *Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining*
- G05-006** - Daniel de Castro Pacheco, *Development of a distributed monitoring system for active radiological protection*
- G05-007** - Cássio Miri Oliveira, *Dose evaluation and suggestion of typical values for head CT headache protocols: a preliminary study*
- G05-015** - Thamyra Cybelle Vieira dos Santos, *Quality assurance in ⁹⁰Y radioembolization: an intercomparison protocol for activity optimization*
- G05-017** - Fábio Lima Guimarães, *Radiometric survey at the Stone House Lookout, Saquarema-RJ*
- G05-019** - Marcela Morais Freitas, *Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection*
- G05-022** - Bianca Rossini Marques, *Validation of analytical methods using gamma spectrometry applied to the radiometric characterization of monazite sand*

G8 - Luminescent materials

- G08-004** - Divanizia do Nascimento Souza, *Development of high-sensitivity CaSO₄:Tm,Mn detectors for thermoluminescent and optically stimulated luminescent dosimetry*
- G08-007** - Roseli Künzel, *Effect of silver doping concentrations on the structural and luminescence properties of CaMoO₄ powders*
- G08-008** - Paula Thays Santos Malachias, *Investigation of the luminescent properties of zirconia-enriched dental resins*
- G08-009** - Bruno Araújo, *Optically stimulated luminescence (OSL) characteristics of seven different color types of quartz samples under 405 nm stimulation*
- G08-010** - Rafael dos Santos Viana, *Promotion of the Compton effect of ¹³⁷Cs photons to enhance the effi-*

ciency of radioluminescent batteries

G08-011 - Anderson Bezerra, *Properties of $\text{CaSO}_4\text{:Eu,Mn}$ for Luminescent Dosimetry*

G08-012 - Patrícia de Lara Antonio, *PTTL and PTOSL of TLD-100 exposed to ^{60}Co and lighting with LEDs*

G08-013 - Daniele Gonçalves Mesquita, *Radiation Detection of Lead-Free BaZrO_3 Perovskite Scintillators*

G08-014 - Claudete Roberta Evangelista, *Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs*

G08-015 - Anderson Manoel Bezerra da Silva, *Structural, Morphological, Luminescent, and Dosimetric Properties of $\text{CaSO}_4\text{:Yb,Mn}$ Phosphors*

G08-016 - André Felipe de Oliveira, *Tailored theranostic nanocomposites: mesoporous silica, gold nanoparticles, and gadolinium complexes for cancer diagnosis and treatment*

G08-020 - Jéssica Pauline Nunes Marinho, *Potential theranostic nanoparticles of HAp-Gd^{3+} with curcumin and folic acid for cancer therapy*

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Ivonne B. L. Rojas

16:00 Invited Speaker – Silvia S. Hidalgo Tobón (UAM-Mexico),

State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

17:00 - 20:00 Welcome & Honors Night - Lumina Auditorium

- For the Ceremony: Bruno M. Mendes, Marco Aurélio de Sousa Lacerda and Telma Cristina Ferreira Fonseca
- Award Ceremony for Honored Researchers: Dr. Tarcísio P. R. de Campos and Dr. Teógenes Augusto da Silva
- Cultural Performance by Aruanda Group
- Welcome Cocktail Gathering

Wednesday, 09/24/2025

Applications of Thermoluminescence
Spectrum Auditorium
Chair: Andrea Mantuano

08:00 Rafael Cogollo

G01-004 - *Dosimetric characteristics and kinetic analysis of pure and cerium-doped alumina (α - Al_2O_3) for low-dose thermoluminescent dosimetry applications*

08:15 Héctor Maya

G01-006 - *Kinetic analysis method for general order TL glow curves*

08:30 Natalia Barbosa Gonzaga Mendes

G01-024 - *Using gamma spectrometry as a tool for assessing residual dose in irradiated topaz*

08:45 Alberto Mizrahy Campos

G01-027 - *Random walk of neutrons in a rotational environment*

09:00 Enrique Alejandro Chávez Camacho

G01-030 - *Analysis of the effects of gamma radiation on amaranth seeds and sprouts: a design of experiments (DoE) approach*

Dosimetry

Lattice Auditorium
Chair: Ana Carolina Costa

08:00 Berenice Guadalupe Olmos Grimaldo

G02-006 - *Medición de radón en la zona del Peñón Blanco, Salinas de Hidalgo, San Luis Potosí, México*

08:15 Renato Monaco Nunes

G02-018 - *A Monte Carlo-based software for photon transport in water using the CUDA platform*

08:30 Iasmin Vieira Nishibayaski

G02-019 - *A new formulation of Fricke gel dosimeter development using salicylic acid*

08:45 Camilla da Silva Sampaio

G02-026 - *Assessing the applicability of TDCR counting for nonpure beta emitters: the case of ^{210}Pb in liquid scintillation cocktails*

09:00 Laura Cardoso Takahashi

G02-040 - *Development of an integrated national database for effective dose calculation from environmental radiation exposure in Brazil*

09:15 Laisa de Almeida Neves

G05-023 - *Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry*

10:00 COFFEE BREAK

Radiation Sources
Spectrum Auditorium
Chair: Denison Santos

10:30 Nayelly Nikole Díaz Gutiérrez

G06-002 - *Estimación del término fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP*

10:45 Ismael Teixeira Vicente

G06-004 - *Alpha particle spectrometry and Monte Carlo evaluation of ^{241}Am sources for use in Elastic Recoil Detection Analysis (ERDA)*

11:00 Estela Maria de Oliveira

G06-006 - *Comparative study of gamma efficiency curves in HPGc detector with different calibration standards*

Medical Physics
Lattice Auditorium
Chair: Lucas Paixão

10:30 Sara Suely Ventura da Silva Lima

G04-004 - *Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures*

10:45 Prycylla Gomes Creazolla

G02-089 - *Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation*

11:00 Juraci Passos dos Reis Junior

G04-011 - *An AI revolution in radiotherapy in SUS: The example of Mário Kroeff Hospital*

11:15 Natasha Briggs e Silva

G04-020 - *Determination of the mass attenuation coefficient of thermoset and thermoplastic resins at energies used in conventional mammography*

11:15 Roos S. de Freitas Dam
G06-012 - *Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning*

11:30 Roger Ferreira da Silva
G04-033 - *Mammography quality assurance based on phantom image with deep learning*

08:00 to 12:30 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-003 - Emilia Suárez, *Development and Assessment of Perovskite-based X-Ray Detectors for In Vivo Dosimetry Applications*

G02-017 - Tales Noujaim Silveira, *^{68}Ga Production with the GE PETtrace-8 Cyclotron: a Monte Carlo study*

G02-020 - Leanderson Pereira Cordeiro, *AI-based segmentation performance and implications for organ-level dosimetry in radionuclide therapy*

G02-021 - José Marques Lopes, *An Environmental Radiometric Assessment Based on Airborne Gamma Spectrometry Data in the State of Rio de Janeiro*

G02-023 - Hugo Leonardo Lemos Silva, *Analytical and simulation methods in determining radiation exposure time of healthcare waste*

G02-027 - Alessandra Vaz dos Anjos Santos, *Automated System for Visual Detection of Personal Dosimeters Using Convolutional Neural Networks*

G02-028 - Tiago Macedo da Cunha e Silva, *Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry*

G02-030 - Heloísa Milena Caldeira Rhodes, *Comparison of Methods for Determining the Half-Value Layer (HVL) in Computed Tomography Equipment*

G02-031 - Nicolle Marques Cardoso Ignacio, *Comparison of the Dosimetric Properties of EBT-3 and EBT-4 Radiochromic Films for Application in Hemocomponent Dosimetry*

G02-032 - Fernanda Rocha Cavalcante, *Comparison of X-ray spectral correction software for a CdTe detector*

G02-033 - Guilherme Gazolla Santana, *Computational Modeling of an Anthropomorphic Phantom for Applications in Aerospace Dosimetry*

G02-034 - José Marques Lopes, *Computational Simulation of a NaI(Tl) Detector and Definition of Its Main Parameters for Application in a Neutron Activation Analysis Laboratory*

G02-035 - Lucas Beloti Silva, *Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms*

G02-043 - João Vinícius Batista Valença, *Dose estimation in organs of a pediatric patient undergoing brain radiotherapy via Monte Carlo simulation*

G02-045 - Guilherme Cavalcante de Albuquerque Souza, *Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams*

G02-046 - Frederico Possato Tagliaferro, *Dosimetry Of Electret Microphones Irradiated By X-Rays*

G02-047 - Josemary A. C. Gonçalves, *Energy Dependence of an Epitaxial Diode for Radioprotection Dosimetry*

G02-048 - Enzo Cortez, *Estimating the dose rate constant for Iodine-131 via Monte Carlo simulation: a first step towards pediatric radioiodine therapy shielding evaluation*

G02-050 - João Vinícius Batista Valença, *Evaluating Radiation Dose in Veterinary Professionals: MCNPX-Based Simulation of Equine Radiography Exposure*

G02-051 - Camila Nunes Carvalho Souza, *Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning*

G02-052 - Maria Rita Pascoal de Souza Barros, *Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry*

G02-054 - Gabriel Gomes do Nascimento, *Evaluation of the reproducibility of a new lens dosimeter holder*

G02-061 - João Victor Ramos Oliveira, *Influence of detector geometry on small-field radiotherapy simulations*

G02-066 - Daniel de Castro Pacheco, *Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6*

- G02-070** - Greiciane de Jesus Cesário, *Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms*
- G02-071** - Kyssylla Monnyelle Araujo Silva, *Multielectrode Array Devices for dosimetry and dose enhancement factor analysis in nanoparticle radiotherapy*
- G02-072** - Mateus de Souza Resende, *^{99m}Tc Radiolabeled Papain Nanoparticle: in silico Dosimetry Evaluation using DM BRA phantom and the experimental biodistribution data*
- G02-074** - Henrique Chaves Gulino, *New Low Cost Process of PIN Diode Fabrication for Radiation Detection*
- G02-077** - Carmen C. Bueno, *Online response of a homemade diode dosimetry system in a mobile 500-700 keV electron beam facility*
- G02-078** - Anderson Manoel Bezerra da Silva, *Optimization of Mn²⁺ and Tb³⁺ Concentrations in CaSO₄:Mn,Tb and Advanced Investigation of its TL/OSL Dosimetric Properties*
- G02-079** - Rômulo Monteiro Lopes, *Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation*
- G02-080** - Hugo Leonardo Lemos Silva, *Assessing Photon and Electron Ranges in Healthcare Waste for Radiation Sterilization*

Note: 10:00 to 12:30 - Referees for evaluation
On-line

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-007** - Lenin Estuardo Cevallos Robalino, *Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with ²⁵²Cf neutron source on the ISO-phantom at LPN-CIEMAT*
- G01-010** - Abel Fernando Martínez Cruz, *Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion Synthesis*
- G01-013** - Hassan Salah, *Assessing Patient Exposure in PET/CT scans: A Head and Neck Dose Parameter Study*
- G01-019** - Mohammed, *Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT Imaging*
- G01-021** - Hassan Salah, *Optimizing Radiation Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety*
- G01-022** - Abdelmoneim Sulieman, *Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols*
- G01-033** - Edwar Alonzzo Canaza Mamani, *Datación OSL-SAR y caracterización tecnológica de las cerámicas arqueológicas de los grupos de tradición ceramista Tupiguarani: Pirajuí I y Arataca II, Pernambuco, Brasil*
- G01-034** - David Alexander Urbina Leal, *Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles*

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

- G02-001** - Alejandro Ferreira Tapia, *Caracterización Física y Computacional de Dispositivos Modeladores de Radiación Electromagnética mediante Simulación Monte Carlo*
- G02-008** - Woody Alem Wanderley Araripe Farias, *A platform for creating computational exposure models: personalized dosimetry with Holmium-166*
- G02-009** - Woody Alem Wanderley Araripe Farias, *All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization*
- G02-012** - Aline Fabiane Gonçalves de Oliveira, *Geospatial assessment of radionuclide dispersion and associated effective dose modeled with ARTM View: a case study at CDTN's radiopharmaceutical unit*
- G02-015** - Jardel Lemos Thalhoffer, *Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park*
- G02-016** - Marcelo Franklin da Silva Santos, *Synthesis and Luminescent Characterization of CaSO₄:Tm,Li via Slow Evaporation, Co-precipitation, and Solid-State Diffusion Routes*
- G02-022** - Akemi Yagui, *Organ Dose Estimation in Pediatric Abdomen and Pelvis Radiographs Using Caldose_X Monte Carlo Simulations*
- G02-025** - Ranulfo da Silva Dias, *Assembly and Geant4-based computational simulation of a Helium-3 de-*

tector for ground-level cosmic radiation monitoring

G02-029 - Daniel Vieira Lamas, *Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin*

G02-037 - Paulo Sérgio de Abreu Junior, *Cytogenetic assessment of ionizing radiation exposure in *Allium cepa* using micronucleus frequency: a comparative study between Antarctic and control conditions*

G02-038 - Ana Laura Burin, *Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations*

G02-039 - Leonardo da Silva Boia, *Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I*

G02-044 - Wallacy Viana, *Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor*

G02-059 - Lucas Fabricio de Araujo, *In Silico Investigation of the Impact of Gold Nanoparticles on the Radiosensitivity of 3D HeLa Cells Using the Monte Carlo Method*

G02-060 - Gabriela Rodé de Assis da Silva, *In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil*

G02-067 - Pedro Henrique Avelino de Andrade, *Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry*

G02-073 - Nadia Rodrigues dos Santos, *Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples*

G02-086 - Marcelo Luís dos Santos Filho, *Study of $\text{CaSO}_4:\text{Tm},\text{Li}$ Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography*

G02-087 - Leonardo da Silva Boia, *MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ^{125}I using the adapted ICRP 110 voxel male phantom*

G02-093 - Daniel V. Lamas, *Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data*

G3 - Ionizing and non-ionizing radiation

G03-005 - Samara Mendes Matos, *Effect of Neutron Activation in the Biodegradable Chitosan polymer associated with Gold Nanoparticles.*

G03-010 - Angel Garcia-Duran, *Simulador de detector de Germanio de alta pureza con fuentes radiactivas, embebido en FPGA*

G4 - Medical Physics

G04-001 - Nicolás Eugenio Martín, *Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications*

G04-007 - Priscila Santos Amorim, *Implementation of free software for managing imaging protocols*

G04-008 - Angelica Viridiana Roman Martinez, *Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients*

G04-009 - Jaime Torres Juárez, *Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies*

G04-012 - Gisell Ruiz Boiset, *Assessment of radiologically tissue-equivalent materials for breast imaging phantoms using X-ray spectrometry*

G04-024 - Jesús Emmanuel Mares Ponce, *Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique*

G04-026 - Camila Hitomi Murata, *Evaluation of the affect on performance of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner*

G04-027 - Carolina Salinas Domján, *Experimental characterization of magnetic fields for dosimetric modeling with Monte Carlo methods in MR-Linac radiotherapy*

G04-029 - Ana Carolina Costa da Silva, *Global Usability of Mobile CT Scanners in Rapid Stroke Diagnosis: A Literature Review*

G04-030 - Raissa Alexia Camargo Guassu, *Impact of Body Mass Index on Radiation Doses in Interventional Radiology*

G5 - Radiological Protection

- G05-002** - Wallacy Viana, *Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the TRIGA IPR-R1 nuclear reactor*
- G05-004** - Hericka O. Kenup-Hernandes, *Computational Modeling for Radiological Risk Assessment: Simulation of Plume Dispersion in Radioactive Waste Fires Using HotSpot*
- G05-008** - Akemi Yagui, *Evaluation of radiation exposure in pediatric interventional cardiology procedures*
- G05-009** - Hericka O. Kenup-Hernandes, *Gadolinium Adsorption by Functionalized Mesoporous Silicas (MCM- 41/SBA-15): A Proposal for Volume and Dose Reduction in Radioactive Waste*
- G05-018** - Watila Lins Silva, *Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ^6LiI (Eu) Detector at the Neutron Metrology Laboratory*
- G05-020** - Fábio Sabará Dias, *Survey of radiation exposure from portable X-ray fluorescence*

G6 - Radiation Sources

- G06-005** - Gláucia de Oliveira Barreto, *Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ -ray spectrometry techniques*
- G06-008** - João Pedro Carvalho Andrade, *Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation*
- G06-011** - Gabriel Tura Echeverria, *Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and γ -ray spectrometry*

G7 - Radiobiology

- G07-004** - Ivana Mara Gomes Andersen Cavalcanti, *Determination of mechanical and physical properties of pig ear cartilage through modeling and simulation using DFT*
- G07-014** - Yasmym Sarmiento Pereira, *Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.*
- G07-016** - Gabriela Rodé de Assis da Silva, *Radiometric characterization of Erythrina verna Vell.: A study of natural radioactivity in plant tissues*

G8 - Luminescent materials

- G08-003** - Marcelo Luis dos Santos Filho, *Development of $\text{CaSO}_4\text{:Li}$ Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry*
- G08-005** - Brunno Florencia de Oliveira, *Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques*
- G08-018** - Caroline Pereira dos Santos, *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy^{3+} y Eu^{3+}) a dos concentraciones 0.1 M y 0.5 M*
- G08-019** - André Luiz Tavares e Silva, *Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping*
- G08-021** - Joel Arturo Rivera-Garcia, *Estudio de las propiedades luminiscentes del aluminosilicato de estroncio dopado con Ce ($\text{SrAl}_2\text{Si}_2\text{O}_8\text{:Ce}$) sintetizado por el método de reacción en estado sólido*

12:30 LUNCH

Applications of Thermoluminescence / Luminescent Materials Virtual Room 1 Chair: Daniel Calheiro/Rodrigo Gadelha	Luminescent Materials / Dosimetry Virtual Room 2 Chair: Marco A. Labarrios/Álvaro M. L. Gómez	Medical Physics / Ionizing and non-Ionizing / Radiobiology Virtual Room 3 Chair: Jaime T. Juárez/Bruno Gallotti
13:30 Katherine G. Delgado G01-005 - <i>Effective dose received by Dentistry students in intraoral dental radiology using TLD dosimetry</i> 13:45 Nayely D. M. Ramirez G01-008 - <i>Synthesis of Y₂O₃ nanophosphors and evaluation of their thermoluminescent response for use in radiation dosimetry</i> 14:00 Abel Fernando Martínez Cruz G01-009 - <i>Thermally Stimulated Luminescence of Chemically Modified Magnesium Oxide with Gd³⁺ and Li⁺, Obtained by a Glycine-based Combustion Process</i> 14:15 Radmila Sabitova G01-016 - <i>Characterization of additional irradiation positions in the IVG.1M research reactor</i> 14:30 E. Cruz-Zaragoza G01-020 - <i>Optically -and thermally stimulated luminescence detection of irradiated oregano (Lippia graveolens) polymineral</i> 14:45 Umme Muslima G01-023 - <i>Thermoluminescence Characterization of Natural Flake Graphite Under 6 MeV Electron Beam Exposure from Medical LINAC for Dosimetric Applications</i> 15:00 André Luiz Tavares e Silva G01-025 - <i>A Data Science Approach on the Study of Al₂O₃ Thermoluminescent Glow Curves Under Varying Heating Rates</i> 15:15 Gabriela Pontes Cardoso G08-001 - <i>Cathodoluminescence behavior of PVDF films doped with CuSO₄ under high-dose gamma irradiation</i>	13:30 Jessica Paola C. Fraga G02-005 - <i>La termoluminiscencia del Grafito para su posible aplicación dosimétrica</i> 13:45 José Edgardo Arellano Hernández G02-004 - <i>Estudio de la respuesta termoluminiscente de NA2O-Y2O3-P2O5-SIO2 irradiados con Electrones</i> 14:00 Marcelo F. da Silva Santos G02-014 - <i>New Approaches in the Synthesis of CaSO₄:Tm,Li Phosphors with Sugars as Chelating Agents for Dosimetry Applications</i> 14:15 Vinícius R. A. A. Martins G02-055 - <i>Experimental Evaluation of the Mechanical and Dosimetric Properties of the Red Perspex 4034 Dosimeter After 15 Years of Aging</i> 14:30 Rayline Leite da Silva G02-084 - <i>Standardized Protocol for the Analysis of Cutting Samples and Lithological Materials by Gamma Spectrometry</i> 14:45 Nadia Rodrigues dos Santos G02-049 - <i>Estimation of the soil-to-plant transfer factor of natural radionuclides in organic crop from Pedra Branca State Park</i> 15:00 Leonardo da Silva Boia G02-036 - <i>Construction of the Glioblastoma scenario in the right optic nerve region using DIP techniques for simulation in the MCNP code with conversion to lattice structure by SCAN2MCNP</i> 15:15 Leonardo Santiago Melgaço Silva G02-041 - <i>Development of an optical sensor prototype for dose readings in Fricke gel dosimeters</i>	13:30 Erika Patricia Azorín Vega G04-005 - <i>Dosimetric and Biological Characterization of a 3D Cell Culture Model for the Evaluation of Radiopharmaceutical Induced Bystander Effect</i> 13:45 Leticia González Zamora G04-010 - <i>Study of Prefractal Antennas for Multiband Radiofrequency Applications</i> 14:00 Maria L. Miranda Maciel G04-014 - <i>Calibration of radiochromic films EBT2, EBT3, EBT4 and RTQA2 and assessment of uncertainties for radiotherapy doses</i> 14:15 Elsa Bifano Pimenta G04-022 - <i>Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Conventional CT</i> 14:30 Estefania Reyes Soto G04-002 - <i>Automated Classification and Manual Segmentation of Pediatric Brain Tumors Using Deep Learning on MRI Data</i> 14:45 Giulia Rita de Souza Faés G04-035 - <i>Radiotherapy Planning with VMAT in Halcyon 2.0: Impacts of Isocenter Configuration on Dosimetric Parameters and Treatment Efficiency</i> 15:00 Natalia Matos Orozco G04-003 - <i>Diseño y cálculo de blindajes para garantizar la protección radiológica en una radiofarmacia productora de radiofármacos</i> 15:15 Kuanysh Samarkhanov G05-016 - <i>Radiological Characterization and Dosimetric Assessment of Liquid Radioactive</i>

15:30 Júlia Carina Orfão Costa
G08-002 - *Characterizations of SrMoO₄:Dy³⁺ powders synthesized by hydrothermal method*

15:45 Daniele da Silva Machado
G08-006 - *Dosimetric Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis*

16:00 Carina Oliva Torres Cortés
G08-023 - *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy³⁺ y Eu³⁺) a dos concentraciones 0.1 M y 0.5 M*

16:15 Caroline Pereira dos Santos
G08-017 - *Uncovering the intrinsic nature of the ternary metallic alloys CuNiPd and Cu₂PPd₂: A DFT study*

16:30 Rodrigo Martínez Baltezar
G08-022 - *Point defects study on Aluminum nitride for applications in luminescence dosimetry*

16:45 Pedro R. González
G08-024 - *Thermoluminescent properties of MgO:Sm,Tb+PTFE dosimeters*

17:00 Sandrith M. Pérez Ramos
G01-003 - *Análisis cinético de la curva de brillo termoluminiscente del Óxido de Berilio (BeO) irradiado con Rayos X en un rango de baja dosis*

15:30 Carolina Salinas Domján
G02-090 - *Assessing Water-Equivalence in Gram-Negative Bacteria-Hydrogel Radiation Dosimeters: A Comparative Analysis with Gram-Positive Bacterial Systems*

15:45 Rayline Leite da Silva
G07-017 - *Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters*

16:00 Teodoro Córdova Fraga
G04-019 - *COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana*

16:15 Teodoro Córdova Fraga
G04-013 - *Automated detection and characterization of renal calculi using U-Net and Hounsfield unit analysis in computed tomography scans*

16:30 Priscila Santos Rodrigues
G04-036 - *Three-dimensional dose mapping of gold-198 nanoparticles: A comparative study of 3D-printed matrices using Fricke xylenol gel and Monte Carlo simulations*

16:45 Arthur Reis Martins
G06-015 - *Production of Radioisotopes in Nuclear Microreactors for Medical, Industrial, and Scientific Applications*

17:00 Jardel Lemos Thalhoffer
G02-010 - *Assessment of the level of natural radioactivity in the region of the future Municipal Environmental Protection Area (APA) Ilha Verde, Muriqui, Mangaratiba - RJ*

Waste from BN-350 Reactor

15:30 Stanislav Svetachev
G06-007 - *Determination of detection limits of elements in sediment and rock standards for INAA using the IGR reactor*

15:45 Jonathan O. dos Santos
G06-017 - *Response Function Validation of HPGe Detector Through GADRAS and MCNP Simulation*

16:00 Roberto Méndez Villafañe
G06-003 - *Neutron measurements around the research reactor RA-6 at Centro Atómico Bariloche with a Bonner Sphere Spectrometer*

16:15 Roberto Méndez Villafañe
G06-001 - *Development of a detailed model with MCNP of the Neutron Standards Laboratory at CIEMAT and validation with a Bonner Sphere Spectrometer System*

16:30 Fatima Castañeda Ortiz
G07-001 - *Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line*

16:45 Daniele G. Mesquita
G07-006 - *Gamma Radiation Modulates Genomic Maintenance Pathways in T-Lymphocytes: An Analysis of Clinical Doses*

17:00 Emilio Romero Muñoz
G07-007 - *Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil*

17:15 Raúl Ken Delgado Velázquez
G07-010 - *Magnetic stimulation as a promoting factor for germination and development in Guazuma ulmifolia L. seeds*

13:30 to 16:00 TECHNICAL VISIT

Chair: Camila Engler

Places to visit at CDTN:

- Gamma Irradiation Laboratory - [More information, access here](#)
- Dosimeter Calibration Laboratory - [More information, access here](#)
- TRIGA IPR-R1 Reactor - [More information, access here](#)

Thursday, 09/25/2025

Invited Speaker - Lumina Auditorium

Chair: Bernardo Dantas

08:00 *Invited Speaker* – Luciana Carvalheira (IEN/CNEN-Brazil),
Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.

09:00 *Invited Speaker* – Bernardo Maranhão Dantas (IRD/CNEN-Brazil),
Diseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clínicas de Medicina Nuclear.

09:25 *Invited Speaker* – Maria Jose Neves (CDTN/CNEN-Brazil),
Range Radiation Exposure Indicator.

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Telma Cristina Ferreira Fonseca

10:30 *Invited Speaker* – Ademar Lugão (IPEN/CNEN-Brazil),
Use of ionizing radiation for the production of nanoparticles and advanced curatives.

11:30 *Invited Speaker* – Richard Hugtenburg (Swansea University-United Kingdom),
New Technologies as a Route to Particle Therapy: Laser-Hybrid Acceleration for Radiobiology Applications – The LhARA Project.

12:30 LUNCH

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

13:30 *Invited Speaker* – Hélio Yoriyaz (IPEN/CNEN-Brazil),
Proton Beam Therapy: Main Characteristics and Challenges.

Dosimetry
Spectrum Auditorium
Chair: Peterson Squair

Medical Physics/Dosimetry
Lattice Auditorium
Chair: Lucas Paixão

13:30 Divanizia do Nascimento Souza
G02-042 - *Dose Distribution in an Anthropomorphic Cervix Phantom Using Two Computer Modeling Approaches*

13:45 Francisco Harley Dantas Hauradou Xavier
G02-057 - *Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP*

14:00 Francisco H. D. Hauradou Xavier
G02-094 - *Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks*

14:15 José Elizeu de Souza Junior
G02-062 - *Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study*

13:30 Ana Beatriz Caldeira de Andrade
G04-025 - *Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing*

13:45 Crislane de Jesus Cesário
G04-028 - *Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using CaSO₄:Dy Dosimetry*

14:00 Nathalia Luzia Aidar Alves
G04-032 - *Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics*

14:15 André Luiz Espindola Fidelis
G04-037 - *Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLDs and Radiochromic Film*

14:30 Leonardo Gualberto Silva Macedo
G02-095 - *Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions*

14:45 Beatriz Diniz de Oliveira Guedes
G02-082 - *Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation*

15:00 Hirys Sales
G02-053 - *Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms Used in the Calibration of In Vivo Monitoring Systems*

13:30 - 17:00 - Poster Session (Main Hall)

Note: 14:30 to 17:00 - Referees for evaluation In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

- G02-081** - Felipe Mamed de Souza, *SMS Notification System for Potential Radiological Risk Scenarios*
G02-085 - Alcilene Cristina da Silva, *Study and Characterization of the EPR Response of New Materials for High Dose Dosimetry*
G02-088 - Ana Letícia Almeida Dantas, *The IRD In Vivo Monitoring Laboratory - LABMIV: Instrumental Capabilities applied in routine and emergency situations*
G02-091 - Maryanna Regina de Souza Roberto, *OSL Tandem Dosimeter for ICRU 95 Operational Quantities*
G02-092 - Thalys Matheus Gama de Oliveira, *Photons Response of an Albedo Monitor*

G3 - Ionizing and non-ionizing radiation

- G03-001** - Paula Regina Rodriguez Coelho, *"Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology"*
G03-002 - José Marques Lopes, *Adjustment Factor for Gamma Spectra Due to the Influence of Sample Density on the Shielding Background Radiation*
G03-003 - Davyd Soares Dunda, *Application of Steam Methodology in Science Teaching: Prototyping a Cloud Chamber*
G03-004 - José Marques Lopes, *Comparison of environmental sampling results using portable and benchtop scintillation detectors*
G03-006 - Adriana de Souza Medeiros Batista, *Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with magnetite*
G03-007 - Paulo Cezar Rocha Silveira, *Gamma Irradiation Effects on the Mechanical and Ballistic Performance of 3D-Printed Polylactic Acid*
G03-008 - Luciana Carvalheira, *Gold nanoparticles with potential for use in multimodal combat of head and neck cancer*

G4 - Medical Physics

- G04-006** - Otávio Gabriel Moreira Silva, *SNR and CTDIvol study as a function of the Noise Index in Computed Tomography exams*
G04-016 - Rute Cristina de Oliveira Soares, *Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment*

G04-017 - Gustavo Freire Pereira da Silva, *Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density*

G04-018 - Rute Cristina de Oliveira Soares, *Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis*

G04-021 - Iasmin Vieira Nishibayaski, *Dosimetric evaluation of EBT3 and EBT4 in superficial HDR brachytherapy for keloids*

G04-023 - Camila Engler, *Evaluation of Breast Positioning Quality and Its Association with Technical Parameters in Screening Mammography*

G04-031 - Douglas Philipe Martinho Dos Santos, *Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)*

G04-034 - Kellen Adriana Curci Daros, *Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: Observational study*

G04-038 - Arícia Ravane Pereira da Cruz, *Validation of an alanine-based postal dosimetry system for quality control in radiotherapy*

G6 - Radiation Sources

G06-013 - Kathleen Martins, *Neutronic Analysis of an Accelerator-Driven Subcritical Reactor Utilizing Thorium Dioxide and Reprocessed Nuclear Fuels*

G7 - Radiobiology

G07-003 - Christiana da Silva Leite, *Comparative Study of Organ Mass in Healthy and Tumor-Bearing Mice for Preclinical Dosimetry Purposes*

G07-005 - Camila Ramos Silva, *Development and validation of a body-shielding device for radiotherapy of breast cancer in mice using a panoramic gamma source*

G07-008 - Luciana Penedo de Melo Teixeira, *Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory*

G07-009 - Olanía Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*

G07-011 - Camila de Almeida Salvego, *Nitric Oxide-Donor Nanoparticles Potentiate Radiotherapy of Triple-Negative Breast Cancer*

G07-018 - Bruno Gallotti, *Growth Inhibition of *S. cerevisiae* BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation*

G07-019 - Maurício Moacir da Silva Borges, *The Role of Caffeine in Reducing TNF- α and Improving Cognitive in Irradiated Mice in the Brain*

G07-020 - Luis Martín García Ortiz, *Cellular response in human leukocytes (in vitro), due to exposure to UVB radiation using apoptotic biomarkers*

Invited Speaker - Lumina Auditorium

Chair: Teógenes da Silva

14:30 *Invited Speaker* – Georgia Joana (CNEN-Brazil) and Ketrim Souza (INB-Brazil), *WiN Brasil - Diversity & Inclusion Management*.

15:30 COFFEE BREAK

16:00 - Mentorship Helen Khoury WiN Brasil (Spectrum Auditorium)

Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership
Moderator: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ketrim Souza and Winners (WiN Brasil)

Roundtable Session (Lumina Auditorium)

16:00 Leadership in Scientific and Technological Research

Moderator: Luciana Carvalheira (IEN/CNEN-Brazil),

Panelists: Adriana Marques and Anna Letícia Barbosa de Sousa

**Radiation Protection
Lumina Auditorium
Chair: Teógenes da Silva**

**Radiation Sources / Radiation Protection
Lattice Auditorium
Chair: Jhonny A. Benavente Castillo**

17:00 Mirella Maturo da Cruz

G05-021 - *Transfer Factors of Natural Radionuclides in Organic Foods from Rio de Janeiro, Brazil*

17:15 Elydio J. D. Soares

G05-010 - *Investigation of radon exhalation from mortars made from niobium tailings and the mitigating role of coating materials*

17:30 Raphael Francisco Gomes dos Santos

G05-005 - *Determination of Radiation Protection Procedures for an Offshore Plant Inspection in Brazil Using Radiotracer Techniques*

17:00 Gabriel do Nascimento Freitas

G06-014 - *Nuclear simulation workflow to correct gamma ray well logs for radioactive contaminated drilling fluids*

17:15 José Alípio dos Santos Filho

G06-016 - *Proposal for the construction of a neutron generator for application in research and industry*

17:30 Welen Nunes de Lima

G05-014 - *Proposal for the use of thorium as fuel in TRIGA[®] research reactors*

Friday, 09/26/2025

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

08:00 *Invited Speaker* – Rolf Behrens (PTB-Germany),
From dosemeter development to routine use – Standards and Uncertainties.

08:40 *Invited Speaker* – Susana Lalic (DFI/UFS-Brazil),
Plant Stem Cells as an Ethical Alternative for Biodosimetry.

09:20 *Invited Speaker* – Thiago Viana Miranda Lima (University of Lucerne-Switzerland),
Current and Future Trends in Nuclear Medicine Dosimetry.

10:00 COFFEE BREAK

Radiological Sources
Spectrum Auditorium
Chair: Carmen C. Bueno

Dosimetry
Lumina Auditorium
Chair: Peterson Squair

10:30 Matheus R. do Nascimento
G03-009 - *Speknife: open-source software for X-ray spectra correction in solid state dosimetry*

10:30 Jhonny A. Benavente Castillo
G02-007 - *Simulación del blindaje del búnker de un acelerador ciclotrón utilizando el código MCNP*

10:45 Eduardo B. de Paula
G05-011 - *Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via γ -ray Spectrometry and Algorithmic Data Processing*

10:45 Natasha Briggs e Silva
G02-011 - *Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project*

11:00 Igor Andrade Machado
G02-083 - *SpecUnPy code: updates and what is coming*

Invited Speaker - Lumina Auditorium

Chair: Bruno M. Mendes

11:30 *Invited Speaker* – Tarcísio P. R. de Campos (DEN/UFGM and CDTN/CNEN-Brazil) and Alberto Mizrahy Campos (Augsburg-Germany),
Challenges in Radioisotope Decontamination and Production through Neutron Spectral Selection.

12:30 LUNCH

Dosimetry
Spectrum Auditorium
Chair: Maria da Penha A. Potiens

Dosimetry
Lattice Auditorium
Chair: Bruno M. Mendes

Applications of Thermoluminescence
Lumina Auditorium
Chair: Teógenes da Silva

13:30 Isabela N. S. Ferreira
G02-064 - *Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais*

13:45 Claudio A. Federico
G02-065 - *Measurements of Ionizing Radiation Dose Onboard*

13:30 Celso G. de Paulo
G02-069 - *Monte Carlo simulation of a HPGe detector*

13:45 Ana G. Leão Ferro
G02-075 - *Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach*

13:30 Tarcísio P. R. Campos
G01-046 - *The reduction of the radiological impact of nuclear fuel waste*

14:00 Júlia B. Severo
G01-039 - *Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban*

*an Aircraft from Southeastern
Brazil to the Antarctic Continent*

14:00 Paulo J. Iack Silva
G02-068 - *Monte Carlo N-Particle
(MCNP) Simulations for
Scintillation Emission in
Quantum Dot-Doped Systems*

14:00 Rafael S. Oliveira
G02-076 - *On the adaptation of a
commercial Bonner Sphere
Spectrometer to be used with
thermoluminescent detectors*

*Quarries in Belo Horizonte,
Brazil*

Invited Speaker - Lumina Auditorium

Chair: Telma Cristina Ferreira Fonseca

14:30 *Invited Speaker* – Carmen Bueno (IPEN/CNEN-Brazil),
Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosimetry.

15:30 COFFEE BREAK

16:00 CLOSING SESSION

- Honorary Award in Memory of Helen Jamil Khoury: Carmen Bueno
- Helen Khoury Award for Best Works: Andrea Mantuano & Ester Andrade
- Final Session: Summary and Perspectives: Telma Cristina Ferreira Fonseca & Bruno M. Mendes

CHAPTER 6

Pre-Symposium Courses

6.1 DEVELOPMENT OF PET RADIOPHARMACEUTICALS

Pre-symposium course speakers

- Dr. Marina Bicalho (CDTN/CNEN-Brazil)
- Special Guest: Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria)
- Special Guest: Dr Emerson Bernardes (IPEN/CNEN-Brazil)

Dr. Marina Bicalho Silveira is an Industrial Pharmacist (UFMG, 2007) with an M.Sc. (2010) and Ph.D. (2016) in Radiation Science & Technology (CNEN). Specialized in Radiopharmacy (SBRAFH, 2023), she works as a Senior Technologist at CDTN/CNEN, leading Quality Assurance for PET radiopharmaceutical production. Her expertise covers GMP, regulatory compliance, and R&D of novel radiopharmaceuticals, including preclinical studies. She also serves as a Professor in CDTN's Graduate Program, bridging industry and academia.

Dr. Isabel Dregely is Head of the Competence Center for Artificial Intelligence and Data Analytics at Fachhochschule Technikum Wien. She previously served as Senior Lecturer at King's College London and was a UKRI Future Leaders Fellow. Her research background includes advanced MRI techniques, particularly in prostate and cardiovascular imaging, with postdoctoral work at UCLA and experience at the Technical University of Munich. She holds a PhD in Physics from the University of New Hampshire, where she focused on hyperpolarized xenon MRI and RF coil development. Dr. Dregely combines expertise in biomedical imaging with AI and data analytics applications.

Dr. Emerson Soares Bernardes is a pharmacist and researcher with over two decades of experience in immunology, molecular biology, and nuclear medicine. He served as Manager of the Radiopharmacy Center at the Nuclear and Energy Research Institute (IPEN), where he led pioneering projects in the development of radiopharmaceuticals. He currently holds a position as a Visiting Researcher at Erasmus Medical Center in the Netherlands, contributing to innovation initiatives and patent development. His career is distinguished by the integration of scientific research, strategic management, and entrepreneurship.

Description: This course covers the basics of PET radiopharmaceuticals-production, clinical applications, and their impact on nuclear medicine-while showcasing CDTN'S cutting-edge technologies and their role in enhancing diagnostics and patient care in Brazil.

Dr. Marina Bicalho - Theoretical Aspects and State of the Art in PET Radiopharmaceuticals.

Dr. Emerson Bernardes - Advances in new PET Radiopharmaceuticals

Dr. Isabel Dregely - Artificial Intelligence Applied to PET Imaging

6.2 INTRODUCTION TO RADIATION DETECTORS

Pre-symposium course speakers

- Dr. Marco Antonio Polo Labarrios (UAM-Mexico)
- Dr. Álvaro M. L. Gómez (DEN/UFGM-Brazil)

Dr. Marco Antonio Polo Labarrios holds a Ph.D. and Master's degree in Energy Engineering from UNAM and a Bachelor's in Energy Engineering from UAM Iztapalapa. He teaches part-time at Universidad Iberoamericana and UNAM, focusing on Transport Phenomena and Renewable Energy. His research involves the simulation of energy processes based on mass, momentum, and energy transport. He has published 21 scientific articles, participated in multiple national and international conferences, and supervised four undergraduate theses. Dr. Polo Labarrios was part of the National Commission on Nuclear Safety and has been a Level 1 member of the National System of Researchers since 2021.

Dr. Álvaro M. L. Gómez holds a Ph.D. and M.Sc. in Nuclear Science and Technology from the Federal University of Minas Gerais – UFGM (2017; 2021). Holds a Bachelor's degree in Physics (diploma validated by UFGM) and two postgraduate specializations: Bioengineering – Universidad Distrital Francisco José de Caldas (2008; 2014) (Colombia), and University Teaching – Universidad Cooperativa de Colombia (2013). Has research experience in ionizing radiation dosimetry, development of tissue-equivalent materials for computed tomography phantoms, material characterization, and teaching experience at the Department of Nuclear Engineering (DEN) at UFGM, lecturing on radiation protection, radiation detection and nuclear instrumentation, radioisotope applications, and introduction to nuclear energy.

Description: Throughout the course, we will explore different types of radiation detectors, their operating principles, and criteria for their proper selection and use. The goal is to provide students with a solid foundation that enables them to identify the most suitable equipment for each situation and use it efficiently, maximizing both measurement accuracy and radiological protection.

6.3 IMPORTANCIA DE LA FÍSICA EN UNA IMAGEN POR RESONANCIA MAGNÉTICA EN EL DIAGNÓSTICO CLÍNICO.

Pre-symposium course speakers

- Dr. Silvia S Hidalgo Tobón (UAM-Mexico)
- Msc. Jaime Torres Juárez (UAM-Mexico)

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.



Descripción: Se presentarán las bases de la física de una imagen por resonancia magnética y su relación con el radio diagnóstico. La seguridad en resonancia magnética que evita complicaciones y accidentes que con la continua capacitación disminuyen notablemente. Así como las técnicas nuevas que están incluyendo inteligencia artificial para un mejor diagnóstico por imagen.

6.4 SOLID STATE DOSIMETRY: PRINCIPLES AND APPLICATIONS & OSL ADVANCED MATERIALS.

Pre-symposium course speakers

- Dr. Ivonne Berenice Lozano Rojas (IPN-Mexico)

Dr. Ivonne, a distinguished Mexican scientist holding a Ph.D. from Centro de Investigación en Ciencia Aplicada y Tecnología Avanzada, Unidad Legaria, conducts pioneering research in TL/OSL dosimetry, radiation detectors, and food irradiation detection, significantly advancing medical physics and safety.

Description: This minicourse focuses on quartz as a natural radiation dosimeter, exploring its defects and their role in optically stimulated luminescence (OSL). Advanced OSL materials, including synthetic and natural quartz, are analyzed for radiation dose assessment in dating and dosimetry applications.

CHAPTER 7

Symposium Speakers

7.1 DR. ADEMAR BENÉVOLO LUGÃO: USE OF IONIZING RADIATION FOR THE PRODUCTION OF NANOPARTICLES AND ADVANCED CURATIVES.

Dr. Ademar Lugão holds Bachelor's in Chemical Engineering from the Polytechnic School of the University of São Paulo – USP (1980), earned a Master's degree in Nuclear Technology from IPEN-USP (1985), and a Ph.D. in Nuclear Materials Technology, also from IPEN-USP (2004). With more than 30 years of experience as a research leader in polymer irradiation, he is currently the Manager of the Center for Chemistry and the Environment at the Nuclear and Energy Research Institute – IPEN. His main research focuses on the synthesis, processing, and characterization of polymers; structural modification of polymers through energetic processes, including crosslinking, grafting, branching, and chain scission; development of hydrogels for health and environmental applications; clean technologies for producing polyolefins with controlled rheology; starch foams for sustainable packaging; advanced low-cost dressings tailored to the needs of the Brazilian Public Health System (SUS); albumin and papain nanoparticles for encapsulating radiotherapeutics and chemotherapeutics; and radioactive gold nanoparticles (Au-198) for applications in radiodiagnosis and radiotherapy (theranostics). He currently serves as the Coordinator of the Nanomaterials and Nanocomposites Coordination Hub of SISNANO (National System of Nanotechnology Laboratories) of the Ministry of Science, Technology, and Innovation (MCTIC).

Description: Use of ionizing radiation for the production of nanoparticles and advanced curatives.

7.2 DR. BERNARDO DANTAS: DISEMINACIÓN DE TÉCNICAS DE MONITOREO IN VIVO OCUPACIONAL EN UNA RED DE CLINICAS DE MEDICINA NUCLEAR

Dr. Bernardo Maranhão Dantas holds a PhD in Sciences (Biology – Nuclear Biosciences) from the State University of Rio de Janeiro (1998). He is a Senior Technologist at the Brazilian National Nuclear Energy Commission (CNEN) and a faculty member of the Graduate Program at the Institute of Radiation Protection and Dosimetry, focusing on Radiation Biophysics. He has published 87 peer-reviewed articles and has supervised numerous students across undergraduate, master's, and doctoral levels. He has led five research and development projects and currently coordinates the CNPq-certified research group "Occupational Internal Dosimetry."



Description: Professionals who handle radiopharmaceuticals for diagnostic and therapeutic purposes in nuclear medicine are subject to the intake of radionuclides through inhalation and ingestion. The radiological protection aspects of this practice are regulated in Brazil by the National Nuclear Energy Commission (CNEN). The International Atomic Energy Agency (IAEA) recommends the implementation of occupational monitoring programs when there is a risk of annual effective doses exceeding 1 mSv. The control of radionuclide intake can be carried out using internal dosimetry techniques. Currently, in Brazil, there are not enough laboratories capable of providing internal monitoring services to meet the entire demand if this requirement were to be enforced by CNEN. This paper presents an overview of the current situation in Brazil and economically viable technical alternatives, aiming to make the implementation of routine internal monitoring programs for workers occupationally exposed to significant risks of radionuclide exposure in nuclear medicine feasible.

7.3 DR. CARMEN BUENO: RADIATION PROCESSING WITH ELECTRON BEAMS: ON THE FEASIBILITY OF USING ELECTRONIC DOSIMETERS FOR REAL-TIME DOSIMETRY

Dr. Carmen holds Bachelor's and Licentiate degrees in Physics from PUC/SP, as well as Master's and Doctorate degrees in Nuclear Instrumentation from the same university. She began her academic career at PUC/SP, where she was appointed as a Full Professor of Nuclear Physics. In 1995, she started working as a researcher at the Nuclear and Energy Research Institute (IPEN-CNEN/SP) and later completed her postdoctoral work at the Instrumentation and Experimental Particle Physics Laboratory (LIP) at the University of Coimbra. She also holds a second Doctorate in nuclear technology from IPEN-USP. She has published nearly 150 full-length articles in international journals and conference proceedings and has supervised numerous master's, doctoral, and postdoctoral students. Currently, she is a senior researcher at IPEN with over 30 years of experience in nuclear instrumentation for radiation measurement and dosimetry. Most of her research focuses on developing semiconductor-based dosimetry systems for medical and radiation processing fields. She currently serves on the Steering Committee of the National Institute of Science and Technology (INCT) project, "Nuclear Instrumentation and Applications in Industry and Health" (INAIS). Additionally, she is a member of the Advisory Boards of the Brazilian Society of Solid-State Dosimetry (SSD) and the Brazilian Association of Nuclear Energy (ABEN).

Description: In this lecture, we will briefly introduce the radiation processing of products, a relatively new industry with wide applications and significant commercial success, primarily involving gamma rays and electron beams (EB). The main features of the irradiation facilities, the doses delivered, and the regulatory requirements for the processed products will be discussed to highlight the role of dosimetry in this field. Several passive solid-state dosimeters, which are well-established in radiation processing dosimetry, will be introduced to emphasize the need for online dosimeters, whether as routine tools or for monitoring the operating conditions of EB facilities. An overview of efforts to develop online dosimeters, with a focus on electronic devices, will be provided. Subsequently, the lecture will explore the operational characteristics and the physics behind the dose and dose rate response of diodes, explaining how our group has addressed issues related to their use as real-time dosimeters in fixed and mobile EB facilities. The latest data obtained with these diodes, both as routine dosimeters and as electron beam monitors, will be presented to assess the feasibility of their application. The current status of online high-dose dosimeters and the challenges of meeting the metrological standards required in radiation processing dosimetry will also be discussed.

7.4 DR. GEÓRGIA SANTOS JOANA: WIN BRASIL - DIVERSITY & INCLUSION MANAGEMENT

Bachelor's degree in Physics from the Federal University of Minas Gerais - UFMG (2003), master's degree in Radiation, Mineral, and Materials Science and Technology from the Nuclear Technology Development Center of the National Nuclear Energy Commission - CDTN/CNEN (2005), and a Ph.D. in Nuclear Science and

Technology from the Postgraduate Program of the Department of Nuclear Engineering at UFMG (2018). She is currently a technologist at the National Nuclear Energy Commission. She previously worked in the field of Medical Physics at the Superintendence of Sanitary Surveillance of the Minas Gerais State Health Secretariat, where she coordinated the State Quality Control Program in Mammography - PECQMamo. She has experience in Medical Physics and Health Surveillance, with an emphasis on Radiological Protection, Regulation, Safety, and Quality in health services. Her research focuses on Radiological Protection, Quality Control, Radiation Dosimetry and Metrology, and Risk Analysis and Management in the field of medical applications of radiation.

Description: WiN Brasil - Diversity & Inclusion Management

7.5 DR. HELIO YORIYAZ: PROTON BEAM THERAPY: MAIN CHARACTERISTICS AND CHALLENGES

Bachelor in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow at the Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge Associated Universities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute (IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN. Bachelor's in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow at the Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge Associated Universities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute (IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN.

Description: Proton therapy is an increasingly relevant form of radiotherapy because it enables a reduction in the total dose compared to photons, preserving critical structures while simultaneously delivering higher and better-conformed doses to the tumor volume. This effect is achieved through the superposition of several weighted Bragg curves with different incident energies, resulting in a broadened Bragg peak, known as the Spread-Out Bragg Peak (SOBP). Although already considered technologically mature, proton therapy still presents important challenges that need to be investigated. Among these, the precise calculation of the relative biological effectiveness (RBE) stands out, defined as the ratio between the doses required for two different types of radiation to produce the same biological effect. This aspect is directly linked to the physical interaction processes between the proton and matter. When a proton with kinetic energy passes through a medium, it interacts with electrons and nuclei of the target, causing its deceleration. These interactions are known as electronic stopping and nuclear stopping, respectively. The stopping power describes the force the medium exerts on the ion, corresponding to the average energy loss per unit path length. In the case of proton-matter interaction, electronic stopping is dominant. This energy loss dynamic is fundamental to understanding the dose deposition profile. As the proton slows down, the energy transferred to biological tissue per unit path length increases, reaching a maximum at a specific depth. This region, known as the Bragg peak, coincides with the maximum stopping power and represents the area of greatest interest for clinical applications. Therefore, its precise positioning is critical during treatment planning. To study these processes in greater detail, advanced approaches have been employed, such as real-time Time-Dependent Density Functional Theory (TDDFT). This is a non-perturbative method based on modern quantum simulations, which has proven particularly useful for investigating electronic stopping in complex systems like water and DNA, bringing computational modeling closer to the real conditions found in biological tissues.



7.6 DR. LUCIANA CARVALHEIRA: NEUTRONS, KNOWLEDGE, AND NEW GENERATIONS: HOW BRAZIL'S ARGONAUTA REACTOR IS SHAPING THE FUTURE OF RADIOCHEMISTRY AND ITS IMPACT ON GLOBAL DOSIMETRY CHALLENGES.

Dr. Luciana Carvalho is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brasil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brasil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

Description: For over 60 years, Brazil's Argonauta research reactor has addressed hands-on education in physics and engineering of reactors, and radiochemistry. As a hub for RD&I, Argonauta drives partnerships with institutions like IME, UFRJ and startups in radiotracer technologies. Modernization efforts, including fuel improvements and new radioisotopes production, exemplify its commitment to advancing radiation science nuclear applications worldwide.

7.7 DR. MARIA NEVES: RANGE RADIATION EXPOSURE INDICATOR

Dr. Maria José Neves is a Senior Researcher at the Brazilian National Nuclear Energy Commission (CNEN), based at the Centre for the Development of Nuclear Technology (CDTN), where she leads the Radiobiology Laboratory. She holds a Bachelor's degree in Medical Biological Sciences from the Faculty of Medicine of Ribeirão Preto, University of São Paulo (USP-RP), as well as a Master's and Ph.D. in Physiology from the same institution, with a focus on metabolism. She completed her postdoctoral training in Biochemistry at the Katholieke Universiteit Leuven (KUL), Belgium. Her main research interests include cancer cell metabolism and the biological effects of ionizing radiation.

Description: A small plastic or glass tube (with a final volume of 1.5mL or smaller) containing a solution, composed of sodium selenite and glucose, can be fixed to the surface of a material and will indicate whether it was itself exposed to moderate doses of range radiation (5 – 100 kGy) in a range irradiator contained in the source of ^{60}Co . This type of irradiator is widely used in the irradiation of foods such as dried grains, fruits, tubers, etc., sterilization of surgical material, among other activities. Given its low cost and ease of preparation, it can be frozen and kept in stock at the time of use, which is why we consider an alternative option to confirm exposure to high-range radiation.

7.8 DR RICHARD HUGTENBURG: NEW TECHNOLOGIES AS A ROUTE TO PARTICLE THERAPY: LASER-HYBRID ACCELERATION FOR RADIOBIOLOGY APPLICATIONS - THE LHARA PROJECT.

Dr Richard Hugtenburg is a medical physics expert and research academic at Swansea University, UK, specialising in radiotherapy physics. Dr Hugtenburg began his career in Christchurch, New Zealand, working as a medical physicist while studying for a PhD on the use of computational methods in radiation oncology. He moved to the UK in 1997 and continued to practice in radiotherapy physics; first at the Queen Elizabeth Medical Centre in Birmingham, then at Singleton Hospital in Swansea. His research examines dosimetry methods associated with emergent radiotherapy technologies, utilising Monte Carlo modelling for a diverse range of radiation physics problems, including the analysis of energy deposition in biological systems, the optimisation of dosimeters and detectors, nanoparticulates, shielding and therapy systems.

Description: Particle therapy is a form of external beam radiotherapy that utilises ions to achieve precise (sub-mm) physical distributions of physical dose with the benefits of high LET to combat radioresistant tumours. Particle therapies are also expected to offer better normal-tissue sparing and reduce long-term side-effects in patients, where the current state-of-the-art in radiotherapy has contributed to significant improvements in the survivability of many cancers in recent years. The LhARA project is a consortium of UK universities that considers three new technologies as a means to reduce the cost, physical footprint and flexibility of particle therapies, where currently treatments, such as carbon ion therapy, are restricted to a few laboratories world-wide. The laser-hybrid method uses high-powered lasers to generate ions with kinetic energies of 10s of MeV and at high dose-rates suitable to examine high dose-rate sparing effects such as FLASH radiotherapy.

7.9 DR. ROLF BEHRENS: FROM DOSEMETER DEVELOPMENT TO ROUTINE USE - STANDARDS AND UNCERTAINTIES

Dr. Rolf Behrens holds a diploma at Technical University of Braunschweig (Germany) and a doctorate (Dr. rer. nat.) in physics from the University of Jena (Germany). He is a senior scientist at the Physikalisch-Technische Bundesanstalt (PTB), Germany's national metrology institute, where he leads the Working Group on Dosimetry for Brachytherapy and Beta Radiation Protection. Dr. Behrens is responsible for quality assurance in the Ionizing Radiation division, serves as Radiation Protection Officer for brachytherapy and beta radiation, and acts as an auditor for the German Accreditation Service (DAkkS). He heads and participates in several national and international standardization committees, including the IEC and ISO working groups on dosimetry and uncertainties, and chairs the German Committee on Radiation Protection Dosimetry. His research focuses on dosimetry of beta and photon radiation for radiation protection and brachytherapy, Monte Carlo simulations, spectrometry, Bayesian data evaluation, measurement uncertainty, and dosimetry for the lens of the eye.

Description: Radiation protection uses protection quantities (not directly measurable) and operational quantities (measured to estimate protection quantities). Standards ensure the quality of dose measurements, established by IEC and ISO, and adopted by regional organizations (CEN, CENELEC, GCC, SARSO, ARSO). Type tests for photon dosimeters use ISO 4037 reference fields; beta and neutron fields follow ISO 6980 and ISO 8529. Standards like IEC 62387 and IEC 61526 address other influencing factors such as angle, dose rate, temperature, and electromagnetic immunity. Uncertainties are calculated by combining the effects of these influences, as described in IEC TS 62461 (updated 2025). Performance is regularly checked using comparison measurements following ISO 14146 (updated 2024, includes EURADOS). [Click here to view the presentation slides.](#)

7.10 DR. SILVIA HIDALGO TOBÓN: STATE OF THE ART OF MAGNETIC RESONANCE IMAGING: FROM FUNDAMENTAL PHYSICS TO CLINICAL APPLICATIONS.

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.

Description: State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

7.11 DR. SUSANA LALIC: PLANT STEM CELLS AS AN ETHICAL ALTERNATIVE FOR BIODOSIMETRY

Dr. Susana de Souza Lalic is Full Professor of Physics at the Federal University of Sergipe. She holds a B.Sc. (1997) and Ph.D. in Physics (2002) from the University of São Paulo and a postdoctoral fellowship at the University of Pisa (2011). She is a faculty member of the Ph.D. Program in Industrial Engineering (Nuclear Engineering and Industrial Safety) at the University of Pisa and on the Board of Directors of the Master Course in CBRNe Protection at Tor Vergata University of Rome. She is Counselor of the Brazilian Physical Society (SBF), Chair of its Medical Physics Committee, and Vice-chair of the International Solid State Dosimetry Organization (ISSDO). She previously served as Secretary-General of SBF and Coordinator of the Graduate Program in Physics at UFS. Her research focuses on medical physics, condensed matter physics, and applied nuclear physics, with emphasis on radiation detectors, material characterization, radiobiology, and radiation metrology.

Description: This study investigates an ethical approach to biodosimetry, utilizing plants as a sustainable alternative to traditional animal models. Plants, known for their sensitivity to environmental stressors such as radiation, provide a valuable platform for identifying genotoxic agents while reducing reliance on animal testing. The research highlights the utility of the onion (*Allium cepa*) as an *in vivo* model for assessing radiation-induced genetic and chromosomal damage. While *Allium cepa* is a widely recognized standard in environmental contamination studies, it remains underexplored as a tool for radiation detection. A comprehensive evaluation of radiation effects and their temporal progression is made possible by analyzing key variables, including linear energy transfer, dose, and dose rate. The study also examines the potential of photobiomodulation to enhance biological recovery in tissues damaged by radiation, offering a novel intervention strategy. To ensure robustness, findings from the *Allium cepa* model are cross-validated against other *in vivo*, *in vitro*, and *in silico* systems, establishing its efficacy as an ethical and effective alternative for radiation biodosimetry.

7.12 DR. TARCÍSIO CAMPOS AND DR. ALBERTO CAMPOS: CHALLENGES IN RADIOISOTOPE DECONTAMINATION AND PRODUCTION THROUGH NEUTRON SPECTRAL SELECTION

Dr. Tarcisio P. R. de Campos is retired full professor from the Federal University of Minas Gerais (UFMG), currently collaborating with both UFMG and the Nuclear Technology Development Center (CDTN). He holds a degree in Engineering from UFMG, an MSc in Nuclear Engineering from UFRJ, a PhD from the University of Illinois at Urbana-Champaign, and a postdoctoral fellowship at MIT as a Fulbright Fellow. His expertise spans nuclear engineering, radiobiology, radiochemistry, radiotherapy, and the development of radioisotopes, simulators, and medical software. He has published extensively, holds 14 patents, and has supervised over 75 graduate theses. He has coordinated over 28 research projects and actively collaborates with national and international research institutions.

Dr. Alberto M. Campos is Postdoctoral researcher in Augsburg, Germany (2024–present), with a PhD in Mathematics from the Institute for Pure and Applied Mathematics (IMPA, 2024). He completed a research stay at Instituto Superior Técnico (IST), Lisbon (2022), holds a Master's and Bachelor's degree in Mathematics from UFMG, and a technical degree in Electrotechnics from CEFET-MG. He has served as a teaching assistant in various probability and analysis courses at IMPA and UFMG, and has presented talks at the Erdős Institute (Budapest) and the Department of Nuclear Engineering at UFMG. His research interests lie broadly in Probability Theory, with a focus on random covering problems, percolation, and random walks.



Description: We present investigations into the production and decontamination of radioisotopes using a selective Spectral Resonant Selector (RSS), focusing on neutron radiative capture reactions in precursor nuclides and transmutation reactions of long-lived radioactive waste. The physical and mathematical aspects of biased Brownian motion are addressed, through which a tailored neutron spectrum can be generated to target specific nuclides. Reaction rates for both precursor nuclides and decontamination processes are presented. Two distinct neutron spectra are considered—one at low energy and another at resonance energy—allowing interaction with a broad range of target nuclides. The production of isotopes such as Sm-153, Lu-177, Re-188, Ho-166, and Yb-169 serves as representative examples of potential radionuclide generation. Additionally, the decontamination of long-lived actinides, including transuranic radionuclides and fission products, through fission or radiative capture reactions, illustrates a promising approach to nuclear waste mitigation. The specific activities and benchmark parameters of various radioisotopes under defined conditions demonstrate the effectiveness of the RSS in both radioisotope production and nuclear decontamination.

7.13 DR. THIAGO LIMA: CURRENT AND FUTURE TRENDS IN NUCLEAR MEDICINE DOSIMETRY

Dr. Thiago holds a BSc in Physics with Medical Physics from UFRJ and a PhD in Medical Physics from University College London. He is the chief diagnostic medical physicist at Lucerne University Hospitals (LUKS) and research physicist at Genolier Hospital. He also lectures at the University of Lucerne. He chairs the education committee of the Swiss Society of Radiobiology and Medical Physics and represents the society in EFOMP working groups. His work focuses on dosimetry, protocol optimization, radioprotection, and research in nuclear medicine and medical imaging.

Description: This lecture delves into the evolution of nuclear medicine dosimetry. It will explore the progression from early rudimentary methods, which relied on basic measurements and empirical estimates, to the sophisticated imaging techniques and computational models employed today. These advancements have significantly enhanced the accuracy of absorbed dose quantification, improving both safety and efficacy in treatments. The lecture will also discuss the future of dosimetry, highlighting the integration of artificial intelligence and personalized medicine. This forward-thinking approach promises even more precise dose calculations tailored to individual patient characteristics, reflecting a continuous effort to optimize therapeutic protocols and improve patient outcomes in nuclear medicine.

7.14 DR. YVONE MASCARENHAS: PERSONAL DOSIMETRY - THE IMPORTANCE OF REGULATION, TECHNOLOGY, AND CULTURE. EVALUATION OF DOSIMETRY SERVICES IN LATIN AMERICA AND THE CARIBBEAN

Dr. Yvone Maria Mascarenhas has a degree in Physics from the University of São Paulo (USP), and a master's, doctorate, and post-doctorate from the USP's Institute of Physics and Chemistry of São Carlos (IFSC), with research in basic physics as well as radiology and dosimetry. She has worked as a coordinator for research projects funded by agencies such as CNPq and FAPESP and has extensive experience in the field of Physics, working primarily in radiological protection, x-ray production, radiodiagnosis, and dosimetry.

Description: Personal dosimetry is a fundamental tool in radiation protection, ensuring that best practices in facilities using ionizing radiation effectively minimize risks to exposed workers. The use of radiation is highly regulated worldwide, and to make personal dosimetry widely accessible, technology plays a key role—not only as a scientific instrument but also as a reliable and reproducible measurement technique. Accurate analysis of population-level dosimetry data is only possible with the correct and appropriate use of personal dosimeters. This can be achieved only when a strong radiation protection culture is promoted and adopted by all stakeholders involved in personal dosimetry. In Latin America and the Caribbean, the evaluation of Individual Monitoring Services (SMIE) varies by country, reflecting differences in regulations, technological infrastructure, and safety culture. This study presents an overview of the current state of personal dosimetry in the region, with emphasis on the 2022 intercomparison exercise organized by REPROLAM. A total of 26 services—both public and private—were evaluated using different irradiation beams and dose levels. The results show that 69.2% of the laboratories complied with ISO 14146:2018. However, some exhibited calibration deviations, particularly with X-ray beams. Notably, OSL (Optically Stimulated Luminescence) dosimetry has seen significant growth, accounting for 34% of the systems evaluated in 2022. In Brazil, the development of Personal Dosimetry Services has been shaped by two decades of intercomparison exercises and over 30 years of a structured certification process. Despite this progress, challenges remain in standardizing methodologies and enhancing laboratory training to improve dosimetry accuracy across the region. This analysis underscores the critical importance of regulation, technological advancement, and the consolidation of a radiological safety culture as pillars for strengthening occupational dosimetry in Latin America and the Caribbean.

7.15 DR. WILSON CALVO: CURRENT STATUS OF STRATEGIC PROJECTS AT CNEN

Dr. Wilson Calvo holds a diploma in Metallic and Ceramic Materials Engineering from the Federal University of São Carlos - UFSCar (1987) and holds a Master's (1997) and a PhD (2005) degrees in Nuclear Technology - Applications, from the University of São Paulo - USP. He began working at the Nuclear and Energy Research Institute (IPEN) in 1988 and went on to pursue a successful career in both administration and technology. He is a Senior Technologist, professor, and postgraduate advisor in the area of Nuclear Technology at IPEN/USP. He published nearly 100 full-length articles in international conferences and indexed magazines, and supervised more than 15 master's, doctoral, and postdoctoral students. He is a leading expert in Nuclear Engineering, with a focus on the applications of Nuclear Techniques in Industry, Health, Agriculture, and the environment. His emphasis is on radioisotope technology (radiotracers and sealed radioactive sources) and ionizing radiation (electron beams, X-rays, and gamma rays). Owing to his great contribution in the nuclear applications field, he was awarded the Nuclear Merit Award from ABDAN (2020). Regarding his administrative career, he served as a Manager of the Radiation Technology Center (2001-2013), Director of Administration and Infrastructure (2014-2016), and Superintendent of IPEN (2017-2023). He was also Vice President of the Board of Directors of the São Paulo Technology-Based Incubator at USP/IPEN-Cietec (2017-2023). He is a Member of the Deliberative Committee of CNEN (2023-), the Advisory Board of ABENDI, and the International Irradiation Association (IIA). Currently, he is the Director of Research and Development at the National Nuclear Energy Commission - CNEN.

Description: The National Nuclear Energy Commission's guidelines for technological innovation focus on three main areas: nuclear technologies—including reactors, instrumentation, materials, and the fuel cycle—applications of ionizing radiation in health, industry, agriculture, and the environment, as well as radioprotection and dosimetry. This lecture will highlight several strategic projects, including CENTENA, GraNioTer, the tracer technique for offshore oil and gas platforms and processing plants, mobile units equipped with electron accelerators, and a multipurpose reactor. The main features of each project, their contributions to addressing major national challenges in developing nuclear technology, and the current progress of each initiative will be discussed. In short, CENTENA aims to design, build, and commission the first nuclear and environmental technology facility in South America for disposing of low-level radioactive waste, thereby establishing a foundation of safety and sustainability for Brazil. GraNioTer's goal is to establish itself as a national technological hub, promoting research, development, and innovation based on graphene, niobium, rare earths, and other materials and minerals with increasing global demand. Another key project is the application of the radiotracer technique to offshore oil and gas platforms and processing plants, which seeks to develop a prototype and method for measuring gas flow in pipelines. Equally important is the Mobile Unit, equipped with an electron beam accelerator, which aims to expand the nation's capacity to treat industrial effluents and improve water quality in urban areas, addressing one of the primary challenges faced by modern society.

7.16 DR. BRUNO MELO MENDES

Dr. Bruno Melo Mendes holds a Bachelor's degree in Biological Sciences and a Master's and PhD in Nuclear Science and Technology from the Federal University of Minas Gerais (UFMG). He is currently a researcher at the Nuclear Technology Development Center (CDTN/CNEN), responsible for the Internal Dosimetry Laboratory, and a professor in the CDTN postgraduate program. He also coordinates the Monte Carlo Modelling Expert Group (MCMEG). His expertise lies in the nuclear field, with a focus on radiopharmacy, computational modeling (Monte Carlo), and internal dosimetry.

7.17 DR. TELMA CRISTINA FERREIRA FONSECA

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMEG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br

CHAPTER 8

Round Tables

8.1 AI'S CHALLENGES AND OPPORTUNITIES: CONNECTING RESEARCH, INDUSTRY, AND SOCIETY

Moderator

- Dr. Telma Fonseca (DEN/UFMG)

Panelists

- Dr. Rita Sebastião (ICEX/UFMG)
- Dr. Ana Carolina (Siemens Healthineers)
- Dr. Andre Henrique Alves Carneiro (USP)

Description: The roundtable is part of the Tech Area on Artificial Intelligence, an innovative space within ISSSD 2025 dedicated to the convergence of technology, healthcare, and innovation. Its main goal is to foster dialogue among startups, researchers, healthcare professionals, and investors, encouraging the use of AI in clinical practice. During the session, strategic topics will be discussed, such as: Challenges and opportunities of AI in clinical radiology Explainability and ethics of diagnostic algorithms Automation and integration of hospital workflows Success stories from innovative startups Trends for the next 5 years in medical AI The roundtable will feature multidisciplinary experts — including radiologists, data scientists, entrepreneurs, and regulators — and will be moderated by professionals with experience in healthcare innovation. There will also be room for audience interaction, with guided questions designed to stimulate debate.

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br



Dr. Rita C. O. Sebastião is a full professor at Universidade Federal de Minas Gerais (UFMG) with experience in Chemistry, working mainly on the topics: inverse problems, artificial neural networks, nuclear magnetic resonance, thermal decomposition, chemical kinetics, combustion processes, and thermodynamics. Extensive experience in education, research, and extension, especially in projects aiming for the popularization of science, educational innovation, and entrepreneurship, connecting the university to basic education. Experience in developing and managing projects on socio-environmental sustainability and education. She is currently a member of the Postgraduate Program in Chemistry (PPGQUI), Postgraduate Program in Technological Innovation (PPGIT) and Professional Master's in Technological Innovation and Intellectual Property (MPITPI) at UFMG.

Dr. André Carneiro is a Senior Data Scientist and Management Consultant with over 6 years of experience working with leading Brazilian companies. Skilled in developing mathematical, statistical, and machine learning models to deliver actionable insights and optimize business decisions. Currently, I am a Senior Data Scientist at Ambev Tech, one of Brazil's largest companies, where I focus on recommendation systems, clustering, predictive modeling, causal inference, and advanced techniques such as double machine learning (e.g., causal forests). In addition to my corporate experience, I have a strong academic background with over 5 years of research experience. I hold a PhD in Production Engineering from Escola Politécnica of the University of São Paulo (Poli-USP), as well as a double degree in Production Engineering from Poli-USP and a Master's degree in Management and Industrial Engineering from the Instituto Superior Técnico of the University of Lisbon. My research focuses on machine learning, statistical control, and project monitoring. I am also a collaborating researcher at LABDAPS (Laboratório de Big Data e Análise Preditiva em Saúde) of the Public Health School of the University of São Paulo, where I pursue postdoctoral research on fairness and social justice in machine learning models, aiming to enhance predictive performance for underrepresented populations.

8.2 RADON IN ENVIRONMENT: TRACER APPLICATIONS AND PUBLIC HEALTH RISKS

Moderator

- Dr. Talita Santos (CDTN/CNEN)

Panelists

- MSc. Elydio Soares (CDTN/CNEN)
- MSc. Laura Takahashi (CDTN/CNEN)
- Dr. Nathan da Silva (CDTN/CNEN)

Description: The two main aspects related to radon will be presented and discussed: the concern regarding the impacts on public health and the applications of radon isotopes as tracers of environmental phenomena.

Dr. Talita de Oliveira Santos holds a degree in Radiology Technology from the Federal Center for Technological Education of Minas Gerais (2007), a master's degree in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2010) and a PhD in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2015). She is currently an adjunct professor at the Federal University of Minas Gerais and a collaborator of the National Commission for Nuclear Energy. She has experience in Physics, with an emphasis on Radiation Physics, working mainly on the following topics: natural radioactivity, application of radiation to the environment, nuclear instrumentation and radioprotection.



8.3 LEADERSHIP IN SCIENTIFIC AND TECHNOLOGICAL RESEARCH

Moderator

- Dr. Luciana Carvalheira (IEN/CNEN)

Panelists

- Adriana Marques (Agência Nacional de Água - ANA)
- Anna Letícia Barbosa de Sousa (CNEN)

Description: This roundtable will convene leading experts to explore the pivotal role of leadership in advancing nuclear scientific and technological research. The discussion will address key challenges, including regulatory frameworks, ethical considerations, nuclear applications, diversity and inclusion, effective communication, and the integration of the next generation into the global nuclear sector. By examining how leadership drives innovation, navigates complex challenges, and shapes the future of nuclear science and technology, this session aims to foster meaningful dialogue. Participants will gain insights into emerging technologies, explore strategic leadership approaches, and strengthen collaboration among researchers, policymakers, and industry leaders.

Dr. Luciana Carvalheira is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brazil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brazil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

CHAPTER 9

Mentorship Helen Khoury WIN-Brasil

The Helen Khoury Mentorship WinBrasil2025

Program: Advancing Diversity, Inclusion, and Female Leadership in the Nuclear Sector

The Helen Khoury Mentorship Program, promoted by Women in Nuclear Brazil (WiN Brasil), was established to foster the professional development of women in the nuclear sector, integrating diversity in age, educational background, and technical expertise. Over seven months, the program combined one-on-one mentoring sessions with collective workshops, encouraging both academic and professional growth.

Mentors and mentees co-developed final projects that reflected the program's multidisciplinary and inclusive approach. Outcomes included: the design of a critical meaningful learning model for future mentorship cycles; the creation of Radioarretadas, a local initiative to engage women from Pernambuco's countryside in nuclear discussions; research on gender and racial equity within Eletronuclear; studies on hybridization of nuclear and renewable energy; structured career development strategies in nuclear physics; participation in scientific proposal submissions (PPSUS/Innovation 2025); scientific dissemination projects on radiation and radioactivity; and professional milestones such as article submissions, approvals in competitive examinations, and postgraduate advancements.

The results demonstrated progress in strengthening both technical and behavioral competencies, expanding professional networks, and integrating diverse regional and cultural perspectives. By connecting institutions such as CNEN, IPEN, IEN, Eletronuclear, INMETRO, and universities across Brazil, the initiative reinforced the importance of knowledge sharing and inter-institutional collaboration. It is concluded that the Helen Khoury Mentorship has been consolidated as a model for professional development aligned with sectoral demands and individual expectations. The program not only contributes to participants' growth but also reinforces the presence and leadership of women in the nuclear sector, leaving a legacy of diversity, inclusion, and innovation.

Keywords: Mentorship, Diversity, Inclusion, Female Leadership, Nuclear Sector.

CHAPTER 10

Honors Awards

The International Symposium on Solid State Dosimetry (ISSSD) is proud to honor outstanding contributions in the field of radiation science and its applications in medicine, industry, and research. The 2025 Awards recognize excellence, innovation, and leadership that have advanced knowledge, safety, and education in nuclear science.

10.1 PROF. DR. TARCISIO PASSOS RIBEIRO DE CAMPOS



We are honored to recognize the exceptional contributions of Prof. Tarcísio Passos Ribeiro de Campos, a distinguished figure in nuclear engineering and radiation science in Brazil. With a career spanning over four decades, Prof. Campos has significantly advanced the fields of medical dosimetry, radiation protection, and nuclear technology.

As a retired full professor at the Federal University of Minas Gerais (UFMG), he continues to collaborate with the university and the Nuclear Technology Development Center (CDTN). Prof. Campos earned his undergraduate degree in Engineering from UFMG (1983), a master's in Nuclear Engineering from the Federal University of Rio de Janeiro (1985) with a focus on Reactor Physics and Nuclear Resonances, and a Ph.D. in Nuclear Engineering from the University of Illinois at Urbana-Champaign (UIUC, USA, 1993). At UIUC, he conducted research at the Beckman Institute in neuroscience, robotics, and visuo-motor coordination. He completed a postdoctoral fellowship at the Massachusetts Institute of Technology (MIT, USA, 1998) under the MIT-Harvard research program as a Fulbright Fellow, focusing on Boron Neutron Capture Therapy (BNCT) for brain tumors.

Prof. Campos has extensive experience across multiple areas of Engineering and Nuclear Energy, with current emphasis on materials, radiation science, radioisotopes, radiochemistry, and radiobiology. His research spans radioactive biomaterials, ionizing radiation effects, radiotherapy, brachytherapy, experimental and computational

dosimetry, radiopharmaceutical production, development of physical radiology simulators, radiotherapy software, neutron capture therapy, radiobiology, radiochemistry, oncology, nuclear instrumentation, automation, and the design of particle accelerators, isotope transmuters, radioisotope generators, and nuclear reactor cores.

He holds 14 patents and 1 software registered with INPI, with 2 additional invention patents submitted to WIPO, and has developed 4 computational systems. Prof. Campos has published 148 articles in indexed journals, 340 full conference papers, and 34 abstracts, and has supervised 57 master's theses, 21 doctoral dissertations, 2 co-supervisions, and currently has 2 ongoing doctoral supervisions. He has also mentored 13 postdoctoral researchers, including 1 senior postdoc.

Prof. Campos has collaborated with numerous institutions, including departments of Surgery, Biology, Pharmacy, Veterinary, and Medicine at UFMG, as well as radiotherapy centers at Santa Casa de Misericórdia, Hospital Mário Pena, Hospital São Francisco, and CDTN. He has coordinated over 28 research projects funded by government agencies.

He is a member of the editorial boards of the Journal of Radiology (Austin Publishing Group) and the Revista Mineira de Radioterapia. Over 30 years of teaching, he has delivered undergraduate and graduate courses in applied radiobiology, applied radiochemistry, medical imaging concepts, biomedical radiation applications, particle accelerators applied to biomedicine, and radiation protection, within the Radiation Sciences research line focused on engineering and health applications.

Prof. Campos's dedication to education, research, and innovation has left a lasting legacy, inspiring generations of scientists and advancing nuclear science and technology both in Brazil and internationally.

10.2 PROF. DR. TEÓGENES AUGUSTO DA SILVA



We are honored to recognize the outstanding contributions of Prof. Teógenes Augusto da Silva, a distinguished researcher and educator in nuclear physics and medical physics in Brazil. With a career spanning over four decades, Prof. Silva has made significant advances in radiation metrology, radiological protection, and dosimetry.

He graduated in Physics from the Federal University of Rio de Janeiro (1976) and earned his Master's (1981) and Ph.D. in Nuclear Engineering from the Programa de Engenharia Nuclear – COPPE/UF (1996). During his doctoral studies, he was a CNPq sandwich fellow at the National Institute of Standards and Technology (NIST, Gaithersburg, USA) from December 1992 to June 1995.

Prof. Silva served as a senior researcher at the Comissão Nacional de Energia Nuclear (CNEN) from July 1977 until his retirement in September 2023. He was a faculty member in the graduate program at the Nuclear Technology Development Center (CDTN, Belo Horizonte) from March 2015 to September 2023 and served as

an adjunct professor at the Postgraduate Program in Nuclear Science and Techniques, DEN/UFMG from March 2000 to June 2017.

His research has focused on Medical Physics, emphasizing radiation metrology, radiological protection, and dosimetry, particularly in the calibration of dosimeters, medical and dental radiodiagnostics, quality control, and radiation safety. He contributed as a Mineiro Researcher (2011–2019) and as a CNPq Productivity Fellowship (PQ 2) awardee (2009–2022). He also served as Vice-Coordinator of the CNPq INCT Radiation Metrology in Medicine project (2008–2017).

Prof. Silva held leadership positions at CDTN, including Chief of the Nuclear and Radiological Safety Division (1988–2010) and Chief of the Reactor and Radiation Division (2014–2018). After his retirement, he continues as a collaborator at CDTN (from October 2024) and currently serves as Executive Secretary and Administrative Director of the INCT for Nuclear Instrumentation and Applications in Industry and Health (INAIS, 2023–present). Through his extensive career, Prof. Silva has left a profound impact on radiation safety, dosimetry, and medical physics in Brazil, mentoring generations of scientists and promoting excellence in research and education.

10.3 PROF. DR. HELEN JAMIL KHOURY - IN MEMORIAM



Prof. Helen Khoury was a leading figure in radiological protection and scientific outreach in Brazil, dedicating over four decades to teaching, research, and development in the nuclear sector. She served as a full professor at the Department of Nuclear Energy (DEN) at the Federal University of Pernambuco (UFPE) and held a Ph.D. in Nuclear Physics from the Pontifical Catholic University of São Paulo.

Her work focused on nuclear dosimetry and instrumentation, with applications in medical imaging, radiation therapy, and radiation metrology. She led the Occupational Radiological Protection Network for Latin America and the Caribbean (Reprolam), fostering professional training and knowledge exchange, and founded the Nuclear Science Museum at DEN/UFPE, a pioneering initiative for popularizing nuclear science in Brazil.

Prof. Khoury's legacy lives on in the countless professionals she trained, the initiatives she helped establish, and the lives she inspired. Her dedication and vision continue to guide the field of radiological protection and nuclear science.

CHAPTER 11

Helen Khoury Award for Best Works

In recognition of excellence in research and innovation, the XXV International Symposium on Solid State Dosimetry is proud to present the Helen Khoury Award. This award honors the legacy of Professor Helen Khoury, whose significant contributions to the field of dosimetry continue to inspire generations of scientists worldwide.

The award is granted in two categories — Panel and Communication — highlighting outstanding contributions in each format. In addition to the winners, the Scientific Committee has selected the five best-ranked works in each category to receive an Honorable Mention, acknowledging the high quality and impact of their research.

11.1 BEST WORK IN COMMUNICATION

ID work code: G02-011

Title of Work: **Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project**

Authors: *Natasha Briggs e Silva, Alexander Camargo Firmino da Silva, Aneli Silva, Sandro Passos Leite, Alexandre Bernardino Pinto Jorge, Camila Salata, Maria Helena Marechal, Ademir Xavier da Silva, Andrea Mantuano Coelho da Silva.*

Presenters: *Natasha Briggs e Silva*

Mini-bio: Master in Nuclear Engineering, graduated from the Military Institute of Engineering (IME). Bachelor's degree in Physics with an emphasis in Medical Physics from Souza Marques College. Occupational Safety Technician from the Celso Suckow da Fonseca Federal Center for Technological Education (CEFET/RJ). Has experience as a teacher in the field of Occupational Safety Technology. Currently working as a Researcher in Dosimetry and Radioprotection at the Laboratory of Radiological Sciences (LCR/UERJ), and pursuing a Ph.D. in Nuclear Engineering at COPPE/UFRJ, focusing efforts on research and development in the field of Dosimetry for HDR brachytherapy.

More info:

<https://www.linkedin.com/in/natasha-briggs-2b722bb3/>

<http://lattes.cnpq.br/2115738852789471>

11.1.1 Honorable Mentions in Communication

ID work code: G05-023

Title of Work: **Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry**

Authors: *Laisa de Almeida Neves, Wander Nascimento de Araújo Júnior, Jonathan Oliveira, Leandro Barbosa*

da Silva, Ademir Xavier da Silva.

Presenters: *Laisa de Almeida Neves*

ID work code: G02-057

Title of Work: **Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP**

Authors: *Francisco Harlley Dantas Hauradou Xavier, Paula Selvatice Pereira Teles, Ademir Xavier da Silva, Mirta Bárbara Torres Berdeguez.*

Presenters: *Francisco Harlley Dantas Hauradou Xavier*

ID work code: G07-002

Title of Work: **Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress**

Authors: *Jhon Brandon Felix Ferreira, Luciana Penedo de Melo Teixeira, Maria Eduarda Gomes Moreira da Silva, Heitor gabriel de Souza Saraiva, Mauricio Moacir da Silva Borges, Bianca Torres Ciambarella, Carlos Eduardo Veloso de Almeida, Samara Cristina Ferreira Machado.*

Presenters: *Jhon Brandon Felix Ferreira*

ID work code: G02-019

Title of Work: **A new formulation of Fricke gel dosimeter development using salicylic acid**

Authors: *Iasmin Vieira Nishibayaski, Anna Luiza Fraga Silveira, Luiz Cláudio Meira Belo.*

Presenters: *Iasmin Vieira Nishibayaski*

11.2 BEST WORK IN PANEL

ID work code: G02-066

Title of Work: **Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6**

Authors: *Daniel de Castro Pacheco, Bruno Melo Mendes, Luiz Cláudio Meira Belo.*

Presenters: *Daniel de Castro Pacheco*

Mini-bio: Graduated in Mechanical Engineering (2020), with over 5 years of experience in the development and coordination of industrial engineering projects for refineries, mining, and steel industries, participating in the preparation of technical and managerial documents, as well as in the coordination and management of people and projects. Holds a Master's degree in Science and Radiation Technology (2023), focused on the development of a computational model for shielding studies in in vivo monitoring, using a CAD tool associated with a Monte Carlo code. Currently a Ph.D. candidate in Science and Radiation Technology at the Nuclear Technology Development Center (CDTN) / National Nuclear Energy Commission (CNEN), working on the development of the dosimetry monitoring and data collection system at CDTN, with an approach based on the Internet of Things (IoT), Internet of Everything, and Digital Twins. Also worked in supporting and developing projects in the field of Information and Communication Technology (ICT), focused on smart cities.

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11.2.1 Honorable Mentions in Panel

ID work code: G02-070

Title of Work: **Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms**

Authors: *Greiciane de Jesus Cesário, Crislane de Jesus Cesário, Linda Viola Ehlin Caldas, Edilaine Honório da*



Silva.

Presenters: *Greiciane de Jesus Cesário*

ID work code: G02-035

Title of Work: **Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms**

Authors: *Lucas Beloti Silva, Lucas Fabricio de Araujo, Telma Cristina Ferreira Fonseca.*

Presenters: *Lucas Beloti Silva*

ID work code: G04-038

Title of Work: **Validation of an alanine-based postal dosimetry system for quality control in radiotherapy**

Authors: *Aricia Ravane Pereira da Cruz, Lucas D. Albino, Ernesto Roesler, Claudio C. B. Viegas, Josemary Angélica Corrêa Gonçalves, Carmen Cecília Bueno, Vinicius Saito Monteiro de Barros, Viviane Khoury Asfora.*

Presenters: *Aricia Ravane Pereira da Cruz*

ID work code: G02-081

Title of Work: **SMS Notification System for Potential Radiological Risk Scenarios**

Authors: *Felipe Mamed de Souza, Alessandra Vaz dos Anjos Santos, Paulo César Rocha Silveira, Wallace Vallory Nunes.*

Presenters: *Felipe Mamed de Souza*



CHAPTER 12

Symposium Papers

Caracterización Dosimétrica y Biológica de un Modelo de Cultivo Celular 3D para Evaluar el Efecto Espectador Inducido por Radiofármacos

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Resumen

Se estableció la metodología para preparar un modelo de cocultivo multicelular tridimensional caracterizado en términos de su morfología y sus propiedades biológicas y dosimétricas. El modelo de cocultivo desarrollado permitirá evaluar los efectos radiobiológicos específicos y no específicos producidos por el tratamiento con radiofármacos. La comprensión de los mecanismos moleculares que subyacen a los efectos no dirigidos fomentará la aplicación de estrategias terapéuticas para aumentar la eficacia de la terapia de radionúclidos dirigidos. La preparación de los cocultivos 3D implica la tinción o tratamiento de las células antes de mezclarlas con una solución de alginato (4 %) la cual se vierte (20 μ l) en una solución de cloruro de calcio (4 %) para formar y gelificar las perlas a través de una incubación de 30 minutos. Mediante el uso de este método optimizado y estandarizado es posible obtener perlas con un diámetro medio de 3,71 mm conteniendo 4×10^4 células con un espacio intercelular de 62 μ m entre ellas. Se comprobó que las células embebidas en la perla presentan un porcentaje de viabilidad superior al 95 % y mantienen intacta su capacidad proliferativa. Para la evaluación de la respuesta radiobiológica se estableció un método basado en la tinción diferencial y el análisis visual por microscopia de fluorescencia que permitió discriminar entre células vivas y muertas. Se obtuvieron parámetros morfológicos y biológicos, como el tamaño de las células, la relación entre el núcleo y el citoplasma y el porcentaje de captación de radiofármacos importantes para el cálculo dosimétrico. Mediante el uso de este modelo de cultivo se demostró que los radiofármacos son selectivos y que producen un efecto terapéutico (reducción significativa de la viabilidad celular, 40 % a las 24 h) exclusivamente en las células blanco. En términos de dosimetría, se demostró que la administración de una actividad inicial de entre 0.05 y 0.1 Bq/célula es suficiente para producir respuestas radiobiológicas en células blanco y no blanco. También se confirmó que la contribución de las células vecinas a la dosis de radiación absorbida por el núcleo de las células blanco es significativa. La distribución espacial de las células, así como su heterogeneidad, son factores que afectan significativamente al efecto biológico. La contribución de las células vecinas a la dosis absorbida por el núcleo de una célula blanco, en un modelo de cultivo 3D en el que las células están separadas (62 μ m), es del 16 % de la dosis total absorbida. La dosis de radiación absorbida por el núcleo de las células no blanco es 24 veces menor que la recibida por las células blanco. Se demostró la idoneidad del modelo de cultivo 3D para el análisis de las respuestas radiobiológicas mediante curvas de supervivencia. La validación de los cálculos dosimétricos con MIRDCell mostró que la estimación de la dosis tiene una diferencia significativa del 10 % debido a una subestimación del transporte de partículas, en contraste con lo que ejecuta el código TOPAS-nBio basado en Montecarlo. Los resultados obtenidos y el establecimiento de un método estandarizado para preparar un modelo de cocultivo 3D completamente caracterizado servirán de base para el análisis de la proteómica y la metabolómica del tratamiento radiofarmacéutico en células blanco y no blanco.

Keywords: ¹⁷⁷Lu-iPSMA, modelo de cultivo celular 3D, sobrevida celular, simulación Monte Carlo

1 Introducción

Los modelos de cocultivo celular tridimensionales (3D) representan un avance fundamental en el campo de la investigación biológica *in vitro*, ya que ofrecen complejidades estructurales y funcionales que reproducen con mayor precisión el microambiente *in vivo* en comparación con los cultivos monocapa bidimensionales (2D) convencionales. Estos modelos, que incluyen esferoides, organoides y sistemas basados en andamiajes, permiten la preparación de cultivos multicelulares en condiciones que emulan una situación patofisiológica controlada dado que se pueden usar de proteínas de matriz extracelular (MEC) y formar gradientes de oxígeno y nutrientes. Esta biomimética mejorada es fundamental para estudiar procesos celulares como la proliferación, la diferenciación y la respuesta a las intervenciones terapéuticas.

En el contexto del desarrollo de radiofármacos, los modelos de cocultivo celular en 3D se han convertido en herramientas esenciales para la evaluación preclínica de las terapias con radionúclidos dirigidos (TRT). La capacidad de estos modelos para simular el complejo microambiente tumoral, incluyendo aspectos clave como las interacciones célula-célula y célula-matriz, las zonas hipóxicas y la expresión heterogénea de receptores, ofrece ventajas significativas para evaluar la eficacia, la especificidad y los perfiles de seguridad de los nuevos radiotrazadores. Además, los sistemas 3D facilitan el uso de técnicas avanzadas de imagen y dosimetría, lo que permite una predicción más precisa de la distribución de los radiofármacos y la respuesta terapéutica.

Fundamentalmente, los modelos de cocultivo celular en 3D también proporcionan una plataforma invaluable para investigar los mecanismos moleculares subyacentes a los efectos no específicos (NTE), tales como el efecto espectador que se producen ante la exposición a la radiación ionizante. Al recrear la organización espacial y las capacidades de señalización intercelular inherentes a los sistemas 3D, es posible estudiar la contribución de la dosis de radiación absorbida, así como la influencia de los factores solubles, las vesículas extracelulares y los contactos directos entre células en la propagación de los NTE. Mediante la preparación de un modelo de cocultivo celular 3D caracterizado dosimétricamente y biológicamente, la evaluación de las respuestas biológicas desencadenadas directa o indirectamente por el tratamiento radiofarmacéutico permite comprender a nivel molecular la relevancia de la heterogeneidad celular en la activación de las vías de señalización celular para establecer nuevas estrategias terapéuticas que modulen las respuestas celulares y tisulares en lugares alejados de la región donde se acumula el radionúclido. Otra ventaja del uso de modelos de cocultivo 3D como herramienta en el estudio de la radiobiología de los radiofármacos es el diseño racional de tratamientos combinados para maximizar la actividad tumoricida y minimizar la toxicidad fuera del tejido de interés. En última instancia, se prevé que la integración de modelos de cocultivo celular 3D en la línea de desarrollo de radiofármacos acelerará la traslación de nuevos agentes y mejorará los resultados clínicos de la TRT como resultado de una comprensión más profunda de los efectos radiobiológicos.

La dosimetría de la radiación a escala celular y subcelular es esencial para comprender las respuestas biológicas a la radiación ionizante, especialmente en el contexto de sistemas *in vitro* avanzados, como los modelos de cocultivo 3D. Estos modelos proporcionan una arquitectura fisiológicamente relevante y condiciones microambientales que imitan mejor el tejido *in vivo*, lo que requiere métodos de estimación de dosis altamente precisos y biológicamente realistas. Las simulaciones basadas en Monte Carlo se han convertido en el estándar de referencia para la dosimetría a microescala y, entre ellas, TOPAS-nBio representa una plataforma de vanguardia especialmente adecuada para esta tarea. La heterogeneidad espacial, la arquitectura intercelular y las densidades celulares variables representan un reto para las herramientas dosimétricas convencionales.

Las distribuciones de dosis no son uniformes debido a la complejidad geométrica, la atenuación y los efectos de dispersión, que deben resolverse con precisión para correlacionar los criterios de valoración biológicos con la exposición a la radiación. TOPAS-nBio es un conjunto de herramientas especializadas de Monte Carlo que permite modelar con precisión estructuras biológicas como células, organelos e incluso cadenas de ADN.

TOPAS-nBio permite la construcción explícita de geometrías complejas de cocultivos celulares en 3D, incluyendo células individuales, núcleos, citoplasma y agregados multicelulares. Esto permite estimar la energía depositada a diversas escalas biológicas y elucidar los mecanismos físicos asociados. A diferencia de otros algoritmos, TOPAS-nBio emplea modelos físicos a escala nanométrica, fundamentales para simular con precisión los eventos de ionización y los patrones de deposición de energía dentro de los compartimentos subcelulares. Los modelos de cocultivo 3D suelen presentar gradientes de oxigenación, densidad y composición que pueden tenerse en cuenta en TOPAS-nBio; la incorporación de la heterogeneidad celular y las propiedades fisicoquímicas de los materiales, así como los datos biológicos reales (morfología, tasa de proliferación y muerte celular) permiten simular entornos más realistas.

Por otra parte, los avances tecnológicos en radiofármacos aumentan nuestra comprensión del funcionamiento de los sistemas en tiempo real con gran precisión permitiendo el diagnóstico precoz de enfermedades mediante la detección de marcadores específicos [5]. En los últimos años, los avances en el desarrollo de radiofármacos han promovido su uso terapéutico dando lugar a un crecimiento significativo del mercado mundial de radiofármacos. Con ello también los requisitos para la aprobación, comercialización y aplicación de nuevos radiofármacos se han reforzado. Es indispensable una selección cuidadosa de vectores que garanticen el reconocimiento del objetivo, la caracterización radioquímica, el control de calidad y los estudios dosimétricos para evitar una exposición indeseable. También es importante aplicar directrices y normas que establezcan las condiciones de uso y manipulación.

Por ejemplo, la reciente normativa internacional establece que los estudios citotóxicos y dosimétricos son una parte esencial del proceso de aprobación de los radiofármacos [2]. Adicionalmente, se ha considerado que algunos radiofármacos no alcanzan la eficacia esperada debido a su biodistribución evidenciando que poco se conoce sobre el destino final y la dinámica de las moléculas radiomarcadas y su relación con la forma en que se administra la dosis de radiación y la heterogeneidad celular. La información recopilada hasta ahora muestra que la respuesta terapéutica depende tanto de la dosis absorbida como de la respuesta radiobiológica general [6]. Esta respuesta está determinada por fenómenos inducidos por la radiación en sí misma, como el efecto espectador, la inestabilidad genómica, la respuesta adaptativa y la hiperradiosensibilidad a dosis bajas de radiación. Investigaciones recientes sobre este tema muestran que la respuesta de las células blanco puede afectar a las células no blanco, y viceversa. Este fenómeno, conocido como efecto espectador, puede producirse tras el tratamiento con radiofármacos y produce respuestas contradictorias que pueden ser protectoras o sensibilizantes, dependiendo de la comunicación entre células (por ejemplo, uniones comunicantes, sistema inmunitario y citoquinas). Otra consideración importante en la terapia radiofarmacéutica es la radioresistencia inducida, que puede ser el resultado de la aplicación de dosis bajas (del orden de MBq) para el diagnóstico por imagen y la dosimetría, seguida de la administración de dosis más altas (del orden de GBq) para el tratamiento. Esto genera en última instancia una respuesta adaptativa al tratamiento [1,7]. Como resultado de la terapia radiofarmacéutica, las células no blanco y sus descendientes se vuelven sensibles al daño genético espontáneo, un fenómeno conocido como inestabilidad genómica que se asocia con un mayor riesgo de que los pacientes desarrollen tumores secundarios [4]. Sin duda, comprender a fondo estos conceptos radiobiológicos y sus mecanismos de acción garantizará el uso seguro y la explotación eficiente de los radiofármacos de blancos moleculares. La preparación y caracterización estandarizada de un modelo de cocultivo 3D es una herramienta indispensable para comprender los mecanismos moleculares que subyacen a la terapia con radionúclidos dirigidos que además permitirá conocer a nivel molecular el papel de la intercomunicación celular, la heterogeneidad tumoral y los efectos no dirigidos en la eficacia del tratamiento radiofarmacéutico.

2 Materiales and Métodos

Preparación del modelo de cocultivo 3D: Tras un análisis factorial, se estableció un procedimiento optimizado y estandarizado para la preparación de lotes reproducibles de perlas de alginato con células. En resumen, el procedimiento consiste en añadir gota a gota 20 μ L de una mezcla de células en alginato a una solución de CaCl_2 con un tiempo de gelificación de 30 minutos.

Caracterización morfológica del modelo: Se tomaron fotografías de diferentes lotes de perlas y se analizaron con el software ImageJ 1.51. Se midieron directamente los ejes de la esfera (mayor y menor), y se determinó la esfericidad de la perla a partir de la relación entre ambas dimensiones. El volumen se estimó con la fórmula $V = (4/3) * \pi * r^3$ y la firmeza del modelo denominada estabilidad mecánica, se midió después de contar el número de fragmentos resultantes tras la exposición de la perla a fuerzas externas controladas entre 0-13 x 10³ rpm).

Caracterización biológica del modelo: La viabilidad celular dentro del modelo de cocultivo 3D se evaluó mediante dos métodos de microscopía (prueba de exclusión con azul de tripano e incorporación de yoduro de propidio). Esta evaluación se realizó en la esfera completa o después de desintegrarla con una solución de citrato de sodio. El mismo procedimiento se repitió durante al menos cuatro días para obtener las curvas de proliferación.

Preparación y caracterización química del ¹⁷⁷Lu-iPSMA: Los radiofármacos se prepararon a partir de formulaciones liofilizadas en solución tampón de acetato de sodio marcadas con 37 GBq de cloruro de lutecio estéril y sin portador (¹⁷⁷LuCl₂; Endolucin Beta, ITG, Alemania), se agitaron vigorosamente y se incubaron a 95 °C durante 30 minutos. El compuesto obtenido se caracterizó químicamente mediante RP-HPLC.

Ensayo de captación celular: Se trataron 5x10⁵ células con ¹⁷⁷Lu-iPSMA (37 KBq) y se incubaron con ligera agitación a temperatura ambiente durante al menos 1 h antes de eliminar el exceso de radiofármaco. La actividad del sedimento celular se midió con un contador gamma (fracción absorbida) antes de añadir 500 µL de una solución de glicina 50 mM en HCl (pH 2,8) para eliminar la fracción del radiofármaco unida a los receptores de la membrana celular. A continuación, tras volver a lavar, se midió la actividad del sedimento celular resultante (fracción internalizada).

Estimación de la dosis de radiación absorbida: Utilizando una simulación de Monte Carlo con el código TOPAS-nBio, se construyó una geometría que consistía en una matriz esférica donde está contenida una mezcla de dos tipos celulares en proporciones iguales, distribuidas aleatoriamente y separadas por 62 µm. El grupo consta de 19 células (esferas concéntricas que representan el núcleo (r = 5 µm) y el citoplasma (r = 8 µm) con una capa adicional rodeando el citoplasma de 5 nm de grosor que representa la membrana celular. Para la fuente de radiación se consideró el espectro de emisión completo del ¹⁷⁷Lu. La ubicación y la distribución se determinaron de acuerdo con datos biológicos. La dosis de radiación absorbida se estimó para toda la célula y para el núcleo de ambos tipos de células. Se realizaron cien mil cálculos de dosis con cuatro repeticiones utilizando diferentes semillas iniciales. El valor final de la dosis absorbida se corrigió con la actividad administrada a cada célula.

Construcción de la curva de supervivencia: Las células LNCaP tratadas con 1.85 KBq y las células PC3 teñidas con Green Cell Tracker se mezclaron con alginato, se gelificaron y se mantuvieron en incubación durante un máximo de 96 h. Se tomaron imágenes cinéticas de las perlas con un microscopio invertido y se midió el número de células positivas para cada fluoróforo. Utilizando el número de células muertas y vivas de cada población en cualquier momento, se construyeron y analizaron curvas de viabilidad para cada línea celular y se ajustaron a una ecuación no lineal según el modelo lineal-cuadrático en el software Graph Pad Prisma 10.

3 Resultados

La caracterización física del modelo de cultivo 3D incluyó la medición directa de las dimensiones físicas de las perlas a partir del procesamiento de imágenes micrográficas (Figura 1) con el software ImageJ. La medición de al menos cuatro lotes de perlas (n = 400) mostró que el modelo de cultivo 3D presenta una forma constante con un diámetro medio de 3.71 ± 0.02 mm.

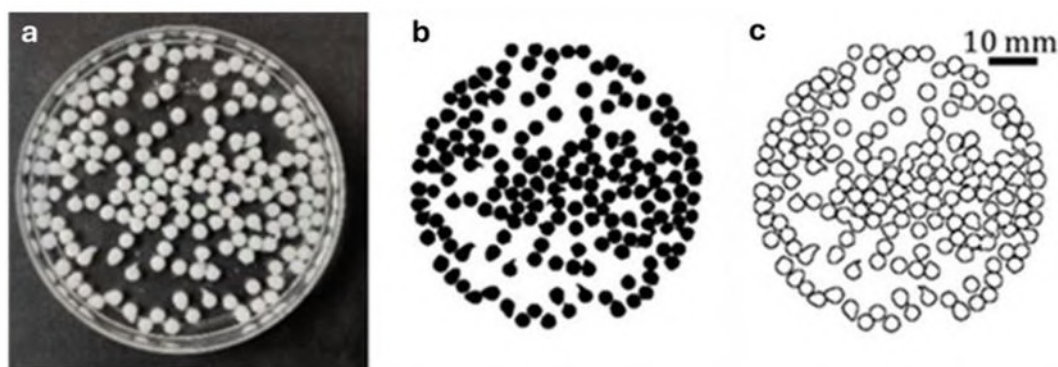


Figura 1. a) Imágenes representativas de un lote de perlas de alginato (b-c) y del procesamiento de imágenes utilizado para medir las dimensiones físicas del modelo de cultivo 3D con el software ImageJ.

Los datos de la longitud del eje mayor y del eje menor de cada perla se utilizaron para determinar el volumen, la esfericidad (relación entre la longitud del eje menor y el eje mayor de una perla) y el espacio entre los núcleos de dos células vecinas embebidas en el alginato (espacio intercelular). La evaluación cinética mostró que el modelo de cultivo 3D se hidrata en las primeras horas, aumentando su diámetro medio de 3.17 a 3.71 mm tras un periodo de incubación de 24 h. El diámetro de las perlas se mantuvo constante a partir de las 24 h y hasta las 96 h. El modelo de cultivo 3D mantuvo una estructura y forma constante (Figura 2a) a lo largo del tiempo, lo que significa que el modelo de cultivo 3D permanece morfológicamente homogéneo.

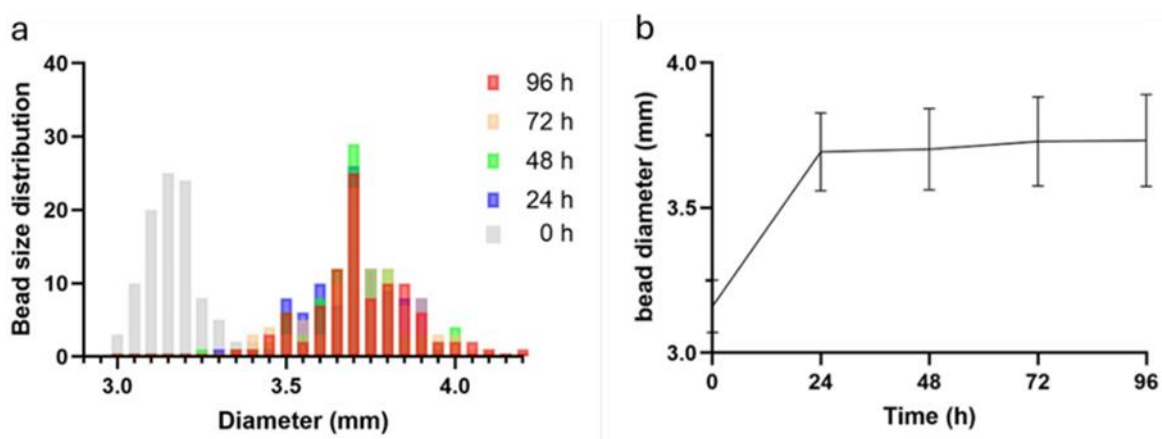


Figura 2. Representación gráfica de a) la distribución del tamaño (diámetro) de las perlas del modelo de cultivo 3D a lo largo del tiempo y b) los valores medios del diámetro de las perlas de alginato mantenidas en condiciones de cultivo estándar durante 0 a 96 h, que muestran el comportamiento cinético de las dimensiones de las perlas de alginato.

Como mostraron los resultados, tras una hidratación inicial de las perlas en las primeras 24 h, el tamaño del modelo se mantiene constante (figura 2b) y las características morfométricas del modelo en promedio (resumidas en la Tabla 1) muestran que se mantienen constantes a partir de las 24 h de cultivo.

Se ha descrito anteriormente [3] que la firmeza y estabilidad del hidrogel está determinada por las interacciones químicas que se establecen entre la matriz polimérica de alginato y los iones de calcio. Como se muestra en la figura 3a, la resistencia mecánica del modelo de cultivo disminuyó de forma constante con el tiempo debido a la presencia de iones fosfato en el medio de cultivo, lo que implica una competencia iónica con los iones de calcio que mantenían la estructura polimérica del alginato; la presencia de otras especies iónicas en el medio de cultivo redujo la estabilidad mecánica del alginato polimérico.

La estabilidad mecánica del modelo de cultivo 3D se redujo a la mitad como resultado de la exposición a una fuerza mecánica de 5735 g en el lote de perlas que permaneció en incubación 24 h. Mientras que para los lotes de perlas que se incubaron durante 48 y 72 h el mismo resultado se produjo con la aplicación de una fuerza de solo 1770 g. Por otra parte, se observó una pérdida significativa de la firmeza de la matriz de alginato en el grupo que se incubó durante 96 h, ya que la resistencia mecánica de estas perlas se redujo a 0.5 tras la exposición a una fuerza relativa de 71 g.

La evaluación de un lote de perlas recién preparadas (0 h de incubación) mostró que, tras ser expuestas a todas las fuerzas evaluadas (Figura 3a, ●), permanecían intactas. Como se mencionó antes, la estructura polimérica de las perlas se reduce por el intercambio de las especies iónicas presentes en los medios de cultivo y, por esta razón, la resistencia mecánica de las perlas disminuye con el tiempo de incubación, como se muestra en la Figura 3a (origen de cada línea, 0 g). Cabe mencionar que, en condiciones de cultivo estándar, la exposición a movimientos bruscos o fuerzas externas es inusual, por lo que no se espera que en condiciones estándar de cultivo y manipulación del modelo se presente la pérdida de integridad de las perlas. Como se muestra en la figura 3b, las condiciones de incubación afectan a la estabilidad, pero la esfericidad permanece prácticamente inalterada y por encima de 0.8.

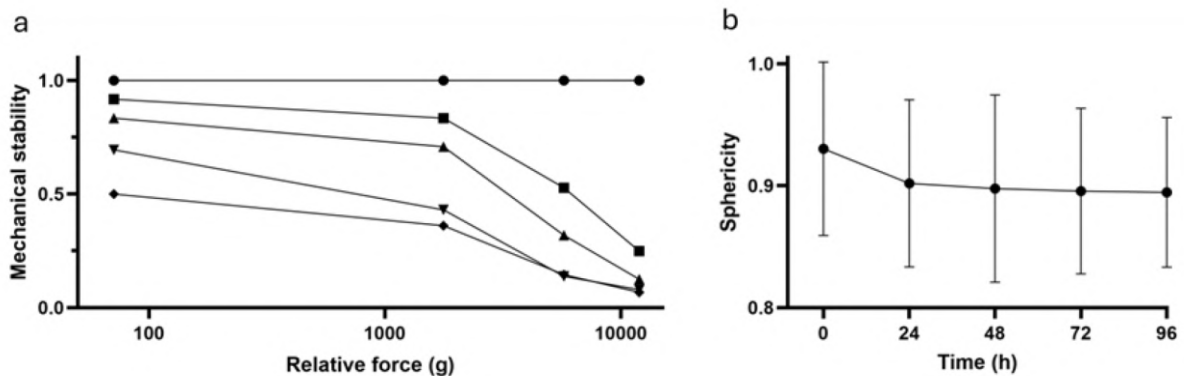


Figura 3. a) Estabilidad mecánica del modelo de cultivo 3D que se ha incubado durante 0 h (●); 24 h (■); 48 h (▲); 72 h (▼) o 96 h (◆) tras aplicar diferentes fuerzas centrífugas relativas de 71 a 11 965 g. b) Esfericidad media del modelo de cultivo 3D tras periodos de incubación de 24 h a 96 h.

La caracterización biológica del modelo de cocultivo 3D incluyó la evaluación de la viabilidad de las células embebidas a lo largo del tiempo. Como se muestra en la figura 4a, el porcentaje de células viables evaluadas por ambas técnicas mostró que, en las condiciones de cultivo del modelo, la viabilidad de las células embebidas se mantiene por encima del 95 % durante al menos el periodo analizado (96 h). Esto demuestra la biocompatibilidad del modelo de cultivo 3D y confirma que este modelo de cultivo es útil para la evaluación cinética de la citotoxicidad de los radiofármacos. La eficacia del modelo de cocultivo celular 3D se confirmó por su capacidad para permitir la proliferación de células sanas (Figura 4b), ya que se observó que las células dentro de la perla tenían un tiempo de duplicación de 36 h, lo que es coherente con la tasa de proliferación de las líneas celulares utilizadas para preparar el modelo de cultivo 3D (PC3 con un tiempo de duplicación de 24-32 h y LNCaP con un tiempo de duplicación de 35-49 h y un tiempo de duplicación medio reportado para ambas líneas celulares de 38 h).

Este valor se estimó ajustando los datos experimentales de la curva de proliferación a un modelo exponencial con un R^2 de 0,9996, lo que dio como resultado una tasa de proliferación $k = 0.03041$.

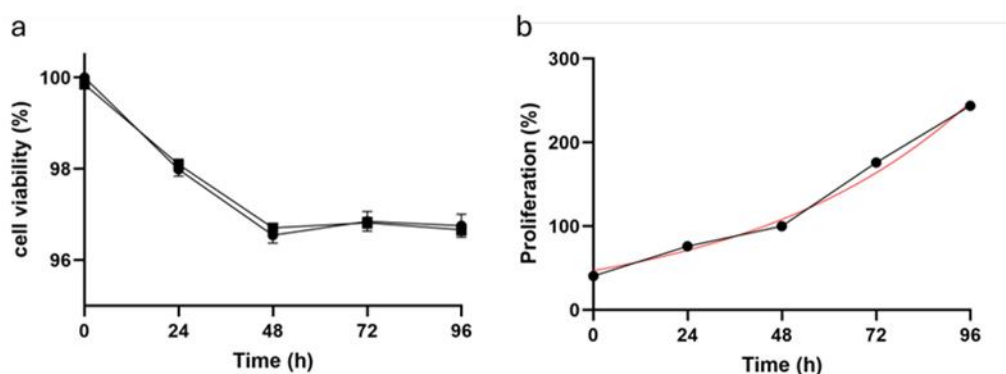


Figura 4. El modelo de cocultivo 3D es adecuado para el mantenimiento de líneas celulares de cáncer de próstata viables y proliferativas. a) Viabilidad de las células PC3 y LNCaP embebidas en perlas de alginato (PC3:LNCaP, 50:50) determinada mediante dos métodos experimentales microscópicos (ensayo de exclusión de toluidina azul (●) y captación de yoduro de propidio por las células muertas (■)) y b) capacidad proliferativa promedio de las líneas celulares de cáncer (PC3:LNCaP, 50:50) población mixta (●) y el ajuste de los datos de proliferación a una ecuación de crecimiento exponencial (línea roja) con un $R^2=0.9889$.

Las células de cáncer de próstata (PC3 y LNCaP) embebidas en el modelo de cocultivo 3D adquirieron una morfología redondeada. Las dimensiones promedio de la población celular mixta embebida en el modelo de cocultivo 3D fueron: diámetro celular promedio de $10 \pm 1.29 \mu\text{m}$ y un diámetro nuclear celular medio de $8 \pm 1.97 \mu\text{m}$.

Dado que el objetivo principal de la preparación y caracterización estandarizada de un modelo de cocultivo 3D es disponer de una herramienta fiable para evaluar las respuestas radiobiológicas resultantes del tratamiento de una población heterogénea con radiofármacos. Entendiendo que una población heterogénea consiste en un grupo de células con diferente capacidad para captar los radiofármacos dirigidos. El ensayo de captación celular in vitro mostró que las células LNCaP (línea celular con sobreexpresión de PSMA) presentaban una captación celular de radiofármacos un $7.02 \pm 0.86 \%$ mayor que las células PC3 (línea celular con baja expresión de PSMA), con una captación celular del $0.99 \pm 0.01 \%$, lo que demuestra que las células de cáncer de próstata utilizadas para la preparación de este modelo de cocultivo 3D (LNCaP como células que absorben radiofármacos frente a PC3 como células que no los absorben) representan una población celular heterogénea similar a la observada en los tumores y, por lo tanto, son adecuadas para la evaluación de los efectos radiobiológicos de los radiofármacos basados en inhibidores de PSMA.

Los datos de la caracterización del modelo de cocultivo 3D, tales como la morfometría de las perlas de alginato, los parámetros físicos y biológicos de las células PC3 y LNCaP embebidas (diámetro celular y nuclear, distribución celular [espacio intercelular], número de células y porcentaje de captación de radiofármacos), que se midieron experimentalmente (Tabla 1) se utilizaron para la caracterización dosimétrica del modelo de cultivo. La representación geométrica del modelo de cocultivo 3D (Figura 6a), se construyó a partir de los parámetros morfométricos y biológicos resumidos en la Tabla 1. La geometría construida, permitió calcular la dosis de radiación absorbida por el cuerpo celular y el núcleo celular de la población de células blanco (captantes) y del grupo celular no blanco, teniendo en cuenta las interacciones de las partículas, las desintegraciones, el transporte y el depósito de energía. Las células se simularon utilizando esferas concéntricas que representaban el núcleo ($r = 5 \mu\text{m}$), el citoplasma ($r = 8 \mu\text{m}$) y, como envoltura del citoplasma con un grosor de 5 nm , la membrana celular. El grupo contenía una disposición esférica de células radiomarcadas (que representaban las células LNCaP captantes) y células no marcadas (correspondientes a las células PC3 no captantes) en una proporción de 50:50 con una separación intercelular de $62 \mu\text{m}$.

Table 1. Morphometric characteristics of 3D culture media based on alginate

Morphometric Characteristic	Time (h)					Average
	0	24	48	72	96	
Radius (mm)	1.58	1.85	1.85	1.87	1.87	1.86 ± 0.02
Diameter (mm)	3.17	3.69	3.70	3.73	3.73	3.71 ± 0.02
Volume (mm ³)	16.60	26.38	26.57	27.15	27.22	26.83 ± 0.42
Sphericity (%)	93.03	90.19	89.76	89.56	89.46	89.74± 0.32
Intercellular space (μm)	73.97	66.72	62.10	59.98	59.21	64.59± 3.37
Cell viability (%)	99.93	98.05	96.84	96.73	96.61	97.63 ± 1.41
Cell number (1x10 ³)	41.23	88.80	110.90	125.85	131.10	99.54 ± 36.61
Doubling time (h)	60.41	67.72	106.01	143.39	143.41	104.27 ± 39.62
LNCaP uptake (%)	5.83	7.87	6.52	7.34	7.52	7.02 ± 0.83
PC3 uptake (%)	0.98	1.01	0.98	0.99	1.00	0.99 ± 0.01
Cell diameter (μm)	11.81	10.24	9.65	8.23	10.21	10.03 ± 1.29
Nucleus diameter (μm)	6.02	10.24	9.65	8.23	10.21	8.00 ± 1.97

El clúster simulado conformado por diecinueve células localizadas aleatoriamente a lo largo de tres planos paralelos longitudinales (cinco células en el plano superior, nueve células en el plano central y las cinco células restantes en el plano inferior) se muestra en la figura 5a.

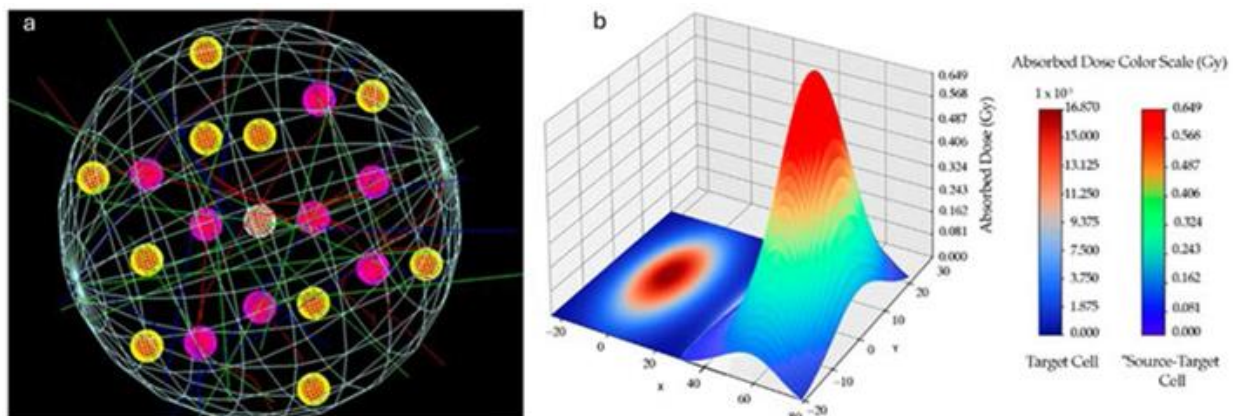


Figura 5. Caracterización dosimétrica del modelo de cocultivo 3D con simulación Monte Carlo mediante el código Topas-nBio a) representación geométrica del modelo de cocultivo 3D y b) visualización tridimensional de la dosis de radiación absorbida en el cuerpo y el núcleo de una célula no blanco (izquierda) y una célula blanco o captante (derecha).

El análisis de la distribución de la dosis dio como resultado una visualización tridimensional de la distribución espacial de la dosis absorbida a las 24 h (figura 5b), en la que la dosis de radiación absorbida se concentra en el centro de la célula (el núcleo como estructura crítica) con un pico máximo que se atenúa hacia el exterior de forma similar en ambas líneas celulares, pero con diferente intensidad. En la figura 6a, la célula en el centro del grupo representa una célula LNCaP que deposita la energía que ella misma desprende y también la contribución de todas las células vecinas. En la simulación ejemplificada esa célula en el centro es sobre la que se realiza el cálculo de dosis absorbida como resultado de su propia contribución y de la de las células vecinas (no captantes), donde todas esas interacciones contribuyen a la dosis total de radiación absorbida en su citoplasma y su núcleo.

Tras 1×10^7 historias con cuatro repeticiones en las que se varió la posición de las semillas iniciales, los valores totales de dosis de radiación absorbida obtenidos para cada célula de una población en cada plano no mostraron diferencias significativas. Esto indica que la dosis final de radiación absorbida, teniendo en cuenta la contribución de la dosis propia y la dosis cruzada, es la misma para todas las células LNCaP que absorben radiofármacos, independientemente de su ubicación. Lo mismo ocurre con la otra población celular presente en los modelos de cocultivo 3D (células PC3 que no absorben). Los valores promediados de dosis absorbida, corregidos por la actividad inicial administrada, se muestran en la tabla 2.

Table 2. Mean radiation absorbed dose values calculated by Monte Carlo simulation for each of the cell lines present in the 3D coculture model

¹⁷⁷ Lu-iPSMA-uptaking cell (LNCaP)			¹⁷⁷ Lu-iPSMA –non uptaking cell (PC3)	
Absorbed Radiation Dose (Gy)				
Time (h)	Cell	Nucleus	Cell	Nucleus
1	0.0288 ± 0.0002	0.0187 ± 0.0022	0.0006 ± 0.0001	0.0008 ± 0.0004
24	0.6578 ± 0.0037	0.4273 ± 0.0509	0.0128 ± 0.0017	0.0185 ± 0.0090
96	2.2679 ± 0.0128	1.4734 ± 0.1754	0.0442 ± 0.0060	0.0636 ± 0.310

El modelo de cocultivo 3D demostró ser eficaz para evaluar la capacidad del ¹⁷⁷Lu-iPSMA para dañar células que no lo absorben mediante un efecto no dirigido, como el efecto espectador (figura 6b). Los tipos de células adquirieron una forma redondeada en el modelo de cocultivo 3D y no fue posible discriminarlas por sus diferencias morfológicas. Por este motivo, desarrollamos una estrategia basada en la tinción diferencial de uno de los tipos de células con un colorante de seguimiento vital (Cell TrackerGreen CMFDA, ThermoFisher Scientific), lo que permitió el reconocimiento y la cuantificación de ese tipo de célula específico y, por sustracción, fue posible deducir el número de células vivas y muertas del otro subtipo celular siguiendo el procedimiento que se muestra en la figura 6a.

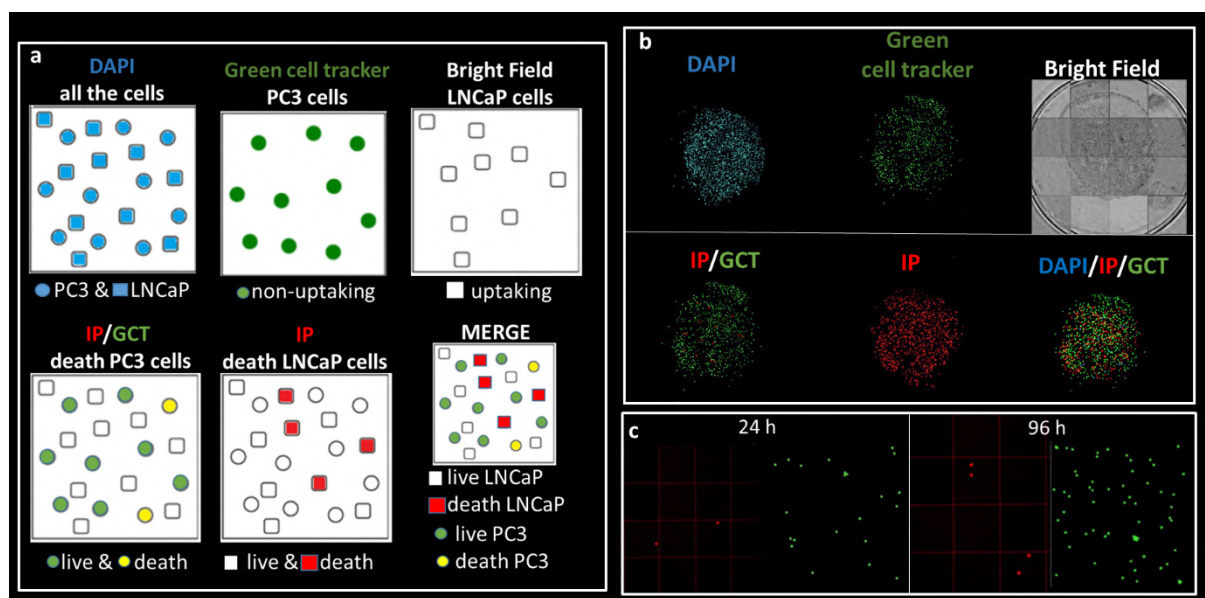


Figura 6. a) Procedimiento para discriminar diferentes tipos de células en un cocultivo celular 3D heterogéneo con el fin de medir las células viables y muertas, b) imagen compuesta de micrografías de fluorescencia las células de cáncer de próstata presentes en modelo de cocultivo 3D mostrando el núcleo de todas las células en azul, las células no captantes PC3 en verde y las células muertas en rojo y c) campos microscópicos representativos del recuento de células vivas y muertas en una cámara Neubauer tras 24 o 96 horas después del tratamiento.

El procedimiento de tinción diferencial utilizado para discriminar entre los dos tipos de células embebidas en el modelo de cocultivo 3D implica la tinción previa de uno de los tipos celulares con un colorante vital de seguimiento, mientras que el otro tipo celular (radiomarcado) se trató con el radiofármaco correspondiente. A continuación, ambos tipos celulares se embebieron dentro de una perla de alginato y se incubaron durante un periodo de tiempo determinado con el fin de administrar una dosis de radiación absorbida al núcleo de ambos tipos celulares (células blanco y no blanco). Una vez finalizado el tiempo de incubación, las perlas se tiñeron con DAPI para visualizar el núcleo de todas las células y con yoduro de propidio para identificar a las células muertas (Figura 6b) mediante su observación con un microscopio invertido de epifluorescencia para obtener micrografías del modelo de cocultivo 3D. La cuantificación de las células positivas para cada fluoróforo y la posterior sustracción con las células que se observan en color amarillo como resultado de la combinación de imágenes (colorante verde y yoduro de propidio) para conocer el número de células muertas y vivas de cada población. Los cálculos de dosis serán relacionados con el efecto radiobiológico (relación dosis-respuesta) para construir una curva de supervivencia para las células LNCaP y PC3 (Figura 7). La curva de supervivencia (Figura 7) para cada uno de los tipos celulares presentes en el modelo de cocultivo 3D se construyó trazando los datos de viabilidad, obtenidos mediante la determinación del número de células muertas y vivas para cada población celular con el método descrito gráficamente en la Figura 6a, frente a la dosis de radiación absorbida estimada mediante simulación de Monte Carlo para los diferentes periodos de incubación (Figura 7).

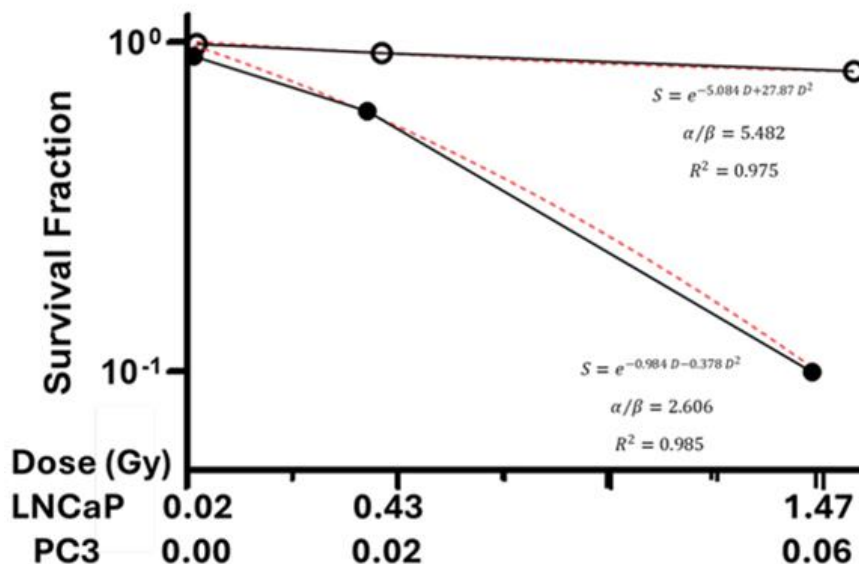


Figura 7. Curva de supervivencia de las células blanco (●) y no blanco (○) presentes en el modelo de cocultivo 3D. Los datos de supervivencia se ajustaron ($R^2 > 0.98$) a la ecuación del modelo lineal-cuadrático (línea punteada roja) y se estimaron los parámetros radiobiológicos.

Estos gráficos ilustran claramente el efecto letal del radiofármaco dirigido ^{177}Lu -iPSMA sobre las células captantes (LNCaP), a diferencia de lo que ocurre con las que no captantes (tejido no dirigido), donde la supervivencia celular prácticamente no varía. Como mostraron los cálculos dosimétricos, la dosis total absorbida por el núcleo de ambos tipos de células varía significativamente, pasando de 0.02 a 1.47 Gy para las células LNCaP y de 0.00 a 0.06 para las células PC3. La dosis de radiación absorbida depositada en la célula blanco fue 24 veces mayor que la recibida por el núcleo de una célula no blanco; sin embargo, la ligera reducción en la supervivencia celular probablemente se deba a efectos no específicos, como el efecto espectador, debido a la secreción de algunos factores solubles que podrían estar afectando la habilidad de las células PC3 para resistir la baja dosis de radiación que reciben.

4 Discusión

Se establecieron las condiciones ideales para preparar el modelo de cocultivo 3D multicelular, constituido por una mezcla de dos tipos celulares 4×10^4 células embebidos en un polímero de alginato (4%). El cual se gelificó en forma de una esfera al entrar en contacto con una solución de cloruro de sodio (4%). Se estableció para formar perlas uniformes y resistentes el volumen óptimo de la gota era de 20 μL y el tiempo de gelificación idóneo de 30 minutos. En cuanto al tratamiento con el radiofármaco dirigido, se estableció que una actividad inicial por célula entre 0.05 - 0.1 Bq es adecuada para evaluar los efectos no dirigidos de la radiación durante un periodo de evaluación de hasta 96 horas.

Se demostró que el modelo de cocultivo 3D alcanza su estabilidad hídrica tras 24 h de incubación y que a partir de ese momento las propiedades reológicas del modelo (diámetro = 3.71 ± 0.01 , volumen = $26.83 \pm 0.42 \text{ mm}^3$) permanecen constantes. La estabilidad mecánica de la perla disminuyó considerablemente durante las primeras 24 h de incubación debido al intercambio iónico y por ello, la estabilidad mecánica del modelo se reduce a medida que aumenta el periodo de incubación; cabe mencionar, que la pérdida de estabilidad mecánica no es una variable crítica del modelo de cocultivo 3D ya que en condiciones de cultivo estándar no se espera que se produzca una exposición a fuerzas externas repentinas de la magnitud de las que fueron analizadas. Como mostraron los resultados, la estabilidad mecánica del modelo se mantiene por encima de 0.86 después de recibir una fuerza centrífuga relativa de 71 g (1000 rpm). Otra característica relevante del modelo de cocultivo 3D es que ofrece la posibilidad de desintegrarse por completo con solo incubarlo en una solución desintegradora de citrato de sodio (200 mM) durante cinco minutos sin afectar la viabilidad celular y con una eficiencia de encapsulado cercana al 99%. La opción de disolver por completo el modelo de cocultivo 3D brinda la posibilidad de realizar análisis más detallados de los componentes solubles y celulares presentes, lo que permitirá estudiar la naturaleza de las sustancias secretadas al microambiente.

La biocompatibilidad del modelo quedó demostrada, ya que las células embebidas mantuvieron una buena viabilidad (superior al 95 %) y su capacidad de proliferar a un ritmo adecuado, lo que indica que las condiciones de cultivo del modelo no afectan el comportamiento biológico de las células. El hecho de que las funciones vitales de las células permanezcan intactas hace que el modelo de cocultivo sea adecuado para la evaluar a nivel celular los efectos citotóxicos del tratamiento. Por otro lado, la evaluación de la eficiencia de encapsulación mostró que, mediante la incubación del modelo en una solución de desintegración, es posible recuperar el $99.88 \pm 0.73 \%$ de las células embebidas en el modelo y el método presentó una alta reproducibilidad. El modelo de cocultivo 3D preparado y caracterizado como parte de este estudio se basa en un andamio de alginato sin matriz extracelular adicional a la que las células pueden secretar por sí mismas durante el análisis. El modelo de cocultivo 3D desarrollado mantiene con éxito la organización celular y un buen intercambio de metabolitos y nutrientes, ya que el modelo mantuvo las células vivas y con su capacidad para realizar sus procesos básicos intacta. El diseño del cocultivo 3D evaluado es sencillo, ya que consiste únicamente en un andamio de alginato sin incluir ninguna proteína extracelular que promueva la adhesión celular, ya que uno de los objetivos de este sistema de cultivo es la evaluación de los efectos radiológicos y bioquímicos del tratamiento radiofarmacéutico dirigido y es deseable la ausencia de información bioquímica adicional. Además, el modelo se mantiene en una microescala, ya que una separación entre las células de alrededor de 100 μm favorece la interacción intercelular además de conferirle mayor estabilidad mecánica. A pesar de no ser un objetivo en este momento, el modelo de cocultivo 3D caracterizado es flexible y permitirá en futuros abordajes experimentales la introducción de proteínas solubles (fibronectina, colágeno, péptidos RGD) para aumentar la adhesión celular y promover la activación de cascadas de señalización mecanosensibles.

Como se mencionó anteriormente, el método propuesto para la discriminación de subtipos celulares, basado en la tinción diferenciada de las células con colorantes fluorescentes (Figura 6a), permite determinar de forma fácil y reproducible el efecto citotóxico de los radiofármacos en dos poblaciones celulares (blanco y no blanco) a lo largo del tiempo y a distancia. Como perspectiva de este estudio, nos interesa la evaluación de la combinación de radiofármacos (dirigidos a ambos tipos de células), la introducción de células del sistema inmunitario (macrófagos o linfocitos) para evaluar la activación de las células que son parte de la respuesta inmunitaria, las vías celulares de excreción y eliminación de los

radiofármacos, y también la participación de células troncales y sus procesos de diferenciación radioinducidos. Toda la información que se puede generar a partir del uso de este modelo de cocultivo 3D permitirá comprender el papel de la comunicación intercelular en la radioresistencia, la diferenciación y la recurrencia. La caracterización morfológica y biológica fue indispensable para el cálculo de dosis y fundamental para determinar la actividad inicial óptima para tratar las células, ya que necesitamos que la dosis de radiación absorbida total administrada en el núcleo de las células blanco sea suficiente para modificar las respuestas celulares (metabólica y proteómica) sin inducir la muerte celular y la pérdida de homeostasis. La dosis de radiación absorbida administrada puede inducir numerosas respuestas celulares (hormesis, hipersensibilidad a dosis bajas, expresión génica, daño en el ADN y activación de los mecanismos de reparación) sin provocar la muerte celular completa permitiendo así estudiar la regulación de esas respuestas radiobiológicas a nivel molecular y comprenderlas con el fin de ejercer una modulación externa que nos permita mejorar la eficacia general de la radioterapia dirigida con radionúclidos.

El tratamiento de las células blanco (LNCaP) con el radiofármaco (^{177}Lu -iPSMA) durante 24 horas produjo una reducción significativa de la viabilidad celular (62.1 %) sin alterar la viabilidad de las células no blanco (PC3) situadas a 62 μm de distancia, lo que demuestra la seguridad radiobiológica de los radiofármacos y la utilidad del modelo de cocultivo para analizar los efectos a distancia debidos al efecto espectador, entre otros. El formalismo MIRD es una herramienta sencilla y accesible para estimar la dosis absorbida. Sin embargo, como demostraron los cálculos de dosis, los programas de código Monte Carlo como TOPAS-nBio permiten una dosimetría más precisa. En este estudio, la diferencia entre los dos cálculos de dosis fue del 10 % (datos no mostrados). Los cálculos de dosis demostraron la importancia de la distribución espacial de las células y la heterogeneidad celular en la dosis de radiación absorbida en el núcleo de la célula blanco.

5 Conclusión

El modelo de cocultivo 3D preparado y caracterizado es adecuado para la evaluación radiobiológica de radiofármacos dirigidos en condiciones tumorales simuladas, teniendo en cuenta la heterogeneidad y la comunicación entre células, así como los efectos no dirigidos y a distancia.

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SNR and CTDIvol study as a function of the Noise Index in Computed Tomography exams

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Abstract

Optimizing image quality in chest computed tomography (CT) is closely related to balancing radiation dose and technical parameters. The Noise Index (NI) is a key quantitative indicator for assessing system noise, and its variation directly affects the signal-to-noise ratio (SNR) and contrast resolution—both essential for detecting subtle pulmonary lesions. The Smart mA tool, which automatically adjusts the tube current (mA) based on patient attenuation, has shown promise in minimizing NI variability while maintaining diagnostic image quality with reduced radiation dose. This study aims to evaluate the impact of NI and the use of Smart mA on image quality and radiation dose in chest CT scans. Scans were performed on a thoracic phantom homogeneously filled with water, using NI values ranging from 0.5 to 70 (in 5-unit increments), with all other acquisition parameters kept constant. The reference protocol was a routine adult chest CT with 120 kVp, 0.9884 pitch, 40 mm detector coverage, 0.7 s rotation time, 0.625 mm helical thickness, and automatic current modulation between 100–300 mA. Examinations were conducted both with and without Smart mA, using a 64-slice GE Lightspeed VCT scanner. Five regions of interest (ROIs) were analyzed in each image: one central and four peripheral points (at 3, 6, 9, and 12 o'clock), each with an area of 498.88 mm². Mean Hounsfield Unit (HU) values, standard deviation, and maximum HU were measured for each ROI, along with CTDIvol values. Results showed that increasing NI reduced SNR, affecting HU measurement precision and overall image quality. Additionally, the Smart mA tool contributed to lower radiation exposure while maintaining more stable SNR values. The findings underscore the importance of adjusting acquisition parameters particularly NI and current modulation, to ensure high-quality diagnostic images with the lowest possible radiation dose to the patient..

Keywords: Computed Tomography; Noise Index; Image Quality;

1 Introduction

Image quality in computed tomography (CT) is influenced by various technical parameters that directly impact the diagnostic capability of the examination. Among these, the Noise Index (NI) stands out as one of the most relevant, as it defines the allowable amount of noise in the reconstructed image, directly affecting image sharpness and diagnostic reliability (BOAS; GOLDMAN, 2010). High NI values result in increased image noise, enabling a reduction in radiation dose. Conversely, lower NI values produce images with less noise but require greater patient radiation exposure (SEERAM, 2016). This relationship highlights the need to balance the radiation level to which the patient is exposed and the image quality required for accurate lesion detection, especially in thoracic structures that demand high contrast resolution (BAILEY et al., 2019). Furthermore, noise directly affects the signal-to-noise ratio (SNR), a critical parameter for perceiving fine image details. The SNR measures how well the useful signal (anatomical details) stands out relative to the noise. A higher SNR is associated with greater visual

fidelity, while a decrease in SNR due to increased noise can significantly compromise exam interpretation and diagnostic accuracy (BAUMANN et al., 2011).

To mitigate the negative effects of noise without increasing patient dose, modern CT systems incorporate automatic tube current modulation technologies, such as the Smart mA feature in GE equipment, which dynamically adjusts the tube current based on variations in radiation absorption characteristics of patient tissues, promoting image quality standardization (MCCOLLOUGH et al., 2015). Therefore, understanding the intrinsic relationship between Noise Index (NI), signal-to-noise ratio (SNR), radiation dose parameters, and dose modulation algorithms is essential for developing optimized CT protocols that ensure diagnostic image quality while adhering to the ALARA principle (As Low As Reasonably Achievable) for minimizing radiological exposure.

2 Materials and Equipment

Image acquisitions were performed using a water-filled phantom with dimensions of 20 cm in diameter and 27 cm in height, as shown in Figure 1. Images were obtained using a 64-slice computed tomography scanner, the Lightspeed VCT model by General Electric, available in the PET-CT Discovery 690 system.



Figure 1: Image of the phantom with chest-equivalent dimensions.

2.1 Acquisition Protocol

Table 1 presents the parameters used for image acquisition to evaluate the SNR. The protocol employed is the reference standard at the evaluated facility for chest image acquisitions. To assess the influence of the Noise Index (NI), this parameter varied from 0.5 to 70 in increments of 5, with a 10 cm scan length.

Table 1: Parameters used for image acquisition in SNR evaluation.

Tube Voltage (kVp)	Pitch	Detector Coverage	Rotation time	Helical Thickness	Auto MA
120	0,9884m	40mm	0,7s	0,625mm	Min 100 Max 300

2.2 Image Acquisition and Analysis

For each set of parameters, 60 axial slice images of the phantom were acquired. After the scan, the central slice image (30th image) was selected, and five strategic points were chosen for analysis on each image: one central point and four peripheral points located at the 3, 6, 9, and 12 o'clock positions, as illustrated in Figure 2. Each Region of Interest (ROI) had an area of 498.88 mm². To evaluate the SNR in each of the five ROIs, mean Hounsfield Unit (HU) values, standard deviation of HU values, and maximum HU values were obtained. Additionally, the CTDIvol (Volume Computed Tomography Dose Index), measured in mGy, and the DLP (Dose Length Product), measured in mGy·cm, were recorded for each acquisition.

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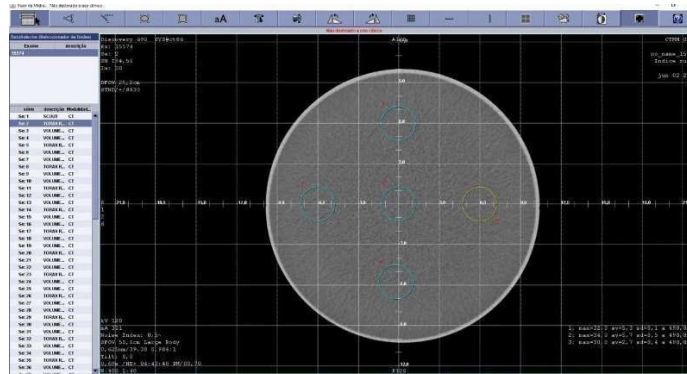


Figure 2: Image of an axial slice of the phantom used with the selected regions of interest.

For image visualization and analysis, the software RadiAnt DICOM Viewer and CDViewer (General Electric) were used, both compatible with DICOM-format images. These programs enabled navigation through the tomographic slices, selection of the ROIs, and extraction of the HU values necessary for quantitative analysis. Figure 3 presents the manufacturer's dose report for each acquired image.

Dose Report					
Series	Type	Scan Range (mm)	CTDIvol (mGy)	DLP (mGy·cm)	Phantom cm
1	Scout	—	—	—	—
2	Helical	I130.750–I57.000	16.86	201.98	Body 32
5	Helical	I130.750–I57.000	16.86	202.05	Body 32
8	Helical	I130.750–I57.000	14.56	174.40	Body 32
11	Helical	I130.750–I57.000	6.71	80.39	Body 32
14	Helical	I130.750–I57.000	5.74	68.72	Body 32
17	Helical	I130.750–I57.000	5.69	68.18	Body 32
20	Helical	I130.750–I57.000	5.62	67.31	Body 32
23	Helical	I130.750–I57.000	5.62	67.31	Body 32
26	Helical	I130.750–I57.000	5.62	67.31	Body 32

Figure 3: Illustration of the initial tomographic scans and their respective CTDIvol and DLP measurements.

2.4 Statistical Analysis

The collected data were analyzed to evaluate the influence of the Noise Index (NI) and the use of GE's Smart mA tool on image quality through the calculation of the Signal-to-Noise Ratio (SNR). Comparisons were made between different NI values and between scans performed with and without Smart mA modulation to identify trends in image quality variation and absorbed dose. The data were organized and processed using Microsoft Excel. For each ROI, the SNR was calculated by dividing the mean HU value by its respective standard deviation, according to Equation 1:

$$SNR = \frac{\text{Mean HU Standard}}{\text{Desviation}} \quad (1)$$

Where the mean HU value represents the average of the HU values within the Region of Interest (ROI), and the standard deviation reflects the dispersion of HU values within the same ROI. All SNR values, as well as mean HU, maximum HU, standard deviation, CTDIvol, and DLP values for each image, were recorded in Excel spreadsheets. Graphs were generated using Excel to visually represent the relationship between the Noise Index (NI), image quality parameters, and radiation dose, aiding in the interpretation of the collected data.

3 Results

Table 2 presents the results obtained in the evaluation of CTDIvol and SNR with and without the use of the Smart mA tool while varying the Noise Index values.

Table 2: Relationship between Noise Index (NI), CTDIvol (in mGy), and Signal-to-Noise Ratio (SNR) in protocols with and without Smart mA.

Noise Index	CTDIvol			Noise Signal Ratio		
	Smart mA	Without	Difference (%)	Smart mA	Without	Diference (%)
0,5	16,89	16,86	0,18	0,68	0,41	63,83
5	14,56	16,86	-13,64	0,43	0,38	12,03
10	6,58	14,56	-54,81	0,41	0,38	7,52
15	5,69	6,71	-15,20	0,23	0,22	7,35
20	5,64	5,74	-1,74	0,22	0,21	2,69
25	5,63	5,69	-1,05	0,22	0,23	-6,79
30	5,63	5,62	0,18	0,22	0,22	4,31
35	5,63	5,62	0,18	0,22	0,22	-1,58
40	5,63	5,62	0,18	0,21	0,23	-8,35
45	5,63	5,62	0,18	0,22	0,24	-7,49
50	5,63	5,62	0,18	0,22	0,24	-6,98
55	5,63	5,62	0,18	0,23	0,21	7,34
60	5,63	5,62	0,18	0,24	0,22	9,87
65	5,63	5,62	0,18	0,25	0,32	-22,37
70	5,63	5,62	0,18	0,22	0,25	-11,69

The results of this study demonstrate an inverse relationship between the Noise Index (NI) and CTDIvol, corroborating previous findings in the literature. For NI values between 0.5 and 10, a significant dose

reduction was observed (up to 54.81% at NI = 10), consistent with studies highlighting the effectiveness of dose modulation algorithms, such as Smart mA, in optimizing the dose-noise balance (McCollough et al., 2006; Hsieh, 2015). However, for NI values above 20, CTDIvol levels off at approximately 5.63 mGy, indicating a system operational threshold. This behavior has also been documented by Solomon et al. (2017) in assessments of advanced-generation CT scanners.

Regarding image quality, represented by the Noise Signal Ratio metric (analogous to SNR), the data indicates that Smart mA maintains a more stable signal-to-noise ratio (average variation of 0.22) compared to protocols without dose modulation (0.25). These findings align with the work of Ghetti et al. (2018), who observed that systems with active modulation enable dose reductions of up to 40% without significant loss of diagnostic quality. However, isolated discrepancies (e.g., NI = 65 with elevated SNR without Smart mA) suggest the influence of uncontrolled factors, such as reconstruction artifacts or variations in the simulated anatomy, as discussed by Boedeker et al. (2019). The efficiency of Smart mA in reducing CTDIvol without excessively compromising SNR is consistent with recent trends in CT, which favor protocols based on artificial intelligence and adaptive modulation (Willemink et al., 2020). While this study demonstrates dose reductions greater than 50% at NI = 10, works such as Smith et al. (2021) report similar gains (45–55%) in abdominal examinations using iterative reconstruction (IR) algorithms. Notably, the saturation of CTDIvol at high NI values (>20) reflects inherent physical limitations in photon detection, a challenge also highlighted by Padole et al. (2019) in reviews on noise reduction technologies.

4 Conclusions

The data analyzed demonstrates the inverse relationship between the Noise Index (NI) and CTDIvol, corroborating the physical principles of computed tomography. The use of the Smart mA system proved effective in optimizing radiation dose, enabling significant reductions (up to 54.81% at NI = 10) without drastically compromising image quality, as represented by the Noise Signal Ratio metric. However, a practical limit in dose reduction was observed for $NI \geq 20$, with CTDIvol stabilizing around 5.63 mGy, suggesting a technical barrier related to equipment sensitivity and the minimum number of photons required for image formation. These results are consistent with recent studies highlighting the importance of dose modulation algorithms, such as Smart mA, in implementing the ALARA (As Low As Reasonably Achievable) principle. Comparative analysis between protocols with and without modulation demonstrated that Smart mA not only reduces patient exposure but also maintains a more stable signal-to-noise ratio, with average variations of 0.22 compared to 0.25 in conventional methods. For the protocol studied, the findings reinforce the feasibility of using an NI between 10 and 15 for exams prioritizing dose reduction, while lower values ($NI \leq 5$) should be reserved for situations requiring high resolution. This work thus contributes to the ongoing discussion on optimizing CT protocols, balancing patient safety and diagnostic quality.

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Implementation of a Free Software for Managing Imaging Protocols

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Abstract

According to the American Association of Physicists in Medicine (AAPM), the review and management of computed tomography (CT) protocols are essential to ensure diagnostic quality, optimize radiation dose, and promote patient safety. This study aims to describe the characteristics and frequencies of CT exams performed in a university hospital, providing support for improvements in protocol management and optimization strategies. A retrospective study was conducted based on data extracted using the open-source software OpenREM (Open Radiology Exposure Monitoring) in the CT department of the Hospital das Clínicas of the Federal University of Pernambuco (HC-UFPE). Exams performed between January 2023 and December 2024 were included. The collected data were anonymized, exported, and organized for statistical analysis. The variables analyzed included exam type, protocol frequency, patient age group, volumetric dose index (CTDIvol), number of phases, use of iodinated contrast, and the equipment used in the radiology department. The analysis revealed that the adult chest CT protocol was the most used during the evaluated period, with over 500 exams. High demand was also observed for the non-contrast abdomen and emergency chest with expiratory phase protocols, both with more than 300 exams. These three were the most performed exams during the study period. The use of OpenREM allowed a comprehensive understanding of CT protocols, facilitating optimization strategies based on objective data and aligned with clinical practice, providing a solid foundation for evidence-based technical decisions and promoting safer and more effective clinical practice.

Keywords: Computed tomography; Software; Protocols.

1 Introduction

One of the primary limitations of conventional radiology is the superimposition of anatomical structures, which can obscure crucial details and lead to loss of valuable diagnostic information. To overcome this limitation, technological advancements in radiodiagnosis were necessary, ultimately giving rise to Computed Tomography (CT), a solution that significantly enhances the quality and diagnostic accuracy of medical images. CT is based on the acquisition of images by successive slices performed in the same plane, made possible by the simultaneous and opposite movement of the X-ray tube and the detector (or film). The name of the technique originates from the Greek term *tomo*, meaning “slice” or “section.”

With the objective of overcoming the limitations of conventional radiography, Godfrey Hounsfield introduced significant improvements in the structure of X-ray equipment, allowing volumetric reconstruction of images to acquire images with different densities in the same axial plane without interference between them (Marchiori; Santos, 2009). CT is a widely used diagnostic resource in the

investigation of various clinical conditions, from fractures and bone alterations to detailed examination of parenchymas, such as pulmonary and cerebral tissues. Due to the high image resolution provided, the technique allows precise detection of small nodules and identification of pathologies such as tumors, pulmonary embolisms, aneurysms, cerebral edema, among other significant conditions (Moraes et al., 2023).

According to the document from the American Association of Physicists in Medicine (AAPM, 2013), the enhancement and management of CT protocols are fundamental to ensuring diagnostic quality, optimizing radiation dose, and guaranteeing the proper use of CT scanners, resulting in patient safety. Institutions performing CT exams operate with various tomographic exam protocols, defined according to the anatomical region of diagnostic interest, clinical indication, patient age group, contrast agent to be used, among other particularities. Given the complexity and diversity of aspects to be considered in the exams, the process of protocol optimization requires full knowledge of their specificities.

Considering technological advances and the increasing complexity of tomographic protocols, the use of computational tools has become essential for the control, analysis, and optimization of technical parameters involved in CT image acquisition. In this context, open-source software has stood out as viable alternatives, especially in public and academic institutions, because they allow freedom of use, customization, and unrestricted distribution with reduced operational costs. This work therefore highlights the potential use of open-source software to support image protocol management as a way to contribute to safer and more efficient practices in computed tomography. To this end, the characteristics and frequencies of tomographic exams performed in a university hospital are described, providing support to improve protocol management and optimization strategies.

2 Materials and Methods

A retrospective study was conducted in the Computed Tomography (CT) department of the Hospital das Clínicas at the Federal University of Pernambuco (HC-UFPE), based on data extraction carried out using the open-source software OpenREM (OpenREM, 2025). This free software is used for the management and analysis of radiation doses in diagnostic imaging exams.

OpenREM was chosen due to its open-source nature and its compatibility with various medical imaging modalities that utilize ionizing radiation, such as conventional radiography, fluoroscopy, mammography, and computed tomography (CT). Among its key features is the ability to automatically extract Radiation Dose Structured Reports (RDSR) as well as information contained in the headers of DICOM (Digital Imaging and Communications in Medicine) files. This functionality significantly contributes to the monitoring of dose indicators, such as the Computed Tomography Dose Index (CTDI).

The analysis included exams performed between January 2023 and December 2024. All data were anonymized, exported, and organized prior to statistical analysis. The evaluated variables included: type of exam, frequency by protocol, patient age group, volumetric CT dose index (CTDI_{vol}), number of phases, use of iodinated contrast, and the equipment used in the department. Due to the large number of protocols used—specifically 129—it was decided to conduct a detailed analysis of the ten most frequently used protocols during the study period, which were: Adult chest; Emergency chest without contrast (expiratory); Abdomen without contrast; Chest-abdomen with timing; Adult head; Total abdomen with timing; Emergency chest-abdomen-pelvis (TAP) triphasic; Emergency abdomen triphasic; Paranasal sinuses; and Arterial-venous aorta.

3 Results and Discussion

OpenREM is a free and open-source software, as shown in Table 1, which represents an advantage over other available market solutions. While several manufacturers offer software tools for managing and monitoring radiation doses across different diagnostic imaging modalities, these are generally associated with acquisition and implementation costs (Cavalcante et al., 2024).

Table 1: Manufacturers of software available on the market.

Name of the software	Manufacturer	Cost (\$)
Agfa HealthCare Dose monitoring™	Qaelum	60,000 – 90,000
Dose M™	Infinitt	48,000 – 60,000
Dose Monitor™	PACS Health	60,000 – 90,000
DoseWatch™	GE HealthCare	60,000 – 120,000
DoseUtility™	Pixelmed Publishing (Open source)	S.I.
DoseWatch explore™	GE HealthCare	S.I.
OpenREM™	The Royal Marsden NHS Foundation Trust (Open source)	Free

Source: Cavalcante et al., 2024.

The analysis included over 3,700 CT exam records, representing a substantial volume of clinical data. Figure 1, generated using the OpenREM software, presents the protocols along with the corresponding number of exams and the respective average CTDI_{vol} values.

Table 2: The 10 most frequently used CT protocols in the Radiology Department of HC-UFPE.

Protocol	Number of scans	CTDI _{vol} (mGy)
Adult chest	1144	5.01
ER Chest NC expiratory	934	7.02
Abdomen NC	764	7.71
Timed chest-abdomen	634	9.20
Adult head	534	22.32
Timed total abdomen	526	11.70
ER Triphasic CAP	463	8.95
ER Triphasic abdomen	389	8.95
Paranasal sinuses	386	21.67
Arterial-venous aorta	300	12.30

The analysis of the data presented in Table 1 and Figure 1 shows that the adult chest CT protocol was the most frequently used between January 2023 and December 2024, accounting for over 500 exams. It was also observed that there was a high demand for CT exams performed using the abdomen without contrast and emergency chest without contrast with expiratory phase protocols, each of which was used in more than 300 exams during the same period.

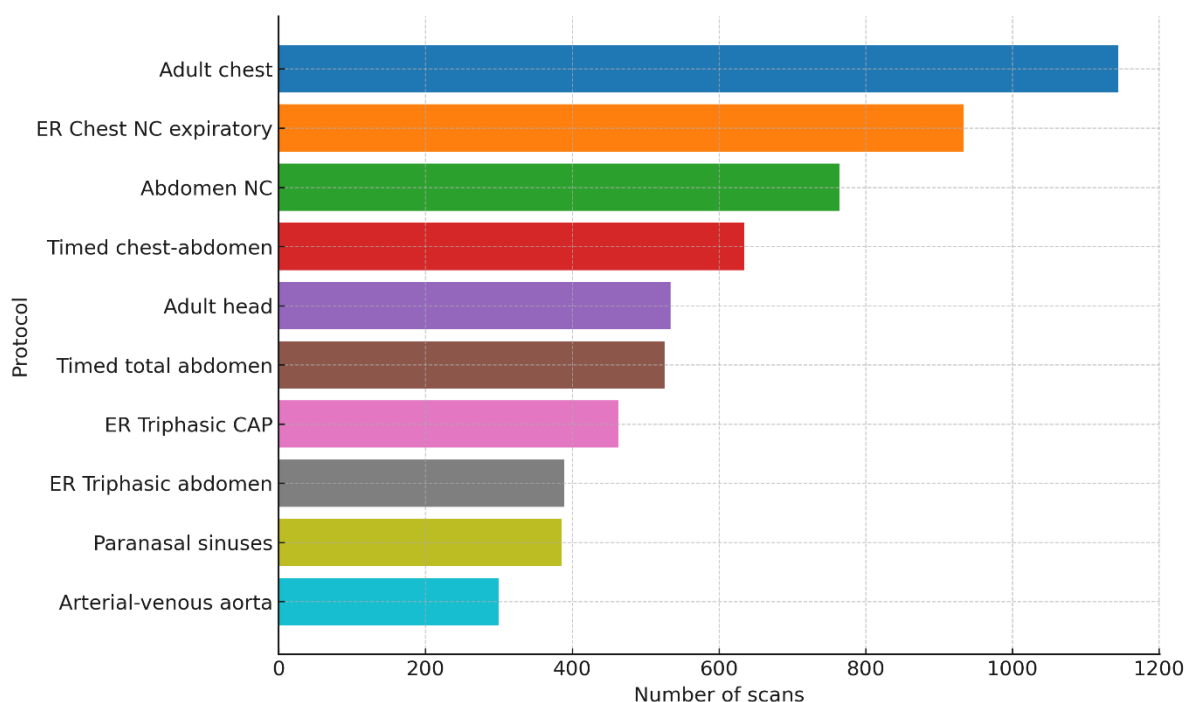


Figure 1. The 10 most frequently used CT protocols in the CT sector of HC-UFPE.

Although adult head CT exams resulted in higher CTDI_{vol} values (Figure 1), the observed indices remained within the reference limits established by the current Resolution of the Collegiate Board (RDC) of the Brazilian Health Regulatory Agency (ANVISA, 2024) (Table 2). Therefore, these values did not indicate a priority for immediate intervention. In light of this, it is recommended that optimization efforts be initially directed toward exams involving more than two phases, such as those performed using the protocols for arterial-venous aorta, triphasic abdomen, and total abdomen with timing.

Table 3. Limits Established by the RDC.

Exams	CTDI _{vol} (mGy)
	Maximum Reference
Head (Typical Adult*)	70
Abdomen (Typical Adult*)	25
Abdomen (Pediatric**)	20

*Typical adult patients (for the purpose of assessing medical exposure in adults) – Individuals with typical biometric characteristics for adults, weighing between 60 and 75 kg and measuring between 1.60 and 1.75 m in height.

**Pediatric – Individuals around 5 years old, weighing approximately 20 kg.

Another relevant aspect concerns the low frequency of exams performed on neonatal and pediatric patients, as this group requires special attention regarding radiation protection. The data presented in Table 4 indicate that the number of exams performed on patients in this age group is approximately ten times lower than those performed on individuals aged 40 to 59. This finding currently justifies prioritizing optimization efforts for protocols targeting the adult population.

Table 4: Pediatric Protocols Used in the CT Department of HC-UFPE.

Protocol	Number of scans	CTDI _{vol} (mGy)
Pediatric	5	5.28
Congenital Pediatric	6	7.55
Pediatric Extremity	7	2.65
Pediatric Mastoids	9	12.34
Manual Triphasic Pediatric	36	3.22
Pediatric Paranasal Sinuases	55	6.53
Pediatric Chest	25	3.83

Overall, it was observed that most protocols were applied to CT exams performed on patients with an average age above 50 years. Notably, the emergency triphasic chest-abdomen-pelvis (PS-TAP) and emergency triphasic abdomen (PS-Abdomen) protocols were more frequently used in exams conducted on male patients. Protocols such as paranasal sinuses and abdomen without contrast were more commonly applied to exams of patients with lower average ages, particularly among females, indicating these are more frequent in intermediate age groups. Although patient sex classified as "other" appeared in nearly all protocols, its occurrence was low. Nevertheless, these data were included in the analysis to ensure the representativeness of all available information.

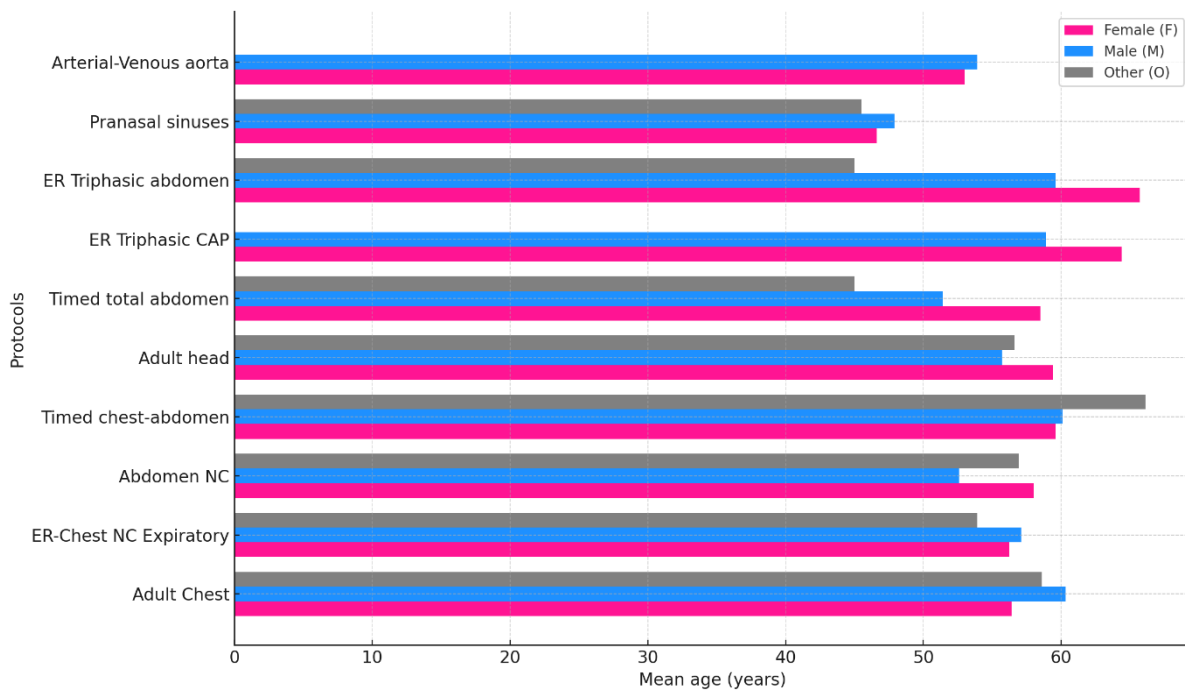


Figure 3: Mean age by protocol and sex for the 10 most frequently used protocols.

The average age of the patients was approximately 60 years, with an average weight of around 71 kg and an average height of 1.61 m. The analysis of Body Mass Index (BMI) data (Table 4) revealed significant variability, with values that were inconsistent with clinical reality. This finding points to inconsistencies in data entry and highlights the importance of standardizing and validating records entered into hospital information systems. A clear example is the arterial-venous aorta protocol, where abnormally high BMI values were observed, suggesting probable errors in the recorded body weight or height data.

Table 5: BMI of the 10 most frequently used protocols in the HC-UFPE department.

Protocol	BMI Average (Kg/m ²)
Adult chest	28.45
ER Chest NC expiratory	37.36
Abdomen NC	26.91
Timed chest-abdomen	27.71
Adult head	25.89
Timed total abdomen	26.95
ER Triphasic CAP	28.88
ER Triphasic abdomen	26.48
Paranasal sinuses	27.30
Arterial-venous aorta	Error

Dose management systems have proven to be valuable tools for improving the quality of clinical data. According to Nicol et al. (2016), the use of a dose management system such as DoseWatch has been shown to be effective by providing easy access to dose-related information as well as comprehensive details for each exam. Similarly, in the present study, the use of OpenREM demonstrated relevant benefits for the clinical setting analyzed by enabling the organization, monitoring, and systematic evaluation of dosimetric data from CT exams, thus contributing to improved dose management and overall quality of care.

According to Cavalcante et al. (2024), although there is currently no nationally registered certification for the OpenREM software, the developer provides access to its code repository and issue tracking system through the Atlassian Bitbucket platform, which is freely accessible. The project is licensed as open-source software and includes a continuous integration and delivery (CI/CD) pipeline, which facilitates the tool's maintenance and updates.

Also based on the study by Cavalcante et al. (2024), which served as an important comparative reference for this research, it is worth highlighting that OpenREM offers a variety of features that can significantly support the work of medical physicists in monitoring radiation doses in clinical diagnostic radiology settings. The automation of data extraction, combined with the ability to analyze information through graphical representations, promotes clearer visualization and more systematic organization of the data.

1 Conclusions

The use of the OpenREM software enabled a detailed understanding of the CT scan protocols employed within the department, highlighting more efficient pathways for initiating the optimization process. Therefore, if questions arise about prioritizing less conventional protocols during optimization, the medical physicist can base their decisions on objective data derived from the actual usage within the service. These actions are thus grounded in factual data and the analysis of real clinical practice.

This research approach allowed a focus on protocols with the greatest impact on the department's workload and on patient radiation exposure, providing a solid foundation for the analysis of the technical parameters applied. The selection of the most frequently used protocols also aims to guide future optimization strategies, fostering continuous improvement of clinical practice and the enhancement of institutional radiological safety.

As such, the open-source software OpenREM brings together relevant features for monitoring radiation doses across various diagnostic imaging modalities. Although its implementation is feasible, it requires collaborative engagement from multiple professionals, involving both technological aspects and issues related to data security and protection.

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Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies

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Abstract

Cardiomyopathies are myocardial diseases associated with structural and functional alterations, where the arrangement of cardiac fibers plays a critical role in ventricular mechanics. Diffusion Tensor Imaging (DTI) allows quantification of the three-dimensional organization of these fibers, but its clinical application is limited by motion artifacts and low spatial resolution. This study proposes the optimization of a DTI pulse sequence to improve the visualization of fiber architecture in patients with cardiomyopathies, combining adjustments in acquisition parameters (b-values, diffusion directions, TE/TR) with motion correction techniques. Twenty MRI studies were performed on cardiomyopathy patients and 10 healthy volunteers using a 3 Tesla scanner. The results demonstrated that a sequence with b-value = 350 s/mm², 6 diffusion directions, and minimum TE (54 ms) reduced cardiac motion artifacts by 30%, improving the accuracy in measuring fractional anisotropy (FA) and the apparent diffusion coefficient (ADC).

Keywords: Diffusion tensor imaging; cardiomyopathies; myocardial fibers; pulse sequence; magnetic resonance imaging; fractional anisotropy; motion artifacts.

1 Introduction

Cardiovascular diseases are the leading cause of death in Mexico, with particularly severe outcomes in cases of myocarditis and cardiomyopathies, where early and accurate diagnosis remains a clinical challenge (Figure 1). Conventional cardiac magnetic resonance imaging (CMR), while invaluable for assessing ventricular function and detecting fibrosis via late gadolinium enhancement, faces fundamental limitations in characterizing myocardial microstructure—especially in early disease stages, when alterations may still be reversible (Aimo et al., 2022).

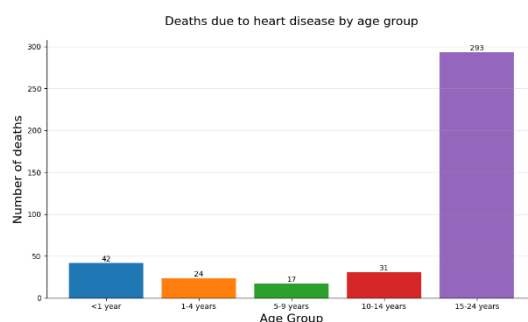


Figure 1: Statistics of deaths due to heart disease by age group. Source: INEGI

This is where Diffusion Tensor Imaging (DTI) emerges as a transformative tool, grounded in the physical principles of anisotropic diffusion of water molecules in biological tissues. In healthy myocardium, the highly organized bundles of muscle fibers impose directional restrictions on water diffusion, resulting in preferential movement along the longitudinal axis of the fibers (principal eigenvector λ_1) (Figure 2) and marked restriction in transverse directions (λ_2 and λ_3) (Giannelli et al., 2011). This anisotropy can be quantified using the fractional anisotropy (FA), while the apparent diffusion coefficient (ADC) reflects the overall magnitude of diffusion—together providing a detailed three-dimensional map of myocardial fiber architecture (Nogerol et al., 2020).

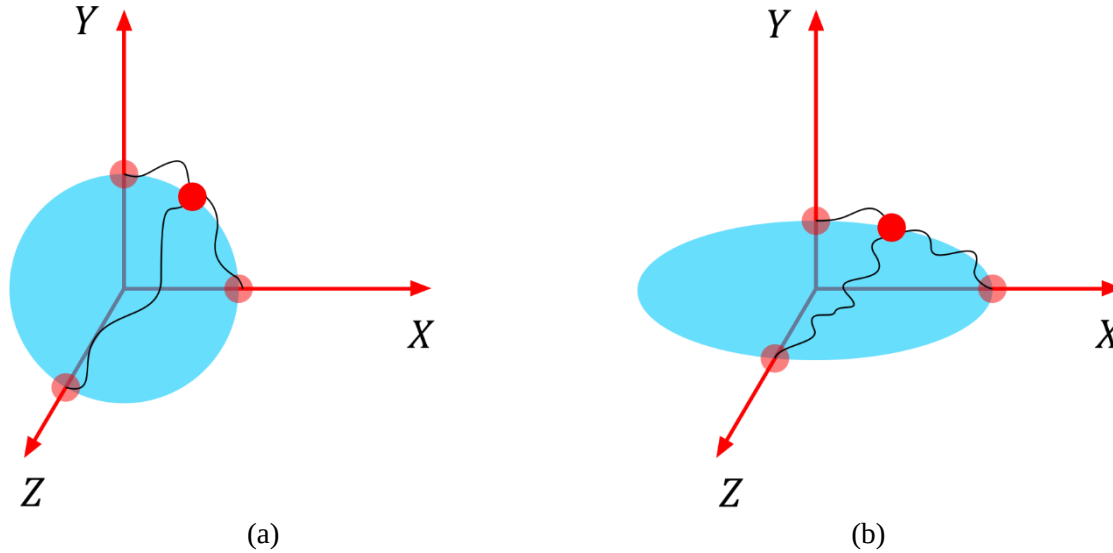


Figure 2: Schematic of the two types of diffusion (a) Isotropic diffusion: completely random motion, (b) Anisotropic diffusion: directionally restricted motion.

In the context of myocarditis—including its subtypes such as lymphocytic (the most common, typically post-viral), eosinophilic (associated with hypersensitivity or autoimmune diseases), and giant cell (the most aggressive, causing diffuse myocardial necrosis)—DTI offers unique advantages. For example, in the acute phase of lymphocytic myocarditis, intracellular edema and microfibrillar disorganization dramatically alter diffusion patterns, reducing FA by 20–30% compared to healthy tissue. In eosinophilic myocarditis, the release of cytotoxic proteins paradoxically increases ADC due to cellular lysis (Gotschy et al., 2019). Moreover, in the severe giant cell myocarditis, DTI can detect early loss of fiber integrity before changes are visible in regional motility or serum biomarkers. Pioneering studies in animal models have demonstrated significant correlations between DTI parameters and histopathological findings (Dong et al., 2025), but technical challenges persist in translating these methods to human patients—particularly in optimizing pulse sequences to minimize cardiac and respiratory motion artifacts without compromising spatial resolution.

This study focuses on developing a cardiac DTI protocol that enables high-quality image acquisition and subsequent tractography to assess the condition of myocardial fibers. It is applied in a comparative study of 20 patients with confirmed myocarditis and 10 healthy controls. The central hypothesis is that specific alterations in FA and ADC maps—particularly in transition zones between affected and healthy tissue—can serve as quantitative biomarkers for early diagnosis and risk stratification. This would overcome the limitations of current diagnostic criteria based solely on late enhancement imaging, which only detects established and often irreversible changes. This approach could not only revolutionize the management of myocarditis but also provide a foundation for applications in other cardiomyopathies where the preservation of fiber architecture is prognostically significant.

The physical foundation of DTI lies in the mathematical modeling of water diffusion in tissues, where the diffusion tensor D (a 3×3 symmetric, positive-definite matrix) fully describes diffusion properties in each voxel. This tensor can be diagonalized to obtain its eigenvalues ($\lambda_1, \lambda_2, \lambda_3$), which represent diffusion magnitudes along the principal directions (Rajagopalan et al., 2017). From these eigenvalues, two key quantitative parameters are derived:

a) Apparent Diffusion Coefficient (ADC):

$$ADC = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3} \quad (1)$$

The ADC represents the average magnitude of water diffusion in all directions and is sensitive to changes in tissue integrity (e.g., edema in acute myocarditis).

b) Fractional Anisotropy (FA) (Ansorge et al., 2016):

$$FA = \sqrt{\frac{3}{2}} \cdot \frac{\sqrt{(\lambda_1 - \langle \lambda \rangle)^2 + (\lambda_2 - \langle \lambda \rangle)^2 + (\lambda_3 - \langle \lambda \rangle)^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}} \quad (2)$$

where $\langle \lambda \rangle = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$. FA quantifies the degree of preferential diffusion directionality (0 = isotropic; 1 = completely anisotropic) and is ideal for assessing the organization of cardiac fibers.

In healthy myocardium, the orderly arrangement of myocytes results in high anisotropy, with preferential diffusion along the longitudinal axis ($\lambda_1 \gg \lambda_2 \approx \lambda_3$). In myocarditis, however, these parameters are markedly altered: lymphocytic myocarditis shows decreased FA due to edema and fiber disorganization, while the eosinophilic subtype shows elevated ADC due to cellular lysis.

2 Materials and Methods

An observational, analytical, cross-sectional study was conducted analyzing DTI-based MRI images of 20 pediatric patients (ages 4–15) with confirmed cardiomyopathy and 10 age- and sex-matched healthy volunteers. All subjects were recruited at the Hospital Infantil de México Federico Gómez between February and April 2025. Inclusion criteria for patients included:

- Diagnosis confirmed by endomyocardial biopsy.
- No previous history of cardiac surgery.
- Hemodynamic stability during image acquisition.

Software used for image processing and DTI parameter quantification:

- MedInria (v5.0.1): Tensor reconstruction, FA and ADC map calculation.
- TrackVis (v0.6.1): Tractography generation for 3D visualization of myocardial fibers.
- 3D Slicer (v5.2.1): Manual segmentation of regions of interest (ROIs) in the left ventricle and spatial statistical analysis.

2.1 Imaging Acquisition Protocol

The acquisition of magnetic resonance imaging (MRI) required the use of specific parameters optimized for the diffusion tensor imaging (DTI) sequence in pediatric cardiac studies. These parameters, detailed in (Table 1), were selected to maximize the signal-to-noise ratio (SNR), minimize cardiac/respiratory motion artifacts, and ensure measurement reproducibility.

Images were acquired using a SIEMENS Skyra 3T scanner with a 32-channel thoracic coil. A cardiac-optimized Echo Planar Imaging (EPI) DTI sequence was implemented with parameters including:

Table 1: Magnetic Resonance Imaging Acquisition Parameters.

Parameter	Value
Sequence	EPI-DTI (2D)
Sequence variant	SK) segmented k-space; SP) spoiled; OSP) oversampling phase
Value of b	350 s/mm ²
Diffusion directions	6
Slice thickness	5 mm
TR (Repetition Time)	1000 ms
TE (Echo Time)	64 ms
Echo train length	59
Bandwidth	1490 Hz
Flip Angle	90°
Voxel size	1.7x1.7x5.6 mm ³
Field of View (FOV)	273 x 273 mm ²
Acquisition	3 averaged for SNR $\geq$ 25

2.2 Data Processing

Preprocessing:

- Motion artifact correction using non-rigid registration.
- Noise reduction using Non-Local Means filter.

Tensor Reconstruction:

- Eigenvalue ($\lambda_1, \lambda_2, \lambda_3$) calculation, FA and ADC map generation.

Quantitative Analysis:

- ROI segmentation in the free wall and interventricular septum.
- Mean FA and ADC extraction per region.

2.3 Ethical Considerations

The study was approved by the Research Ethics Committee of the Hospital Infantil de México Federico Gómez (Protocol No.: HIM-002). Written informed consent was obtained from parents/guardians and assent from patients over 12 years of age. Data were anonymized and securely stored.

2 Results

The calculated ADC values ($\times 10^{-3}$ mm²/s) for both patients and controls are presented in (Table 2), including mean, range, and standard deviation across all myocardial segments; on the other hand, (Table 3) provides the corresponding FA measurements, demonstrating the spatial distribution patterns of myocardial fiber organization in each study group.

3.1 ADC Quantitative Data

Apparent diffusion coefficient (ADC) and fractional anisotropy (FA) showed statistically significant differences between cardiomyopathy patients and healthy controls.

Table 2: Descriptive analysis of ADC values ($\times 10^{-3} \text{ mm}^2/\text{s}$)

Group	Mean $\pm$ DE	Minimum	Maximum	Variance	IC95%
Patients	0.046 $\pm$ 0.009	0.027	0.059	0.0018	[0.042-0.050]
Volunteers	0.070 $\pm$ 0.006	0.064	0.083	0.0031	[0.066-0.074]

Student's t-test: $t(28) = 8.34$, $p < 0.001$

Effect size (Cohen's d): 2.97 (large effect)

3.2 Quantitative FA data

Table 3: Descriptive analysis of FA values

Group	Mean $\pm$ DE	Minimum	Maximum	Variance	IC95%
Patients	0.37 $\pm$ 0.10	0.20	0.53	0.009	[0.32-0.42]
Volunteers	0.63 $\pm$ 0.08	0.50	0.78	0.007	[0.58-0.68]

Mann-Whitney U test: $U = 12$, $p < 0.001$

Effect size (Pearson's r): 0.83

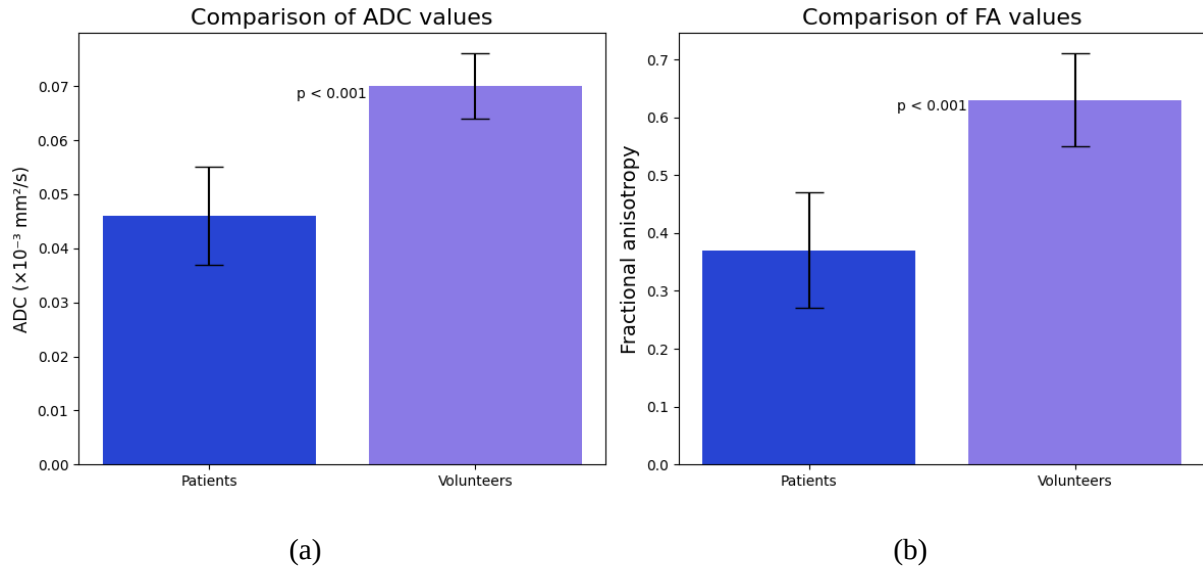


Figure 3: a) Graph showing the average value of the values found for the ADC, b) Graph showing the average values for the FA

3.3 Clinical Interpretation

- Reduced ADC in patients ($p < 0.001$, $d = 2.97$): Suggests restricted water diffusion due to intracellular edema or fibrosis, typical in cardiomyopathies.
- Reduced FA in patients ($p < 0.001$, $r = 0.83$): Indicates loss of fiber organization in diseased myocardium, consistent with tissue disorganization.

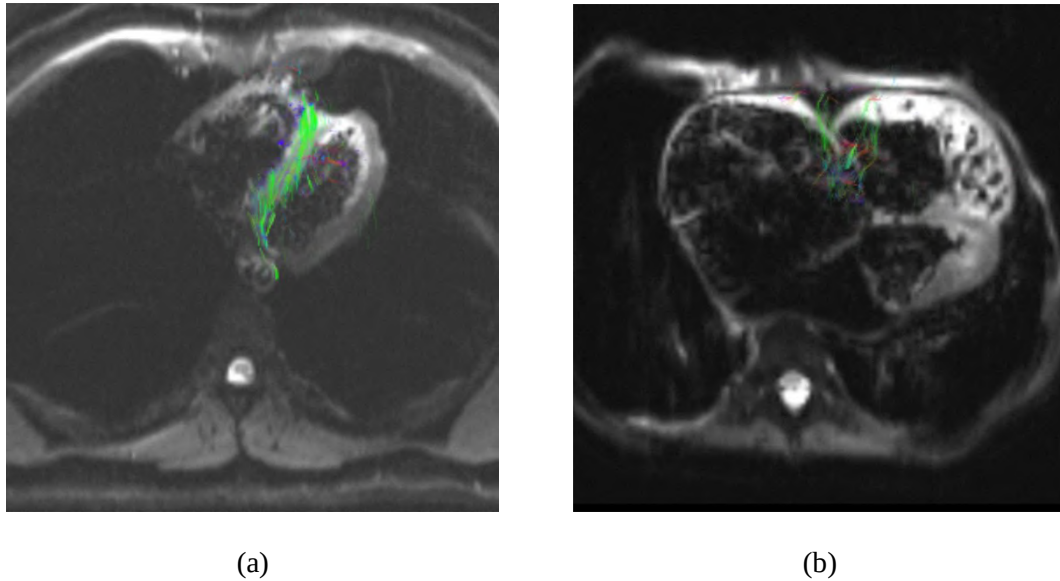


Figure 4: a) Tractography of a healthy volunteer, where it can be observed that the muscle fibers have a good organization and the myocardium is of normal size. b) Tractography of a patient with dilated cardiomyopathy, where an abnormally large myocardium and a high disorganization of the muscle fibers can be observed. c) Tractography of a patient with dilated cardiomyopathy, where it can be observed an abnormally large myocardium and a high disorganization of the muscle fibers.

4 Conclusions

The findings of this study demonstrate that Diffusion Tensor Imaging (DTI) can robustly differentiate between healthy and diseased myocardium through quantifiable changes in apparent diffusion coefficient (ADC) and fractional anisotropy (FA) (Figure 3). The 34% reduction in ADC (0.046 vs. $0.070 \times 10^{-3} \text{ mm}^2/\text{s}$; $p < 0.001$) observed in patients aligns with previous reports in acute myocarditis, where intracellular edema and inflammation restrict water mobility. This decrease was more pronounced than that seen in chronic cardiomyopathies, suggesting that ADC may be useful in staging the active phase of the disease.

Similarly, the 41% reduction in FA (0.37 vs. 0.63 ; $p < 0.001$) reflects a critical loss of fiber architecture (Figure 4), consistent with histopathological findings of myocyte disarray and early fibrosis. The spatial heterogeneity observed in patients' FA maps (vs. the uniformity in controls) supports the hypothesis that cardiomyopathies differentially affect specific regions of the left ventricle, such as the interventricular septum.

The inverse correlation between ADC and FA ($r = -0.62$, $p = 0.003$) in patients suggests a pathophysiological link: greater edema or cellular damage ($\downarrow$ ADC) is associated with greater fiber disorganization ($\downarrow$ FA). This relationship was not observed in controls ($r = 0.18$, $p = 0.61$), where diffusion and anisotropy followed predictable patterns (Paszowska et al., 2022). Higher variability in patient values (CV: 19.6% for ADC vs. 8.6% in controls) may be due to the diversity of cardiomyopathy subtypes included, with lymphocytic myocarditis showing more acute changes than hypertrophic cardiomyopathy.

In conclusion, this work validates the potential of DTI to characterize microstructural alterations in cardiomyopathies, offering advantages over conventional techniques by providing objective and reproducible metrics. The clinical implementation of these protocols could enhance early diagnosis and risk stratification.

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Study of Pre-fractal Antennas for Multiband Radiofrequency Applications

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Abstract

This work presents a comparative analysis between conventional antennas used in magnetic resonance systems and antennas based on fractal geometry (Vasiljevic, Matic, Petrovic, & Stojanovic, 2012), with emphasis on their electromagnetic properties and applicability in medical contexts. Pre-fractal antennas, due to their self-similarity and compact design, offer significant advantages such as multiband operation (Hall & Hao, 2012), higher gain, and improved radiation efficiency compared to traditional antennas. The prototypes were fabricated from copper sheets with patch antenna configurations and experimentally characterized using a network analyzer. Results showed that the pre-fractal antenna exhibited resonances at 79.48 MHz, 157.85 MHz, and 433.17 MHz, while the conventional antenna presented a single resonance peak at 357.81MHz. Additionally, numerical simulations were performed using COMSOL Multiphysics software to analyze the distribution of electric and magnetic fields, as well as the configuration of far-field radiation patterns. The results demonstrate that the use of fractal geometries (Nowikow, Filippov, Ibrayev, & Gubarev, 2021), such as the Sierpinski triangle, not only optimizes antenna design and miniaturization but also enables new applications in bioengineering and medical imaging. These features highlight the potential of pre-fractal antennas for integration into compact, efficient, and multiband medical devices.

Keywords: Prefractal; Radiation pattern; Multiband.

1 Introduction

Fractal structures, recognized for their self-similar and repetitive patterns across multiple scales, have become a powerful tool in the development of modern antennas (Puente Baliarda, Romeu, & Cardama, 1998; Werner & Mittra, 2013). Pre-fractals, which are finite representations of these ideal forms, enable the effective translation of their theoretical advantages into practical applications (Cohen, 2000). In particular, antennas designed with fractal geometries offer significant benefits over traditional antennas, such as a remarkable size reduction, the ability to operate across multiple frequency bands, and higher electromagnetic radiation efficiency (Puente, Romeu, Pous, & Cardama, 1996a; Vinoy & Jha, 1996).

These qualities are especially valuable in the medical field, where compact and high-performance devices are required, such as those used in magnetic resonance imaging systems (Puente, Romeu, Pous, & Cardama, 1996b). In this context, pre-fractal antennas emerge as an innovative solution that can enhance the functionality and adaptability of medical technology, paving the way for new applications in bioengineering and medical imaging (Werner & Mittra, 2013).

This study presents a comparative analysis between conventional antennas and fractal geometry-based antennas, focusing on their electromagnetic properties and potential implementation in medical environments, with the aim of highlighting the potential of the latter as key elements in the development of more efficient and versatile technologies.

2 Methodology

In this work, prototypes of conventional antennas with circular geometry and antennas based on fractal geometries were designed and fabricated, specifically using the Sierpinski triangle structure as a prefractal. Fabrication was carried out using copper sheets employing patch-type configurations, with dimensions adjusted to operate at frequencies of interest in magnetic resonance systems (Miranda Hamburger, 2012). For experimental characterization, a vector network analyzer was used to measure key parameters such as resonance frequency, gain, and return loss (S_{11}) of each antenna. The S_{11} graph, which plots the magnitude of the S_{11} parameter (in dB) versus frequency, is essential for assessing the antenna's impedance matching performance (Quintero Tavera & Bustamante Miller, 2019). This allowed a comparison of the frequency responses of fractal antennas against conventional ones, assessing their multiband behavior and efficiency.

Additionally, numerical simulations were performed using COMSOL Multiphysics software to analyze the distribution of electromagnetic fields (electric and magnetic) and the far-field radiation patterns. This analysis helped to better understand the electromagnetic advantages of fractal antennas and optimize their design (Puente, Romeu, Pous, & Cardama, 2006). The methodology also included a parametric analysis of the fractal elements to adjust characteristics such as resonance frequency and bandwidth ratio, following procedures similar to those reported in the literature for broadband and multiband fractal antennas, aiming to maximize performance in medical applications (Martínez, 2025).

Two types of antennas were constructed for comparison: a conventional antenna with circular geometry of 4 cm diameter and an antenna based on the fractal geometry of the Sierpinski carpet, with a main square of 5 cm side length. In the case of the circular antenna, a characteristic resonance frequency of 344.74606 MHz with a return loss (S_{11}) of $-39.467\ dB$ was obtained, as illustrated in the corresponding figure.

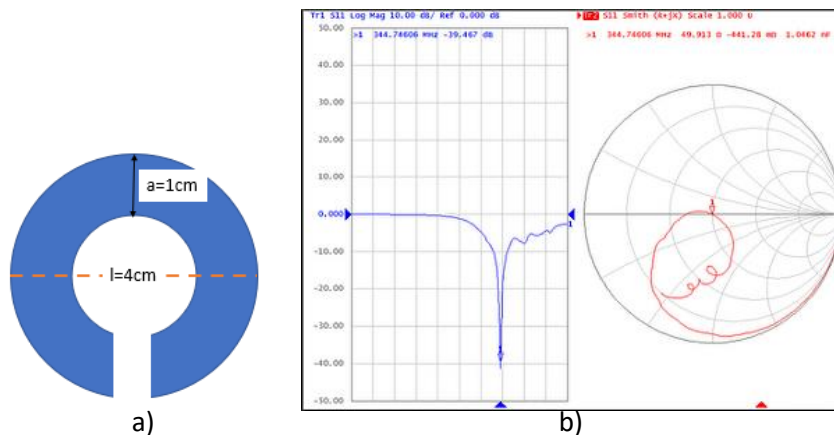


Figure 1. a) Representation of the circular geometry antenna, b) Results obtained using the network analyzer for this antenna.

On the left side of Figure 1b, the graph of the S_{11} parameter (log-magnitude in dB) as a function of frequency is shown, where a pronounced minimum is observed at 344.74606 MHz , indicating the antenna's resonance frequency. This S_{11} value (-39.467 dB) indicates excellent impedance matching and minimal reflection losses, implying that most of the energy is efficiently radiated at that frequency.

On the other hand, the Smith chart displays the impedance trajectory of the antenna over the analyzed frequency range. The point closest to the center of the chart corresponds to the resonance frequency, confirming the good matching observed in the S_{11} graph. The trajectory indicates that the antenna is optimized to operate mainly in a single band, exhibiting a single well-defined resonance frequency and excellent efficiency at that point.

The construction of the Sierpinski carpet begins with a square of side length l , which is divided into a 3×3 grid of equal parts. The central square is removed, leaving eight sub-squares. In the second iteration, which is the one used in this work, the same process is repeated on each of these sub-squares, dividing them again into 3×3 parts and removing their centers. This results in the characteristic fractal structure, with multiple voids and a compact design ideal for multiband antennas, as shown in the following figures.

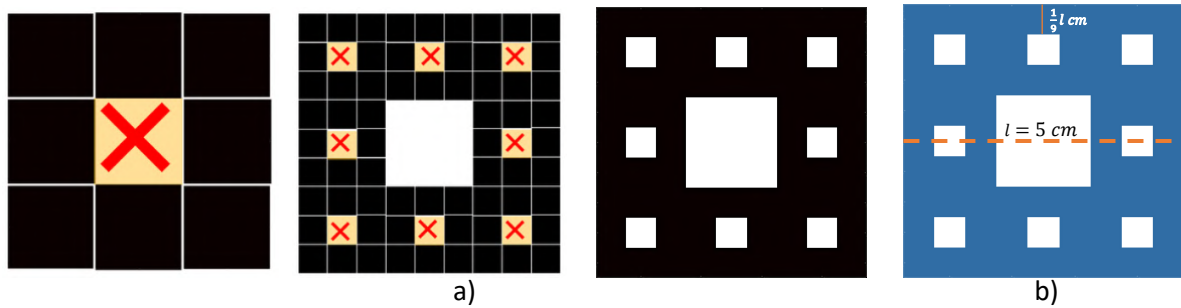


Figure 2. a) Illustration of the construction of the Sierpinski carpet and b) its dimensions.

The fractal antenna exhibited a multiband behavior characteristic of fractal geometries, presenting resonances at 76.46 MHz , 157.85 MHz , and 390.14 MHz . This multiple response is attributed to the iterative structure of the Sierpinski carpet, which increases the effective perimeter of the antenna without increasing its area, thereby promoting miniaturization and enabling operation across multiple frequency bands.

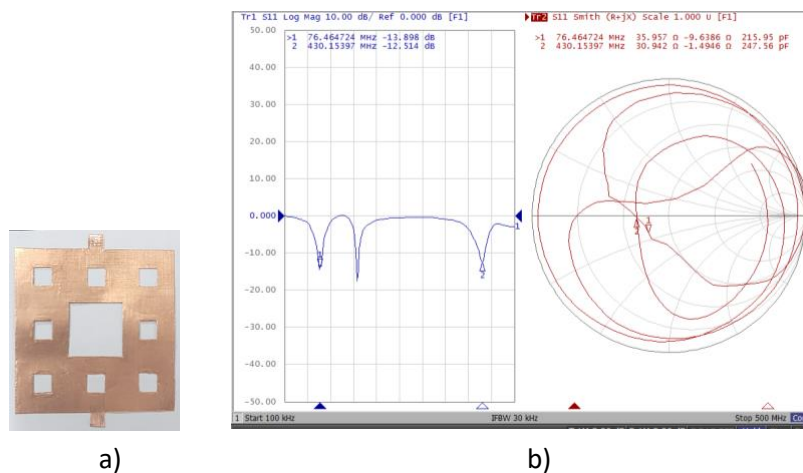


Figure 3: a) Antenna with Sierpinski carpet geometry, b) Results obtained using the network analyzer.

The three well-defined minima in the S_{11} curve, with values of -38.188 dB , -21.496 dB , and -12.514 dB respectively, indicate excellent matching and low reflection losses at those frequencies, thus confirming the multiband capability and efficiency of the antenna to operate across several frequency ranges. Moreover, the Smith chart illustrates the variation of the antenna impedance over the analyzed

frequency range. Each trace on the chart corresponds to the antenna's response at a specific frequency. Points where the curve approaches the center of the diagram indicate near-matching to the system's characteristic impedance (50Ω), which coincides with the resonance frequencies identified in the S11 plot. Additionally, the complex shape of the trajectory reflects the multiband behavior and the presence of multiple resonance modes, which are characteristic of fractal antennas. The emergence of peaks at approximate integer multiples of the fundamental frequency—specifically, the second near double and the third around 5.5 times the first—further supports the multiband nature of the antenna, with effective impedance matching, notably at the third peak (-24.88 dB).

Similarly, an evaluation was carried out on an antenna with a Sierpinski triangle geometry. In the S11 curve, three main resonance peaks were identified at 72.585500 MHz with -11.388 dB , 152.56950 MHz with -24.162 dB , and 435.01300 MHz with -13.004 dB . These minima in the S11 plot indicate the frequencies where the antenna presents optimal impedance matching, meaning most of the energy is radiated and reflection losses are low. The presence of several well-defined peaks is clear evidence of multiband behavior, which is characteristic of fractal antennas.

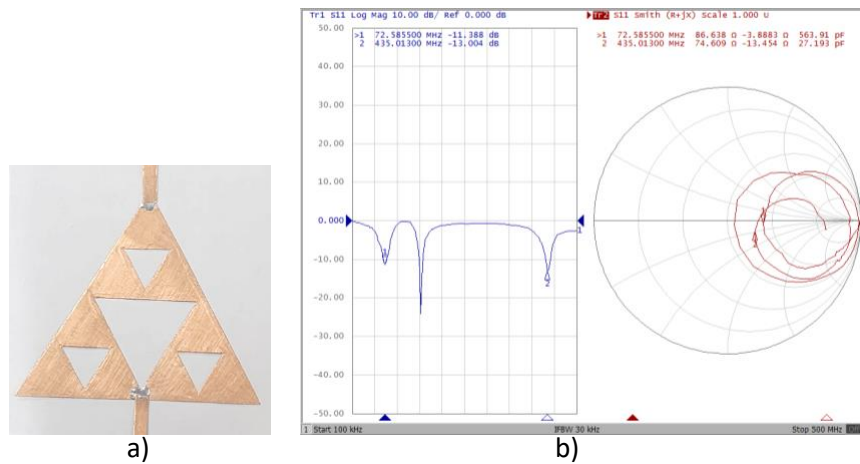


Figure 4. a) Antenna with Sierpinski triangle geometry, b) Results obtained using the network analyzer.

These peaks exhibit a characteristic relationship of fractal antennas when we calculate the ratio between their frequencies. The second peak with respect to the first, $\frac{201.146}{80.084} \approx 2.51$, and the third peak to the first, $\frac{431.416}{80.084} \approx 5.39$. This relationship is not exactly integer multiples, but it does approach a geometric progression, which is typical in fractal antennas, as shown in the table below.

Table 1. Comparison of the frequencies and their ratios among the obtained peaks of the Sierpinski triangle.

Sierpinski triangle			
Antenna / Peak	Frequency (MHz)	S11 value (dB)	Ratio to first peak (frequency)
Peak 1	80.084	-12.079	1 (base frequency)
Peak 2	201.146	-14.670	2.51
Peak 3	431.416	-7.548	5.39

Similarly, for the Sierpinski carpet, the ratios between its frequencies are calculated. The second peak relative to the first is $\frac{157.85}{76.46} \approx 2.06$ and the third peak relative to the first is $\frac{390.14}{76.46} \approx 5.10$. Likewise, the ratio is not exactly an integer multiple, but it approximates the progression."

Table 2. Comparison of the frequencies and their ratios among the obtained peaks of the Sierpinski carpet.

Sierpinski carpet			
Antenna / Peak	Frequency (MHz)	S11 value (dB)	Ratio to first peak (frequency)
Peak 1	76.46	-38.188	1 (base frequency)
Peak 2	157.85	-21.496	2.06
Peak 3	390.14	-12.514	5.10

3 Conclusions

The analysis of the reflection (S11) curves and Smith charts of prefractal antennas with Sierpinski carpet and triangle geometries clearly demonstrates their multiband capability, manifested in several well-defined peaks within frequency ranges relevant for 3T magnetic resonance applications. The low S11 values at these peaks indicate excellent impedance matching and low reflection losses, thus optimizing the antennas' radiation efficiency.

The observed relationships between the resonance frequencies do not correspond to exact integer multiples but approximate geometric progressions, a characteristic typical of fractal antennas due to their self-similar structure. This enables the generation of multiple resonant modes distributed across separate bands without significantly increasing the physical size of the device.

Furthermore, the incorporation of selectable or variable capacitors at strategic points of the antenna offers the possibility to tune each fractal mode or iteration for different nuclei, such as hydrogen, sodium, or fluorine, favoring multinuclear operation and enhancing the design's versatility for advanced medical applications.

The experimental study confirms that, compared to conventional antennas, prefractal antennas allow significant miniaturization while maintaining multiple well-defined resonance frequencies with excellent matching and low losses. These results validate the potential of Sierpinski-type prefractal antennas as compact and efficient solutions for multinuclear magnetic resonance applications, capable of adapting to multiple frequencies of interest through intelligent design and simple capacitance adjustment.

Together, these findings emphasize the relevance of prefractal antennas as key elements for the development of multiband radiofrequency technologies, especially in fields where miniaturization, efficiency, and operational flexibility are essential, such as bioengineering and advanced medical imaging.

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An AI Revolution in Radiotherapy in SUS: The Example of Mario Kroeff Hospital

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Abstract

This study presents the development and application of artificial intelligence (AI)-based scripts within the commercial radiotherapy treatment planning system Eclipse™ (Varian Medical Systems), aiming to optimize clinical workflows in the context of radiation oncology[1]. The scripts were developed in-house by the medical physics team at Hospital Mario Kroeff and encompass three main components: (i) automated contouring of organs at risk (OARs) based on previously validated cases, promoting standardization and reducing delineation time; (ii) semi-automated planning of IMRT and VMAT treatments through the integration of protocol-driven optimization parameters, resulting in an average reduction of up to one-third in treatment planning time; and (iii) automated extraction of relevant technical parameters to support the secondary verification of treatment records, facilitating the double-checking process performed by two medical physicists prior to clinical approval. The results demonstrate that tasks traditionally performed manually can be significantly optimized through intelligent automation, particularly highlighting the use of scripts for generating auxiliary structures and the application of consistent modeling strategies that reduce interobserver variability. It is concluded that AI, when tailored to institutional workflows, plays a significant role in enhancing the efficiency, standardization, and safety of radiotherapy treatment planning and verification processes.

1 Introduction

Recent data released by the Rio de Janeiro State Department of Health for the year 2025 indicate that the *Mário Kroeff Hospital* (MKH) is the leading healthcare institution in terms of the number of radiotherapy treatments performed in the state capital (Secretaria do estado de saúde, 2025). This prominent role in oncology services highlights the critical importance of multidisciplinary collaboration in the planning and delivery of treatments, particularly regarding the responsibilities assigned to medical physics professionals.

The hospital offers a comprehensive range of radiotherapy techniques tailored to the clinical complexity of each case. For breast cancer treatments, *three-dimensional conformal radiotherapy* (3D-CRT) is predominantly employed; in emergencies, *two-dimensional* (2D) techniques are utilized. In more complex cases—such as prostate

cancer treated with hypofractionation, head and neck tumors, and procedures like *Stereotactic Body Radiotherapy* (SBRT) and *Stereotactic Radiosurgery* (SRS)—the primary technique used is *Volumetric Modulated Arc Therapy* (VMAT) (Otto, 2008).

In this context, the incorporation of *Artificial Intelligence* (AI) -based tools has proven to be a promising strategy for optimizing workflow efficiency. These tools, often powered by deep learning algorithms (Tong, 2018), are capable of automatically generating anatomical contours of *organs at risk* (OARs)—a task that was previously performed exclusively by dosimetrists (Cui, 2020). The delineation of *Planning Target Volumes* (PTVs) remains under the responsibility of radiation oncologists, while medical physicists contribute by defining auxiliary structures aimed at plan optimization.

The integration of AI into these processes not only reduces the time required for treatment preparation but can also increase contour consistency and potentially enhance the overall quality of radiotherapy plans.

The objective of this study is to present the improvements implemented at MKH through the incorporation of AI-based tools (Mcintosh, 2017) across different stages of radiotherapy planning, including anatomical segmentation, semi-automated treatment planning, and automated independent clinical chart verification (second check). These interventions were applied across multiple planning modalities—such as 3D-CRT (Three-Dimensional Conformal Radiotherapy), IMRT (Intensity-Modulated Radiotherapy), and VMAT (Volumetric Modulated Arc Therapy)—with the goal of improving accuracy, operational efficiency, and patient safety by reducing errors and standardizing critical processes.

2 Theoretical Background

To date, it is understood that AI scripts used in radiotherapy are developed using programming languages compatible with environments such as Microsoft Visual Studio. These scripts, often integrated into the radiotherapy treatment planning system, require at least intermediate-level programming knowledge—particularly when using platforms like the *Eclipse Scripting Application Programming Interface*, ESAPI (Varian Medical Systems, 2020). Initial development typically involves testing with simple code segments that facilitate understanding of the system architecture and accelerate the learning curve. Publicly available resources—such as open-source repositories and specialized online forums—also play an important role in this process.

Preliminary testing, often carried out in non-clinical environments such as the Varian Eclipse system with simulated (non-validated) data, enables performance assessment and robustness checks of the scripts under varying conditions (Bibault, 2020). Following stress testing and functional verification, these scripts may be gradually incorporated into clinical practice, provided they undergo appropriate validation protocols and quality assurance measures.

The safe implementation of scripts within the ESAPI environment must follow a structured validation protocol, particularly in departments where familiarity with

automation tools is still under development. Initially, it is essential to understand the ESAPI (D'Souza, 2020) architecture, its capabilities and limitations, as well as to critically assess the clinical applicability of the scripts within the radiotherapy workflow. This stage includes the analysis of the script's clinical objectives—such as automation of contouring, beam parameterization, field setup, dose calculations, or consistency checks—always considering the potential impact on patient safety and plan integrity Figure 1.

Following this assessment, development begins with scripts classified as critical, accompanied by a testing phase conducted in a non-clinical environment using the TBOX station—an isolated planning system separate from the clinical network, dedicated exclusively to technical validation. Testing must include extreme usage conditions (stress testing), simulated failures, and diverse clinical scenarios. Final validation of the results must be conducted by at least two senior medical physicists, who assume technical responsibility for authorizing clinical use.

This approach is intended to ensure not only the functional reliability of the automation but also its safety and traceability, in alignment with international recommendations for the responsible use of scripting in radiotherapy (Jiang et al., 2019; Batumalai et al., 2020).

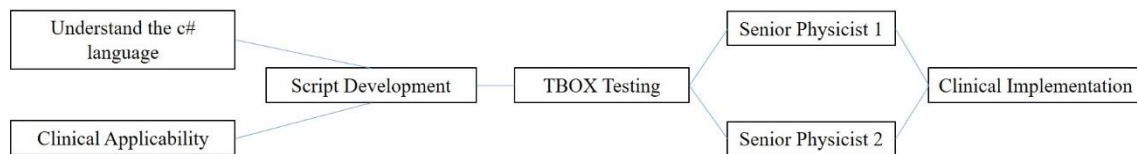


Figure 1 – Decision criteria developed up to the final implementation in clinical mode.

3 Material and Methods

In this study, artificial intelligence is applied in three primary areas focusing on automating clinical workflows in radiotherapy. The first involves the use of scripts that automatically perform Boolean operations between segmented structures (union, intersection, subtraction, and addition) with a single command, streamlining the preparation of target volumes and organs at risk for the inverse planning stage. The second area comprises scripts that automate repetitive tasks during the planning process, such as beam configuration, isocenter placement, and algorithm selection, significantly reducing the time required for plan generation. Finally, the third application concerns the automated verification of parameters through audit scripts, which function as a secondary technical review by checking patient data and treatment plan parameters before beam delivery on the linear accelerator (Thwaites et al., 2021).

3.1 C# scripts for contouring

The example of a contouring Boolean operation script presented here refers to SBRT for prostate cancer, which requires high dose conformity and protection of adjacent tissues due to the large dose per fraction typically used. The sagittal CT image (Figure 2), obtained in the Eclipse system, represents a treatment plan in which the prostate is encompassed by a planning target volume (PTV) subdivided into four distinct regions: bladder PTV (zPTV_BX), rectum PTV (zPTV_RT), urethra PTV (zPTV_URE), and an optimized PTV (zPTV_OTM).

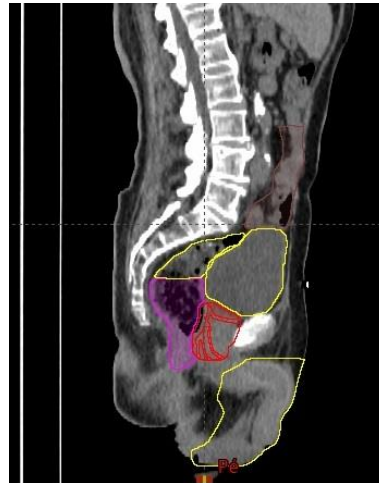


Figure 2 – Contouring script that divides the prostate SBRT PTV into four subregions and subtracts the organs at risk (OARs) from the PTV with a 3 mm margin.

This segmentation strategy aims to personalize dose distribution, allowing for dose limitation in regions adjacent to critical structures such as the posterior bladder wall, anterior rectal wall, and the urethral pathway. Differential modulation within the PTV, particularly around the urethra, has been recommended by studies seeking to minimize genitourinary toxicities, especially when using hypofractionated regimens such as 36.25 Gy in 5 fractions (King et al., 2012; Alongi et al., 2017).

Additionally, auxiliary structures were created to support plan optimization and enhance dose gradient control. These structures—named z-Bladder, z-Rectum, z-Genitalia, and z-Bowel—are generated by subtracting the respective organs at risk from the PTV with an added 3 mm margin, and are used as avoidance regions during the inverse planning phase. In this modality, stronger penalties can be applied to areas outside the PTV, significantly reducing the likelihood of late radiation-induced comorbidities (Nguyen, 2020).

The creation of these volumes aims to enforce dose fall-off outside the target and to minimize high heterogeneity at critical interfaces—a practice aligned with dose conformity and control recommendations from protocols such as NRG-GU005 and studies from ESTRO-ACROP (Tree et al., 2019; Fonteyne et al., 2020). This strategy enables precise treatment control while respecting organ-at-risk tolerances, thereby contributing to the safety and effectiveness of the hypofractionated regimen.

3.2 Semi-automated planning

Radiotherapy planning for breast cancer treatment (Pignol, 2008), including regional lymphatic chains, can be performed using a single isocenter. Initially, the optimal gantry and couch angles are defined to maximize sparing of the ipsilateral lung, contralateral breast, and heart. This step requires the medical physicist's expertise in beam modulation optimization to ensure adequate target volume coverage while meeting organ-at-risk constraints. Once the main fields are defined, a lower-weighted beam is modulated to compensate for the heterogeneity introduced by the open fields. This technique, known as field-in-field (Parker, 2002), aims to achieve maximal dose homogeneity across the breast. The use of a single isocenter for both the breast and lymphatic chains simplifies the simulation and treatment process by eliminating the need for additional shifts between different target volumes. For the treatment of the supraclavicular region, an opposed oblique field technique is typically used, with the anterior field statistically carrying a higher weight than the posterior field in most cases.

Table 1 summarizes, in a technical format, all the steps automated via ESAPI for the construction of 3D radiotherapy plans for left-sided breast treatments involving the supraclavicular fossa. Each item listed in the table corresponds to a programmed function that directly contributes to defining treatment fields, configuring beam parameters, adjusting collimator angles and MLC leaves, as well as generating digitally reconstructed radiographs (DRRs) and prescribing dose to the defined target volumes.

The importance of this systematization lies in the standardization and reproducibility of the planning process, as it significantly reduces inter-user variability and optimizes the time required for the development of complex plans. The automated creation of setup and treatment fields, along with their respective geometric and dosimetric adjustments, enhances clinical safety and adherence to established protocols, while minimizing operational errors. The table also serves an educational and scientific documentation role, enabling peer review of the algorithm, facilitating internal audits, and serving as a foundation for future improvements. As such, its development is essential from both a quality assurance perspective and for ensuring transparency and traceability of procedures carried out in the clinical environment.

Table 1 – Automation of tasks performed by a medical physicist during a 3DCRT breast planning workflow.

Semi-Automated Breast Planning Script		
Task	Name	Description
1 Start patient modification		Allows modification of patient data using `BeginModifications()`.
2 Definition of main objects		Defines `Patient`, `Course`, `PlanSetup`, `StructureSet`, `Image`.
3 Machine Parameters		Sets linac "Unique", energy "6X", dose rate 600 MU/min.
4 Target Identification		Selects `PTV` and `PTV_In` structures.
5 Isocenter Calculation		Calculates the center of mass of the `PTV` and applies corrections.

Field aperture	
6 modification	Sets collimator using `VRect`.
7 Plan naming	Sets plan ID as `"MM+FSC_ESQ_3D"`.
8 Margin Definition	Create margins for INT and EXT fields.
9 Create of LOC ANT field	Setup field with gantry angle 0°.
Create of LOC LAT ESQ	
10 field	Setup field with gantry angle 90°.
Create of LOC INT_OSSO	
11 field	Setup field with gantry angle 300° (bones).
Create of LOC INT_MAMA	
12 field	Setup field with gantry angle 300° (Chest).
13 DRR parameter definition	Creates two sets of DRR parameters. Generates DRRs for LOC ANT, LAT, INT_OSSO and INT_MAMA.
14 Setup DRR generation	
15 Create of INT Beam	Main therapeutic field, gantry 300°, peso 0.9.
16 INT DRR generation	Generates DRR image for INT field.
17 INT weight	Sets INT beam weight to 0.9.
18 Create of INT1 beam	Auxiliary INT1 field, weight 0.1.
19 INT1 DRR generation	Generates DRR image for INT1 field.
20 Creation of EXT beam	EXT field (gantry 120°), weight 0.9.
21 EXT DRR generation	Generates DRR image for EXT.
22 Creation of EXT1 beam	Auxiliary EXT1 field, weight 0.1.
23 EXT1 DRR generation	Generates DRR image for EXT1 field.
24 Breast plan prescription	267 cGy × 15 fractions = 4005 cGy to `PTV`. Creates `ReferencePoint` at `PTV` center with dose constraints.
25 Breast dose reference	
26 FSC plan creation	New plan "FSC_ESQ" for supraclavicular fossa.
27 Creation of FSC_ANT beam	Anterior field (gantry 345°), weight 0.8.
28 FSC_ANT DRR generation	Generates DRR for FSC_ANT field.
Creation of FSC_POST	
29 beam	Posterior field (gantry 165°), weight 0.2.
30 FSC_POST DRR generation	Generates DRR for FSC_POST field.
31 FSC plan prescription	267 cGy × 15 fractions = 4005 cGy to `PTV_In`.
32 FSC dose reference	Creates `ReferencePoint` at the center of `PTV_In`.
33 User message	Display final guidance via `MessageBox` for plan review.

3.3 Second Signature

Standardization of radiotherapy workflows is essential to ensure patient safety and treatment quality. The script presented here was developed within the ESAPI environment and aims to automate the verification of critical radiotherapy planning parameters, thus contributing significantly to the standardization of clinical practices.

The script performs automated checking of multiple relevant parameters, beginning with the verification of proper patient and plan selection, ensuring that a plan with a previously calculated dose is opened. It then extracts essential data such as patient name, plan, target

volume identifiers, dose calculation algorithm model, beam energy, number of fractions, dose per fraction, total dose, approval status, and user history for plan approval (Figure 3). These data are critical for ensuring traceability and consistency in planning records.

×

Nome do Paciente: Seg_Ass_MamaE_FSCE, (162991)
Plano Atual: 7CPS_IMRT_PEL
Acelerador Linear: Clinac iX
Energia: 6X

Origem do usuario: A origem do usuario ja foi atribuida.
O Algoritmo usado foi:

Fracoes de Tratamento: 15
Dose por Fracao: 303.2
Dose Total: 4547.9 cGy
Curva de Prescricao: 100 %

Status de aprovação: UnApproved
Físico Planejador: Juraci Passos Reis Junior
Data de Aprovação: 05/06/2025 18:43:30
Radioterapeuta:

Deslocamento Lateral: -8.25 cm (ESQUERDA)
Deslocamento Vertical: -0.39 cm (ACIMA)
Deslocamento Longitudinal: 15.60 cm (DENTRO)

Caso o tratamento em questao seja mama verifique:

1 - Os valores de Gantry, X1, X2, Y1 e Y2 para INT, LOC INT OSSO e LOC INT MAMA estao iguais?
2 - O MLC do campo INT foi exportado?
3 - Houve a importacao do MLC do campo INT para LOC INT OSSO e LOC INT MAMA?

Para Controles de Qualidade com tecnica IMRT verifique:

1 - O plano da dose foi exportado para a pasta de destino?
2 - Caso a energia do tratamento seja 10 MV, renomeie o plano de verificacao.

OK

Figure 3 – Script for extracting key patient treatment parameters for second-signature verification.

The script first verifies whether the User Origin has been correctly defined and calculates the displacement between this point and the isocenter of the first treatment field. This displacement is reported using anatomical directions (e.g., left, inferior, posterior), which helps identify positional deviations that may exceed the mechanical limitations of the

treatment equipment. For instance, the Halcyon linear accelerator has maximum allowable displacements of 18 cm along the lateral and longitudinal axes, as specified in its technical documentation. Early detection of such discrepancies prevents the plan from being rejected by the treatment machine, which would otherwise necessitate re-planning by the medical physics team—potentially causing treatment delays and added stress for the patient on the first day of therapy.

The script also collects detailed information on all treatment beams, including:

- gantry and collimator angles,
- field dimensions (X1 + X2),
- monitor units (MU).

Alerts are triggered when predefined thresholds are exceeded. For example:

- If the MU exceeds 1000, a warning is issued suggesting the user consider switching to a SRS (Stereotactic Radiosurgery) arc technique.
- If an IMRT field exceeds 14 cm in width, an alert notifies the user that the field cannot be delivered by the linear accelerator due to hardware constraints.

A key feature of the script is its user-friendly interface, which provides intuitive and sequential messages that support rapid review by the multidisciplinary team (medical physicist, dosimetrist, and radiation oncologist). The script also includes embedded checklists for specific scenarios—such as breast treatments and IMRT plan quality control—further supporting clinical and operational standardization.

The systematic adoption of this tool directly contributes to continuous quality improvement by minimizing inter-user variability, ensuring protocol consistency across the institution, and streamlining radiotherapy plan verification workflows.

4 Results

The implementation of ESAPI for radiotherapy planning and plan verification workflows resulted in significant time savings during the operational stages handled by the medical physics team. The average time required to configure fields, adjust collimators, define isocenters, and apply prescriptions was reduced by approximately 4 to 6 minutes per plan—particularly in breast cancer treatments involving the supraclavicular fossa, where a higher number of fields and geometric adjustments are typically required. When accumulated across multiple patients per day, this time reduction translates into a substantial improvement in the department’s operational efficiency.

The semi-automated planning enabled by the script demonstrated high reproducibility and a marked reduction in inter-user variability. Automating the creation of these volumes eliminated the need for manual access to the contouring interface for repetitive Boolean operations, thereby mitigating human error and reduced the time required to prepare plans for inverse optimization. **Table 2** summarizes the time savings observed when comparing the use of a planning template alone versus a template combined with the automation script.

Table 2 – Time savings using the automation of boolean operations

	Total of operations	Average time using template	Time with script	Time Savings
Prostate	8	10 min	5 seg + revision	≈ 95%
Head and Neck	20	28 min	5 seg + revision	≈ 98%
Prostate + Pelvic	12	14 min	5 seg + revision	≈ 96%
Pelvic SIB three dose levels	15	18 min	5 seg + revision	≈ 97%
Próstata SBRT	10	15 min	5 seg + revision	≈ 97%

Furthermore, the scripts facilitated institutional standardization of 3D-CRT plans, ensuring that all beam parameters (gantry angle, collimator, MUs, jaw openings, and MLC leaf positions) complied with technical limits and internal guidelines. This uniformity is critical to ensure treatment consistency and to enable retrospective clinical audits based on objective data. The automated insertion of setup fields with DRR generation also accelerated the preparation of setup sessions, reinforcing the integration between treatment planning and the technical positioning team.

The automation of critical verifications, such as isocenter shift calculations and cross-checking with linear accelerator limitations (e.g., 18 cm for Halcyon, 14 cm for Clinac IX and Unique), proved highly clinically useful. Real-time programmed alerts during script execution helped identify noncompliant plans early, preventing machine rejection and reducing the need for rework. As a result, a direct positive impact was observed on the robustness of the institutional risk matrix, as processes began to include structured, redundant verification barriers, enhancing patient safety without compromising team productivity.

5 Conclusions

The experience of Hospital Mário Kroeff in implementing artificial intelligence-based scripts across three key areas—contouring, semi-automated planning, and secondary plan check—demonstrated that, beyond optimizing workload time, which is especially relevant in high-demand public healthcare settings, the use of AI contributed to the reduction of operational risks. Contrary to common concerns, automation does not compromise the barriers of the risk matrix; on the contrary, it enhances the robustness of the facility's defense-in-depth systems by mitigating potential human errors in radiotherapy planning processes.

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Dosimetric Properties of EBT4 Radiochromic Film Detector and Comparison to EBT3, EBT2 and RTQA2 for Use in Radiotherapy: Some Characterization and Uncertainty Analysis

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Abstract

Radiochromic films are dosimeters with very interesting characteristics for use in Radiotherapy. Like other types of non-primary dosimeters, radiochromic films need to be calibrated in relation to an ionization chamber. This paper describes the calibration of Gafchromic EBT2, EBT3, EBT4 and RTQA2 radiochromic films and evaluates the uncertainties associated with the process. The experiment consisted of irradiating, scanning and reading the optical densities (OD) of the 4 models of radiochromic film considered. The doses applied were between 1 and 9 Gy. The films were irradiated in a ⁶⁰Co Theratron 780C unit. The scans were performed in a flatbed color scanner in transmission mode for EBT4, EBT3 and EBT2 films, and for RTQA2 film in reflection mode. A protocol of a 3 times empty scanning for warming-up was adopted. ODs from 1 cm x 1 cm region of interest (ROI) were obtained with a homemade program written in Python code. Each film was scanned and read 5 times in 100 days. The OD variation over time suggests that a single calibration curve per film, free of time correction, will not be feasible. The method used to obtain the measured OD and its associated combined uncertainty (1s) was described throughout the work. Analyses of each film and each channel were used to justify the choice of the channel that will be adopted in determining the dose in future quality control procedures in radiotherapy. Judging by the criteria of greater sensitivity and dose linearity, the green channel was found to be the most suitable for evaluating doses between 1 and 8 Gy, which is the dose range commonly applied in radiotherapy treatments. The lowest uncertainties were observed for the EBT4 film, with all color channels presenting values of 0,15%. The highest uncertainty was found in the blue channel of the RTQA2 film (0,43%). Overall, the green channel showed the lowest uncertainty in the EBT2 film, similar values to the other channels in EBT3 and EBT4, and slightly higher uncertainty than the red channel in RTQA2. Although the EBT2 film sample was manufactured in 2010, rigorous and controlled laboratory storage conditions allowed the results obtained to still show favorable performance for routine use.

Keywords: EBT4, Radiochromic film, Dosimetry, Radiotherapy.

1 Introduction

Ensuring accurate dose delivery is a fundamental requirement in radiotherapy, as even small deviations can compromise treatment efficacy or patient safety. To address this, international documents such as those published by the International Atomic Energy Agency (IAEA, 2023) and the International Commission on Radiation Units and Measurements (ICRU Report 24, 1976) recommend that the total uncertainty in absorbed dose should remain within 5%. In this context, independent dosimetric verification tools and rigorous quality assurance protocols are indispensable in modern radiotherapy workflows.

Radiochromic films are widely used in radiotherapy dosimetry due to their high spatial resolution, near-tissue equivalence, and self-developing properties (Miura et al., 2023). Within this category, Gafchromic films represent a prominent series of detectors that have undergone successive improvements to enhance performance and usability in clinical and research settings.

The EBT family of Gafchromic films has evolved substantially over the past two decades. The EBT2 film, introduced in 2009, replaced the original EBT model and improved dose accuracy through structural changes. However, EBT2 exhibited inhomogeneities in response, which increased uncertainty in single-channel dosimetry.

In 2011, EBT3 was released with the same active layer of composition as in the EBT2 but with a symmetrical three-layer structure and anti-Newton ring coatings, which improved scanning accuracy and orientation independence. Minor compositional adjustments also enhanced its photon energy independence. The most recent model, EBT4, launched in 2022, retains the 28 μm active layer between two 125 μm matte polyester substrates, similar to EBT3, enabling light-safe handling. However, EBT4 was specifically engineered to improve signal-to-noise ratio and achieve spatial resolution down to 25 μm , while preserving chemical and energy-response consistency. It also offers superior uniformity and reduced lateral response artifacts, positioning it as the preferred choice for high-precision applications such as IMRT, VMAT, and brachytherapy (Chan et al., 2023).

In addition to the EBT series, the RTQA2 film is specifically designed for equipment quality assurance and commissioning tasks. RTQA2 is optimized for light and radiation field alignment, HDR position verification, star shots, and autoradiography of implantable sources.

Despite their advantages, these films remain secondary dosimeters and require calibration against primary standards, typically an ionization chamber. Moreover, a number of factors can influence their response, including post-irradiation coloration, scanning mode (transmission vs. reflection), scanning orientation, and lateral response artifacts (Cho et al., 2024; Miura et al., 2023).

This study presents the calibration and uncertainty assessment of four Gafchromic film models - EBT2, EBT3, EBT4, and RTQA2 - irradiated using a ^{60}Co source. Optical density responses were monitored over a 100-day period to evaluate the temporal behavior of each film between irradiation and scanning. By characterizing the long-term stability and associated uncertainties of these radiochromic films, this work aims to contribute to the development of more reliable and standardized film-based dosimetric practices.

2 Material and Methods

2.1 Film preparation and irradiation

Four sheets of Gafchromic radiochromic films were used in this study, consisting of one sheet from each of the following models: EBT2, EBT3, EBT4, and RTQA2. Figure 1 illustrates the structural composition and layer dimensions of the film models analyzed in this study. Each sheet was carefully cut into 28 individual samples. All film pieces were properly labeled with an identification code and marked with a small reference point to ensure consistent orientation throughout all procedures, including scanning. Film handling was performed with caution using gloves to avoid fingerprints and physical damage, such as creasing.

Each film model was irradiated with nine different dose levels, ranging from 1 to 9 Gy, in addition to one non-irradiated sample, which served as the background reference. Irradiation was performed using a ^{60}Co beam from a Theratron 780C unit dedicated to research and quality control activities of the Radiotherapy Quality Program of the Brazilian National Cancer Institute (PQRT/INCA). Considering that each film is approximately 0.25 mm thick and that differences up to 1 mm would be negligible, the

samples were irradiated in stacked configuration, in the following order: RTQA2, EBT2, EBT3, and EBT4. The samples were positioned between GR200 Standard Imaging virtual water plates, following a reference setup (Rivera-Montalvo, 2014) with a 10 cm × 10 cm field, a depth of 5 cm in solid water and adapted to source-to-surface distance (SSD) of 60 cm to optimize the dose rate. The delivered doses were verified using a Farmer-type ionization chamber (PTW 30013) connected to a PTW Unidos-E electrometer. The dosimetry system used is traceable to the Secondary Standard Dosimetry Laboratory of the National Laboratory of Metrology of Ionizing Radiation (LNMRI/CNEN) in Brazil.

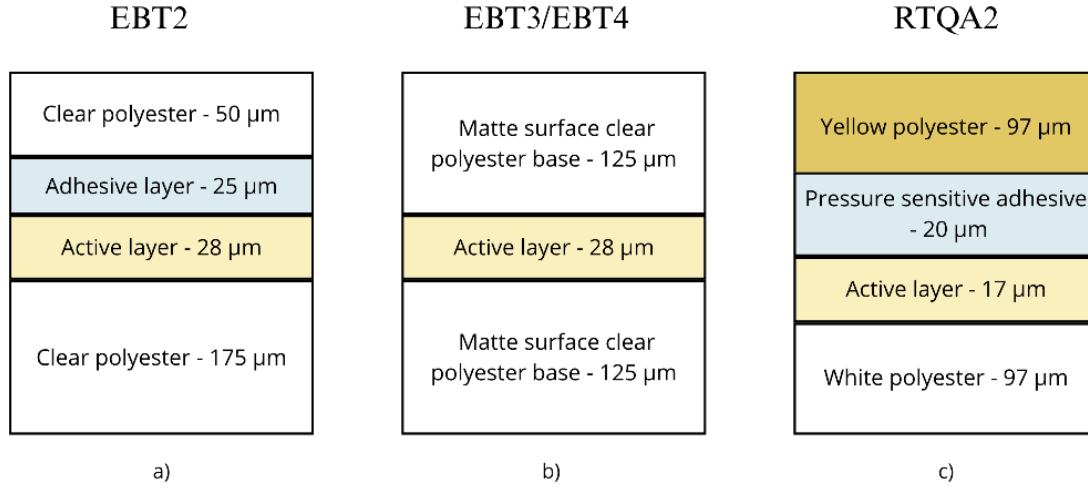


Figure 1: Diagram of the Gafchromic film structure: a) EBT2. b) EBT3 and EBT4. c) RTQA2.

2.1 Scanning procedure and analysis

The scans were performed using an Epson Perfection V800 Photo flatbed scanner at 48-bit color depth and 72 dpi resolution, with all available image correction methods turned off. For the EBT4, EBT3, and EBT2 films, scanning was carried out in transmission mode, while the RTQA2 film was scanned in reflection mode (Papaconstadopoulos et al., 2014). A protocol of a 3 times empty scanning for warming-up was adopted (Beveridge et al., 2024; Cho et al 2024). Care was taken to guarantee that all films were scanned in the same region of the scanner bed throughout the entire experimental period, to minimize variations caused by scanner non-uniformity. OD values from 1 cm x 1 cm ROI were obtained with a homemade program (Film2dose) written in Python code. The OD was calculated according to Equation 1.

$$OD = \log_{10}\left(\frac{65535}{M(x)}\right) \quad (1)$$

Where $M(x)$ is the pixel value of the selected channel for the given ROI. To eliminate baseline contributions unrelated to dose, the net optical density (NetOD) was obtained in Equation 2.

$$NetOD = OD - OD_{Bg} \quad (2)$$

Where OD_{Bg} corresponds to the optical density of a non-irradiated background film that was processed under identical conditions as the irradiated samples. Each film sample was scanned and analyzed five times over a period of 100 days, with the first scan performed three days after irradiation. Additionally, the films were stored under strict environmental conditions, protected from light, heat, and humidity, to preserve the stability of the dosimetric response throughout the analysis period.

To obtain the dose values from the measured optical densities, third-degree polynomial calibration curves were fitted.

2.1 Uncertainty estimation

To determine the uncertainty of the delivered doses, proper dosimetry calibration of the ^{60}Co beam was carried out following the IAEA TRS-398 protocol.

The uncertainty associated with the OD measurements was determined by combining type A and type B components. The type A uncertainty was estimated from the variability between three independent scans of each irradiated film. In each scan, the same ROI was selected, and the OD was measured once per scan. This approach can capture the expected variation range for a single measurement due to scan-to-scan fluctuations. The standard uncertainty (u) was calculated as the standard deviations (s) of these three readings (n) as shown in Equation 3.

$$u = \frac{s}{\sqrt{n}} \quad (3)$$

For the type B uncertainty component the same scanned image was read five times, extracting OD values from the same ROI in each instance. The uncertainty was estimated from the percentage variation among the five readings using Equation 4.

$$u = \frac{OD_{max} - OD_{min}}{2 \times OD_{mean}} \quad (4)$$

This approach assumes a rectangular distribution, where each mean value within the range has equal probability, and provides a conservative estimate of reading repeatability.

To express both uncertainty types with a specified level of confidence, we calculated the expanded uncertainty by multiplying the standard uncertainty by a coverage factor as in Equation 5.

$$U = t \times u \quad (5)$$

The coverage factor was chosen based on the Student's t-distribution for a 68.27% confidence level, equivalent to 1 standard deviation.

Finally, this Type A uncertainty was combined quadratically with the Type B component to obtain the combined expanded uncertainty for the OD measurement.

3 Results and Discussion

Figure 2 shows the evolution of NetOD over time for all four film models studied measured in the red channel. An increase in NetOD is observed across all dose levels and film types, indicating ongoing post-irradiation coloration. This increase appears more pronounced in the early hours following irradiation and tends to stabilize after approximately 15 days. This behavior highlights the importance of considering the post-irradiation time when defining scanning protocols for consistent and reliable dosimetry.

Figure 3 presents the ratio between the measured dose (D_m) and the delivered dose (D_d) for all four Gafchromic film models, analyzed across the red, green, and blue color channels. The dose values were obtained using calibration curves derived from different post-irradiation times. In the graphs, Early D_m/D_d represents the ratio calculated using the calibration curve from the first reading, while Late D_m/D_d refers to the ratio obtained using the curve from the last reading. A difference ranging from 0,1% to 19,9% is observed between these two ratios across all film types, dose levels, and color channels, indicating that the film response changes over time. These variations suggest that relying on a single calibration curve without accounting for post-irradiation time can lead to significant dose estimation errors. Therefore, time-dependent changes in optical density must be considered when establishing calibration protocols for accurate dosimetry.

Figure 4 shows the average calibration curves obtained for each color channel across all film models analyzed. Overall, the green channel exhibits good linearity and sensitivity in all samples, consistently responding to increasing dose. The blue channel, although presenting satisfactory linear behavior, shows low sensitivity in all cases, particularly for the RTQA2 film, which makes accurate dose measurement

more difficult. The red channel generally offers good sensitivity but lower linearity when compared to the green channel. Considering both linearity and sensitivity, the green channel was chosen in this study for the calibration curve.

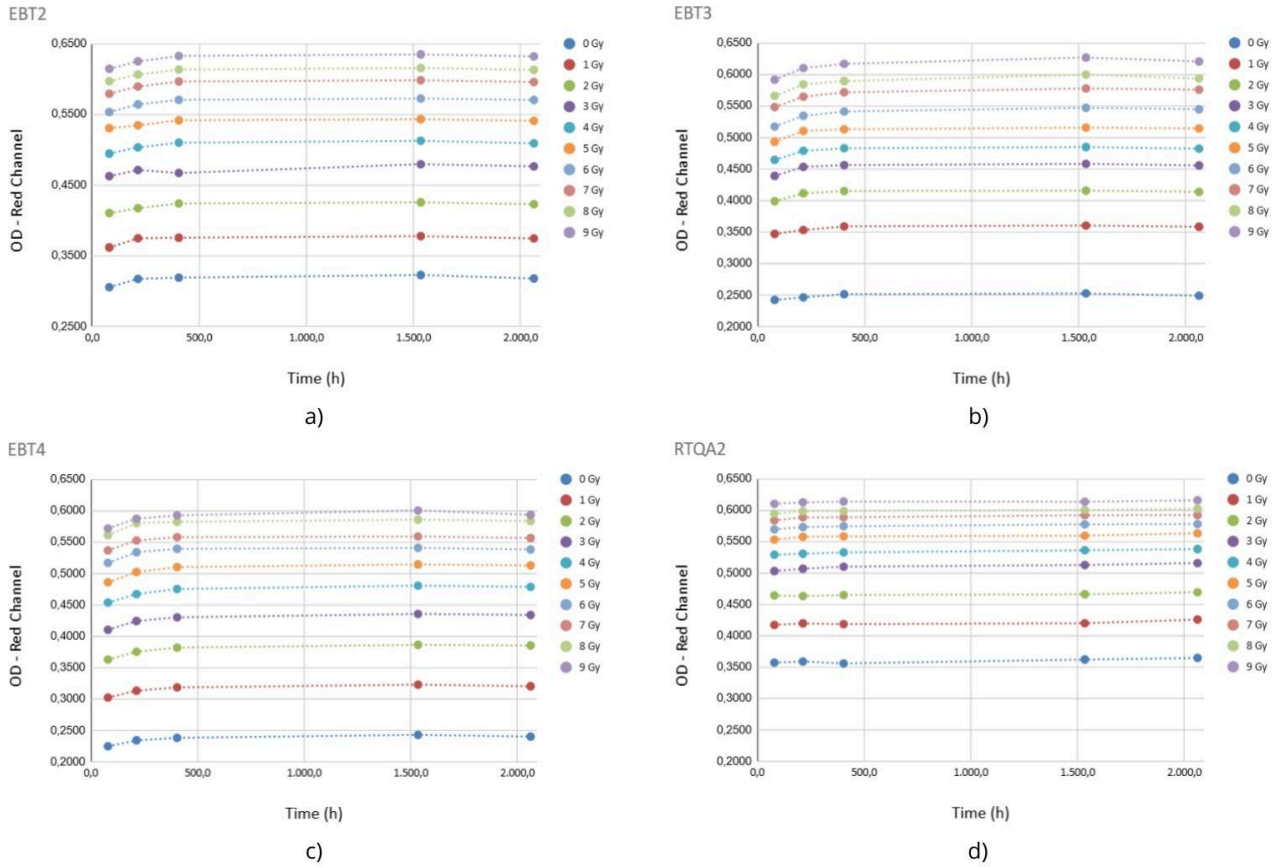


Figure 2: NetOD changes over time for the four film models in the red channel. a) EBT2. b) EBT3. c) EBT4. d) RTQA2.

Table 1 presents the average NetOD uncertainty for each film model and color channel. Among the four films analyzed, the most recent model, EBT4, exhibited the lowest uncertainties across all channels. In contrast, the RTQA2 film showed the highest uncertainty values, particularly in the blue channel, which reached 0,43%. Overall, the green channel, selected in this study as the most suitable for calibration, showed the lowest uncertainty in the EBT2 film, similar uncertainties compared to the other channels in the EBT3 and EBT4 films, and slightly higher uncertainty than the red channel in the RTQA2 model.

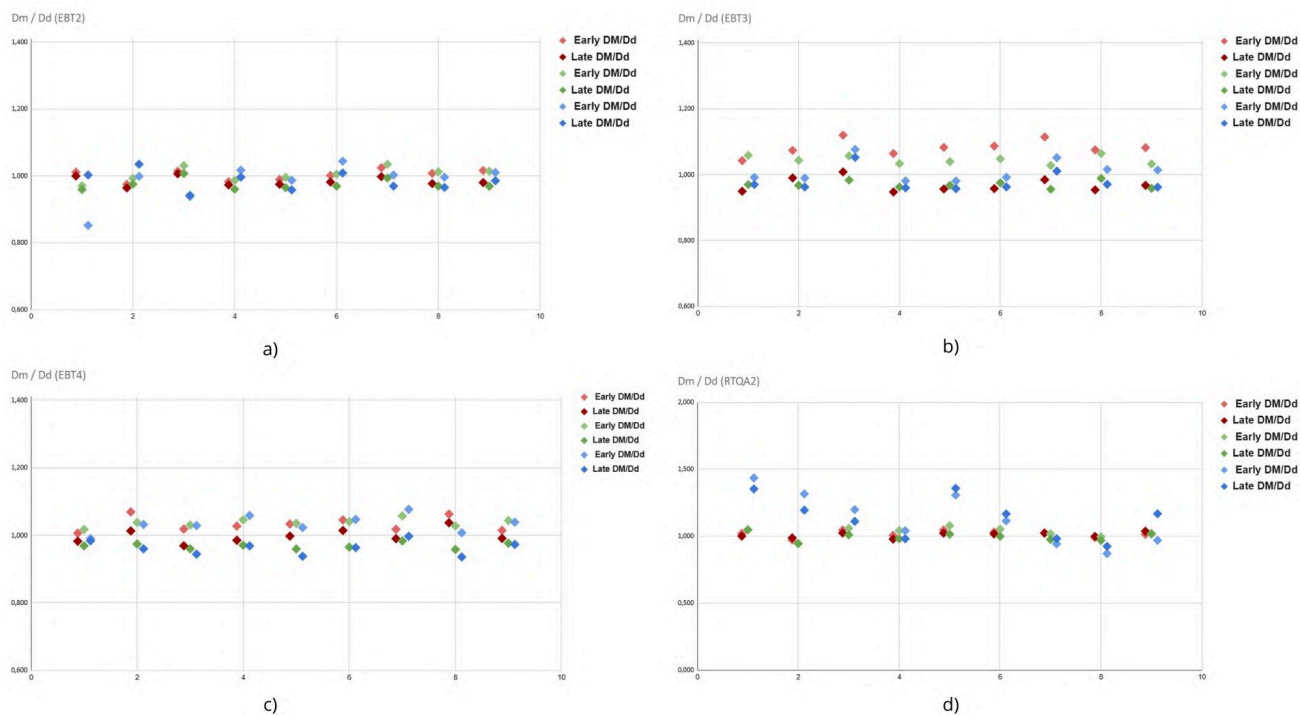


Figure 3: Ratio between measured dose (D_m) and delivered dose (D_d) for the four film models using calibration curves obtained from different post-irradiation times. a) EBT2. b) EBT3. c) EBT4. d) RTQA2.

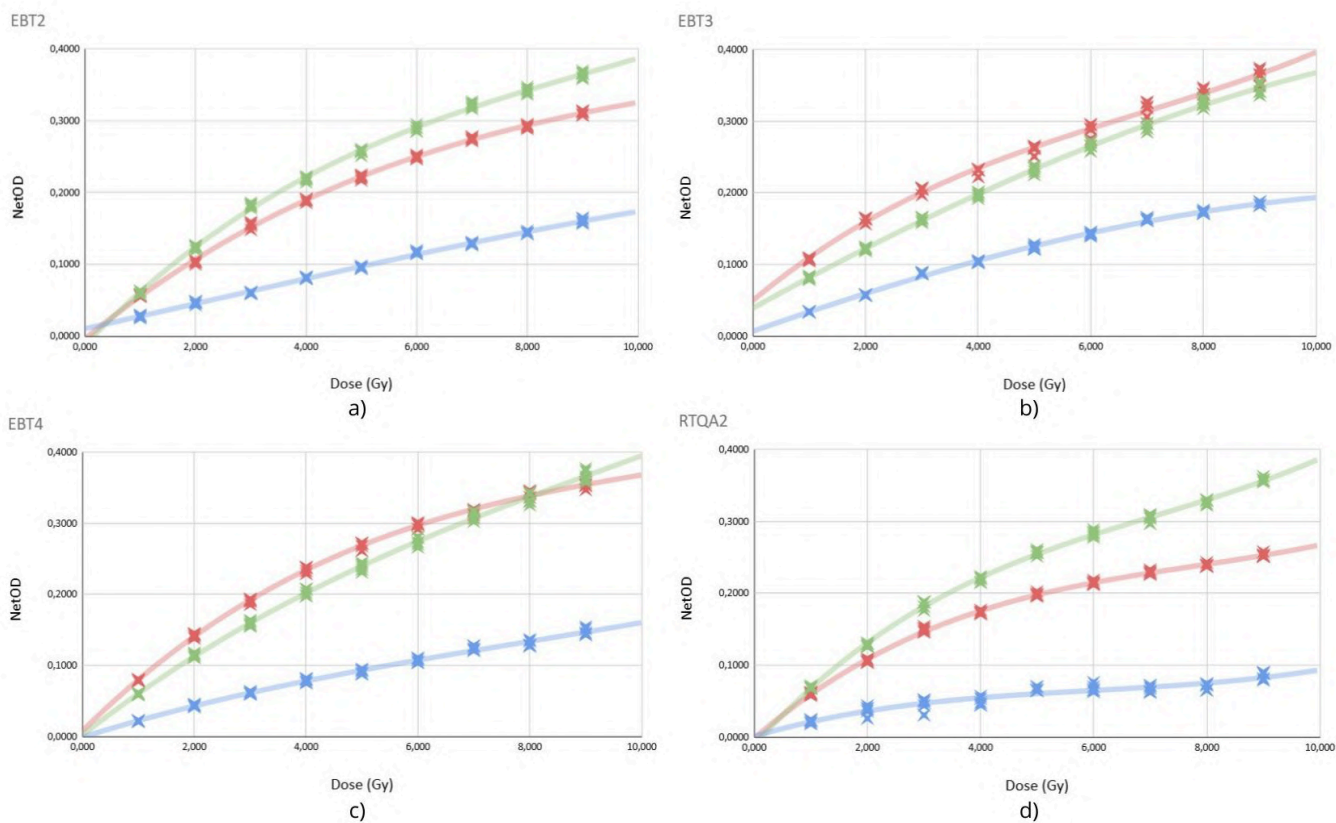


Figure 4: Average calibration curves for each film model and color channel. a) EBT2. b) EBT3. c) EBT4. d) RTQA2.

Table 1 presents the average NetOD uncertainty for each film model and color channel. Among the four films analyzed, the most recent model, EBT4, exhibited the lowest uncertainties across all channels. In contrast, the RTQA2 film showed the highest uncertainty values, particularly in the blue channel, which reached 0,43%. Overall, the green channel, selected in this study as the most suitable for calibration, showed the lowest uncertainty in the EBT2 film, similar uncertainties compared to the other channels in the EBT3 and EBT4 films, and slightly higher uncertainty than the red channel in the RTQA2 model.

Table 1: Average NetOD uncertainty for each film model and color channel.

Film model	Color channel	Average NetOD uncertainty
EBT2	Red	0,26%
	Green	0,23%
	Blue	0,24%
EBT3	Red	0,20%
	Green	0,20%
	Blue	0,20%
EBT4	Red	0,15%
	Green	0,15%
	Blue	0,15%
RTQA2	Red	0,30%
	Green	0,32%
	Blue	0,43%

4 Conclusions

This study evaluated the post-irradiation behavior, sensitivity, linearity, and NetOD uncertainty of four Gafchromic film models (EBT2, EBT3, EBT4, and RTQA2) across the red, green, and blue color channels. The results demonstrate that NetOD increases over time following irradiation, with the most significant changes occurring in the first hours and gradually stabilizing after approximately 15 days. This time-dependent behavior highlights the necessity of carefully defining the post-irradiation scanning interval to ensure consistent and reliable dosimetric results.

The analysis of D_m/D_d ratios obtained from early and late calibration curves revealed clear discrepancies, confirming that using a single calibration curve regardless of scanning time may introduce significant errors in dose estimation. Therefore, calibration protocols must account for the evolution of optical density with time to improve accuracy.

Among the three color channels, the green channel showed the most favorable combination of linearity and sensitivity, making it the most suitable choice for calibration purposes. Additionally, uncertainty analysis indicated that the EBT4 film models showed the lowest NetOD uncertainties across all

channels. On the other hand, the RTQA2 film exhibited the highest uncertainties, particularly in the blue channel.

Despite being manufactured in 2010, the EBT2 film still showed satisfactory performance in all analyses. Suggesting that, when stored under controlled conditions, older film batches may remain suitable for use in dosimetric applications.

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Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment

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Abstract

Lung cancer holds the highest global incidence and mortality rates[IARC,2014]. Tobacco use remains the primary risk factor for various cancers[NIH,2013]. According to INCA[2025], for every BRL 1 of profit generated by the tobacco industry, Brazil spends BRL 5 on related diseases, leading to annual losses of BRL 153 billion, of which BRL 67.2 billion are direct medical costs and BRL 86.3 billion result from productivity loss due to premature death, disability, and informal care[INCA,2025]. Computed tomography (CT) is indicated for diagnosis, treatment planning, staging, follow-up, and evaluation of cancer extent[NIH,2023]. Evidence suggests that low-dose CT screening in high-risk populations can significantly reduce lung cancer mortality[NIH,2013]. The TNM staging system is widely used to evaluate the size and extent of the primary tumor (T), spread to regional lymph nodes (N), and presence of metastases (M)[AJCC,2025]. CT involves exposure to ionizing radiation, and its dosimetry is primarily based on the Computed Tomography Dose Index (CTDI)[ICRP,2000]. Examination parameters and dosimetric data are recorded and accessed through the international standard DICOM (Digital Imaging and Communications in Medicine). The effective dose is important for radiological protection, especially in sensitive populations. This study aims to analyze the application of CT in the context of oncological neoplasms in a public university hospital, based on the principles of justification and optimization of radiological practice, in line with radiological protection guidelines. This study is part of the project “Radiology Scenario at the SUS University Assistance Hospital” (CAAE 71737417.9.0000.5149), and includes a survey of chest CT scans from outpatients conducted in the first quarter of 2025. The clinical profiles analyzed showed that 55.43% of patients were female and 44.57% male, with a mean age of 61.38 ± 11.79 years. Dosimetric parameters were collected from two CT scanners: PHILIPS INCISIVE 64-slice (CT A) and CANON AQUILION 64-slice (CT B). The chest CT parameters were: 120 kV (both A and B), pitch of 0.80 (A) and 0.83 (B), with a mean current-time product of 74.25 ± 18.48 mAs (A) and 723.01 ± 316.17 (B). For CT A, the mean DLP was 267.06 ± 86.10 mGy.cm, CTDI_{vol} was 8.64 ± 2.56 mGy, and effective dose was $23.44 \pm$

9.77 mSv. For CT B, the mean DLP was 263.95 ± 130.27 mGy.cm, $CTDI_{vol}$ was 37.87 ± 34.75 mGy, and effective dose was 31.42 ± 19.47 mSv. Both examinations remained below the reference level established by the ICRP[2000] of 650 mGy.cm. CT A presented a $CTDI_{vol}$ lower than the UK's Diagnostic Reference Level (DRL) of 13 mGy[CAMPELO,2019]. However, CT B showed higher doses compared to CT A, possibly due to a higher current-time product. Therefore, considering the clinical justification it can be concluded that dosimetric assessment is essential for optimizing imaging protocols based on Diagnostic Reference Levels (DRLs), while ensuring and enhancing radiological protection for patients.

Keywords: Computed Tomography (CT); lung cancer; oncology; Dosimetry.

1 Introduction

In Cancer Today, presented by the Global Cancer Observatory (GCO/IARC), an estimated 19,976,490 new cancer cases were reported globally in 2022, with 9,743,832 cancer-related deaths. Lung cancer is the type of cancer with the highest incidence and mortality rate for both sexes and all ages (IARC, 2024).

Tobacco use is the most important risk factor for lung cancer, as well as for several other types of cancer (NIH, 2023). According to a report by INCA (2025), for every R\$1 in revenue generated by the tobacco industry, Brazil incurs R\$5 in healthcare and social costs related to tobacco-induced diseases. Of the country's annual economic losses, estimated at R\$153 billion, direct medical expenses account for R\$67.2 billion—representing approximately 7% of total health expenditures—while R\$86.3 billion are attributed to indirect costs from lost productivity due to premature deaths, disability, and informal caregiving (INCA, 2025).

Computed tomography (CT) is indicated for the diagnosis, treatment planning, staging, and monitoring of oncological diseases (NIH, 2023). According to the National Lung Screening Trial (NLST), low-dose CT screening for lung cancer in people at high risk for the disease demonstrated a reduction in CT-related deaths compared to chest radiography screening (NIH, 2013). Chest CT is also instrumental in assessing the extent of the disease, including tumor size, lymph node involvement, and the presence of metastases in other organs. These parameters are critical for determining treatment strategies and prognostic outcomes (KING et al., 2021). The TNM staging system, developed collaboratively by the Union for International Cancer Control (UICC) and the American Joint Committee on Cancer (AJCC), remains the standard framework for cancer staging. It classifies disease based on the size and local extent of the primary tumor (T), involvement of regional lymph nodes (N), and the presence of distant metastasis (M) (AJCC, 2025).

Radiographic image data are obtained through the Picture Archiving and Communication System (PACS), and dosimetric parameters were derived from metadata embedded in Digital Imaging and Communications in Medicine (DICOM) files, an international standard for medical imaging. Computed Tomography Dose Index (CTDI) values served as the basis for CT dosimetry calculations. This study followed the dosimetric parameter guidelines established by the American Association of Physicists in Medicine (AAPM), which emphasize that the biological effects of ionizing radiation depend not only on the absorbed dose but also on the radiosensitivity of the irradiated tissue or organ. Accordingly, the effective dose was estimated as a radiological protection metric, primarily developed for occupational exposure, while also accounting for variations across different age groups (AAPM, 2008).

This study aims to evaluate the use of chest computed tomography in the context of oncological neoplasms at a public university hospital in Belo Horizonte/MG/Brazil. The analysis includes a survey of the number of examinations performed, analyzing the clinical indications and patient demographics, and assessing dosimetric assessment based on Diagnostic Reference Levels (DRLs), in accordance with radioprotection principles.

2 Methodology

This is one of the research lines of the project “Radiology Scenario at the University Hospital Assistance of the SUS” (CAAE 71737417.9.0000.5149) and is linked to the action research of the extension project “Assistance in Radiology: Bringing the Experiences of the University, the Public Hospital and the Community Closer Together” (SIEX-UFMG 402801).

This study surveyed the number of chest CT scans performed at the Imaging Diagnostic Unit (IDU) of a university hospital in Belo Horizonte (MG/Brazil). The analysis focused on elective outpatient scans with clinical indications of neoplasia, including breast and uterine cancer. Data collection was limited to a 90-day period during the first half of 2025 and excluded hospitalized (inpatient) cases

Body Mass Index (BMI), a parameter used to assess body mass in relation to height and to classify obesity into subcategories, was calculated for patients with available height and weight data.

To estimate DRLs in CT, the criteria established by the International Atomic Energy Agency (IAEA) were followed, which recommends collecting data from at least 30 patients per examination protocol. For patient selection, the IAEA guidelines recommended to include adult patients weighing between 50 kg and 90 kg, regardless of sex (IAEA, 2024).

Anonymized data were collected from two CT scanners: a 64-channel PHILIPS INCISIVE (CT A) and a 64-channel CANON AQUILION (CT B). The DICOM information provides the following data: CTDIvol (mGy), DLP (mGy-cm), voltage (kV), average current-time product (mAs), pitch, and series. CTDIvol represents the average absorbed radiation dose in the x, y, and z directions, estimating the average radiation dose within the irradiated volume for an object, reflecting the total absorbed energy, that is, the potential biological effect, attributable to the complete scan acquisition, from the ratio of the weighted CTDI (CTDIw) to the pitch.

To mitigate the influence of variability in the number of series acquired per exam, the DLP, CTDIvol, and cumulative mAs values extracted from the DICOM headers were normalized by the number of series corresponding to each exam. This approach allowed for standardization of the data for calculating the medians, means, and standard deviations of these relative parameters per series, favoring more equitable comparisons between different protocols and equipment.

The effective dose E (mSv) is normalized by the dose-length product (DLP) for adults (standard physicist) and the conversion factor K ($\text{mSv mGy}^{-1} \text{cm}^{-1}$), as in Equation 1.

$$E = \approx k \cdot DLP \quad (1)$$

The effective dose was calculated using a conversion factor K equal to $0.014 \text{ (mSv mGy}^{-1} \text{cm}^{-1})$, and was applied, according to the AAPM table (2008), which assumes the use of a CT body phantom with a diameter of 32 cm. The calculation was estimated based on the accumulated DLP, according to the recommended conversion coefficients.

3 Results and Discussions

Among the results collected, the number of CT scans performed in the service during the analyzed quarter was 3,768, considering both inpatients and elective outpatients. Chest CT scans of elective outpatients accounted for 4.67% of the total number, in a sample of 175 CT scans.

Among the exams studied, 25.57% had clinical indication of breast neoplasms, followed by 12.50% of lung neoplasms, with liver neoplasms in third place with 5.68%, as shown in Figure 1. A higher prevalence of neoplasms in the breast region is observed among females, while the digestive system is

commonly affected in both sexes. The data highlight significant differences in incidence based on sex and anatomical location.

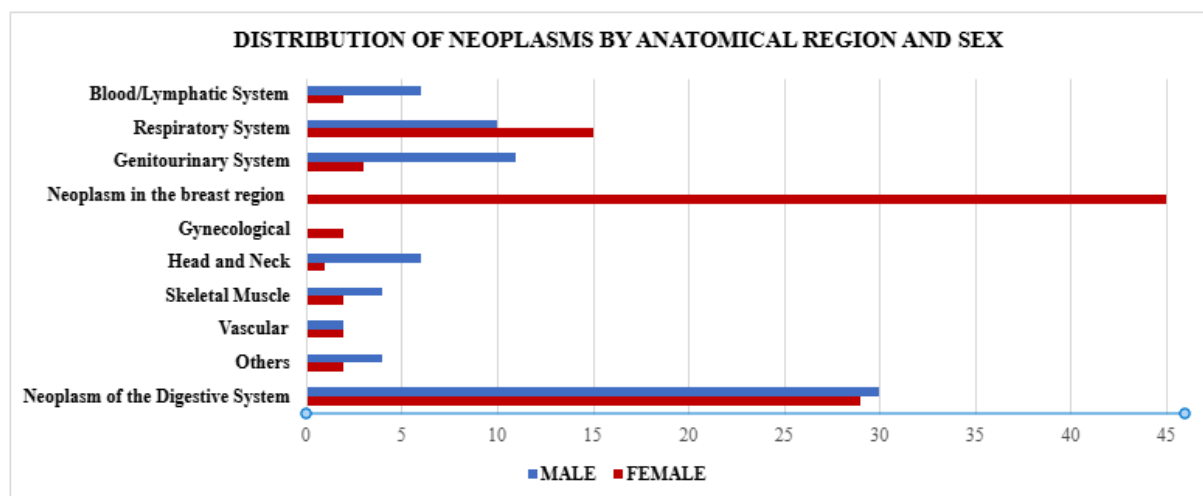


Figure 1: Distribution of neoplasm cases by anatomical region and patient sex.

Among the patients evaluated, 55.43% were female and 44.57% were male. The overall mean age was 61.38 ± 11.79 years, with a mean of 62 ± 12.52 years for women and 66 ± 13.47 years for men. Regarding anthropometric characteristics, the overall mean Body Mass Index (BMI) was 24.79 ± 3.91 kg.m^{-2} , with women presenting a mean BMI of 26.76 ± 3.68 kg.m^{-2} and men 24.16 ± 3.85 kg.m^{-2} . The WHO (2025) defines an adult as overweight if they have a BMI greater than or equal to 25 kg.m^{-2} and obese if they have a BMI greater than or equal to 30 kg.m^{-2} . The overall average in this study was below the obesity threshold, as shown in Figure 2.

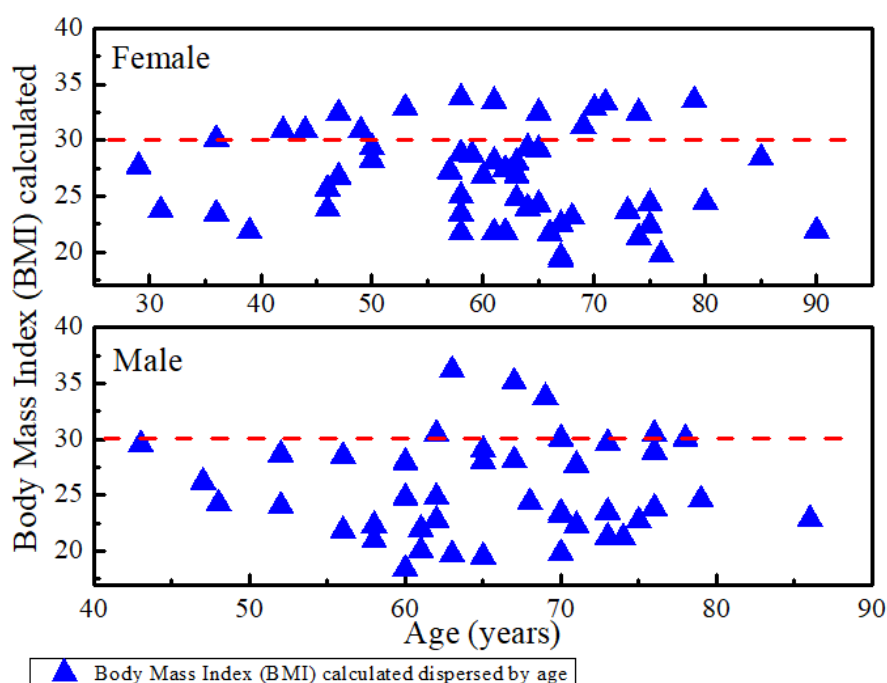


Figure 2: Body Mass Index (BTI) distribution for age patients surveyed, male and females, with BTI reference 30 for obesity.

The likelihood of developing cancer is influenced by multiple factors. Primary prevention involves limiting exposure to carcinogenic agents or risk factors such as tobacco use, a sedentary lifestyle, and an inadequate diet. Although the precise role of diet in cancer development remains uncertain, evidence suggests that a diet rich in fruits, vegetables, legumes, whole grains, and cereals, and low in fats especially of animal origin, helps reduce the risk of cancer and other chronic noncommunicable diseases. High total caloric intake and the consumption of red meat, fats, and milk are not recommended, while the consumption of fruits, vegetables rich in carotenoids (such as tomatoes and carrots), and legumes (such as beans, peas, and soybeans) may exert a protective effect due to the presence of vitamins (A, D, and E) and minerals (selenium). Certain substances produced during food preparation, such as heterocyclic amines and polycyclic aromatic hydrocarbons, as well as factors like insulin-like growth factor, excessive alcohol consumption, tobacco use, and vasectomy, have been identified as potential dietary or lifestyle-related cancer risks. Secondary prevention focuses on procedures that enable early diagnosis or the detection of precancerous lesions (INCA, 2002; BACELAR JÚNIOR et al., 2015). In this context, in a secondary prevention, knowing the importance of diagnosis of neoplasms in the early stages, computed tomography becomes an important tool in the early detection and diagnosis of neoplasms. In this context, within the scope of secondary prevention, computed tomography is a valuable tool for the early detection and diagnosis of neoplasms, underscoring the importance of identifying cancer at initial stages.

The parameters used for chest CT were: 120 kV (CT A and CT B), Pitch Ratio varied from 0.80 (A) and 0.83 (B), and the mean current product was 74.25 ± 18.48 mAs (A) and 723.01 ± 316.17 mAs (B). For CT A, the mean DLP was 267.06 ± 86.10 (mGy.cm), $CDTI_{vol}$ 8.64 ± 2.56 (mGy), and effective dose 2.93 ± 1.50 (mSv). For CT B, the mean DLP was 263.95 ± 130.27 (mGy.cm), $CDTI_{vol}$ 37.87 ± 34.75 (mGy), and effective dose 31.42 ± 19.47 (mSv). Table 1 shows a detailed comparison of the parameters with the medians of DLP and $CDTI_{vol}$.

Table 1: Comparison between the parameters of CT A and CT B of the exams studied.

Parameters	CT A	CT B
Voltage (kVp)	120	120
Mean Tube Current-Time Product (mA.s) $\pm$ Desvpad	$74.25 \pm 18.48\%$	723.01 ± 316.17
Pitch	0.80	0.83
DLP (mGy.cm) Median/Mean $\pm$ Desvpad	$244.51/267.06 \pm 86.10$	$209.97/251.58 \pm 142.81$
$CDTI_{vol}$ (mGy) Median/Mean $\pm$ Desvpad	$8.96/8.64 \pm 2.56$	$19.92/32.68 \pm 25.76$
Effective dose (mSv) $\pm$ Desvpad%	2.93 ± 1.50	31.42 ± 19.47

Considering the reference levels (ICRP, 2000) for CT examinations in adult patients DLP by routine chest examination of 650 (mGy.cm), the values found in this study, in both equipment, were below this threshold. According to Campelo (2019), chest CT scans have an effective dose of approximately 8 mSv, approximately 2.93 times higher than that observed for CT A in this study. This article also notes that the European Commission recommends $CTDI_{vol}$ values below 10 mGy for chest CT in 4- and 16-channel devices. Additionally, the Diagnostic Reference Levels (DRLs) for multislice devices in the United Kingdom recommend $CTDI_{vol}$ of 13 mGy for chest scans performed on adults for lung studies, which is 1.57 times higher than that found in CT A. International references for 64-channel CT are still limited. However, this study identified the need to optimize CT B protocols compared to CT A, possibly influenced by the higher average current product applied. According to the service studied, CT B is being deactivated, justifying the replacement due to difficulty in replacing maintenance parts. Further studies to optimize parameters are necessary.

In the study by Kumsa et al. (2023), for the chest CT study, the median CTDI_{vol} was 7.58 and the median DLP was 300.50 mGy.cm, and the defined DRLs were CTDI_{vol} 13 mGy and DLP 63 mGy.cm. The medians found in this study, comparatively, were 18.19% higher for CTDI_{vol} and 18.63% lower for DLP.

A dispersion study was performed using the median CTDI_{vol} (Figure 4) and DLP (Figure 5) values for CT A and CT B. Each figure displays the sampling distribution, including the mean and interquartile range (25th to 75th percentiles), with color coding to distinguish between devices. Figure 4 shows a greater difference in the dispersion of the CTDI_{vol} data, making it necessary, for better observation of the parameters, to place the X-axis with different maximum values, with CT B being in a range up to 3 times larger. Figure 5 reveals the DLP was in similar ranges, allowing both datasets to be plotted on the same X-axis. Although CTA data exhibit slightly greater dispersion, the interquartile ranges around the medians and means are similar between both devices.

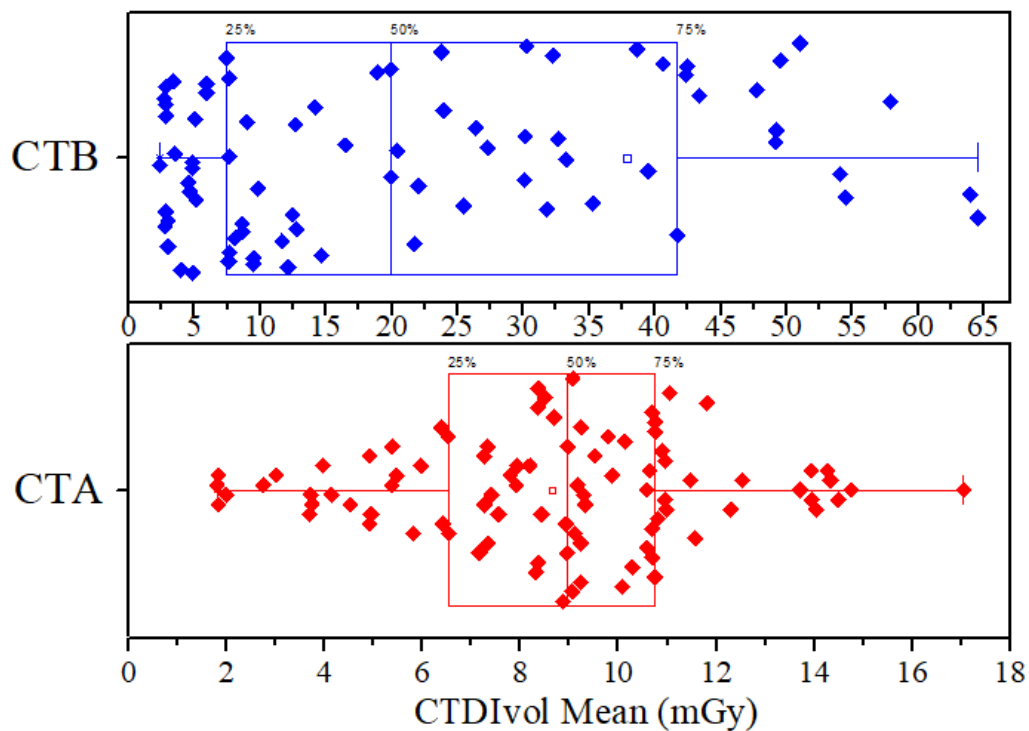


Figure 4. Box and whisker plot of median CTDI_{vol} values for each TC model.

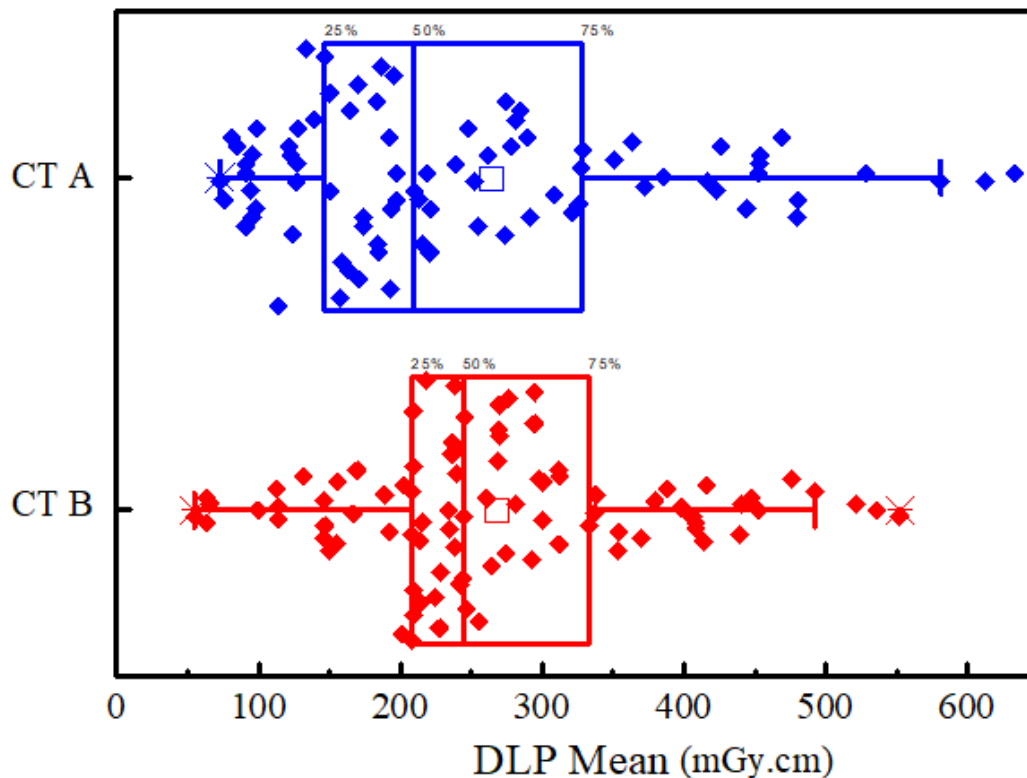


Figure 5. Box and whisker plot of median DLP values for each TC model.

4 Conclusions

In conclusion, chest computed tomography (CT) is an important tool in oncology, both in the treatment, staging, and follow-up of lung cancer and other neoplasms, reinforcing their clinical and social relevance. This study evaluated the clinical indications for chest CT exams and the profiles of the patients undergoing these procedures. Among the sample, 55.43% were female and 44.57% were male. The mean age was 61.38 ± 11.79 and the mean overall BMI was $24.79 \pm 3.91 \text{ kg.m}^{-2}$, below the threshold for obesity-related risk. In the DRL study, the parameters of adult patients were assessed. The DLP found in both equipment investigated had values below the ICRP (2000) threshold. CT A presented lower values of DLP, CTDIvol, and effective dose. However, CT B is already in the process of discontinuing the service. In the context of justification for medical imaging, where no strict dose limits are applied to patients, the adherence to DRLs are necessary. The dosimetric assessments of CT protocols is essential to determine the need for parameter optimization and ensure patient radiation protection.

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Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density

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Abstract

The physical quantities electron density (ρ_e) and effective atomic number (Z_{eff}) are essential in dose calculations for radiotherapy and play a key role in image formation and material characterization in diagnostic imaging. Their reference values can be estimated by weighting the chemical elements composing a sample when its composition is known. In this work, two methodologies were employed to estimate ρ_e and Z_{eff} : DEEDZ, based on Dual Energy CT images, and TEMPy, based on photon interaction cross-sections, and implemented in a web-based application. The results were evaluated and compared with reference values. Two phantoms were used: one for calibration (employed in the DEEDZ method), and a test phantom composed of water, paraffin, and polyethylene, which was used to evaluate the methods. A measured 50 kV X-ray spectrum from a tungsten target was adopted. DECT image projections were obtained through Monte Carlo simulations (GEANT4 code) considering an ideal photon-counting detector, and reconstructed using the Filtered Back Projection (FBP) method, with a 30 keV energy threshold. The same parameters were also applied in TEMPy. The Mean Absolute Percentage Error (MAPE) for comparison between reference values and the DEEDZ/TEMPy methodologies were better than 10%/1.4% for ρ_e and 7.1%/7.1% for Z_{eff} , respectively. The best estimate of ρ_e was obtained for water using TEMPy (MAPE = 0%), while the best estimate of Z_{eff} was achieved for polyethylene using DEEDZ (MAPE = 0.0(1)%). When comparing DEEDZ and TEMPy directly, the differences ranged from 0.79(9)% for the Z_{eff} of water to 13.1(3)% for the Z_{eff} of paraffin. The DEEDZ methodology produced better results for estimating Z_{eff} , while TEMPy provided a more accurate estimation of ρ_e . However, both methods demonstrated good accuracy for these quantities and can be used complementarily for tissue and material studies.

Keywords: Effective Atomic Number, Electron Density, PCCT, TEMPy, Monte Carlo Simulation, DEEDZ.

1 Introduction

Electron density (ρ_e) and effective atomic number (Z_{eff}) are fundamental parameters that describe how materials interact with ionizing radiation. Electron density influences Compton scattering, while the effective atomic number, a weighted average of atomic numbers, governs photoelectric absorption. These quantities are essential for dose calculations in radiotherapy and for distinguishing materials in diagnostic imaging (Torikoshi et al.; 2003). Traditional methods to estimate ρ_e and Z_{eff} rely on known material compositions, but advanced techniques like dual-energy CT (DECT) (Badalova et al., 2008a; Badalova et al., 2008b) and photon-counting CT (PCCT)

(Saito M., 2019) enable their derivation from imaging data (Landry et al., 2011), offering non-invasive and accurate alternatives. In this work, was used the electron density relative to water and, for simplicity, it was referred as ρ_e .

An alternative approach is implemented in the TEMPy toolkit (Boiset et al., 2024), which uses photon interaction cross-sections, based on Manohara method (Manohara et al, 2008), to estimate ρ_e and Z_{eff} . TEMPy employs a weighted least-squares optimization to match the linear attenuation coefficients of tissue-equivalent materials (TEMs) to reference values across a specified energy range. Its integration with spectral modeling tools like SpekPy enhances TEMPy's versatility, enabling tailored TEM formulations for specific imaging applications. The Manohara method provides a comprehensive set of formulas for Z_{eff} and ρ_e , applicable to a wide range of materials and energies. Other alternative is the method introduced by Saito and Sagara (Saito M. and Sagara, 2017; Saito M. and Saito T., 2025), called DEEDZ, which refers to a way of using dual energy CT to estimate electron densities and effective atomic numbers. This method simplifies the estimation by using a single linear relationship for ρ_e and a power-law model for Z_{eff} , which leverages dual-energy data to derive these quantities. This approach reduces computational complexity while maintaining high accuracy, as validated through simulations and experimental data.

This study explores the use of a photon-counting detector (PCD) as an alternative PCCT technology for estimating ρ_e and Z_{eff} through Monte Carlo simulations using GEANT4 (Agostinelli et al, 2003; Arce et al., 2025; Allison et al., 2006; Allison et al., 2016). The results were compared with TEMPy estimative, which implements a well-established methodology for estimating these quantities, and its useful in characterizing materials. All methodologies were compared with reference values, and agreement metrics were calculated.

2 Methods

2.1 Simulation Setup

A Monte Carlo simulation was developed using the GEANT4 toolkit (version 11.1.2) to model a photon-counting detector. Figure 1 presents the simulation environment inspired by previous CT geometry (Carramate et al., 2019), which was filled with air, and includes both a calibration phantom and a test phantom, simulated independently under identical beam and detector configurations.

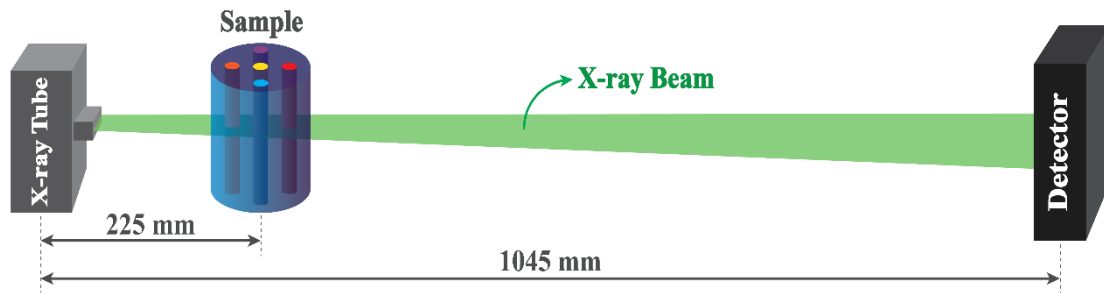


Figure 1: Simulated experimental setup for the cross-section acquisition. The sample was rotated from 0° to 180° angle at 1° steps, and the primary (without the phantom) and transmitted X-ray spectra was acquired.

The calibration phantom consisted of a cylindrical water volume (6 mm radius, 10 mm height). Five cylindrical inserts (1 mm radius), composed of different materials (polypropylene, nylon-6/6, silicon dioxide, polycarbonate, and Teflon) were embedded within the phantom to provide material contrast. The test phantom used the same external water cylinder as the calibration phantom and

contained two cylindrical inserts, one made of polyethylene with a 1.5mm radius, and another made of paraffin with a 2.5 mm radius. Both phantoms are illustrated in Figure 2.

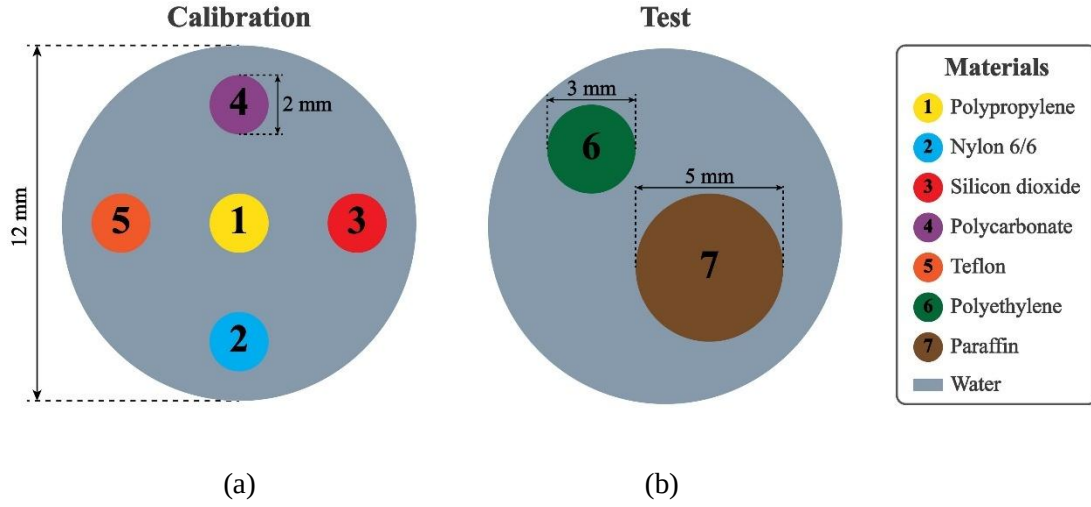


Figure 2: Sketch of the water phantom for (a) calibration with inserts of polypropylene, nylon 6/6, silicon dioxide, polycarbonate and Teflon, and (b) test with inserts of polyethylene and paraffin, in order.

An experimental X-ray spectrum was measured from a 50 kV tungsten target, using an MCN 421 (Philips Inc, Netherlands) and a CdTe spectrometer (model XR-100T, Amptek Inc., USA), and was subsequently employed as input to simulate the parallel X-ray beams in GEANT4. A planar photon-counting detector with an effective area of 6.5 mm x 6.5 mm and ideal efficiency was modeled, generating an image matrix of 256 x 256 pixels, and positioned to record the transmitted photons. To generate the projection data, the phantom was rotated from 0° to 180° in 1° steps, resulting in a total of 181 projections. For each angular projection, 3 million primary photons were simulated. All projection data was reconstructed using a filtered back projection (FBP) algorithm. A threshold of 30 keV was defined, allowing splitting the projection data into two datasets, relative to photons with energy below and above 30 keV, respectively, which each dataset was then used to reconstruct a CT image. For each image, the CT numbers of the materials were calculated using regions of interest (ROIs). Due to the presence of salt-and-pepper noise, a median filter with a 3 x 3 kernel was applied.

For materials with known chemical compositions, the electron density relative to water, referred to here only as electron density to simplify, and effective atomic number can be estimated using the following Equations (1)-(2) (Hünemohr et al, 2014):

$$\rho_e = \frac{\rho N_A \sum_i \omega_i \frac{Z_i}{A_i}}{\rho_e^W}, \quad (1)$$

and

$$Z_{\text{eff}} = \left(\frac{\sum_i \omega_i \frac{Z_i}{A_i} Z_i^m}{\sum_i \omega_i \frac{Z_i}{A_i}} \right)^{1/m}, \quad (2)$$

where ω_i , A_i and Z_i represent the elemental mass fraction, atomic mass and atomic number of the i -th element in the compound, respectively. ρ denotes the mass density of the compound, ρ_e^W the electron density of water and m represents the contribution of Compton and photoelectric to the

total photon attenuation, varying between 2.8 to 3.8 in the diagnostic X-ray energy range. A $m = 3.3$ was adopted as optimal value based on Landry (2011). With these expressions, the nominal ρ_e and Z_{eff} were established for the test and calibration phantom materials and inserts, and are shown in Table 1.

Table 1: References values for electron density and effective atomic number for each phantom material.

	Nominal			Nominal	
Sample	ρ_e	Z_{eff}	Sample	ρ_e	Z_{eff}
Water	1.000	7.478	Silicon dioxide	2.087	11.685
Polyethylene	0.966	5.501	Nylon 6/6	1.125	6.18
Paraffin	0.960	5.484	Teflon	1.902	8.461
Polypropylene	0.925	5.501	Polycarbonate	1.139	6.308

2.2 TEMPy Methodology

The main goal of TEMPy as a toolkit is to improve the formulation of Tissue Equivalent Materials (TEMs). However, when the chemical formulation of a sample is known, this toolkit can also be used to estimate the electron density and effective atomic number, based on photon interaction cross-sections method, developed by Manohara, and the composition from NIST XCOM database (Berger and Hubbell, 1987), both implemented in a web application (Boiset et al, 2024).

In this study, the input parameters included a tungsten anode tube, without filtration, and a tube voltage of 50 kV. Each material from phantom composition was entered as additive material, one by one, and its electron density and effective atomic number were estimated individually. To obtain the electron densities relative to water, a ratio between the samples and water electron density was calculated. Figure 3 illustrated the TEMPy toolkit graphical user interface (GUI) for paraffin estimation.

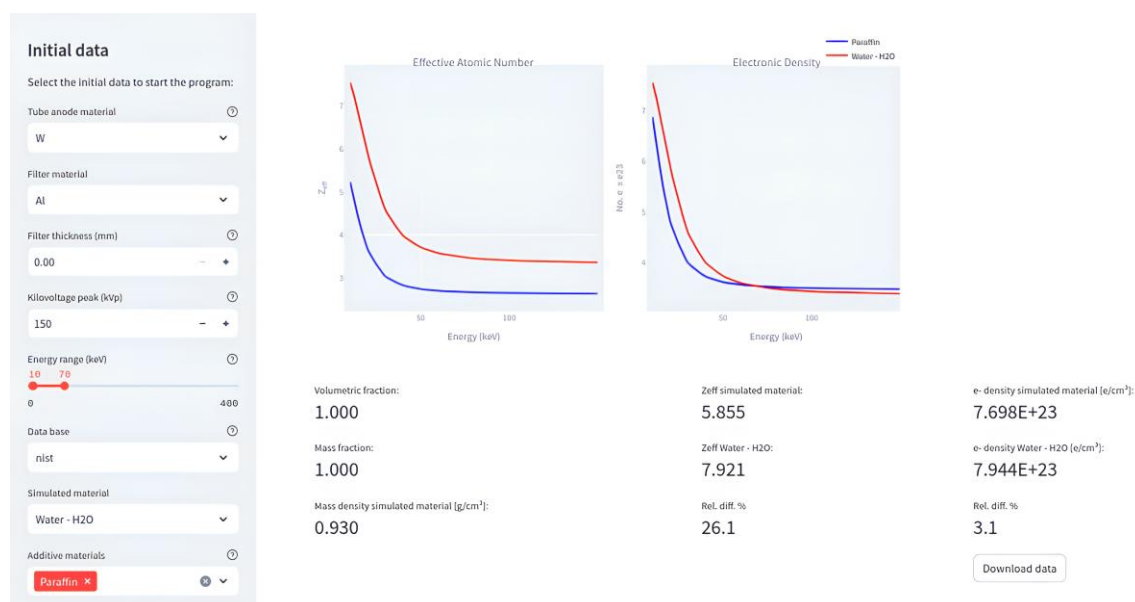


Figure 3: TEMPy GUI platform. The inputs parameters are entered in the left column, and the results are displayed on the right for a paraffin example.

2.3 DEEDZ Methodology

Resuming the methodology proposed by Saito and Sagara (2017), the composition in the image domain from CT data, referred to as DHU, is estimated for both high and low energy bands, pondered by an α parameter. This estimation considers average region-on interest (ROI) CT numbers in Hounsfield Units (HU) for high (CTN_{high}) and low (CTN_{low}) energy spectral ranges

$$DHU = (1 + \alpha)CTN_{high} - \alpha CTN_{low}. \quad (3)$$

This expression exhibits a linear relationship with ρ_e , as expressed in Equation (4):

$$\rho_e = a \left(\frac{DHU}{1000} \right) + b. \quad (4)$$

By fitting this function, it is possible to estimate the ρ_e from CT images, obtained at two energy bands. To achieve this, the calibration phantom was used to fit the equation, determining the slope (a) and intercept (b), parameters that must be one as estipulate by Saito M. (2012). When applying this fit to the test phantoms images, a simulated ρ_e can be calculated, which was compared with nominal values. The α parameter was numerically estimated by selecting the value that maximize the R^2 of the fit.

Particularly for low energy image, the reduced CT number (η) for a CT image, which is related to CTN and the linear attenuation coefficient of the pixel ($\mu(E)$) relative to water ($\mu_w(E)$), is expressed in Equation (5):

$$\eta_{low} = \frac{CTN_{low}}{1000} + 1 = \frac{\mu(E_{low})}{\mu_w(E_{low})}. \quad (5)$$

An approximation of the linear attenuation coefficient as the sum of contributions of Compton scattering ($p(E)$) and photoelectric absorption ($q(E)$), is given in Equation (6):

$$\mu(E) \approx \beta_e (p(E) - q(E) Z_{eff}^m), \quad (6)$$

where β_e is the electron density per unit volume, not relative to water.

After some algebraic manipulations combining, Equations (5) and (6) yields Equation (7), which establishes the relationship between Z_{eff} and η :

$$\left(\frac{Z_{eff}}{Z_{eff,W}} \right)^m - 1 = \gamma_{low} \left(\frac{\eta_{low}}{\rho_e} - 1 \right), \quad (7)$$

here, γ_{low} is a proportionality parameter, and $Z_{eff,W}$ the effective atomic number of water. In this study, m was estimated in the same way as α parameter, by maximizing the linear fit between the left-hand and right-hand sides of Equation (7), with γ_{low} as the slope, and the intercept allowed to vary as a free parameter.

3 Results

According to the setup described in section 2.1, the dataset obtained using GEANT4 was used to generate the CT images for both calibration and test phantoms, showed in Figure 4.

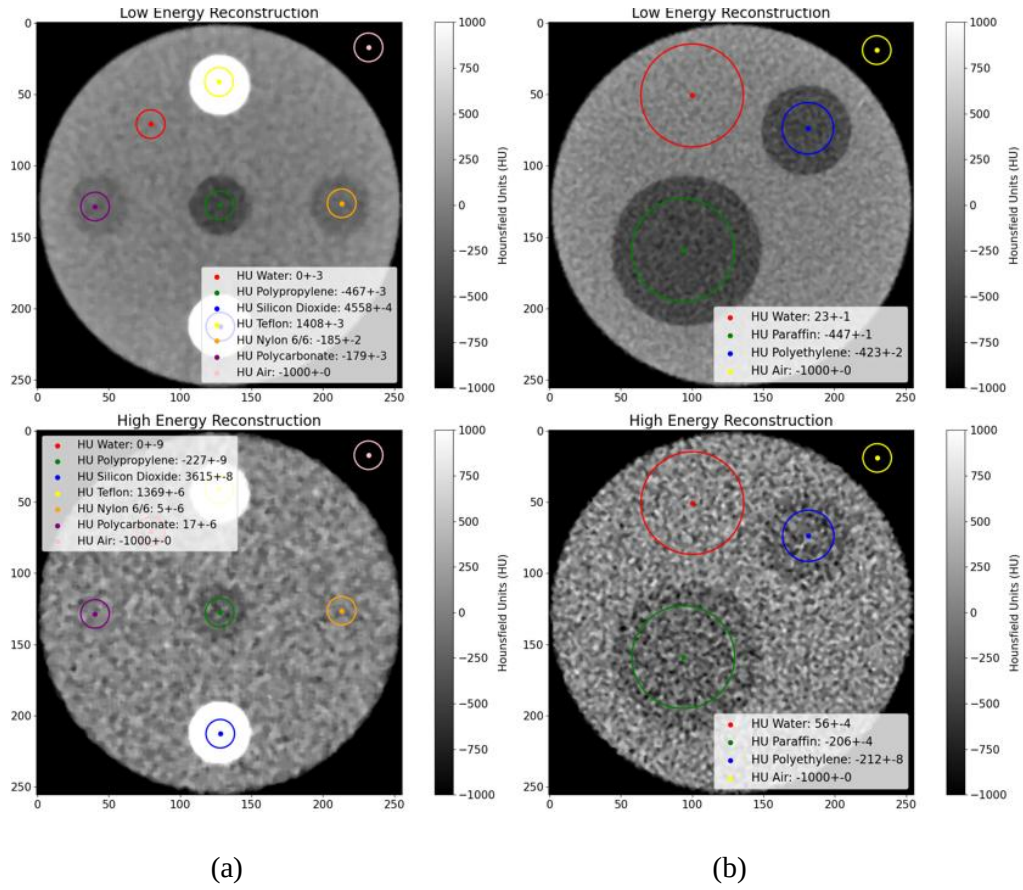


Figure 4: CT images for low and high energy bins, determined by a 30 kV threshold, of a cross section of (a) the calibration and (b) test phantoms. The colored circles represent the region of interest where the CT number was calculated for each material.

Mean CT numbers, and respective standard deviations of the calibration phantom materials, was used to establish the α fitting parameter (Equations (3) and (4)), yielded the highest R^2 value, as illustrated in Figure 5(a). Figure 5(b) presents the plot used to determine the electron density, resulting in $\alpha = 1.99$, $R^2 = 0.9843$, and the fitting parameters $a = 0.842(7)$ and $b = 0.843(7)$.

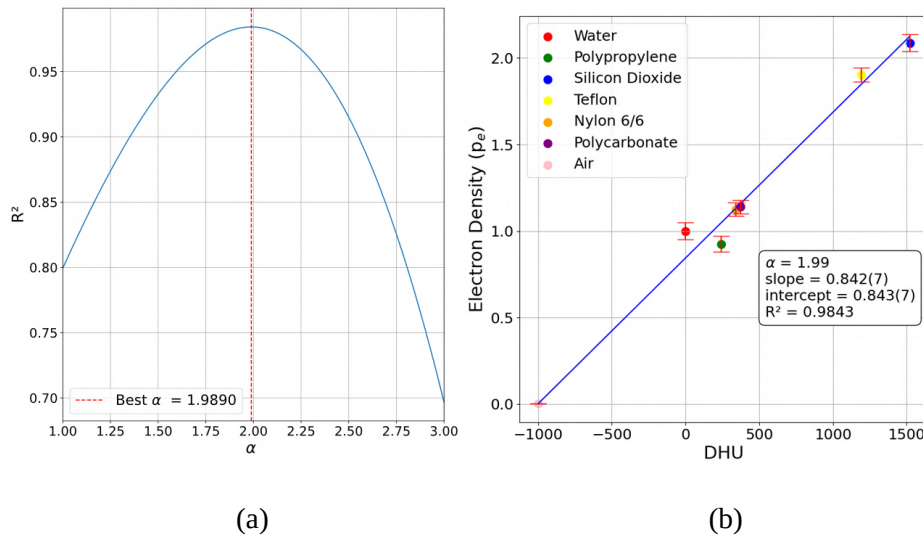


Figure 5: (a) Determination of parameter α that yields to highest R^2 of the fit. (b) Linear fit of electron density in function of DHU.

By applying this linear fit pixel-by-pixel to the subtracted CT image (DHU) of the test phantom, an electron density map can be generated, as shown in Figure 6. Using the defined ROIs, the electron density of each material can be estimated (Table 2).

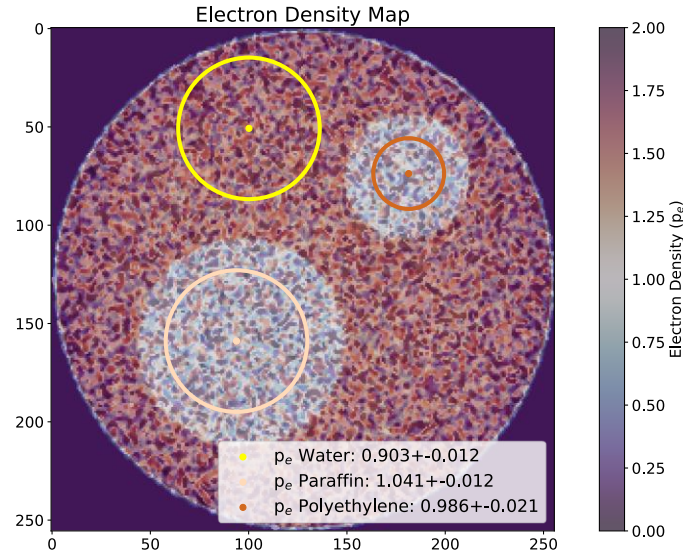


Figure 6: Electron density image obtained from the DHU estimation pixel-by-pixel.

Subsequently, Equation (7) was fitted for the Z_{eff} estimation. First, coefficients η_{low} for each material (Equation (5)) were calculated based on the ROIs shown in Figure 4, and the corresponding nominal electron densities were used to estimate the right-hand of the fitting function. Figure 7 shows the fit where $m = 2.70$, yielding the best $R^2 = 0.9997$. The slope of the fit was $\gamma_{\text{low}} = 1.399(2)$, and the intercept $0.019(2)$.

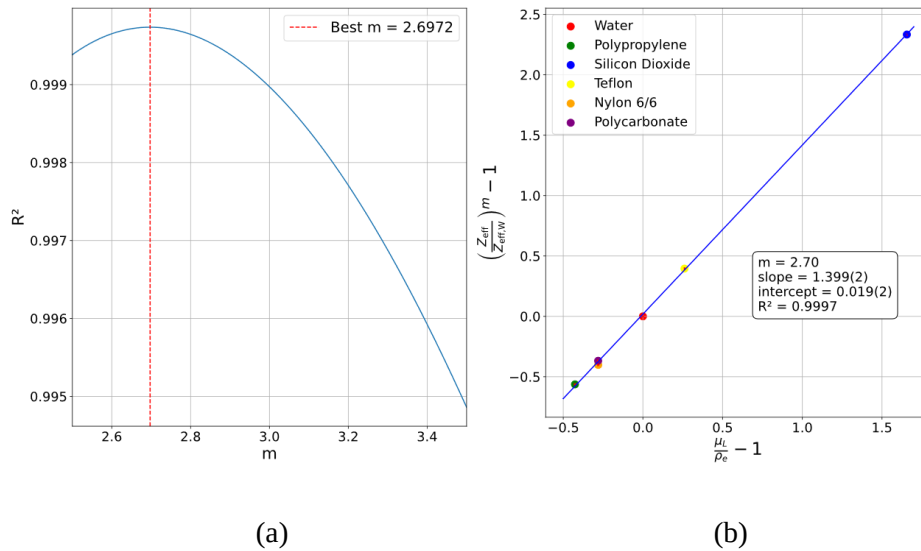


Figure 7: (a) Determination of parameter m that yields to highest R^2 of the Z_{eff} fit. (b) Linear fit to Z_{eff} estimation.

Using this fit, by calculating η_{low} for each material in each ROI and using the electron density values obtained in Figure 6, the Z_{eff} fit was applied pixel-by-pixel to the test phantom image, allowing the construction of an effective atomic number map, as shown in Figure 8.

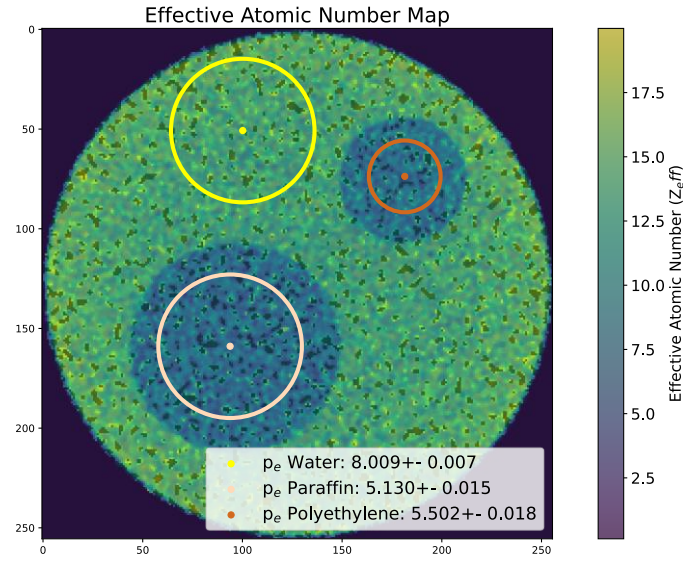


Figure 8: Effective atomic number map calculated by applying the linear fit from Equation (7), using the parameters of Figure 7, pixel-by-pixel.

The results obtained using TEMPy are presented in Table 2, alongside those obtained from DEEDZ methodology. The mean absolute percentage errors (MAPE) were calculated for each material and method, using the nominal values from material compositions taken as reference, shown in Table 1. The estimation using TEMPy yielded better results for determining the electron density of the materials, whereas DEEDZ was more accurate in estimating the effective atomic number. The MAPE for ρ_e using DEEDZ methodology showed higher values, as this quantity was obtained from a combination of the two CT images, thus summing the effects of low photon statistics.

Table 2: Comparison of the estimative for ρ_e and Z_{eff} between TEMPy and DEEDZ with nominal values, using the mean absolute percentage error as the metric.

Sample	TEMPy				DEEDZ			
	ρ_e	MAPE (%)	Z_{eff}	MAPE (%)	ρ_e	MAPE (%)	Z_{eff}	MAPE (%)
Water	1.000	0.0	7.946	5.9	0.90(1)	10(1)	8.009(7)	7.10(9)
Polyethylene	0.954	1.2	5.904	6.8	0.99(2)	2(2)	5.50(2)	0.0(1)
Paraffin	0.974	1.4	5.901	7.1	1.04(1)	8(1)	5.13(2)	6.5(1)

Table 3 presents the comparative results for each material considering the two methodologies, aiming to assess the agreement between them. For paraffin, a great disagreement between the methodologies is observed. The metrics show similar values for both approaches when compared to the nominal values. Despite the methods not being equivalent, each appears to be efficient for material characterization independently and could be used complementarily.

Table 3: Comparative evaluation of ρ_e and Z_{eff} considering TEMPy and DEEDZ approaches, using the mean absolute percentage error as the metrics.

Sample	TEMPy - DEEDZ	
	MAPE (%)	
	ρ_e	Z_{eff}
Water	10(1)	0.79(9)
Polyethylene	3(2)	6.8(3)
Paraffin	7(1)	13.1(3)

4 Conclusions

This work demonstrated the feasibility of using a photon-counting CT approach to determine the electron densities and effective atomic numbers for some materials, based on DEEDZ method, supported by Monte Carlo simulations using GEANT4.

According to Equation (4) and mentioned in section 2.3, a and b must be one, however, as illustrates in Figure 5, was obtained $a = 0.842(7)$ and $b = 0.843(7)$. In this case, two main factors affect this result. First, the image is influenced by noise due to the low photon statistics, which impacts the CT number of the materials and, consequently, the fitting. Second, as shown by Saito (2012), the model assumes that α is a single value, which is not a good approximation to low energies (<25 keV).

However, the comparison with TEMPy showed similar estimation when compared with reference values. TEMPy and DEEDZ methodologies do not seem to be equivalent, but both approaches proved to be good alternatives for material characterization. Future efforts will focus on increasing the number of photons in the simulation and implement spectra with higher energies.

Acknowledgements

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Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis

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Abstract

Breast cancer is the second most incident malignancy and the fourth leading cause of cancer-related death worldwide [IARC, 2024]. While mammography remains the recommended screening method [INCA, 2015], computed tomography (CT) is indicated for guiding biopsy procedures, as well as during staging and follow-up phases of the disease [NIH, 2024];[NIH, 2024]. The TNM staging system is widely used to assess the size and extent of the primary tumor (T), regional lymph node involvement (N), and the presence of distant metastases (M) [AJCC, 2025]. Since CT involves exposure to ionizing radiation, its dosimetry is based on the Computed Tomography Dose Index (CTDI), which can be retrieved through the international standard DICOM (Digital Imaging and Communications in Medicine). Effective dose estimation is essential parameter in radiological protection, particularly for sensitive populations [AAPM, 2008]. This study aims to analyze the use of chest CT in the context of breast neoplasms at a public university hospital, grounded in the principles of justification and optimization of radiological practices, in accordance with radiological protection guidelines. The study is part of the project “Radiology in the University Hospital Scenario of the Brazilian Unified Health System (SUS)” (CAAE 71737417.9.0000.5149). It involved a survey of chest CT scans of elective outpatients conducted during the first quarter of 2025, specifically with clinical indications related to breast pathology. Patient profiles and dosimetric parameters were analyzed. Among the results collected during the study period, the total number of CT exams was 3,768, with clinical indications relevant to this study representing 1.30% of this total, in a sample of 49 exams. All patients were female, with mean age of 60.19 ± 10.25 years; 6.25% were between 50 and 69 years, the recommended age range for screening [INCA, 2015], while 18.75% were below this threshold. DICOM data were extracted from two 64-slice CT scanners: PHILIPS INCISIVE (CT A) and CANON AQUILION (CT B). Technical

parameters included: 120 kV, pitch of 0.80 (A) and 0.83 (B), with mean tube currents of 500.25 ± 163.48 mAs (A) and $6,454.63 \pm 4,075.47$ mAs (B). Dosimetric results were as follows: for CT A, mean DLP of 270.91 ± 80.15 mGy·cm, CTDI_{vol} of 8.26 ± 1.92 mGy, and estimated effective dose of 3.0 ± 4.20 mSv; for CT B, mean DLP of 251.58 ± 142.81 mGy·cm, CTDI_{vol} of 32.68 ± 25.76 mGy, and effective dose of 31.28 ± 20.53 mSv. Both scanners yielded DLP values below the 650 mGy·cm diagnostic reference level established by the ICRP (2000), and CT A also recorded a CTDI_{vol} below the UK's DRL of 13 mGy [4]. Nevertheless, CT B demonstrated significantly higher dose values, likely attributable to the increased tube current-time product. Therefore, it is concluded that dosimetric surveys are necessary to optimize protocols, considering the justification of exam practices through diagnostic reference levels (DRLs), thereby safeguarding patient radioprotection during the procedure.

Keywords: breast cancer metastasis; chest CT exam; computed tomography (CT); Dosimetry.

1 Introduction

According to the Global Cancer Observatory (GCO/IARC), breast cancer ranks as the second leading cause of incidence and the fourth leading cause of cancer-related mortality worldwide (IARC, 2024). Age and body weight are recognized risk factors, with studies indicating that high serum estrogen levels, along with increased peripheral estrogen production, have been seen as the main contributors to the development of breast cancer in overweight postmenopausal women (MOHANTY AND MOHANTY, 2021). Hereditary breast and ovarian cancer (HBOC) syndrome occurs in approximately 5% to 10% of breast cancer cases, caused by mutations in the BRCA1 and BRCA2 genes. (KAZMI *et al.*, 2008; Amirikia *et al.*, 2013).

Mammography is the screening modality recommended by the Brazilian Ministry of Health and the National Cancer Institute (INCA/MS) for women aged 50 to 69 (INCA, 2015). However, for guiding biopsy procedures, staging breast cancer, and conducting follow-up assessments, computed tomography (CT) serves as a valuable tool for evaluating both female and male breast anatomy, according to the United States National Cancer Institute (NIH) (NIH, 2024; NIH, 2025). In the study by SON *et al.* (2016), the prevalence of breast cancer among incidental lesions detected by CT ranged from 24% to 70% and may include primary and secondary malignancies. In addition to benign lesions, including benign calcifications, fibroadenomas, and lipomas, radiologists are advised to carefully evaluate breast tissue during routine chest CT examinations (SON *et al.*, 2016).

Staging refers to the assessment of disease extent, including tumor size, involvement of regional lymph nodes, and the presence of distant metastases, factors that are critical in guiding treatment decisions and determining prognosis. Follow-up involves systematic monitoring after the completion of treatment to detect possible recurrences or the emergence of new pathological findings. The Union for International Cancer Control (UICC) and the American Joint Committee on Cancer (AJCC) jointly develop and periodically revise the TNM staging system, which is widely employed for classifying various malignancies. It assesses the size and extent of the primary tumor (T), regional lymph node involvement (N), and the presence of distant metastases (M) (AJCC, 2025).

According to a 2018 NIH study, women routinely underwent pelvic or abdominal CT scans within six months of a breast cancer diagnosis to ensure the cancer had not spread before starting any active treatment (NIH 2018). Breast cancer metastasis to the pleura ipsilateral to the primary tumor is one of the three main causes of malignant effusion, most likely resulting from lymphatic dissemination. (MIRANDA *et al.*, 2022).

In medical imaging, dose limits are not applied to patients; instead, the concept of Diagnostic Reference Levels (DRLs) is utilized, as the potential diagnostic benefit of the radiological procedure outweighs the associated radiation risk. Therefore, the radiation protection principle of justification for

the practice and ALARA (As Low As Reasonably Achievable) are applied, with dosimetric control in CT examinations using DRL. DRL assesses the amount of ionizing radiation applied in a medical imaging procedure collected from a representative sample of patients at a given facility. According to the International Commission on Radiological Protection (ICRP), recommended DRL metrics for CT include the volume CT dose index (CTDIvol) per acquisition and the total dose-length product (DLP) for the complete examination (ICRP, 2017).

Radiographic image data are obtained through the Picture Archiving and Communication System (PACS). Based on metadata provided by the Digital Imaging and Communications in Medicine (DICOM) standard, it is possible to calculate computed tomography (CT) dosimetric parameters. According to the guidelines of the American Association of Physicists in Medicine (AAPM) the potential biological effects of ionizing radiation depend not only on the absorbed dose delivered to a specific tissue or organ, but also on the tissue's biological radiosensitivity. This radiosensitivity is accounted for in the calculation of the effective dose, a dosimetric quantity used primarily for radiological protection purposes, particularly in occupational exposure assessments and across different age groups (AAPM, 2008).

Breast cancer is primarily associated with mammography and radiation therapy, wherefore, the role of computed tomography (CT) often receives less attention from the general public. This study aims to highlight the significance of CT in the context of breast cancer management. Specifically, it seeks to evaluate the use of chest CT for the detection and assessment of metastases in breast cancer patients treated at a public university hospital in Belo Horizonte, Minas Gerais, Brazil. The analysis will encompass a review of the number of chest CT examinations performed, the clinical indications related to breast cancer, patient demographic characteristics, and a dosimetric assessment based on Diagnostic Reference Levels (DRLs), in alignment with radiation protection principles.

2 Methodology

This is one of the research lines of the project “Radiology Scenario at the University Hospital Assistance of the SUS” (CAAE 71737417.9.0000.5149) and is linked to the action research of the extension project “Assistance in Radiology: Bringing the Experiences of the University, the Public Hospital and the Community Closer Together” (SIEX-UFMG 402801).

This study surveyed the number of chest CT scans performed at the Imaging Diagnostic Unit (IDU) of a university hospital in Belo Horizonte (MG/Brazil). The analysis focused on elective outpatient scans with clinical indications of neoplasia, including breast and uterine cancer. Inpatients were not included in the samples. CT scans of other regions of the body were not included in the sample, only chest CT scans of elective patients, because their data antropometrics were more better registered. It must also be considered that hospitalized patients undergo frequent examinations for purposes other than the clinical indication studied. Data collection was limited to a 90-day period during the first half of 2025 and excluded hospitalized (inpatient) cases.

Body Mass Index (BMI), a parameter used to assess body mass in relation to height and to classify obesity into subcategories, was calculated for patients with available height and weight data.

To estimate DRLs in CT, the criteria established by the International Atomic Energy Agency (IAEA) were followed, which recommends collecting data from at least 30 patients per examination protocol. For patient selection, the IAEA guidelines recommended to include adult patients weighing between 50 kg and 90 kg, regardless of sex (IAEA, 2024).

Anonymized data were collected from two CT scanners: a 64-channel PHILIPS INCISIVE (CT A) and a 64-channel CANON AQUILION (CT B). The DICOM information provides the following data: CTDIvol (mGy), DLP (mGy-cm), voltage (kV), average current-time product (mAs), pitch, and series. CTDIvol represents the average absorbed radiation dose in the x, y, and z directions, estimating the average radiation dose within the irradiated volume for an object, reflecting the total absorbed energy,

that is, the potential biological effect, attributable to the complete scan acquisition, from the ratio of the weighted CTDI (CTDI_w) to the pitch.

To mitigate the influence of variability in the number of series acquired per exam, the DLP, CTDI_{vol}, and cumulative mAs values extracted from the DICOM headers were normalized by the number of series corresponding to each exam. This approach allowed for standardization of the data for calculating the medians, means, and standard deviations of these relative parameters per series, favoring more equitable comparisons between different protocols and equipment. The effective dose E (mSV) is normalized by the dose-length product (DLP) for adults (standard physicist) and the conversion factor K (mSv mGy⁻¹ cm⁻¹), as in Equation 1.

$$E = \approx k \cdot DLP \quad (1)$$

The effective dose was calculated using a conversion factor K equal to 0.014 (mSv mGy⁻¹ cm⁻¹), was applied, according to the AAPM table (2008), which assumes the use of a CT body phantom with a diameter of 32 cm. The calculation was estimated based on the accumulated DLP, according to the recommended conversion coefficients.

3 Results e Discussions

Among the results collected, the total number of CT scans performed in the service during the three-month study period analyzed was 3,768.00, including both outpatients and inpatients. Outpatients with clinical indications relevant to this study represented 1.30% of this total, in a sample of 49 exams. It should be noted that this small percentage pertains to preliminary data collected over a 90-day period in the first half of 2025, excluding inpatients.

Of the selected exams, 38 had clinical indications for breast cancer, 8 for breast neoplasia, 2 for ovarian cancer, and 1 for squamous cell carcinoma (SCC), the most common histological type of cervical cancer. Hereditary breast and ovarian cancer (HBOC) syndrome, caused by pathogenic variants in the BRCA1 and BRCA2 genes, significantly increases the risk of developing breast and ovarian cancers (KAZMI et al., 2008; Amirikia et al., 2013). Therefore, ovarian cancer was included in this study due to its established association with breast neoplasms.

All patients who underwent CT scans in this study were female, with a mean age of 60.19 ± 10.25 years. Among them, 6.25% were exactly 50 years old, the minimum age recommended for breast cancer screening (INCA, 2015), and 18.75% were younger than the recommended age.

Regarding anthropometric characteristics, the mean BMI was 27.32 ± 3.64 kg.m⁻². According to the WHO (2025), an adult is classified as overweight when BMI is greater than or equal to 25 kg.m⁻² and as obese when BMI greater than or equal to 30 kg.m⁻². MOHANTY & MOHANTY (2021) reported a significant association between breast cancer risk and obesity, with postmenopausal obesity contributing to increased risk due to the role of adipose tissue as the primary site of estrogen biosynthesis after menopause. Of the patients studied, only 16.33% had a BMI greater than 30, and 75.51% were over the age of 50, as illustrated in Figure 1.

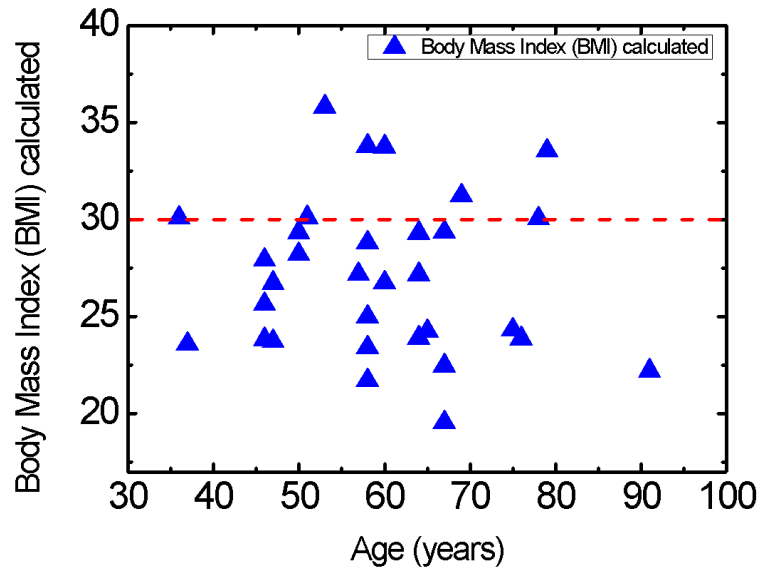


Figure 1: Body Mass Index (BTI) distribution for age patients surveyed with BTI reference 30 for obesity.

Table 1 shows the parameters used (Voltage, Pitch Ratio and average current product), and the dosimetric measurements (average DLP, CTDI_{vol} and Effective Dose) for chest CT A and CT B.

Table 1: Comparison between the parameters of CT A and CT B of the exams studied.

Parameters	CT A	CT B
Voltage (kVp)	120	120
Mean Tube Current-Time Product (mA.s) $\pm$ Desvpad	18.26 $\pm$ 18.26	678.30 $\pm$ 283.00
Pitch	0.80	0.83
DLP (mGy-cm) Median/Mean $\pm$ Desvpad	253.35/270.91 $\pm$ 80.15	170.90/251.58 $\pm$ 142.81
CTDI _{vol} (mGy) Median/Mean $\pm$ Desvpad	8.61/8.26 $\pm$ 1.92	20.42/32.68 $\pm$ 25.76
Effective dose (mSv) $\pm$ Desvpad	3.0 $\pm$ 4.19	31.28 $\pm$ 20.53

Considering the reference levels (ICRP, 2000) for routine Chest CT examinations in adult patients, the DLP is 650 (mGy-cm), and the values found in both equipment were lower. According to Campelo (2019), chest CT scans have an effective dose of around 8 mSv; in this study, CT A had a value 2.67 times lower. The dosimetric findings in this study are consistent with recommendations from the European Commission for chest CT scans, which recommend CTDI_{vol} values below 10 mGy for 4- and 16-slice devices. CT A also has a value 1.57 times lower than the DRLs for multislice devices in the United Kingdom, where the reference CTDI_{vol} index value is 13 mGy for chest scans performed in adults for lung studies. Although international reference levels for 64-slice CT scanners are still limited. The results highlight the need to optimize CT B in comparison to CT A, possibly influenced by a higher average tube current-time product applied in CT B (CAMPELO, 2019).

Considering the diagnostic reference levels (DRLs) established by the International Commission on Radiological Protection (ICRP, 2000) for adult CT scans—specifically a DLP of 650 mGy-cm for routine chest examinations—the values obtained from both scanners were below this threshold. According to Campelo (2019), the effective dose for a standard chest CT is approximately 8 mSv, while

in CT angiography (CTA), this value is approximately 2.67 times lower. The dosimetric findings in this study are consistent with recommendations from the European Commission, which suggest CTDIvol values below 10 mGy for 4- and 16-slice scanners. Additionally, Scanner A presented a CTDIvol value 1.57 times lower than the DRL for multislice scanners in the United Kingdom, where the reference CTDIvol for adult chest CT scans in lung imaging is 13 mGy. Although international reference levels for 64-slice CT scanners are still limited, the results highlight the need to optimize Scanner B in comparison to Scanner A, likely due to the higher average tube current-time product applied in Scanner B (CAMPELO, 2019).

In the chest CT study by Kumsa *et al.* (2023), the median CTDIvol was 7.58 and the median DLP was 300.50 mGy.cm, and the defined DRLs were CTDIvol 13 mGy and DLP 63 mGy.cm. The medians found in this study, comparatively, were for CT A 9.35% higher in CTDIvol and 18.61% lower in DLP.

According to the report from the service studied in this paper, CT B is being deactivated, replacing the equipment due to difficulty in replacing maintenance parts. Therefore, only a dispersion study will be performed using the median CTDIvol (Figure 2) and DLP (Figure 3) for CT A in relation to the examination indications found.

Figures 2 and 3 represent the sampling incidence and mean, with the 25th and 75th percentile range of the median dispersion from the median, for breast neoplasia, ovarian cancer, and breast cancer, with each clinical indication represented by its respective core. It can be seen that in all cases, there was a greater dispersion concentrated closer to the median in both figures.

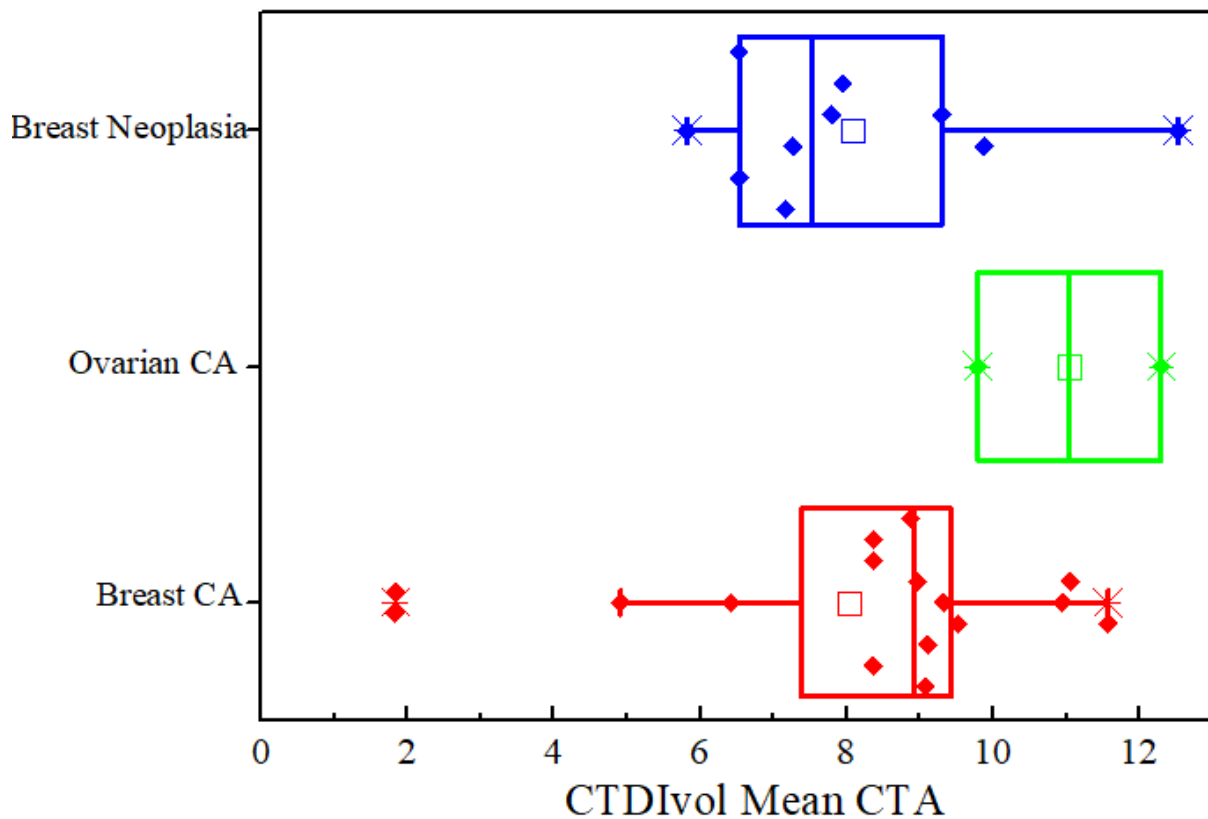


Figure 2. Box and whisker plot of median CTDIvol values from CT A for each exam.

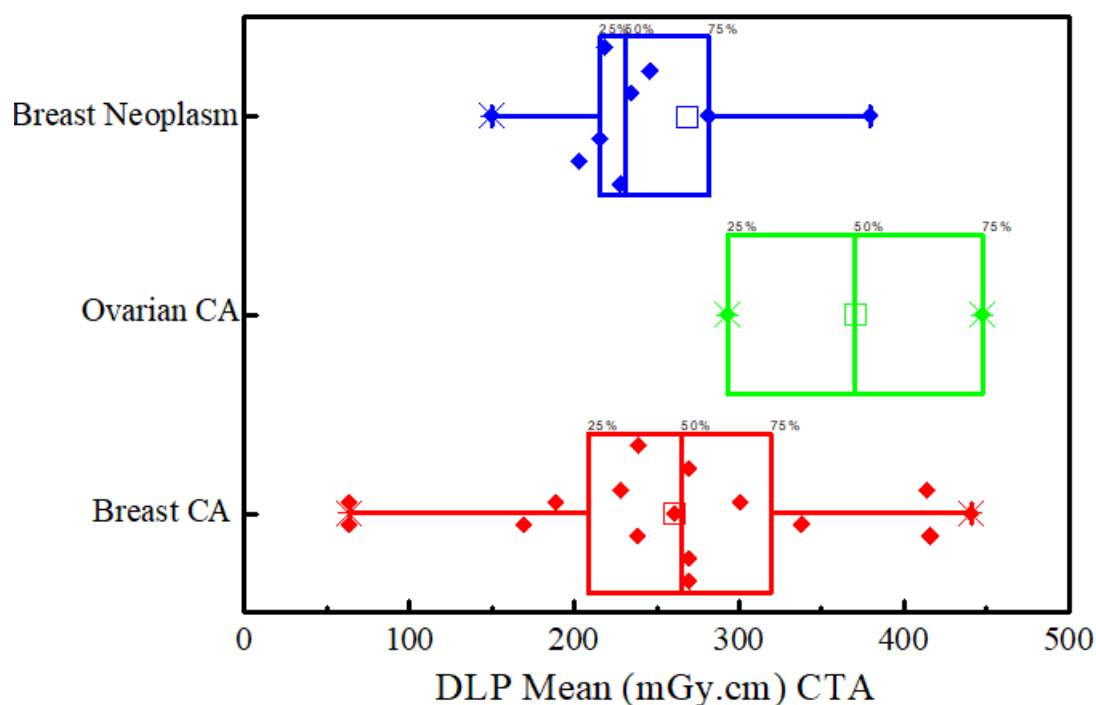


Figure 3. Box and whisker plot of median DLP values form CT A for each exam

Although breast cancer is predominantly associated with mammography and radiation therapy, the role of computed tomography (CT) remains underrecognized. This study investigated the use of chest CT for metastatic assessment in breast cancer patients at a public university hospital in Belo Horizonte, Minas Gerais, Brazil. The analysis included the number of chest CT examinations performed, their clinical indications, and patient demographic profiles. Given its relevance, a dosimetric evaluation based on Diagnostic Reference Levels (DRLs) was also conducted to ensure adherence to radiation protection principles.

4 Conclusions

Chest computed tomography (CT) is an important examination for staging and monitoring of breast cancer treatment, supporting the justification for using this radiological modality. This study evaluated the clinical indications for chest CT scans performed in patients with breast neoplasms as well as the demographic and anthropometric profiles of the study population. All patients were female, with a mean age of 60.19 ± 10.25 years, and a mean BMI of $27.32 \pm 3.64 \text{ kg.m}^{-2}$. Among them, 16.33% had a BMI within the obesity risk range, and 75.51% were over 50 years old, corresponding to the recommended age group for breast cancer screening. The diagnostic reference level (DRL) analysis was conducted for adult patients, and the DLP values found in both CT scanners were below the thresholds established by the ICRP (2000). CT A presented lower values for DLP, CTDIvol, and effective dose. However, it is noteworthy that CT B is currently in the process of being discontinued. Therefore, it is concluded that it is essential to promote compliance with DRLs, a fundamental requirement for optimizing protocol parameters, ensuring patient radiation protection, and reinforcing the clinical and social relevance of CT in breast cancer management.

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COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana

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Resumen

Este trabajo se enfoca en los efectos prolongados de la COVID-19, en particular la fibrosis pulmonar, mediante la clasificación automática de imágenes de tomografía computarizada de pacientes mexicanos. Para ello, se implementó una red neuronal convolucional basada en la arquitectura VGG16, complementada con filtros especializados como Meijering y Roberts, que optimizan la detección de las regiones de interés en las imágenes. Dado que la COVID prolongada se ha asociado con complicaciones respiratorias graves, como la fibrosis pulmonar, se resalta la importancia de una detección temprana y precisa. El modelo propuesto fue entrenado y validado con un conjunto de datos clínico exhaustivo, alcanzando una precisión de clasificación superior al 97%, logrando distinguir de manera efectiva entre pacientes con y sin signos de fibrosis pulmonar. Los resultados obtenidos demuestran el sólido desempeño del modelo y su potencial como herramienta de apoyo en la práctica clínica, permitiendo diagnósticos más oportunos y una mejor atención a pacientes afectados por secuelas pulmonares asociadas a la COVID-19.

Keywords: COVID largo; Fibrosis pulmonar; Tomografía computarizada; Clasificación automática

1 Introducción

La pandemia de COVID-19, provocada por el virus SARS-CoV-2, ha sido un fenómeno global con un impacto profundo y duradero en la salud pública. Para mediados de 2023, se han registrado más de 769 millones de casos a nivel mundial desde la aparición del virus en 2020 (Aguilar 2013, Orús 2023). Aunque la enfermedad se caracteriza principalmente por provocar manifestaciones respiratorias agudas, se ha identificado en numerosos pacientes una evolución persistente conocida como "COVID largo". A medida que los estudios avanzan en la comprensión de sus secuelas a largo plazo, ha surgido una creciente preocupación en torno a la fibrosis pulmonar, una condición en la que el tejido pulmonar sano es reemplazado por tejido cicatricial.

Desde la epidemia de SARS-CoV-1 ya se habían documentado algunas secuelas respiratorias similares, siendo la fibrosis pulmonar una de las más frecuentes. Esta patología es una respuesta del organismo a una lesión pulmonar crónica y se ha observado en un número significativo de pacientes convalecientes de

COVID-19, incluso hasta un año después de la infección inicial. La fibrosis compromete seriamente la función respiratoria al reducir la elasticidad pulmonar, dificultando el intercambio de oxígeno. Se estima que alrededor de un tercio de los pacientes hospitalizados por COVID-19 desarrollan alteraciones fibróticas pulmonares con el tiempo (Amin et al., 2022) .

Para quienes enfrentan esta secuela, la calidad de vida se ve notablemente afectada. La disminución de la capacidad respiratoria limita la realización de actividades cotidianas, genera fatiga crónica, trastornos del sueño y un deterioro general en la salud física y emocional. El manejo de esta condición crónica representa un desafío significativo que requiere atención médica especializada a largo plazo (Achenbach et al., 2010).

Dado que una proporción considerable de personas que han padecido COVID prolongado desarrollan fibrosis pulmonar tras un año de la infección, se hace evidente la necesidad urgente de desarrollar métodos eficaces para su detección temprana y tratamiento oportuno. Este contexto resalta la importancia de la investigación en herramientas diagnósticas que permitan identificar esta complicación de manera no invasiva, rápida y precisa. En la Figura 1 se muestra una imagen de pulmones con presencia de fibrosis.

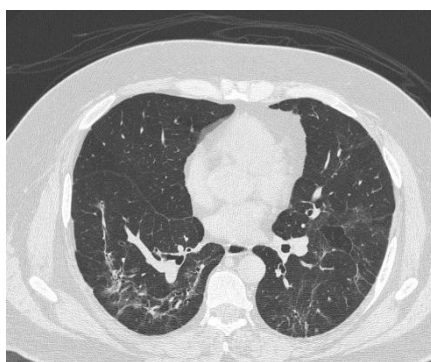


Figura 1: Pulmones con hallazgo de fibrosis pulmonar.

En este contexto, el presente trabajo se enfoca en abordar la problemática de la fibrosis pulmonar asociada a COVID prolongado mediante la implementación de tecnologías avanzadas, específicamente la clasificación automatizada de imágenes de tomografía computarizada (TC). El objetivo principal es mejorar la detección temprana y el manejo clínico de esta complicación en pacientes mexicanos, a través del uso de redes neuronales convolucionales (CNN), concretamente la arquitectura preentrenada VGG16, combinada con filtros de mejora de imagen. Con ello, buscamos aportar herramientas eficaces que contribuyan tanto al conocimiento científico como a la práctica clínica, facilitando diagnósticos más oportunos y precisos.

Para ello, se experimentó con un conjunto de tomografías axiales computarizadas (con y sin indicios de fibrosis pulmonar atribuible a COVID largo), aplicando diferentes filtros de procesamiento, con el fin de identificar cuál de ellos resalta con mayor claridad las cicatrices formadas en los pulmones. Posteriormente, las imágenes procesadas fueron clasificadas automáticamente utilizando la CNN VGG16, a fin de detectar con precisión aquellas que presentaban señales de fibrosis.

Como se ha documentado, la fibrosis pulmonar es una afección progresiva cuyo tratamiento se complica conforme avanza, y cuya detección temprana es difícil debido a que en fases iniciales las cicatrices pulmonares no son lo suficientemente visibles, lo que puede llevar a que se pasen por alto. Tradicionalmente, su diagnóstico se basa en tomografías axiales y pruebas de función pulmonar, como las de aliento. No obstante, la investigación en detección temprana continúa avanzando; por ejemplo, investigadores de la Universidad Eberhard Karls de Tübingen han propuesto el uso de un agente de imagenología PET, específicamente el ^{64}Cu -GPVI-Fc, como herramienta no invasiva para detectar la actividad fibrosa en etapas iniciales, ya que se asocia directamente con la presencia de tejido cicatricial pulmonar (Chérrez et al., 2021).

En cuanto a las tomografías axiales computarizadas, estas son herramientas indispensables en el diagnóstico médico, utilizadas para detectar traumatismos, accidentes cerebrovasculares, diversos tipos de cáncer y enfermedades respiratorias. Su resolución espacial puede alcanzar los 0.3 mm, y su tiempo de adquisición varía entre 0.5 y 10 segundos por volumen completo. Para mejorar la calidad de estas imágenes, se han implementado filtros de procesamiento, los cuales ayudan a reducir el ruido y realzar las regiones de interés. Por ejemplo, en (Christe et al., 2019) se señala que el filtro bilateral anisotrópico permite una mejor relación entre ruido, resolución espacial y detección de contrastes bajos, lo que contribuye a obtener resultados más claros y útiles en la interpretación clínica.

La aplicación de estos filtros y técnicas de procesamiento está ampliamente respaldada por la literatura. En (Amin et al., 2022), se argumenta que “estos conceptos permiten ir más allá de donde puede llegar el ojo y la percepción humana, lo que ayudará de manera significativa a los especialistas en su toma de decisión”. Diversos estudios han demostrado el uso efectivo de herramientas como el algoritmo Snake para la detección de tumores o tejido dañado, el filtro gaussiano para suavizar contornos, y el filtro de mediana para reducir ruido en las imágenes. Estas técnicas, empleadas según los requerimientos clínicos específicos, pueden mejorar significativamente la precisión diagnóstica (Christe et al., 2019, Coca et al., 2008, Marcos, 2020).

2 metodología

Se evaluarán los conjuntos de imágenes médicas aplicando filtros específicos, con el objetivo de analizar las características visuales particulares que se manifiestan en cada tomografía. Esto permitirá determinar si las cicatrices pulmonares —producto de la fibrosis— se reflejan de manera eficiente en las imágenes, lo cual facilitará la detección automatizada y la clasificación de hallazgos por parte de la red neuronal convolucional utilizada en este estudio, que se describe más adelante (Nayak & Bushomg 2012). A continuación, se detallan los filtros aplicados:

Filtro Roberts: Es un operador de detección de bordes eficaz en imágenes binarias, ya que utiliza únicamente dos coeficientes en su matriz de convolución. Su aplicación permite resaltar significativamente los bordes originales de la imagen. Sin embargo, presenta baja tolerancia al ruido, lo que lo hace susceptible a errores en la detección de contornos (Pajares, 2015). En la figura 2 se muestra el efecto de este filtro sobre la imagen de pulmón.

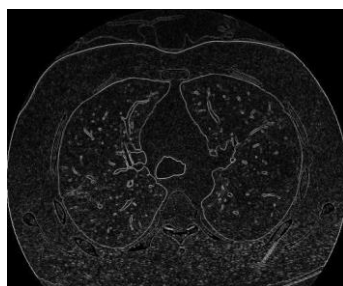


Figura 2: Filtro Roberts en tomografía axial con hallazgo de fibrosis pulmonar

Filtro Meijering: Este filtro de procesamiento de imágenes se utiliza principalmente para la detección de picos o puntos de alta intensidad, como ciertas arrugas o prolongaciones neuronales (neuritas). En el contexto de imágenes médicas, es especialmente útil para identificar estructuras con morfología tubular, lo que lo convierte en una herramienta adecuada para el análisis de tejidos complejos y vasos sanguíneos (Ramírez et al., 2010). En la figura 3 se muestra el efecto de este filtro sobre la imagen de pulmón.



Figura 3: Filtro Meijering en tomografía axial con hallazgo de fibrosis pulmonar.

Filtro de dilatación: El filtro de dilatación es una técnica comúnmente utilizada en procesamiento de imágenes para suavizar contornos y eliminar pequeñas imperfecciones o elementos no deseados. Su operación se basa en el uso de una matriz denominada elemento estructurante, sobre la cual se aplica una suma vectorial entre dos conjuntos de píxeles. Este proceso permite "expandir" las regiones brillantes de la imagen, iluminando áreas oscuras adyacentes y facilitando así el relleno de huecos o discontinuidades entre píxeles contiguos (Nayak & Bushomg 2012). En la figura 3 se muestra el efecto de este filtro sobre la imagen de pulmón.



Figura 4: Filtro de dilatación en tomografía axial con hallazgo de fibrosis pulmonar

En este trabajo se empleó la red neuronal convolucional VGG-16, perteneciente al conjunto de técnicas de aprendizaje profundo, una subdisciplina del aprendizaje automático. Las redes neuronales constituyen la base fundamental de los algoritmos de aprendizaje profundo; estas se estructuran en capas de nodos, compuestas por una capa de entrada, una o varias capas ocultas, y una capa de salida. Cada nodo está interconectado con otros y cuenta con pesos y umbrales asignados. La activación de un nodo se produce cuando su valor de salida supera el umbral definido, lo que permite la propagación de la información hacia la siguiente capa; de lo contrario, la señal no se transmite (Pajares, 2015). La arquitectura VGG-16 consiste en 16 capas y ha sido preentrenada con más de un millón de imágenes, permitiéndole realizar tareas de clasificación en múltiples categorías visuales con alta precisión (Christe et al., 2019). Para los fines de este trabajo, se conformó una base de datos con tomografías axiales de tórax obtenidas del Hospital IMSS UMAE T-1, ubicado en la ciudad de León, Guanajuato, México. Se recopilieron 96 imágenes de tomografía computarizada, de las cuales 39 correspondían a casos con hallazgos de fibrosis pulmonar, y 57 a casos sin dicho hallazgo. Durante el proceso experimental, se aplicaron tres filtros de procesamiento de imágenes —Meijering, Roberts y dilatación— siguiendo un procedimiento uniforme para cada uno. Utilizando el lenguaje de programación Python, se procesaron las imágenes y se etiquetaron en dos categorías: "con hallazgo" y "sin hallazgo" de fibrosis pulmonar. Posteriormente, se generó una matriz de vectores característicos para ambos conjuntos, que fue almacenada en una base de datos. Finalmente, estos vectores fueron procesados por la red neuronal VGG-16, la cual, con base en el conocimiento adquirido durante su entrenamiento, clasificó las imágenes según la presencia o ausencia de fibrosis. La validación cruzada, junto con el clasificador KNeighborsClassifier, mostró un rendimiento superior en comparación con los resultados obtenidos mediante la tradicional división entre conjunto de entrenamiento y conjunto de prueba.

3 Resultados

Durante la fase de clasificación, se aplicó el método de validación cruzada al conjunto de tomografías axiales computarizadas, utilizando previamente el filtro de Meijering. Los resultados obtenidos mediante el clasificador k-NN (KNeighborsClassifier) fueron satisfactorios, alcanzando un 92% de exactitud y precisión, métricas clave para evaluar el desempeño del modelo. En contraste, al emplear el método tradicional de separación por conjunto de entrenamiento y prueba, el rendimiento fue inferior, alcanzando solo un 89% de exactitud con el mismo clasificador. Por otro lado, el conjunto de imágenes sin filtros aplicados presentó un desempeño aún más limitado, con un 87% tanto en exactitud como en precisión, por debajo de las expectativas iniciales.

Ante estos resultados, se implementó un nuevo filtro de máscara sobre las imágenes del conjunto validado, el cual permite delimitar específicamente el contorno de cada pulmón. Esta estrategia mejoró significativamente el rendimiento del modelo, elevando la exactitud al 95% y la precisión al 97%, al focalizar la clasificación únicamente en las regiones de interés. En conclusión, entre los filtros aplicados —como Roberts o el filtro morfológico de dilatación—, el filtro Meijering combinado con la máscara pulmonar demostró ser el más eficaz para resaltar las lesiones y cicatrices características de la fibrosis pulmonar, contribuyendo así a una clasificación más precisa y confiable de las imágenes médicas analizadas.

Tabla 1: Resultados del primer experimento.

Imagen	Exactitud	Precisión
Original	0.87 ± 0.14	0.87 ± 0.16
Roberts	0.81 ± 0.12	0.87 ± 0.12
Meijering	0.92 ± 0.12	0.92 ± 0.12
Dilatación	0.87 ± 0.14	0.87 ± 0.14

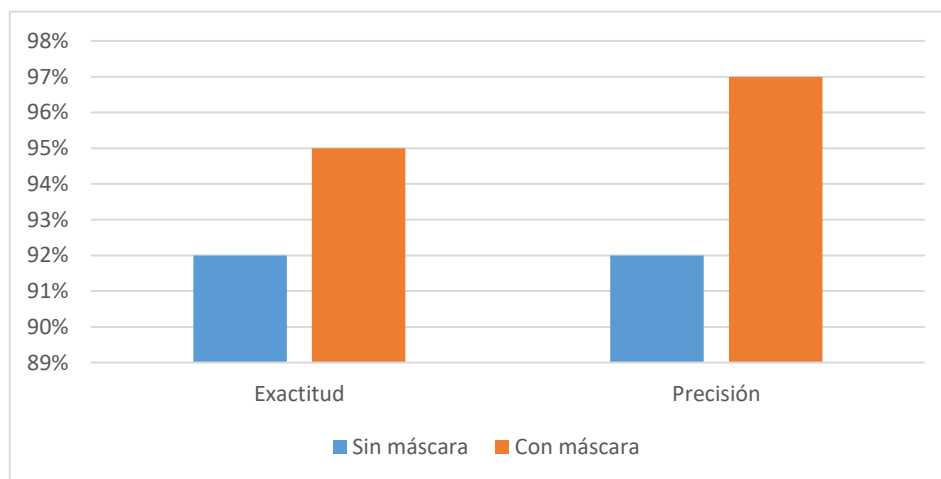


Figura 5: Resultados del segundo experimento vs primer experimento.

Conclusiones

El desarrollo de nuevas tecnologías aplicadas al ámbito médico ha facilitado la detección temprana de diversas anomalías en el cuerpo humano. En este estudio, se demostró que el rendimiento diagnóstico de las tomografías computarizadas puede optimizarse mediante el uso de técnicas avanzadas de procesamiento de imágenes. En particular, la aplicación del filtro Meijering permitió resaltar indicios de lesiones en el tejido pulmonar, facilitando su identificación por parte de una red neuronal convolucional,

la cual logró clasificarlas con alta precisión. Cabe destacar que esta herramienta no pretende sustituir el juicio clínico del profesional médico, sino actuar como un apoyo visual que mejore la interpretación de las imágenes, permitiendo una visualización más clara de posibles lesiones en estudios de tomografía axial computarizada.

Mediante la implementación de la arquitectura VGG16, combinada con técnicas de mejora de imágenes, el modelo alcanzó una precisión de clasificación superior al 97%, lo que evidencia su alto potencial en la detección temprana de fibrosis pulmonar en pacientes con COVID prolongado. Estos hallazgos pueden traducirse en una mejora significativa en la gestión clínica de esta condición persistente, particularmente en la población mexicana. Este trabajo representa una contribución relevante al campo de la inteligencia artificial aplicada a las imágenes médicas, y ofrece una herramienta confiable que puede facilitar intervenciones oportunas, mejorar la calidad de vida de los pacientes y optimizar los recursos del sistema de salud.

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Determination of the Mass Attenuation Coefficient of Thermoset and Thermoplastic Resins at Energies Used in Conventional Mammography

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Abstract

Advances in mammography have increasingly demanded more realistic physical phantoms for quality control (QC) and dosimetry. These models require materials with attenuation coefficients similar to breast tissues, in addition to being cost-effective. Although PMMA is widely used as a homogeneous phantom, its thickness does not accurately represent the equivalent thickness of a 50/50 breast, necessitating the use of spacers in QC tests. In this context, this study evaluated the mass attenuation coefficient of different commercially available thermoset and thermoplastic resins, aiming to investigate their feasibility for manufacturing homogeneous breast phantoms. Measurements were performed using a CdTe spectrometer and a tungsten target X-ray tube with a 0.05 mm rhodium filter, operating at 30 kV. For each material, five samples were produced in the form of rectangular plates with average thicknesses of 3 mm. The mass attenuation coefficients were determined from measurements of the corrected spectra transmitted through different thicknesses. To validate the methodology, reference materials such as PMMA and Al were analyzed. The investigated samples included thermoplastic resins such as acrylonitrile butadiene styrene (ABS), and thermoset resins such as: (I) PMMA with polyurethane, (II) ultraviolet (UV) photopolymerizable resin, (III) soft dental relining resin, (IV) a mixture of UV resin with 15% soft resin, and (v) 3D printing resin. The mass attenuation coefficients of PMMA and Al showed average percentage differences of 4.6% and 0.75% between 12 and 23 keV, respectively. The results presented the mass attenuation coefficients determined for each material at energies between 12 and 23 keV, along with their respective deviations compared to the mass attenuation coefficient of breast tissue.

Keywords: Mammography, Breast Phantoms, Mass Attenuation Coefficient, Thermoset and Thermoplastic Resins.

1 Introduction

Breast cancer represents the most prevalent neoplasm among women worldwide and remains a major public health concern due to its high incidence in the female population (IARC, 2025). Mammography is the imaging modality most widely used for screening and early diagnosis, and its effectiveness depends directly on the implementation of rigorous quality assurance protocols. Within this context, breast phantoms play a central role, as they are indispensable tools in routine quality control programs and in the comparative evaluation of mammography performance (Hatamikia, 2022).

The design of mammographic phantoms requires materials capable of accurately mimicking the radiological properties of breast tissues to ensure reliable assessment of image quality and patient dose. Polymethyl methacrylate (PMMA) is commonly used as a standard material due to its homogeneous composition; however, its thickness does not precisely correspond to that of a 50% glandular/50% adipose breast, which is considered representative of an average breast composition. Consequently, the use of additional spacer plates is often necessary during quality control testing. To overcome such limitations, researchers have explored alternative materials and manufacturing methods, particularly focusing on polymers and resins with different structural properties. Thermoset resins, for example, are three-dimensional polymers that irreversibly degrade when exposed to heat, while thermoplastic resins are two-dimensional molecules that soften under heat and can be reshaped without degradation (Carraher 2003).

The development of innovative and cost-effective technologies has further expanded the possibilities for phantom fabrication. Additive manufacturing methods such as fused deposition modeling (FDM) 3D printing enable the production of realistic and customizable physical phantoms derived from computed tomography data. This technique allows for fine spatial resolution (e.g., 0.1 mm) and complete infill density, using widely available filaments such as acrylonitrile butadiene styrene (ABS), polylactic acid (PLA), or polyethylene terephthalate (PET) to emulate glandular, adipose, and skin tissues (Varallo, 2022). Such advances have encouraged the exploration of new resin formulations for phantom development, but their application requires careful characterization of radiological properties.

One of the most reliable approaches to evaluate phantom materials is the determination of their mass attenuation coefficients, which can be obtained numerically through Monte Carlo simulations or experimentally using X-ray spectrometry (Santos 2019, Frimaio 2020, Tousi 2022). Experimental methodologies have provided important contributions in this field. For instance, Santos (2017) investigated the absorbed dose distribution in different mammographic beam qualities using calibrated detection systems at low energies, thereby establishing a methodological basis for measuring linear attenuation coefficients in narrow-beam geometries. More recently, studies have emphasized that the experimental determination of the linear attenuation coefficient for novel breast tissue equivalent materials, particularly under monoenergetic X-ray beams, enables direct comparison with both real breast tissues and computational simulations. This approach not only validates the feasibility of new phantom materials but also widens the range of available options for the design of advanced phantoms, including resins adapted for 3D printing (Mohammad, 2023).

In this context, this study evaluated the linear mass attenuation coefficient by X-ray spectrometry for different commercially available thermoset and thermoplastic resins (soft dental relining resin, UV resin with 15% soft resin, UV resin, PMMA with polyurethane, acrylonitrile butadiene styrene (ABS), 3D printing resin) aiming to investigate their feasibility for manufacturing homogeneous mammographic phantoms.

2 Materials and Methods

This study employed a CdTe spectrometer (Amptek) and a tungsten target X-ray tube (Pantak Seifert – Isovolt 160 HS), with a 0.05 mm rhodium filter and operated at 30 kV, Figure 1 illustrates the

experimental setup. Spectral measurements were performed for each material listed in Table 1, with five samples per material, produced at an average thickness of 3 mm.

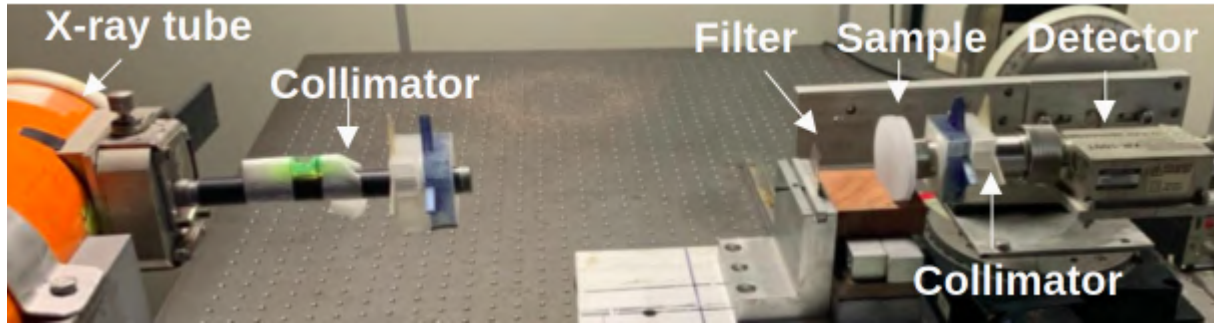


Figure 1: Experimental setup.

Table 1: Materials and their densities used for mass attenuation coefficient measurements.

Materials	Density (g/cm ³)
Aluminum	2.70 ± 0.05
PMMA	1.18 ± 0.05
Soft dental relining resin	1.01 ± 0.05
UV resin with 15% soft resin	1.17 ± 0.05
UV resin	1.20 ± 0.05
PMMA with polyurethane	1.24 ± 0.05
Acrylonitrile butadiene styrene (ABS)	0.97 ± 0.05
3D printing resin	1.13 ± 0.05

To validate the method, measurements were performed for aluminum (Al) and PMMA, as these materials can be directly compared with simulations from the NIST XCOM database. Five aluminum plates with an average thickness of 1 mm and a PMMA sample with an average thickness of 5 mm were used.

2.1 Determination of the Mass Attenuation Coefficient

The mass attenuation coefficient (μ/ρ) was determined through X-ray transmission measurements using a calibrated spectrometer system. The spectrometer's energy calibration was initially performed using an americium-241 source. The spectra obtained for each sample set were plotted, and the methodology for determining the linear attenuation coefficient (μ) at each energy was based on the Lambert-Beer law (Equation 1) (Santos et al., 2019).

$$\mu(E_i) = \frac{1}{x} \ln \ln \left[\frac{I_0(E_i)}{I(E_i)} \right] \quad (1)$$

Where $I_0(E_i)$ intensity of incident photons with energy E_i , and $I(E_i)$ intensity of transmitted photons with energy E_i after passing through a material of thickness x .

A Python program was developed to calculate the mass attenuation coefficient for each energy. In addition to computing μ , the program also performs spectrum corrections, based on the stripping method, and accounts for air attenuation when the sample thickness is included (Santos 2019, Tomal 2015, Nascimento 2021). The resulting linear attenuation coefficient (μ) is divided by the density (ρ) of each material to obtain the mass attenuation coefficient (μ/ρ), as listed in Table 1.

The experimentally determined μ/ρ values for aluminum and PMMA were compared with simulations from the NIST XCOM database. For the comparative analysis of materials with breast tissue, the following compositions were simulated using the NIST XCOM database: PMMA, adipose tissue (Hammerstein et al., 1979), and glandular breast tissue with 25%, 50%, and 75% glandularity (Chang et al., 2020).

3 Results and discussion

3.1 Validation

To validate the method, the experimentally measured mass attenuation coefficients for PMMA and aluminum were compared with reference values, following the approach described by Santos (2019). The μ/ρ values for PMMA and Al showed average percentage deviations of 4.6% and 0.75%, respectively, compared to the NIST XCOM database (Berger et. al, 2010) in the energy range of 12–23 keV. The comparison of μ/ρ for aluminum exhibited better agreement with the reference data; however, the average discrepancy for both materials remained below 5%. This supports the validity of the method. In the case of PMMA, the data points showed greater variation, probably due to low counts at the extremes of the considered spectral energy range.

3.2 Comparison of mass attenuation coefficients

Figure 2 presents the experimental mass attenuation coefficients for the evaluated materials: soft dental relining resin, UV resin with 15% soft resin, UV resin, PMMA with polyurethane, acrylonitrile butadiene styrene (ABS), and 3D printing resin. Each material was compared with reference data for breast tissues of different glandularities, enabling an assessment of their potential application in mammographic phantom fabrication.

In Figure 2(a), the soft dental relining resin demonstrated strong agreement with 75% glandular breast tissue in the 17–20 keV energy range, with percentage differences of 1.96%, 1.69%, 0.33%, and 3.31% at 17, 18, 19, and 20 keV, respectively. Figure 2(b) shows that the UV resin with 15% soft resin closely approximated 25% glandular breast tissue between 17–22 keV, with percentage differences ranging from 0.64% to 2.32%. Additionally, in the lower 14–16 keV range, this material matched adipose tissue more closely, with deviations of 2.69%, 1.97%, and 1.43% at 14, 15, and 16 keV. By contrast, the UV resin without additives, shown in Figure 2(c), consistently exhibited attenuation values below those of adipose tissue.

The PMMA with polyurethane, represented in Figure 2(d), displayed a curve significantly higher than that of 75% glandular breast tissue, indicating an overestimation of attenuation when compared to actual breast compositions. Figure 2(e) illustrates that ABS achieved good correspondence with 50% glandular breast tissue in the 15–19 keV range, with percentage differences of 3.93%, 0.07%, 0.04%, 2.36%, and 2.81% at 15, 16, 17, 18, and 19 keV, respectively. Finally, Figure 2(f) shows that the 3D printing resin is closely aligned with 50% glandular breast tissue between 19–23 keV, with deviations of 2.85%, 2.49%, 0.36%, 1.22%, and 1.64% at the respective energies.

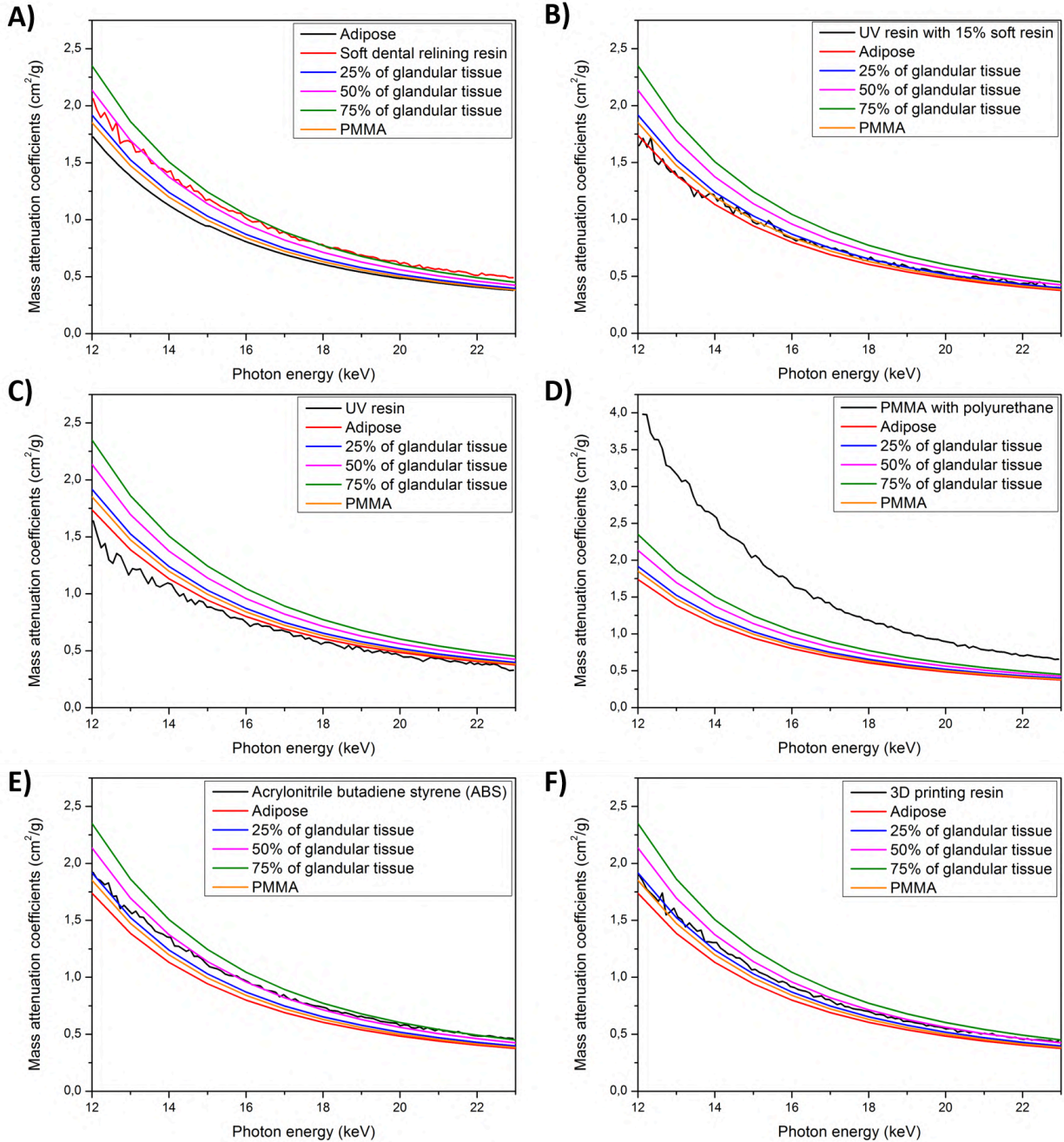


Figure 2: Experimental mass attenuation coefficients for (a) Soft dental relining resin, (b) UV resin with 15% soft resin, (c) UV resin, (d) PMMA with polyurethane, (e) Acrylonitrile butadiene styrene (ABS) and (f) 3D printing resin, compared with simulations from the NIST XCOM database: PMMA; adipose tissue composition (Hammerstein et al., 1979); and glandular breast tissue compositions 25%, 50%, and 75% glandularity (Chang et al., 2020).

The μ/ρ curve of conventional PMMA was also analyzed for comparison. It was observed that its attenuation coefficients lie above those of adipose tissue and below those of 25% glandular breast tissue. This behavior indicates that PMMA simulates breasts with a predominantly adipose composition, which limits its applicability for phantoms representing higher-density tissues.

Overall, the results suggest that some of the investigated materials exhibit attenuation properties similar to breast tissues within the average energy range of mammography. In particular, UV resin with 15% soft resin can be used to construct homogeneous phantoms simulating 25% glandular breast tissue, or

even serve as a substitute for PMMA given their similar μ/ρ behavior in the analyzed energy range. For 50% glandular breast tissue, both ABS and 3D printing resin proved suitable alternatives for phantom fabrication. Meanwhile, soft dental relining resin demonstrated the best agreement with 75% glandular breast tissue, supporting its potential use in simulating denser breast compositions.

4 Conclusions

This study characterized the mass attenuation coefficients of materials for mammography phantom applications. Experimental measurements suggest good agreement (0.75% deviation) between aluminum's μ/ρ values and NIST reference data, validating our methodology. While PMMA showed variation of 4.6%.

Alternative materials with attenuation properties similar to different breast tissue compositions in the conventional mammography average energy range were identified. The material UV resin with 15% soft resin could be an optimal PMMA alternative for 25% glandular tissue simulation, showing nearly μ/ρ values. Both 3D printing resin and ABS polymer provide appropriate attenuation matching for 50% glandular tissue phantoms, and Soft dental relining resin could be qualified for dense breast simulations (75% glandularity).

These results contribute to the development of more accurate breast phantoms, across the full clinical spectrum of breast densities, which can be used in dosimetry and quality control tests.

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Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Energy Integrating CT

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Abstract

Radiologist reader studies are costly and time-consuming, requiring substantial datasets and multiple readers for adequate statistical power. However, the detectability index (d') has been adopted to support image quality assessment. The detectability of a synthetic ground-glass opacity (GGO) nodule in an anthropomorphic lung phantom was assessed by evaluating the median d' values with interquartile ranges (IQR) across different CT protocols at various dose levels (CTDI_{vol}: 0.2 to 2.8 mGy). The study compared an energy-integrating CT (EICT, Aquilion One PRISM Edition) and a prototype CdZnTe photon-counting CT (PCCT), both by Canon Medical Systems. Three reconstruction methods were evaluated: deep learning reconstruction (DLR), model-based iterative reconstruction (MBIR) with the LUNG kernel, and hybrid iterative reconstruction (HIR, AIDR 3D) with FC51, FC52, and FC86 kernels. Results at 0.2–0.3 mGy, PCCT/DLR ($d' = 5.2$, IQR: 2.0) and EICT/DLR ($d' = 5.4$, IQR: 1.4) demonstrated superior performance compared to EICT/HIR with FC52 ($d' = 3.6$, IQR: 2.5) and FC86 ($d' = 4.3$, IQR: 3.1) (Kruskal–Wallis, $p < 0.001$). At 0.4–0.5 mGy, PCCT/DLR ($d' = 7.6$, IQR: 2.2) demonstrated superior performance compared to all EICT reconstructions ($p < 0.001$). At 1.4 mGy and 2.7–2.8 mGy, PCCT/DLR achieved the highest detectability ($d' = 13.8$ and 12.0; IQR: 2.8 and 2.3, respectively), with highly significant differences compared to EICT/HIR ($p < 0.0001$). The findings indicate that DLR demonstrated superior performance then other reconstruction methods for both EICT and PCCT. In addition, PCCT demonstrated superior performance over EICT at CTDI_{vol} ≥ 0.4 mGy.

Keywords: Detectability index; Anthropomorphic lung phantom; Photon-counting CT;

1 Introduction

When evaluating a pulmonary nodule identified on computed tomography (CT) images, radiologists must determine whether to initiate surveillance through follow-up examinations. This decision is guided by the assessment of nodule's features—including size, morphology, location, and apparent growth dynamics—in the context of the estimated malignancy risk. The complex pulmonary anatomy, characterized by an intricate network of vasculature and bronchioles, presents a significant diagnostic challenge. These normal anatomical structures can mimic the radiographic appearance of pulmonary nodules, increasing the risk of both false positives and false negatives in nodule detection. Nodule features depend on image details, so reducing spatial resolution can hinder assessment and decision-making (Cai et al., 2025). Image quality also has a significant impact on the detectability of nodules in a pulmonary background particularly spatial resolution and noise levels (Saunders Jr et. al., 2004). Moreover, the high workload

burden on radiologists has been shown to adversely affect diagnostic accuracy, primarily due to fatigue and limited time available for image interpretation. (Krupinski et al., 2012).

Reader studies are fundamental to the evaluation of radiologist perception and performance in medical image analysis. They often employ receiver operating characteristic (ROC) or forced-choice approaches based on the signal detection theory (Burgess, 2011; Szczykutowicz et al., 2022). Despite their status as the gold standard for evaluating detection performance, radiologist reader studies are costly, time-consuming, and require substantial datasets and multiple readers to achieve sufficient statistical power (Gallas and Brown, 2008; Alexander et al., 2022; Obuchowski, 2016; Chakraborty, 2017).

Statistical power is critical in diagnostic radiology research, with 80% generally considered the minimum acceptable threshold to detect true differences and minimize type II errors (Beam, 1992). Chakraborty (2017) observed that detecting subtle variations in diagnostic accuracy between imaging modalities typically requires the interpretation of at least 5–10 readers interpreting between 100 and 300 cases to achieve statistical significance with 80% power. However, in critical scenarios such as the detection of small pulmonary nodules, even higher power levels may be required (Das et al., 2006).

Image fidelity metrics such as root mean square error, peak signal-to-noise ratio, and structural similarity index are frequently used in simulations and in the training of neural networks, such as those employed in AI-based CT reconstruction techniques. Convolutional neural networks (CNNs), U-Net, and ResNet have also shown promise in the field of pulmonary nodule detection (Prabhavalkar and Arthi, 2023). Although valuable in controlled settings, these metrics fail to adequately capture essential aspects of image quality, such as spatial resolution, noise level, and noise texture. As a result, optimizing reconstruction algorithms solely based on these metrics can lead to over-smoothed images that compromise the visibility of subtle but clinically important structures (Sidky et al., 2021).

Further complexity arises with the adoption of advanced CT technologies such as iterative reconstructions and deep learning-based image reconstruction. These innovations offer enhanced spatial resolution, noise suppression, and improved contrast-to-noise ratio, which are critical for low-dose imaging (McLeavy, et al., 2021; Fletcher et al., 2024). However, they also introduce nonlinearities on the image reconstruction processes and noise textures not fully characterized by traditional metrics. This complexity increases the demand for careful optimization of imaging protocols to ensure diagnostic reliability (Lee and Kim, 2014), particularly in dose reduction scenarios.

The detectability index (d') can be used as a potential alternative solution. It has been adopted to support image quality assessment (Pimenta and Costa, 2025). Derived from model observers, d' estimates how well a signal, such as a pulmonary nodule, can be distinguished from noise background under varying imaging conditions (Burgess, 1994). This metric is indicative of key image quality components, including contrast, noise, and spatial resolution (Myers and Barrett, 1987; Richard and Samei, 2010). The application of d' is particularly valuable in low-dose CT, where optimizing the trade-off between radiation exposure and lesion detectability is essential. By integrating clinically relevant dimensions of image quality into a single quantitative metric, d' facilitates a systematic assessment of CT imaging protocols. This enables evidence-based protocol optimization and provides a robust framework for comparing image reconstruction algorithms (Pimenta and Costa, 2025).

The present work adopted the d' to compared a conventional energy-integrating CT and a photon-counting CT considering three reconstruction methods: deep learning reconstruction (DLR), model-based iterative reconstruction (MBIR) with the LUNG kernel, and hybrid iterative reconstruction (HIR, AIDR 3D) with FC51, FC52, and FC86 kernels.

2 Materials and Methods

2.1 Detectability Index (d') and Image Quality Assessment

The detectability index (d') was calculated from 90 repeated measurements within a single region of interest (ROI) placed at the center of the synthetic GGO nodule on the slice showing its largest diameter. All measurements were performed on the same acquisition, with the ROI maintained in the same location to ensure reproducibility. The Task Transfer Function (TTF) was estimated from these ROIs, while the Noise Power Spectrum (NPS) was calculated from four ROIs placed in the lung parenchyma surrounding the nodules across three different slices. The approach enables the analysis of median d' values under different CT systems, image reconstruction methods, reconstruction kernels, and dose levels. Figure 1 illustrates the interdependent components that influence the d' in CT imaging. It is determined by three main elements: *TTF*, which describes the effect of contrast and noise on spatial resolution properties; *NPS*, which reflects the magnitude, texture and correlation of noise; and the task function (W), which models the detection task by incorporating the size, shape and contrast of the signal. In the analysis using the anthropomorphic lung phantom, regions of interest (ROIs) were extracted from areas containing synthetic GGO nodule to compute the TTF, and from regions of lung parenchyma to evaluate the NPS.

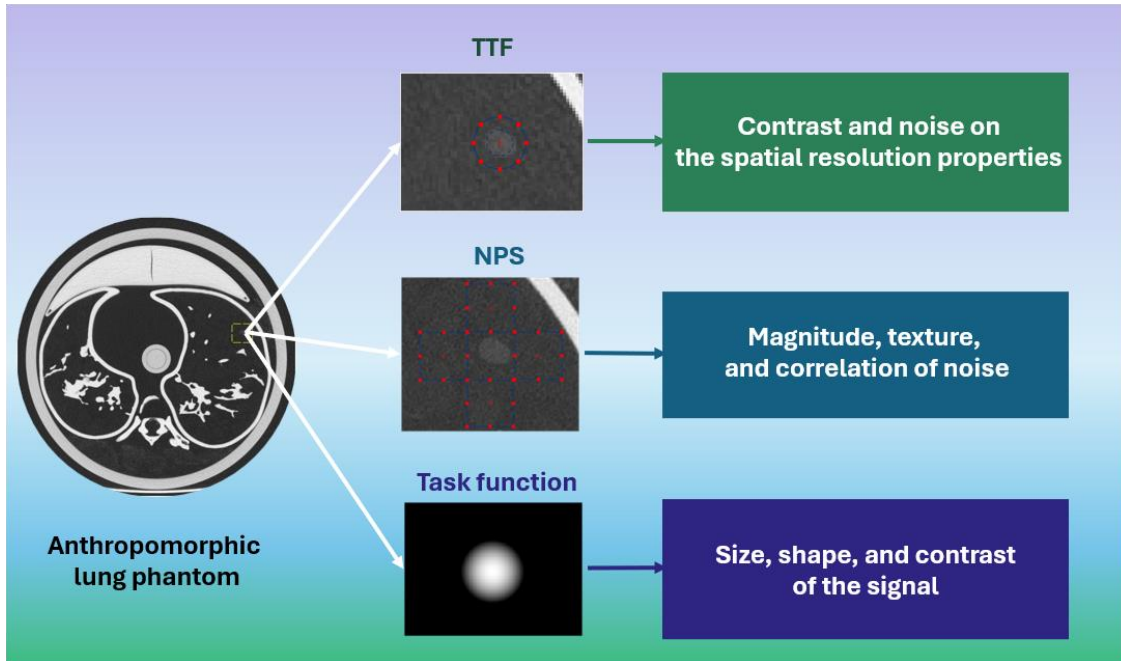


Figure 1: The detectability index (d') is determined by three main elements: the Task Transfer Function (TTF), which describes the effect of contrast and noise on spatial resolution; the Noise Power Spectrum (NPS), which reflects the magnitude, texture, and correlation of noise; and the task function (W), which models the detection task by incorporating the size, shape, and contrast of the signal.

In accordance with the recommendations of the AAPM Task Group 233, phantom images were analyzed using imQuest software to assess image quality metrics, including the detectability index (d'), *TTF*, and *NPS*. The detectability index was computed using the non-prewhitening observer model with an eye filter (NPWE), as defined in Equation 1.

$$d' = \frac{[\int W(f)^2 TTF^2(f) E^2(f) df]^2}{\int W(f)^2 TTF^2(f) E^4(f) NPS(f) df} \quad (1)$$

The task function W was defined through the modeling of the signal, simulating a non-spiculated lung nodule. This approach was taken to provide a clinically relevant task for thoracic imaging. The NPWE model was implemented with a zoom factor of 1.5, a viewing distance of 500 mm, and the Eckstein eye filter function $E(f)$.

2.2 Anthropomorphic Phantom and CT Acquisition Parameters

A patient-based anthropomorphic 3D-printed lung phantom was developed and validated to simulate the thoracic region and anatomical background in a realistic scenario (Costa et al., 2025). The internal structure of the phantom consists of a patient-based 3D-printed model of the bronchial tree and lung walls. The bronchus/bronchioles stem and the lung wall were fabricated using PolyJet VeroBlue photopolymer (Stratasys Inc., USA), while the parenchymal region was filled with fractionated foam chips. The materials were selected to replicate the Hounsfield Unit (HU) values observed in actual patient lung tissues. Synthetic GGO nodule were embedded into the lung parenchyma of the phantom to replicate clinically relevant detection tasks. This nodule replicate low-contrast, non-spiculated pulmonary lesion commonly encountered in thoracic imaging (Figure 1).

The phantom was imaged using a energy-integrating CT (EICT, Aquilion One PRISM Edition) and a prototype CdZnTe photon-counting CT (PCCT, TSX-501R), both from Canon Medical Systems, employing different clinical protocols typically used in chest CT imaging. Scans were acquired at CTDI_{vol} levels of 0.2 mGy, 0.3 mGy, 0.5 mGy (ultra-low dose), and 1.4 mGy, 2.8 mGy (low dose). All acquisitions used 120 kV, 0.5 mm slice thickness, a copper beam filter, pitch of 0.813, and rotation time of 0.275 s.

The image matrix size was 512² for EICT and 1024² for PCCT. Tube current was adjusted in each case to achieve the desired dose levels. Three image reconstruction methods were evaluated: model-based iterative reconstruction (MBIR) using FIRST (Forward projected model-based Iterative Reconstruction SoluTion), hybrid iterative reconstruction (HIR) using AIDR 3D (Adaptive Iterative Dose Reduction 3D) with FC51, FC52, and FC86 kernels, and a deep learning-based reconstruction (DLR) using AiCE (Advanced intelligent Clear-IQ Engine) developed by Canon Medical Systems (see Table 1).

Table 1: CT Acquisition and Reconstruction Parameters

Parameter	Specification
CT Scanner	EICT (Aquilion One PRISM Edition, Canon Medical Systems) PCCT (TSX-501R, Canon Medical Systems)
Tube Potential	120 kV
Filter	Copper
Pitch	0.813
Slice Thickness	0.5 mm
Image matrix size	512 ² (for EICT); 1024 ² (for PCCT)
Tube Current	Adjusted for each acquisition (10 - 100) mA
Rotation time	0.275 s
Reconstruction Techniques	HIR (AIDR 3D), MBIR (FIRST), DLR (AiCE)
Reconstruction Kernels	FC51, FC52, FC86 (for HIR); LUNG (for MBIR and DLR)
Dose Levels (CTDI _{vol})	0.2 mGy, 0.3 mGy, 0.4 mGy, 0.5 mGy, 1.4 mGy, 2.7 mGy, 2.8 mGy

3 Results

Figure 2 shows a boxplot analysis of the detectability index (d') for various CT acquisition and reconstruction protocols, grouped by four CTDI_{vol} dose levels: 0.2–0.3 mGy, 0.4–0.5 mGy, 1.4 mGy, and 2.7–2.8 mGy. Each protocol combines one CT system (EICT or PCCT), a reconstruction method (DLR, HIR, MBIR), and a reconstruction kernel (LUNG, FC51, FC52, FC86).

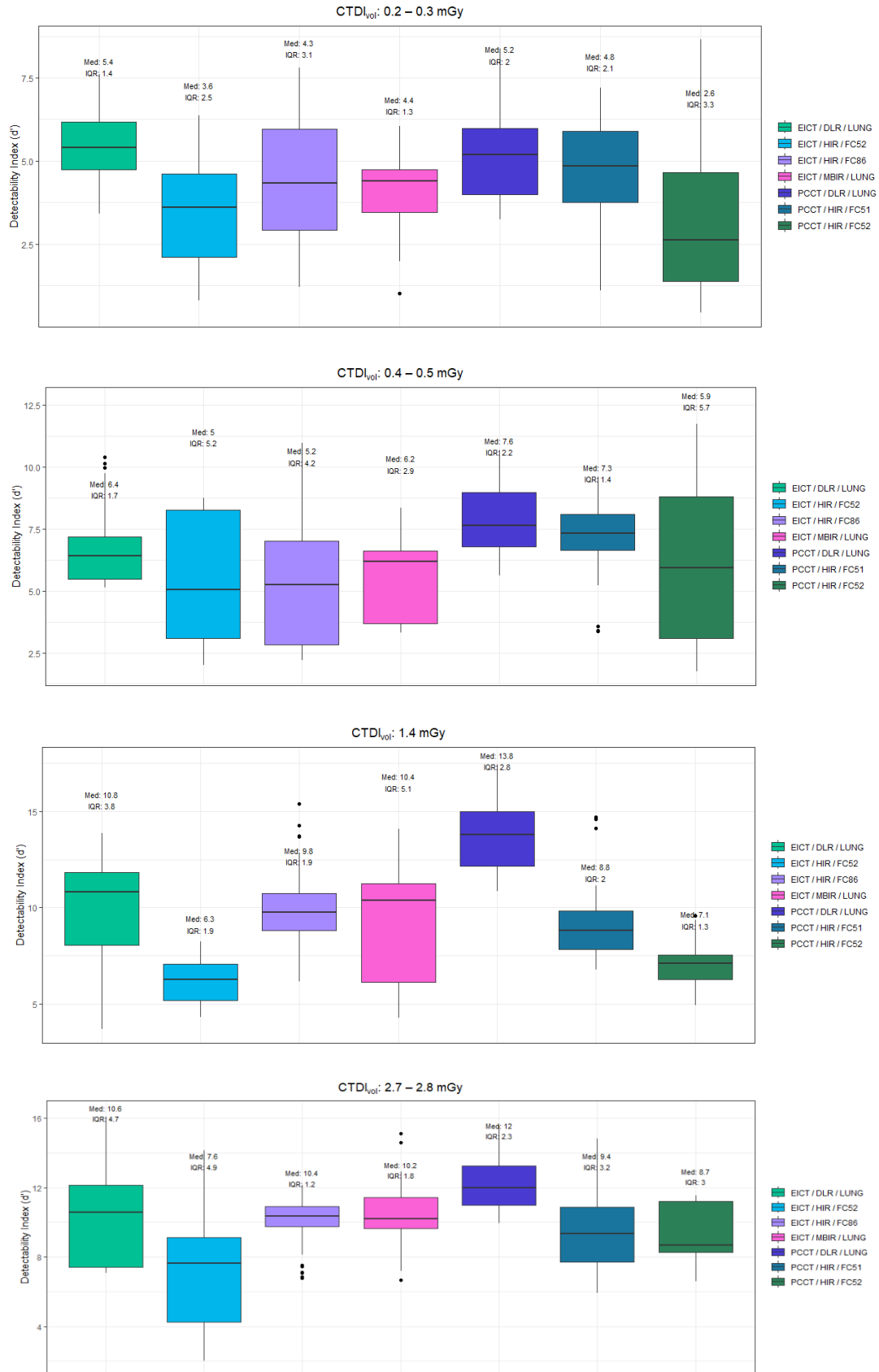


Figura 2: The detectability index (d') distribution for different reconstruction algorithms at four dose levels: 0.2–0.3 mGy, 0.4–0.5 mGy, 1.4 mGy, and 2.7–2.8 mGy. Each boxplot shows the distribution of d' values measured for synthetic ground-glass opacity (GGO) nodules in an anthropomorphic lung phantom. The median (Med) and interquartile range (IQR) are annotated for each protocol.

Across all configurations, higher CTDI_{vol} levels result in increased detectability index values. This increase is especially notable in protocols using advanced reconstruction algorithms and appropriate kernel selection. The highest detectability values are observed for protocols employing DLR reconstruction and the LUNG kernel, regardless of CT technology. In particular, the PCCT combined with DLR and the LUNG kernel achieved the highest detectability index values across all dose level. At 1.4 mGy, this protocol reached a median d' of 13.8 (IQR: 2.8), indicating both high detectability and low variability. At the highest dose level (2.7–2.8 mGy), it sustained its superior performance with a median d' of 12.0 (IQR: 2.3), suggesting robustness in high-dose conditions. Even at a low dose of 0.4 - 0.5 mGy, it maintained a median d' of 7.6 (IQR: 2.2), suggesting robustness at reduced radiation levels. At the lowest dose shown (0.2 - 0.3 mGy), it achieved a median of 5.2 (IQR: 2.0), demonstrating superior performance to all hybrid techniques.

EICT combined with DLR and the LUNG kernel also demonstrated competitive performance across all dose levels, though slightly inferior to PCCT-based DLR reconstructions. At 2.7–2.8 mGy, it achieved a median d' of 10.6 (IQR: 4.7), and at 1.4 mGy, a slightly higher median of 10.8 (IQR: 3.8). Notably, at the low-dose range of 0.4–0.5 mGy, the method maintained high detectability with a median d' of 6.4 (IQR: 1.7), demonstrating superior detectability compared most hybrid approaches. Even under the most constrained dose conditions, at 0.2–0.3 mGy, it achieved a median of 5.4 (IQR: 1.4), indicating good detectability in highly dose-limited scenarios. However, despite these results, EICT/DLR/LUNG remained slightly behind PCCT/DLR/LUNG, which consistently led across all dose levels. These findings highlight the benefits of photon-counting detector technology in enhancing detectability, especially when combined with deep learning-based reconstruction methods. Additionally, the use of MBIR with EICT (EICT/MBIR/LUNG) provided a modest improvement over hybrid reconstruction, achieving a median d' of 10.4 (IQR: 5.1) at 1.4 mGy, yet still not matching the performance observed with DLR-based techniques.

Across all dose levels, LUNG kernel reconstructions outperformed sharper kernels such as FC52 and FC86. This trend was especially pronounced at low-dose levels, where sharper kernels led to increased image noise, potentially compromising low-contrast detectability. For example, at 0.4–0.5 mGy, PCCT/DLR/LUNG achieved a median d' of 7.6 (IQR: 2.2), while PCCT/HIR/FC52 reached only 5.9 (IQR: 5.7). Within the EICT configurations, reconstruction kernel and algorithm choice had a substantial impact on detectability performance. At 0.2–0.3 mGy, EICT/DLR/LUNG achieved a median d' of 5.4 (IQR: 1.4), outperforming both EICT/HIR/FC86 at 4.3 (IQR: 1.3) and EICT/HIR/FC52 at 3.6 (IQR: 2.5). This trend continued at 0.4–0.5 mGy, with EICT/DLR/LUNG reaching 6.4 (IQR: 1.7), again higher than FC86 at 5.2 (IQR: 4.2) and FC52 at 5.0 (IQR: 5.2). Similarly, at 1.4 mGy, the LUNG kernel maintained superior performance with $d' = 10.8$ (IQR: 3.8), compared to 6.3 (IQR: 1.9) for FC52 and 9.8 (IQR: 1.9) for FC86. These findings emphasize the importance of kernel selection in optimizing image quality, especially when dose reduction is a clinical priority, as LUNG reconstructions offer a more favorable balance between noise suppression and detectability.

Statistical comparisons using the Kruskal–Wallis test confirmed the superiority of DLR methods across all dose levels. At the lowest doses (0.2–0.3 mGy), both PCCT/DLR and EICT/DLR showed significantly higher detectability than EICT/HIR with FC52 or FC86 kernels ($p < 0.001$). This trend persisted at 0.4–0.5 mGy and at higher doses (1.4 and 2.7–2.8 mGy), where PCCT/DLR consistently achieved the highest overall d' values, with statistically significant advantages over HIR-based reconstructions ($p < 0.0001$).

4 Conclusions

The findings clearly demonstrate that DLR demonstrated superior performance over all other reconstruction techniques across multiple dose levels and CT technologies. When combined with PCCT systems, DLR consistently yielded the highest detectability values and the lowest variability, even at ultra-low doses (≤ 0.3 mGy). This robustness at reduced radiation levels underscores the clinical potential of DLR to maintain image quality while minimizing patient exposure. Reconstructions using the LUNG kernel also outperformed sharper kernels (e.g., FC52 and FC86) across all dose levels, particularly under

dose-limited conditions, highlighting its role in preserving low-contrast detectability. EICT systems paired with DLR also performed well, but their detectability metrics were generally inferior to those obtained with PCCT, particularly at doses above 0.4 mGy. These differences were statistically significant, as confirmed by Kruskal–Wallis tests, further reinforcing the advantage of advanced reconstruction techniques, especially when paired with state-of-the-art detector technology.

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Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique

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Abstract

The magnetic properties of materials allow for the highly sensitive evaluation of certain physical characteristics related to their elemental composition. In this study, the magnetic susceptibility technique was used to estimate the iron concentration in four commercial pharmaceutical formulations (designated as L1, L2 and L3), each with a theoretical elemental iron content of 65.74 mg per tablet, as specified by the manufacturer. An experimental system was configured to measure the magnetic interaction between the pharmaceutical sample and a permanent magnet, using a high-precision analytical balance (Radwag, model AS 220.R2; resolution: 0.1 mg) and a neodymium magnet with a constant magnetic field of 370×10^{-3} T. The system was calibrated based on the axial distance (Z-axis) from the base of the magnet up to 10 cm. At a distance of $Z = 0.516$ mm, a magnetic dipole moment of $m = 0.15 \text{ A} \cdot \text{m}^2$ was determined, corresponding to a magnetic induction of $B = 245 \times 10^{-3}$ T. The calibration curve was obtained using a reference drug (ferrous fumarate: 115 mg of Fe in 1.3 g of $\text{C}_4\text{H}_2\text{FeO}_4$), with powdered mixtures prepared at elemental iron concentrations of 24.86, 33.14, 41.46, 49.72, 66.60, 82.83, and 146.43 mg. The resulting model exhibited an exponential fit with a coefficient of determination of $R^2 = 0.992$. The evaluation of the commercial tablet samples yielded the following average elemental iron concentrations: $L1 = 69.54 \pm 0.10$ mg, $L2 = 74.49 \pm 0.30$ mg, $L3 = 69.42 \pm 0.25$ mg, and $L4 = 69.61 \pm 0.37$ mg. Preliminary results suggest that magnetic susceptibility could serve as a viable alternative technique for quality control and quantification of elemental iron in pharmaceutical formulations. It is recommended to conduct complementary studies using other elemental analysis techniques to validate and strengthen the findings obtained.

Keywords: Magnetic susceptibility; iron concentration; pharmaceutical formulation; quantitative analysis.

1 Introduction

Magnetic materials has been in focus as an interdisciplinary topic for several decades. Scientists from physics, material science, chemistry, biology and engineering are working on various research, The properties of a material are defined not only by the magnetization or the magnetic induction, but the way in which these qualities vary with the applied magnetic field (Spaldin, 2010).

The susceptibility indicates how responsive a material is to an applied magnetic field (diamagnetic, paramagnetic or ferromagnetic). Diamagnetism is a property of all matter and arises from the interaction of paired electrons with the magnetic field. the diamagnetism is especially important in the consideration of materials with completely filled electronic shells, that is, systems which do not contain any unpaired electrons (Carlin, 1986). This cannot be taken as an operational definition because certain paramagnetic materials can become diamagnetic under certain conditions.

Ferromagnetism is a phenomenon whereby a substance can become a permanent magnet or strongly reacts to a magnetic field. The term “ferromagnetism” is derived from the magnetism detected iron, as Fe was the first element in which ferromagnetism was observed, (Chikazumi et al. 1997). Pierre-Ernest Weiss discovered the structural alignments of dipoles, the compounds exhibiting ferromagnetism show the spontaneous parallel alignment of permanent dipoles due to the movement of electrons around atomic orbitals (Coey, 2010).

The use of magnetic susceptibility as a parameter to detect ferromagnetic metals has already been applied in various studies, particularly in the field of geophysics. López et al. (2000) found a negative correlation between magnetic susceptibility and the amount of heavy metals in a coastal environment. On the other hand, Orosun et al. (2020) used magnetic susceptibility to detect heavy metal contamination at a car service station. In both studies, magnetic susceptibility was used as an indirect method to assess the presence of heavy metals. This same principle is applied in the present study to detect the concentration of iron in various commercial pharmaceutical products.

The use of linear regressions for the determination of metal concentrations through magnetic susceptibility has already been investigated by Ayoubi et al. (2019). In their study, they constructed semi-quantitative calibration curves to determine metal concentration based on the magnetic susceptibility of a solid sample.

Using the principle of metal detection through magnetic susceptibility, this work proposes the use of an experimental setup like that used by Davis (1992), consisting of a neodymium magnet and an analytical balance to detect the concentration of iron in samples of various supplements containing ferrous fumarate, based on the methodology of Cano et al. (2008) and Sosa et al. (2006).

2 Materials, methods and procedures

System calibration

A neodymium magnet measuring $2.5 \times 2.5 \times 2$ cm was used, and its magnetic field was mapped using a teslameter. The magnetic flux density (B) was recorded at incremental distances ranging from 0.1 to 10 cm, in order to characterize its spatial field distribution.

The implemented system is shown in Figure 1. An analytical balance (Radwag Model AS 220.R2, precision 0.1 mg) was used, on which a cylindrical support was placed to hold the neodymium magnet with a permanent magnetic field. To measure the magnetic susceptibility of the sample, a Petri dish was positioned above the balance top glass surface.

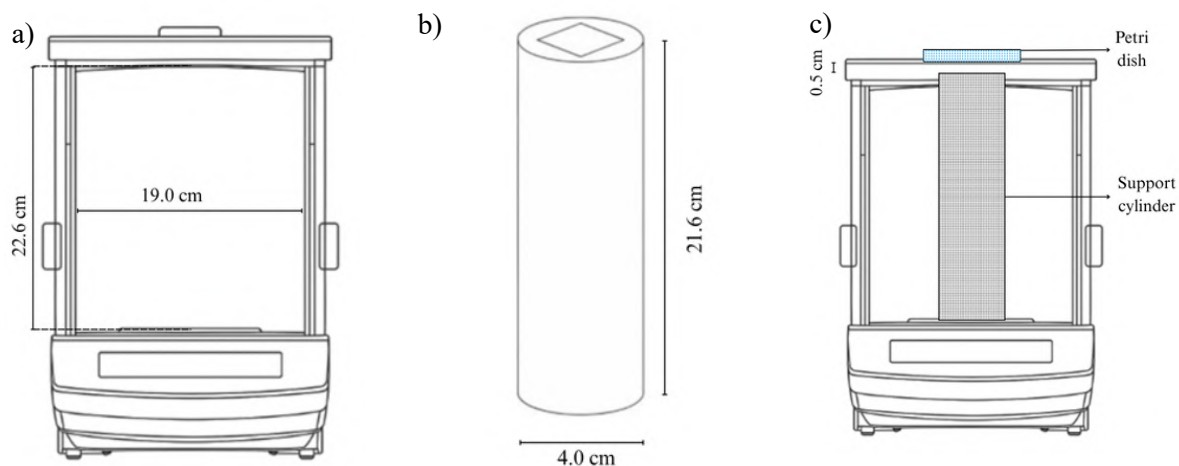


Figure 1. Experimental setup. a) Dimensions of the analytical balance; b) Dimensions of the support cylinder; c) experimental arrangement

Sample Preparation and Calibration Curve Construction

"Five tablets of the evaluated pharmaceutical product (ferrous fumarate) were pulverized into a fine powder. From this, aliquots ranging from 0.008 g to 6.5 g were accurately weighed. Using the manufacturer's specifications as control, the elemental iron content of each sample was calculated. A calibration curve was constructed by plotting iron concentration on the X-axis and the instrument response (balance reading) on the Y-axis, facilitating quantitative determination of iron in unknown samples.

Measurement of Test Samples (Pharmaceuticals)

Three pharmaceutical tablets of ferrous fumarate from different brands were evaluated to compare the labeled iron content with the experimentally determined amount. One tablet from each brand was pulverized into a fine powder, and an aliquot was accurately weighed using an analytical balance. The elemental iron concentration in each sample was then determined using the previously established calibration curve.

3 Results and discussion

Using the teslameter, magnetic induction (B) was measured at distances from 0.1 cm to 10 cm, with each point recorded in triplicate showing an exponential behavior where the closer the distance to the magnet, the greater the magnetic induction, Figure 2.

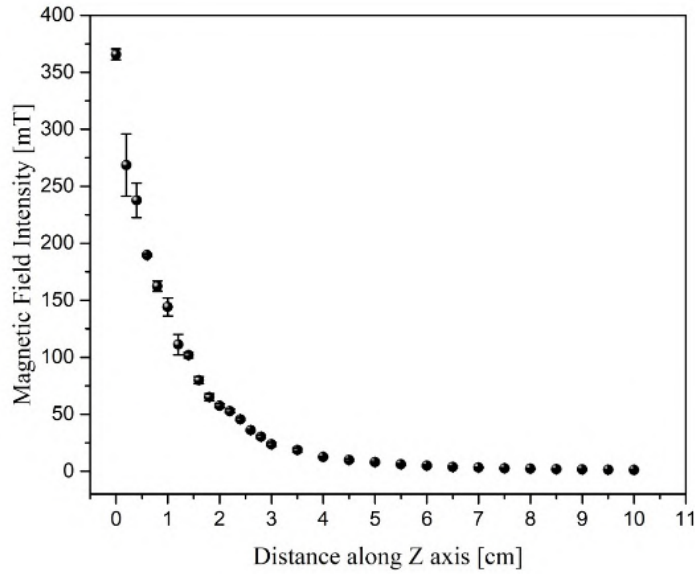


Figure 2. Behavior of magnetic induction at different distances from the magnet

The distance (r) between the neodymium magnet and the sample is approximately 0.516 mm along the Z-axis. This corresponds to a magnetic induction (B) of 245×10^{-3} T (figure 2). Using Sosa's equation (2006), a dipole moment (m) of $0.15 \text{ A} \cdot \text{m}^2$ is obtained.

$$B = \frac{2m \times 10^{-7}}{r^3} (T) \quad (1)$$

The calibration curve to determine the concentration of iron used a drug whose elemental iron concentration is 115 mg per tablet (1.3 g of $\text{C}_4\text{H}_2\text{FeO}_4$). A powder mixture was prepared at elemental iron concentrations ranging from 24.86 to 146.43 mg. The resulting model showed an exponential fit with a coefficient of determination of $R^2 = 0.992$ (figure 3).

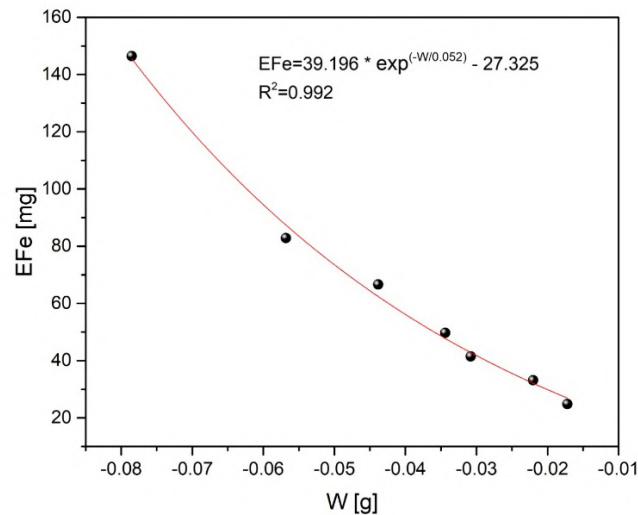


Figure 3. Calibration curve of elemental iron with a drug

This graph allows us to observe an exponential model (equation 2), which enables us to identify different samples containing elemental iron.

$$EFe = 39.196e^{\frac{-W}{0.052}} - 27.325 \quad (2)$$

To verify this model, different drugs (L1, L2 and L3) that contained ferrous fumarate were used. These drugs report a concentration of elemental iron of 65.74 mg. Five tablets of each medication were ground and measured separately in the experimental arrangement (Table 1).

Table 1: Measurements of the drugs.

Sample	tablet 1 (g)	tablet 2 (g)	tablet 3 (g)	tablet 4 (g)	tablet 5 (g)	Average of the tablets (g)
L1	-0.0616	-0.0630	-0.0628	-0.0649	-0.0587	-0.0622
L2	-0.0591	-0.0618	-0.0598	-0.0595	-0.0540	-0.0588
L3	-0.1054	-0.1039	-0.1088	0.1035	-0.1000	-0.1043

A significant variation can be seen in the drug L3 where the result (-0.1043 g) is much higher than the rest when substituting into equation 2, it is possible to determine the concentration of elemental iron in the drugs (Table 2).

Table 2: Concentration of elemental iron of each drug.

Sample	Average of the tablets (g)	Elemental iron (mg)
L1	-0.0622	102.3103
L2	-0.0588	78.4043
L3	-0.1043	263.9721

These results can be compared both with the reported composition (65.74 mg per tablet) of each drug or through an elemental analysis of each sample. Additionally, it must be taken into account the compositions of the excipients of each drug, since if these samples contain any other materials that interfere with the magnetic field, it will be noticeable in this system. This is done in order to provide validity to this procedure. Excipients are used to convert a drug into a pharmaceutical form that the patient can use to achieve a therapeutic benefit. Excipients are included in a formulation due to their properties that, along with a process, allow for the manufacture of a medication with the required specifications. The desired properties of an excipient refer to its functional performance or functionality (Moreton, 2006).

Classification of the excipients is based on their role in the pharmaceutical formulation and on their interactions influencing drug delivery, based on their chemical and physico-chemical properties. The main classes are the antioxidants, coating materials, emulgents, taste- and smell-improvers, ointment bases, conserving agents, consistency-improvers and disintegrating materials. Some of the excipients may serve multiple purposes; for example, methylcellulose is a coating material, is applied in the preparation of suspensions, to increase viscosity, as a disintegrating agent or binder in tablets (Kalasz et al. 2006).

4 Conclusions

In this experiment, we measured the magnetic susceptibility of iron using a simple experimental setup that is easy to implement in a laboratory. The measurement involves determining, through an analytical balance, the magnetic interaction between the sample and a small magnet. This system allowed us to work with different samples of ferrous fumarate and obtain the expected behavior; although the model is simple, further tests with samples of known exact composition need to be conducted in order to give it greater validity. Finally, this technique has potential for applicability in the field of radiology, especially in advanced magnetic resonance imaging such as SWI (Susceptibility-Weighted Imaging) and QSM (Quantitative Susceptibility Mapping), to detect microhemorrhages, iron or calcium deposits, and quantify biomarkers such as gadolinium or nanoparticles.

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Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing

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Abstract

Positron Emission Tomography (PET) is a functional imaging modality that evaluates various pathologies, primarily oncological, but also those related to the central nervous system and cardiac conditions^[Alessio 2004; Christian 2013; Kolle 2023]. This technique plays a significant role in oncology treatments, being widely used for diagnosis, staging, treatment response assessment, recurrence detection, and prognosis^[Alessio 2004; Christian 2013]. To perform this exam, hardware, software, and image quality verification are necessary^[Kolle 2023; IAEA 2004]. These evaluations follow international standards such as NEMA (National Electrical Manufacturers Association), IAEA (International Atomic Energy Agency), and ACR (American College of Radiology)^[IAEA 2004; NEMA 2018]. To ensure proper verification of equipment and components, many quality control procedures require highly specific phantoms recommended by these organizations^[NEMA 2018]. The Jaszczak phantom, recommended by the ACR, allows assessment of many parameters, including spatial resolution for PET and SPECT equipment^[Albus 2025]. However, acquisition cost, low supplier availability, and bureaucratic customs obstacles make it challenging for nuclear medicine services to obtain phantoms like this. 3D printing technology can facilitate access to these phantoms more easily and at a lower cost, while also enabling the customization of phantoms for specific equipment^[Aimar 2019; Filippou 2018; Valchanov 2023]. This study aims to assess the feasibility of evaluating the spatial resolution of a PET device using a phantom manufactured through additive manufacturing. A prototype was developed utilizing 3D printing techniques, including FDM and MSLA, with PETG filament and photopolymer resins. Solid rods with diameters of 3mm, 5mm, 6.5mm, 8mm, 9.5mm, and 11mm were positioned within a sealed cylindrical chamber that measures 15cm in height and 25cm in internal diameter. The equipment used was the PET/CT Discovery 690 by GE (General Electronics), located at the CTMM at UFMG. The injected activity was 2.90 mCi (107.3 MBq) of FDG in a total volume of 4.1L water. PET images of 2 minutes and 10 minutes of acquisition time were acquired. The acquired images revealed that the equipment exhibited a spatial resolution between 5mm and 6.5mm for images with 2-minute acquisition and between 3mm and 5mm for 10-minute acquisitions. These values align with IAEA recommendations^[4], which indicate a spatial resolution range of 4mm to 6mm for PET equipment. The production cost of components was below USD 150,00. This represents a significantly lower cost compared to purchasing a commercial phantom. The study concluded that it is possible to evaluate a PET device using a Jaszczak-analog phantom produced via additive manufacturing and to determine spatial resolution variations according to different imaging acquisition protocols. This approach may enhance access to such simulators for nuclear medicine services at a lower acquisition cost.

Keywords: *Phantom, Additive Manufacture; PET; Spatial Resolution; Quality Control.*

1 Introduction

Positron Emission Tomography (PET) imaging modality assesses the metabolic and physiological activity of organs, tissues, or systems, depending on the radiopharmaceutical used. The most commonly used is ^{18}F -FDG (fluorodeoxyglucose), which is widely applicable due to its glucose analog properties, making it suitable for oncological, cardiac, and neurological evaluations. In oncology, this technique allows for the analysis of possible metastases and disease staging, thus playing a significant role in patient treatment. Given these factors, to ensure accurate reports and examinations, periodic quality checks are required for the exam and all associated systems, such as quality control of the imaging equipment, the dose calibrator, and the radiopharmaceutical. These quality controls follow guidelines from international organizations (IAEA, ACR, NEMA) and reference parameters provided by the equipment manufacturer. [Alessio *et al.*, 2004; Christian, 2013; Kolle *et al.*, 2023; IAEA 2009].

International organizations such as NEMA recommend simulator objects with specific materials and geometries to ensure proper equipment evaluation during quality control procedures. Each phantom has a distinct purpose and should behave in a manner that simulates organs, tissues, and lesions according to the imaging equipment it is intended for, replicating radiation absorption and scattering similar to the human body. Given these specifications and the limited availability of manufacturers, acquiring such phantoms involves high costs, making access difficult for many nuclear medicine services. [IAEA, 2009; NEMA, 2018]

The Jaszczak phantom, recommended by the ACR, is a versatile simulator suitable for use not only with PET equipment but also with SPECT systems. This phantom includes a set of spheres of varying diameters and six groups of rods with different diameters, as shown in Figure 1. These rods and spheres, with variable diameters, are designed to simulate hot or cold lesions, depending on the type of Jaszczak phantom used. This phantom enables the evaluation of multiple parameters, such as overall image quality, accuracy of attenuation and scatter correction, uniformity testing, and hot/cold lesion detectability and spatial resolution. [Albus, 2025]

The spatial resolution of imaging equipment is defined by its ability to distinguish between two adjacent, hyperactive or hypoactive lesions. In addition to the intrinsic characteristics of the device itself, spatial resolution also depends on image acquisition protocols, reconstruction techniques, and proper acquisition setup. [IAEA 2009]

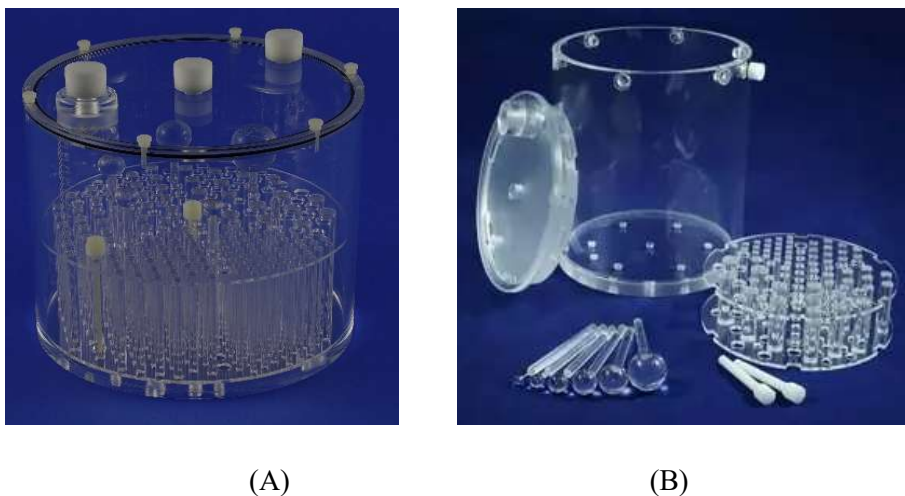


Figure 1: Commercially available Jaszczak QC phantom; (A) assembled and sealed phantom, (B) view of all internal parts in disassembled state. [DATASPECTRUM, 2025]

The Jaszczak phantom is highly versatile and particularly important in nuclear medicine services. However, it is associated with a high acquisition cost and low supplier availability. According to quotations from import companies in Brazil during the first quarter of 2025, the acquisition cost for the Jaszczak phantom, applicable to PET and SPECT equipment, was approximately USD 16,000.00. This high cost makes it unfeasible for many services to acquire this phantom. 3D printing technology can increase accessibility to this type of phantom for various nuclear medicine services at a lower cost, while also allowing customization according to the available equipment. [Aimar, 2019; Filippou, 2019; Valchanov *et al.*, 2023]

Additive manufacturing, or 3D printing, is a volumetric construction method that builds components from digital models designed in computer-aided design (CAD) software, for example (ISO/ASTM 52900, 2021). Several types of printers and materials are available. For this prototype, the chosen technologies were Material Extrusion (FDM - Fused Deposition Modeling) and Vat Photopolymerization (MSLA - Masked Stereolithography).[ISO/ASTM 52900, 2021; Xu *et al.*, 2021]

The objective of this study was to evaluate the feasibility of assessing the spatial resolution of a PET device using a phantom developed in-house and produced through additive manufacturing.

2 Methodology

A prototype analogous to the Jaszczak phantom was constructed using additive manufacturing to carry out this study. Two construction methods were employed: Fused Deposition Modeling (FDM) and Masked Stereolithography (MSLA). Solid rods were fabricated to simulate cold lesions in the interior part of the phantom. The commercially available Jaszczak phantom features rods with diameters of 4.8 mm, 6.4 mm, 7.9 mm, 9.5 mm, 11.1 mm, and 12.7 mm, with a height of approximately 9 mm. The prototype, in contrast, was built with rods measuring 3 mm, 5 mm, 6.5 mm, 8 mm, 9.5 mm, and 11 mm in diameter. Smaller diameters were used to evaluate the feasibility of construction and resolution testing across different dimensions. Each rod was spaced apart by a distance equal to its diameter. Additionally, the lid, sealed cylindrical chamber (approximately 15 cm in height and 25 cm in internal diameter), and eight screws for securing the lid to the chamber were also manufactured.

Each set of rods was arranged in a layer and connected using interlocking puzzle-style fittings. These components were printed using three different materials. Rod sets of 3 mm, 5 mm, and 6.5 mm were manufactured via MSLA using the *Crystal Resin* by 3D Cure®. The 9.5 mm and 11 mm rods were printed via MSLA using *Pro High-Performance Resin* from 3Dlab®. The 8 mm rod set, the eight screws, the sealed cylindrical chamber, and the lid were produced using PETG filament from 3Dlab® using the FDM method.

Shenzhen Creality 3D Technology Co., Ltd manufactured the printers used. The FDM printer was the *Creality Ender 5 Plus*, and the MSLA printer was the *Creality Halot Mage Pro*. The time and cost required to fabricate each component will be compared to assess the economic feasibility of prototype construction.

2.1 Prototype Assembly and Acquisition Protocols

The internal assembly of the rod sets is consistent with Figure 2. The translucent components were manufactured using Crystal Resin, the gray parts were produced with Pro High-Performance Resin, and the white component was printed with PETG filament. The rod sets were inserted into the cylindrical chamber and sealed to prevent water leakage.

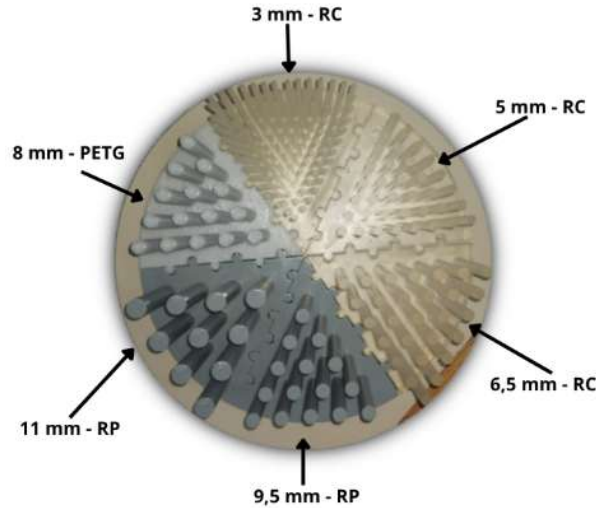


Figure 2: Location of rod sets used for testing. RC = components manufactured with Crystal Resin. RP = components manufactured with Pro High-Performance Resin. PETG = components manufactured with PETG filament. (author's archive)

The prototype was filled with approximately 4.1 liters of water about 50 hours before image acquisition to reduce air bubbles, as the sealed cylindrical chamber material is not transparent. An activity of 107.3 MBq (26 KBq/mL) of ^{18}F -FDG was injected approximately 50 minutes before the imaging procedure. The phantom was positioned at the center of the equipment's field of view (FOV).

The image acquisition protocols were based on IAEA specifications; one aligned with oncological scanning parameters and the other with brain imaging protocols. The CT scan was performed using the following parameters: tube voltage of 120 kV; tube current of 400 mAs; helical acquisition mode; reconstruction window of "standard" or "PETAC"; slice thickness of 3.75 mm; pitch of 0.984:1; full reconstruction mode; and tube rotation time of 0.7 s. PET imaging protocols used two acquisition times (2 minutes and 10 minutes) and two sizes of the reconstruction matrices: (192×192) and (256×256). All imaging procedures were carried out using a GE Discovery 690 PET/CT system at the Medicine School of the *Universidade Federal de Minas Gerais* - UFMG.

2.2 Image Analysis

Image analysis was performed to qualitatively determine the spatial resolution of the equipment using this prototype to identify and differentiate the rods visible in the acquired images. Accordingly, the smallest set of distinguishable rods indicates the range in which the spatial resolution of the equipment lies.

As image acquisition parameters influence spatial resolution, a comparative assessment will be conducted between images acquired at different acquisition times (2 minutes and 10 minutes) and reconstructed with varying matrix sizes (192×192 and 256×256). Additionally, data about printing times and the cost of the prototype components using 3D printers will be presented to evaluate the economic feasibility of this additive manufacturing approach.

3 Results and Discussion

The phantom prototype is shown in Figure 3(A), and its positioning within the equipment is illustrated in Figure 3(B), where the laser beams indicate the center of the equipment's Field of View (FOV).

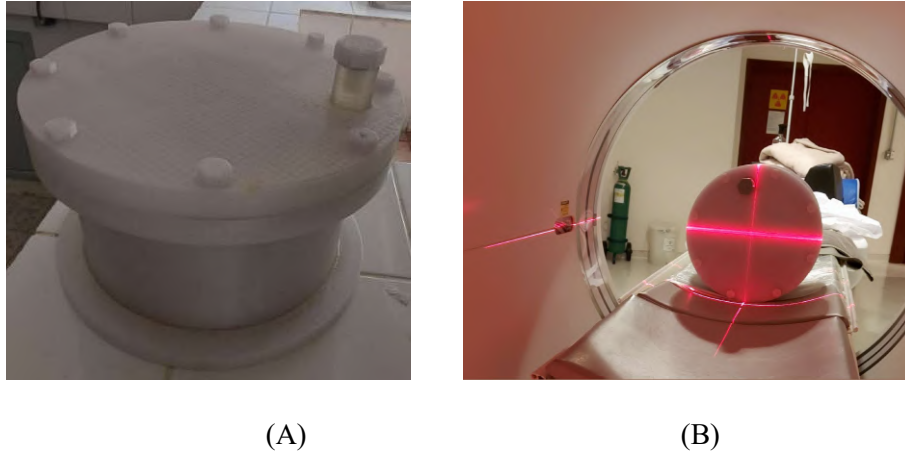


Figure 3: (A) Closed prototype after rod insertion and water filling. (B) Prototype positioning within the equipment, centered at the Field of View (FOV). (author's archive)

Figure 4 displays the CT scan of the prototype, where the position and arrangement of the internal rods are visible. This layout remains consistent across all subsequent PET images, thus serving as a reference for spatial resolution assessments in the rest of this work.

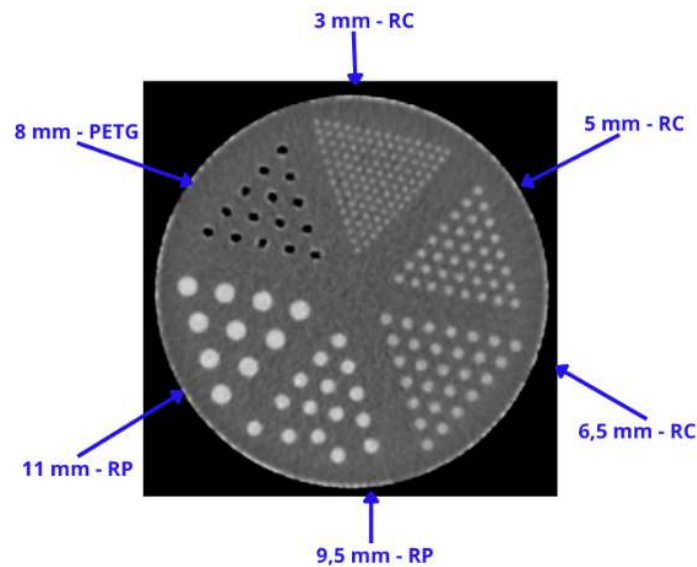


Figure 4: CT scan image used to verify and identify the positioning of internal components. This scan provides a reference image depicting the arrangement of the rods inside the prototype, facilitating the evaluation of the PET images. (author's archive)

Figure 5 illustrates the differences between acquisitions with varying durations and reconstruction matrices. The 2-minute acquisitions demonstrated rod differentiation within the 5 mm to 6.5 mm range for the spatial resolution analysis. Among these, reconstruction with the 192×192 matrix provided superior resolution, allowing more precise visualization of the rods than the 256×256 matrix.

The images acquired with longer acquisition times demonstrated superior spatial resolution, particularly when reconstructed using the 192×192 matrix. Despite generating larger voxels, this matrix yielded better performance due to enhanced count statistics. The highest resolution, ranging from 3 mm to 5 mm, was observed in the 10-minute acquisition, the only instance where the 5 mm rods were individually

distinguishable. In contrast, all the 2-minute acquisitions exhibited a spatial resolution between 5 mm and 6.5 mm. These findings were compared with the results of the spatial resolution test conducted on the equipment using point sources, in accordance with NEMA standards, in early 2025. The results showed transaxial spatial resolutions of 5.07 mm at 1 cm from the center and 6.06 mm at 10 cm. Both parameters are within the NEMA-established limits of 6.2 mm and 6.9 mm, respectively. Based on these results, it can be observed that the values obtained using the phantom prototype produced in this work were compatible with the values obtained using point sources, according to the NEMA standards.

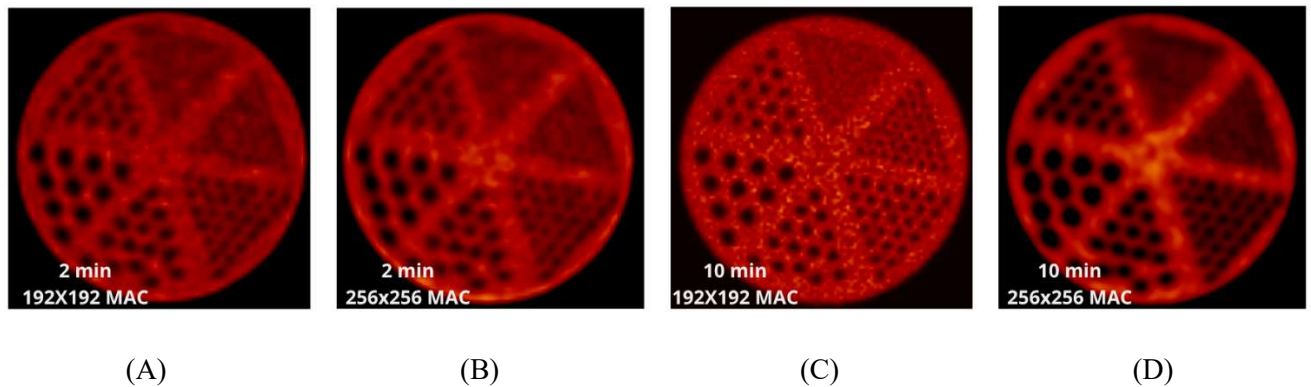


Figure 5: PET Image Acquisition with Different Times and Reconstruction Matrices. (A): 2-minute acquisition using a 192×192 matrix.(B): 2-minute acquisition using a 256×256 matrix. (C): 10-minute acquisition using a 192×192 matrix. (D): 10-minute acquisition using a 256×256 matrix. (author's archive)

Table 1 contains the construction time, material quantity, and material cost for each component used in this experiment, as well as the total prototype construction cost, considering only the material expenses.

Table 1: Construction time, material quantity, and material cost for each component used in this experiment.

Part	Material used	Printing time (hh:mm:ss)	Mass of the parts (g)	Estimated cost (USD*)
3mm	RC	09:48:12	148.55	6.68
5mm	RC	09:48:12	145.93	6.56
6.5mm	RC	09:48:12	197.66	8.89
8mm	PETG	07:02:15	176.90	3.81
9.5mm	RP	09:48:12	172.25	12.39
11mm	RP	09:48:12	163.95	11.79
Sealed cylindrical chamber	PETG	45:16:37	1417.50	30.59
Seal	PETG	41:00:14	1342.50	28.97
8 x Fixing Screws	PETG	03:35:36	44.80	0.97
Total	-	145:55:42	3810.04	110.65

*Dollar exchange rate on June 8, 2025

(Produced by the author)

Based on the data in Table 1, the total cost of constructing the prototype was approximately USD 111.00. This amount is significantly lower than the acquisition cost of a commercially available Jaszczak phantom, highlighting the economic advantage of additive manufacturing for nuclear medicine services.

4 Conclusions

The tests demonstrated that the spatial resolution values for the equipment evaluated using the 3D printed phantom were in accordance with the parameters required by NEMA standards, and the values measured using the point source methodology.

The results also demonstrated the influence of parameters such as image acquisition time and reconstruction matrix size on the spatial resolution of the equipment.

The production phantoms for image quality evaluation using additive manufacturing proved technically viable and economically advantageous. The production costs were significantly lower than those of commercial models such as the Jaszczak phantom.

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Evaluation of the affect on Quality Control tests of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner

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Abstract

The increasing number of magnetic resonance imaging (MRI) scanners in dense urban settings introduces challenges in mitigating external interferences from nearby large metallic structures. This study evaluates the performance (QC tests) of a 3.0 T MRI scanner since its installation, and assesses potential external interferences resulting from the installation of another MRI scanner in an adjacent room and the construction of nearby subway tracks. The Magphan 170 phantom was scanned according to ACR standards. A total of 108 MR image acquisitions were collected from the 3T MRI Philips scanner (from its installation in March 2016 until December 2023). QC tests evaluated were: geometric distortion, slice thickness accuracy (STA), signal-to-noise ratio (SNR), percent image uniformity (PIU), percent signal ghosting, low contrast detectability, and high contrast spatial resolution. Geometric distortion, SNR and PIU tests presented statistically significant results from the first intervention due to renovation of shielding, which led to better results in all three tests. In the STA test, a significant increase was observed after the beginning of subway operations, which may have been influenced by the gradient field non-uniformity generated by the subway train passing near the MRI scanner, producing eddy currents. Percent signal ghosting, low contrast detectability and high contrast spatial resolution tests did not have significant changes for both interventions. The performance evaluation showed that the installation of another MRI scanner did not interfere. However, geometric distortion, SNR and PIU tests presented better results due to shielding renovation efficiency. Only the STA test showed a statistically significant change following subway construction.

Keywords: magnetic resonance imaging, quality control tests, interference, subway line.

1 Introduction

The increasing number of magnetic resonance imaging (MRI) scanners in densely populated urban areas introduces challenges in mitigating external interferences from nearby large metallic structures. These interferences, referred to as quasi-DC electromagnetic disturbances, are caused by moving direct current voltage sources that generate magnetic field lines, such as those produced by subways. Even at distances up to 100 meters from MRI scanners, subway stations can generate magnetic field lines that impact the strength of the static magnetic field, consequently affecting the quality of acquired images (PROADESS, 2025), (Kellogg, 2025), (Hellerbach, 2013).

For reliable MRI diagnosis, regular quality control (QC) tests following national and international protocols are essential (ACR, 2015), (ANVISA, 2021), (NEMA, 2008).

An MRI scanner in a specific imaging diagnosis department is positioned just 28 meters from a subway track (Takeuchi, 2018). Additionally, this department houses two 3.0 Tesla MRI scanners. One, with a gradient strength of 40 mT/m, and another with 45 mT/m, are positioned approximately 14 meters apart from their magnetic isocenters.

This study evaluates the performance (QC tests) of a 3.0 T MRI scanner since its installation, and assesses potential external interferences resulting from the installation of another MRI scanner in an adjacent room and the construction of nearby subway tracks.

2 Materials and Methods

The Magphan 170 phantom was scanned according to ACR standards, specifications in Table 1. The phantom, featuring a 20 cm external diameter and containing a 10 cm test cube with known distances, groups of line pairs, acrylic ramps, and circles of varying acrylic thicknesses, was utilized. The 8-channel head coil of the scanner was selected to position the phantom and acquire the images (ACR, 2015), (AAPM, 2010).

Table 1. Sequence parameters used in image acquisitions for quality control tests.

Sequence	TR (ms)	TE (ms)	FOV (mm)	MATRIX (pixels)	NEX	Bandwidth (Hz/pixel)	Flip angle	Slice thickness (mm)	Time sequence (min)
Spin Echo	500	20	256 x 256	256 x 256	1	364	90	5	03:12

A total of 108 MR image acquisitions were collected from the 3T MRI scanner (from its installation in March 2016 until December 2023). These acquisitions were categorized into three periods: the first period (n = 21) extended from March 2016 (installation of the MRI scanner) to December 2017; the second period (n = 15) encompassed May 2018 (installation of the second scanner) to December 2018, and the third period (n = 69) extended from January 2019 (beginning of subway operations) to December 2023.

QC tests evaluated were: geometric distortion, slice thickness accuracy (STA), signal-to-noise ratio (SNR), percent image uniformity (PIU), percent signal ghosting (PSG), low contrast detectability (LCD), and high contrast spatial resolution (HCSR).

Data analysis was performed using Microsoft Excel software. For MR image manipulation and region of interest (ROI) measurements, RadiAnt Dicom Viewer software was utilized. The statistical analysis was conducted using R 4.0.3 software. The Wilcoxon rank-sum test was employed to assess significant changes associated with two interventions: first, in December 2017, following the installation of an additional scanner in an adjacent room; and second, in January 2019, coinciding with the beginning of nearby subway station operations.

3 Results

Table 2 presents the detailed statistical outcomes, including p-values and other numerical findings, obtained from the Wilcoxon rank-sum test. The analysis was divided into three periods: the first represents the "before" phase, preceding the installation of the second MRI scanner (first intervention); the second period represents the time after the second scanner's installation but before subway operations commenced (second intervention); and the third period captures the data once subway activities began.

Table 2. Statistical analysis of QC test results for each period.

QC tests	Period	Mean $\pm$ SD	p-value for interventions	
			1st	2nd
Geometric Distortion (%)	1st	0.58 ± 0.2	p<0.05	-
	2nd	0.467 ± 0.149		0.603
	3rd	0.483 ± 0.187	-	
STA (%)	1st	-1.976 ± 10.362	0.773	-
	2nd	1.75 ± 2.688		p<0.05
	3rd	3.967 ± 2.3	-	
PSG (%)	1st	0.05 ± 0.027	0.177	-
	2nd	0.04 ± 0.031		0.252
	3rd	0.044 ± 0.027	-	
SNR (normalized)	1st	0.359 ± 0.197	p<<0.05	-
	2nd	0.706 ± 0.129		0.252
	3rd	0.746 ± 0.124	-	
PIU (%)	1st	88.229 ± 4.685	p<0.05	-
	2nd	92.281 ± 2.521		0.559
	3rd	92.557 ± 2.796	-	
LCD	1st	34.678 ± 12.958	0.060	-
	2nd	43.517 ± 7.936		0.301
	3rd	40.185 ± 10.77	-	
HCSR	1st	1.014 ± 0.222	0.817	-
	2nd	0.983 ± 0.158		0.945
	3rd	0.985 ± 0.156	-	

3 Discussion

Table 2 presents statistically significant results ($p < 0.05$) from the geometric distortion, SNR, and PIU tests following the first intervention. This intervention involved reinstalling additional shielding in an adjacent room to accommodate a new MRI scanner. This renovation led to better results in all three QC tests, indicating the effectiveness of the new shielding despite the higher magnetic field (3.0 T) in the adjacent room. All measured values remained within acceptable tolerance levels (below 2%). Despite significant differences observed after the new MRI scanner's installation (2nd period), there's no evidence that image distortion compromised diagnostic quality (Price, 2010), (Kellogg, 2021), (Farias et al, 2018).

Variations in magnetic field strength can influence the Signal-to-Noise Ratio (SNR) in MRI systems. SNR is crucial for evaluating hardware performance, with issues like RF coil failure or compromised RF shielding leading to reductions. A significant increase in SNR values was observed after the first intervention (2nd period), indicating that the previous shielding in the adjacent room was ineffective or due to reduced interference with RF pulses. Subsequently, SNR values slightly increased in 3rd period, though this change was not statistically significant (Table 2, $p = 0.252$), suggesting subway passage did not affect magnetic field strength or hardware. (Kellogg, 2021), (ACR, 2015), (ANVISA, 2021), (NEMA MS 1, 2008), (AAPM, 2010), (O'Callaghan et al, 2014), (Price et al, 1990), (Domingues, 2020), (Keenan et al, 2018), (Murata et al, 2023), (Mazzola et al, 2005), (Milidonis et al, 2016).

The PIU test directly reflects B0 homogeneity and coil functionality. Low PIU values may indicate coil intensity variation, eddy currents, or RF system problems. Table 2 shows a significant difference ($p < 0.05$) in PIU measurements during the first intervention, suggesting ineffective shielding in the old adjacent room. After the shielding renovation (2nd period), PIU values increased. Even after the second intervention (3rd period), values did not increase significantly, demonstrating that subway activities are not influencing this parameter (ACR, 2015), (ANVISA, 2021), (NEMA MS 3, 2008), (AAPM, 2010), (O'Callaghan et al, 2014), (Price et al, 1990), (Domingues, 2020), (Keenan et al, 2018), (Murata et al, 2023), (Mazzola et al, 2005).

The STA test assesses RF amplifier malfunctions, gradient field non-uniformity, and poor gradient performance. During the 1st period, measurements showed a high standard deviation, possibly due to compromised shielding in the adjacent room interfering with the RF signal. However, these measurements did not show a statistically significant change after the MRI scanner's installation in 2nd period. Upon analysis of the second intervention, a significant increase was observed after the beginning of subway operations in STA test accuracy ($p < 0.05$). This increase in measurement after the beginning of subway operations may have been influenced by gradient field non-uniformity generated by the subway train passing near the MRI scanner, producing eddy currents. Eddy current compensation is ineffective due to the train's continuous movement. Despite the Faraday cages incorporated into the scanner room's shielding, the magnet remained susceptible to external magnetic field interferences, indicating that the shielding did not provide complete attenuation (Price, 2010), (ACR, 2015), (ANVISA, 2021), (NEMA MS 5, 2008), (AAPM, 2010), (O'Callaghan et al, 2014), (Price et al, 1990), (Domingues, 2020), (Keenan et al, 2018), (Murata et al, 2023), (Mazzola et al, 2005).

PSG, LCD, and HCSR tests did not have significant changes for both interventions. The PSG test evaluates phantom movement and vibration. Special damping systems on the subway track attenuated vibrations, preventing phantom movement. The tolerance for this test varies from 1.0 to 3.0%, but the observed data was consistently very low. This suggests the standard's recommended tolerance is significantly higher for this equipment under these specific conditions (adjacent MRI scanner, nearby subway construction, 24-hour service). Therefore, a stricter baseline will be used for future tests. (ACR, 2015), (ANIVSA, 2021), (AAPM, 2010), (O'Callaghan et al, 2014), (Price et al, 1990), (Domingues, 2020), (Keenan et al, 2018), (Murata et al, 2023), (Mazzola et al, 2005).

The LCD test evaluates the system's ability to distinguish objects in noise. Factors degrading LCD performance are similar to those affecting SNR, including magnetic field strength variations and RF shielding integrity. Despite this, mechanical vibrations and magnetic field influences had minimal impact. All low-contrast objects in the phantom were visible. While a change in the scanner operator during the third measurement period increased outlier measurements, CNR values consistently ranged between 30 and 50, which will serve as a baseline for future tests. (ACR, 2015), (ANIVSA, 2021), (AAPM, 2010), (O'Callaghan et al, 2014), (Price et al, 1990), (Domingues, 2020), (Keenan et al, 2018), (Murata et al, 2023), (Mazzola et al, 2005).

The HCSR test measures the MRI scanner's ability to resolve small objects. No significant differences were observed across the three measurement periods. As this test involves a visual assessment, it might have influenced the evaluation. It is believed that a Modular Transfer Function (MTF) analysis should

have been conducted for a more accurate assessment of differences between intervention periods. (ACR, 2015), (AAPM, 2010), (Price et al, 1990).

5 Conclusion

The performance evaluation showed that the installation of another MRI scanner in an adjacent room did not interfere. However, geometric distortion, SNR and PIU tests presented better results due to shielding renovation efficiency. Only the STA test showed a statistically significant change following subway construction. This fact may have been influenced by the gradient field non-uniformity generated by the subway train passing near the MRI scanner, producing eddy currents.

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Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using $\text{CaSO}_4\text{:Dy}$ Dosimetry

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Abstract

The use of 3D-printed materials for constructing physical phantoms has emerged as a viable and cost-effective alternative in mammography dosimetry. This study experimentally evaluates the thermoplastics acrylonitrile butadiene styrene (ABS) and polyethylene terephthalate glycol (PETG)—commonly used in 3D printing—for their suitability as tissue-equivalent materials in homogeneous breast phantoms. The materials were characterized in terms of density and linear attenuation coefficient, and their radiation response was assessed at different thicknesses and energies under simulated clinical conditions. $\text{CaSO}_4\text{:Dy}$ thermoluminescent dosimeters (TLDs) were used after being characterized for reproducibility, uniformity, and calibration over the 0.2 mGy to 10 mGy dose range. Samples were irradiated with X-ray beams at 28 kVp and 35 kVp, typical in mammography, and doses were measured at various depths within the printed blocks. The experimental data enabled the construction of depth-dose profiles and the analysis of attenuation as a function of material thickness. Results showed that PETG, with a density of approximately 1.17 g/cm³, exhibited attenuation behavior similar to glandular breast tissue, while ABS, with a density of 0.90 g/cm³, was closer to adipose tissue. Both materials demonstrated increasing attenuation with thickness and showed linear attenuation coefficients consistent with literature values for the respective energies. These findings support the effective use of ABS and PETG in the fabrication of homogeneous breast phantoms, offering advantages in cost, availability, and customization through 3D printing. The reliability of $\text{CaSO}_4\text{:Dy}$ dosimeters validated the dose measurements. In conclusion, these materials represent a promising alternative for applications in quality control, training, and experimental research in mammography, enabling the development of customizable phantoms that replicate the anatomy and composition of real breast tissues.

Keywords: 3D printing; ABS; PETG; Breast phantom; Mammography; Thermoluminescent dosimetry.

1 Introduction

Breast cancer is one of the leading causes of mortality among women worldwide, and mammography remains the most effective technique for the early detection of this disease (Vrbaški et al., 2023). The quality of mammographic imaging, as well as the control of the radiation dose received by the patient, are crucial aspects for the effectiveness of the exam. To ensure that imaging systems operate within acceptable standards of quality and safety, physical phantoms are employed. These phantoms simulate the characteristics of breast tissues and allow for system calibration and analysis of absorbed dose (Poletti et al., 2001).

Traditional phantoms used in mammography are typically made from materials such as polymethyl methacrylate (PMMA), due to their availability and ease of handling. However, PMMA does not exhibit

radiological properties similar to those of actual breast tissues—such as adipose, glandular, and skin tissues—particularly in terms of density and effective atomic number (Ivanov et al., 2018).

With advances in additive manufacturing technology, 3D printing has emerged as a promising alternative for the fabrication of customizable, affordable phantoms with controlled composition. Materials such as ABS and PETG have been widely used in industrial and biomedical applications. Recent studies have shown that, when printed with suitable infill, these polymers exhibit density and attenuation characteristics comparable to those of breast tissues, making them promising candidates for use in simulators (Mettivier et al., 2022; Almeida et al., 2017).

Furthermore, the use of TLDs based on calcium sulfate doped with dysprosium ($\text{CaSO}_4\text{:Dy}$) has proven effective for dose measurements in radiological phantoms. These dosimeters are highly sensitive, reusable, and exhibit a linear response over a wide dose range (Rocha et al., 2019). By combining 3D printing and TLD dosimetry, it is possible to develop accessible experimental solutions for dose assessment in mammography and for improving clinical protocols.

Therefore, this study aims to experimentally evaluate the physical and radiological properties of 3D - printed ABS and PETG using $\text{CaSO}_4\text{:Dy}$ dosimeters, focusing on their application in homogeneous breast phantoms. The approach includes characterization of density, attenuation coefficients, and dosimetric response at different thicknesses and energies, and comparison with the properties of breast tissues reported in the literature (Vrbaški et al., 2023; Ivanov et al., 2018; Mettievier et al., 2022).

2 Materials and Methods

2.1 Evaluation of $\text{CaSO}_4\text{:Dy}$ Thermoluminescent (TL) Dosimeters

$\text{CaSO}_4\text{:Dy}$ thermoluminescent dosimeters (TLDs) embedded in a Teflon matrix, developed by the Nuclear and Energy Research Institute (IPEN), were used in this study. These dosimeters are known for their high sensitivity and reproducibility (Campos and Lima, 1986, 1987). Prior to use, they were thermally treated at 300 °C for 1 hour to stabilize the crystal structure. Irradiations were performed using a Pantak-Seifert X-ray unit (Isovolt HS model, 160 kV) equipped with a tungsten target. The exposure conditions, including the radiation qualities used, are summarized in Table 1. The focus-to-detector distance was maintained at a constant 1 m for all measurements.

Table 1: Radiation quality parameters used in the experimental setup.

Radiation Quality	Tube Voltage (kV)	Tube Current (mA)	Half-Value Layer (mmAl)	Additional Filtration (mm)	Air Kerma Rate (mGy/min)
RQA-M2	28	10	0.61	0.07 Mo+2.0 Al	0.68
RQA-M4	35	10	0.93	0.07 Mo+2.0 Al	1.48

Readings were carried out using a TL/OSL-DA-20 reader (Risø, Denmark), with heating up to 400 °C at a rate of 10 °C/s and employing a Schott BG-39 optical filter. The emitted light, proportional to the absorbed dose, was recorded and corrected using background readings.

To evaluate reproducibility, 80 dosimeters were irradiated with a 5 mGy dose and underwent five consecutive readings. The coefficient of variation (CV) was calculated for each dosimeter, and only those with a CV below 6.2% (80% of the total) were selected. Uniformity was then assessed among the selected dosimeters, yielding a mean TL response of $(1.19 \pm 0.13) \times 10^7$ arbitrary units and a CV of 6.9%.

The calibration curve of $\text{CaSO}_4\text{:Dy}$ dosimeters was established with an RQA-M2 beam in the dose range of 0.2–10 mGy, showing a linear response ($R^2 = 0.999$). For the experimental measurements in the

phantom with the RQA-M4 beam, an energy dependence correction was applied considering RQA-M2 as the reference.

2.2 Printing of the Phantom Objects

The phantoms were developed to replicate breast tissues in density and attenuation using fused deposition modeling (FDM) additive manufacturing. The 3D design was created in Fusion 360 Autodesk®, with modular geometries and cavities for TLDs, allowing configurations with different thicknesses.

Printing was performed on a Raise 3D Pro 2® printer with a 0.4 mm nozzle, using a raft with skirt for better adhesion. For ABS (Raise3D, 2025a), the parameters were: 250 °C nozzle temperature, 110 °C bed temperature, 90% grid infill, two solid outer layers, 50 mm/s speed, and 0.1 mm layer height. For PETG (Raise3D, 2025b), the settings were: 240 °C nozzle temperature, 80 °C bed temperature, 100% grid infill, same outer layers, 80 mm/s speed, and 0.1 mm layer height.

2.3 Determination of Density and Radiation Dose Assessment in 3D-Printed ABS and PETG Samples

The density and radiation attenuation of ABS and PETG materials were evaluated using step-shaped samples, consisting of four steps with increasing heights (0.5 cm to 3.0 cm) and a base of $1 \times 1 \text{ cm}^2$. The sample dimensions were measured with a digital caliper, and their masses were obtained using an analytical balance, allowing the calculation of density as the ratio between mass and volume.

For the radiation dose assessment, the same samples were irradiated under controlled conditions using previously calibrated $\text{CaSO}_4:\text{Dy}$ TLDs. Dose measurements were performed for each thickness and material in duplicate to ensure reproducibility.

2.4 Determination of the Linear Attenuation Coefficient (μ)

The linear attenuation coefficient of ABS and PETG materials was determined using the mixture rule, based on the elemental compositions described by Alssabbagh et al. (2017), Poletti et al. (2001), and Hammerstein et al. (1979), as shown in Table 2. Calculations were performed for an energy of 28 keV to compare the printed polymers, and for comparison with glandular and adipose tissues, values at 15, 20, 28, 30, and 35 keV were also considered, according to data from Tomal et al. (2010), Poletti et al. (2001), Hammerstein et al. (1979), and Vrbaški et al. (2023).

Table 2: Elemental composition of materials/tissues (mass fractions).

Material/Tissue	H	C	N	O	P	S	Source
Adipose Tissue	0.112	0.619	0.017	0.251	0.001	–	Hammerstein et al. (1979)
	0.124	0.765	0.004	0.107	–	–	Poletti et al. (2001)
Glandular Tissue	0.102	0.184	0.032	0.677	0.005	–	Hammerstein et al. (1979)
	0.093	0.184	0.044	0.679	–	–	Poletti et al. (2001)
Skin	0.098	0.178	0.050	0.667	0.007	–	Hammerstein et al. (1979)
ABS	0.075	0.855	0.053	0.016	–	0.001	Alssabbagh et al. (2017)
PETG	0.075	0.652	–	0.271	–	0.002	Alssabbagh et al. (2017)

Mass fractions were used to obtain the mass attenuation coefficients (μ/ρ) of the elements from the XCOM database of the National Institute of Standards and Technology (NIST) (Hubbell and Seltzer, 2004). The total mass attenuation coefficient was calculated as a weighted average, considering the element fractions, and the linear attenuation coefficient (μ) was obtained by multiplying by the experimental or reference density value.

The uncertainty of the coefficients was estimated by error propagation, taking into account uncertainties in density and, when available, in chemical composition. The results were presented in comparative tables for analysis of the polymer materials and breast tissues.

2.5 Irradiation of Homogeneous ABS and PETG Simulator Objects

The evaluation of the depth dose distribution was carried out using calcium sulfate doped with dysprosium ($\text{CaSO}_4\text{:Dy}$) thermoluminescent dosimeters (TLDs) inserted into homogeneous modular phantoms fabricated via 3D printing with ABS and PETG. The blocks were stacked within containment boxes made of the same material ($5 \times 4 \times 4 \text{ cm}^3$), ensuring alignment and stability during irradiation.

Each homogeneous phantom consisted of 1 cm^3 cubic pieces with a central hole (6.5 mm diameter $\times$ 1.1 mm depth) for TLD placement, followed by additional blocks measuring ($2 \times 1 \times 1 \text{ cm}^3$) to allow variation in total material thickness between the source and detector. The irradiations, illustrated in Figure 1, were performed using X-ray beams at RQA-M2 and RQA-M4 radiation qualities, with initial doses ranging from 1.5 mGy to 7 mGy. The TLD readings enabled the generation of attenuation curves as a function of thickness for each material and exposure condition.

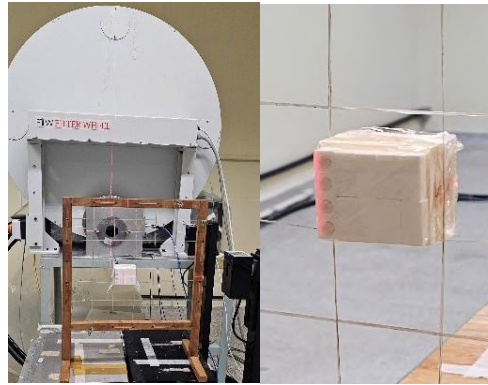


Figure 1: Assembly of the homogeneous phantoms, with modular blocks including the cubic piece for TLD insertion and additional blocks used to adjust the irradiated thickness.

3 Results and Discussion

3.1 Determination of the Density of ABS and PETG Materials Used in 3D Printing

The densities of ABS and PETG materials were compared with the nominal values provided by the manufacturer (Raise3D, 2025a; 2025b), which are 1.12 g/cm^3 for ABS and 1.30 g/cm^3 for PETG. The measured values, shown in Table 3, highlight the influence of infill on the effective density, a crucial aspect for the accurate characterization of phantom objects. Although PETG has full infill, microcavities created by the extrusion geometry can cause internal variations, emphasizing the importance of direct density measurements, as pointed out by Mettievier et al. (2022).

Table 3: Comparison between the measured densities of 3D-printed materials and the densities of actual breast tissues.

Material/Tissue	Density $\pm$ uncertainty (g/cm^3)	Source
ABS	0.899 ± 0.002	This work
PETG	1.169 ± 0.003	This work
Adipose Tissue	0.900 ± 0.020	Vrbaški et al. (2023)
Glandular Tissue	1.070 ± 0.030	Vrbaški et al. (2023)
Skin	1.190 ± 0.020	Vrbaški et al. (2023)

The similarity between the densities of ABS and PETG and those of breast tissues supports the potential of these materials as substitutes in breast phantoms. ABS exhibits a density compatible with adipose tissue, while PETG approaches the typical values for glandular and skin tissues. This correspondence makes both materials suitable for constructing physical models with adjustable radiological properties, as demonstrated in previous studies (Mettivier et al., 2022; Varallo et al., 2022; Ivanov et al., 2018).

3.2 Linear Attenuation Coefficient (μ)

The linear attenuation coefficients (μ) of PETG and ABS materials were experimentally determined and compared with theoretical values calculated based on various literature sources, considering specific densities. These data are presented in Figure 2.

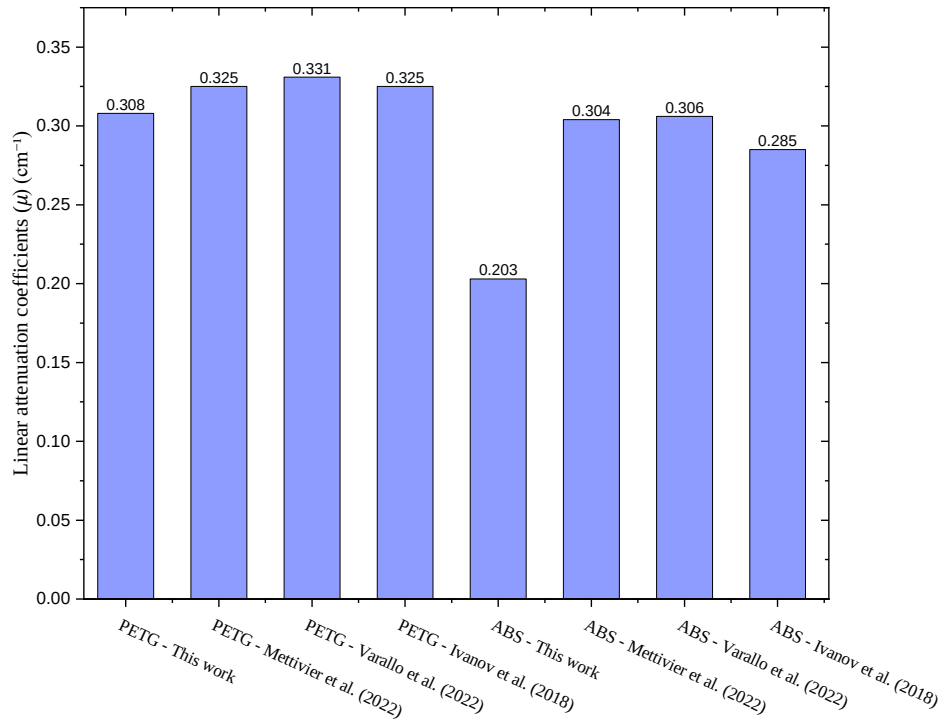


Figure 2: Linear attenuation coefficients (μ) for PETG and ABS under RQA-M2 radiation quality.

The experimental results show good agreement with theoretical values calculated via the mixture rule, particularly for PETG, whose coefficient (0.308 ± 0.003) cm^{-1} closely matches literature data. This indicates that the 100% infill structure preserves the attenuation properties of the pure material, supporting the reliability of PETG as a breast tissue substitute in dosimetry.

For ABS, a lower experimental value (0.203 ± 0.002) cm^{-1} was observed, likely due to the 90% infill, which reduces the effective density and, consequently, the attenuation capacity. This effect is widely discussed by Dancewicz et al. (2017), Mettivier et al. (2022), and Hatamikia et al. (2023), who highlight the importance of controlling printing parameters such as infill and conditions.

Additionally, the linear attenuation coefficients at different energies were compared with values for actual breast tissues (glandular, adipose, and skin), as shown in Table 4.

Analysis of these data indicates that PETG's attenuation coefficients closely match those of glandular tissue, particularly at energies between 25 and 30 keV, typical of mammography. For instance, at 28 keV, PETG's μ (0.308 cm^{-1}) aligns well with values reported by Vrbaški et al. (2023) and Tomal (2007), confirming its suitability as a glandular tissue surrogate.

Conversely, ABS, with its lower density and 90% infill, exhibited attenuation coefficients nearer to adipose tissue. At 28 keV, ABS's μ (0.203 cm^{-1}) differs by less than 10% from adipose tissue values reported by Poletti et al. (2001), supporting its use as an adipose tissue simulator in mammographic phantoms.

However, both materials showed lower μ values than those of skin tissue, which maintains higher attenuation across all energies. This limits their applicability for simulating breast skin, despite their suitability for internal tissues. Therefore, the choice between PETG and ABS depends on the target tissue, with PETG being more appropriate for mammary glands and ABS better suited for adipose regions.

Table 4: Linear attenuation coefficients (μ) for materials and breast tissues at various energies (keV).

Material/Source	15 keV	20 keV	25 keV	28 keV	30 keV	35 keV
PETG – This work	1.241	0.931	0.501	0.308	0.294	0.229
ABS – This work	0.724	0.422	0.253	0.203	0.187	0.158
Glandular – Vrbaški et al. (2023)	1.431	1.288	0.656	0.326	0.323	0.234
Glandular – Tomal (2007)	1.535	1.381	0.703	0.349	0.346	0.251
Glandular – Poletti et al. (2001)	1.550	1.395	0.710	0.353	0.350	0.254
Adipose – Vrbaški et al. (2023)	0.927	0.694	0.380	0.240	0.230	0.181
Adipose – Tomal (2007)	0.957	0.717	0.393	0.248	0.238	0.187
Adipose – Poletti et al. (2001)	0.799	0.530	0.308	0.223	0.211	0.173
Skin – Vrbaški et al. (2023)	1.606	1.439	0.733	0.366	0.362	0.263
Skin – Hammerstein et al. (1979)	1.621	1.452	0.740	0.369	0.366	0.265

3.3 Depth Dose Analysis for ABS and PETG Simulators under Different Radiation Qualities and Radiation Doses

The depth dose profiles obtained for ABS and PETG materials demonstrate the typical exponential attenuation behavior of radiation as the material thickness increases, under different energies and initial dose conditions. These results enable the evaluation of each material's performance in simulating breast tissues in clinical contexts.

For PETG, a significant dose reduction was observed within the first few centimeters of depth, especially under RQA-M2 radiation quality. This behavior is related to the higher interaction of radiation with the material's atoms at lower energies, where the photoelectric effect predominates (Bushberg and Boone, 2011). Under RQA-M4 quality, attenuation was less pronounced, indicating greater beam penetration, resulting from a lower probability of interaction with matter at higher energies. These patterns are illustrated in Figure 3, which shows the depth dose profiles for PETG at the two analyzed radiation qualities.

Similarly, the ABS material also exhibited more pronounced attenuation under RQA-M2 quality and a gentler dose reduction under RQA-M4 quality. The difference in profiles between the two radiation qualities reflects the energy dependence of the linear attenuation coefficient, a behavior expected in materials with composition similar to soft tissues, as highlighted by Tomal et al. (2010). The results obtained for ABS are illustrated in Figure 4, with curves following the behavior predicted by classical radiation-matter interaction theory, including both photoelectric and Compton effects (Bushberg and Boone, 2011).

Regardless of the nominal dose used (ranging from 1.5 mGy to 7 mGy), the attenuation profiles showed proportionality among themselves, reinforcing the reproducibility of the obtained data and the suitability of PETG and ABS materials for simulating dose deposition in mammographic radiology. The observed

attenuation pattern is characteristic of homogeneous materials (Heggie, 2008) and supports their potential use in phantoms for dose and imaging assessment.

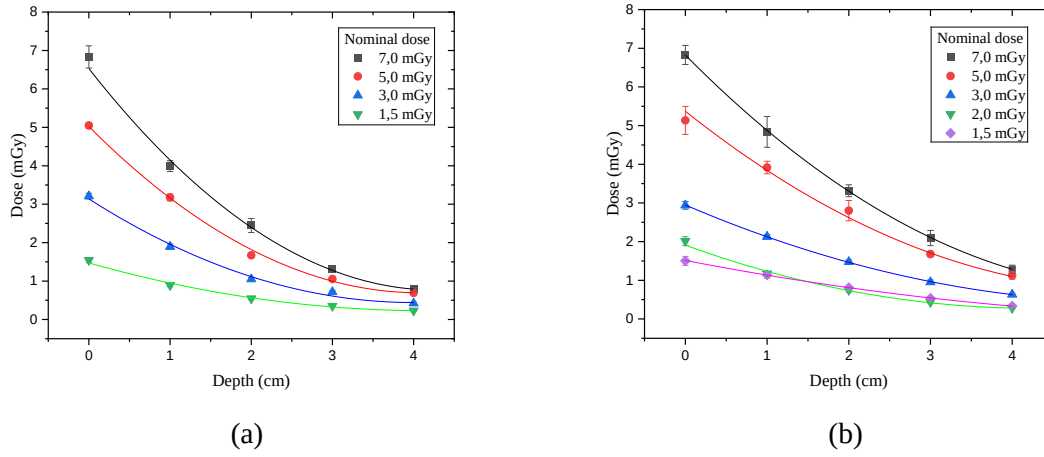


Figure 3: Depth dose distribution for PETG material at radiation qualities of a) RQA-M2 and b) RQA-M4, for different material thicknesses. Radiation attenuation is observed as the radiation penetrates the PETG, with dose variations according to the thickness and radiation quality used.

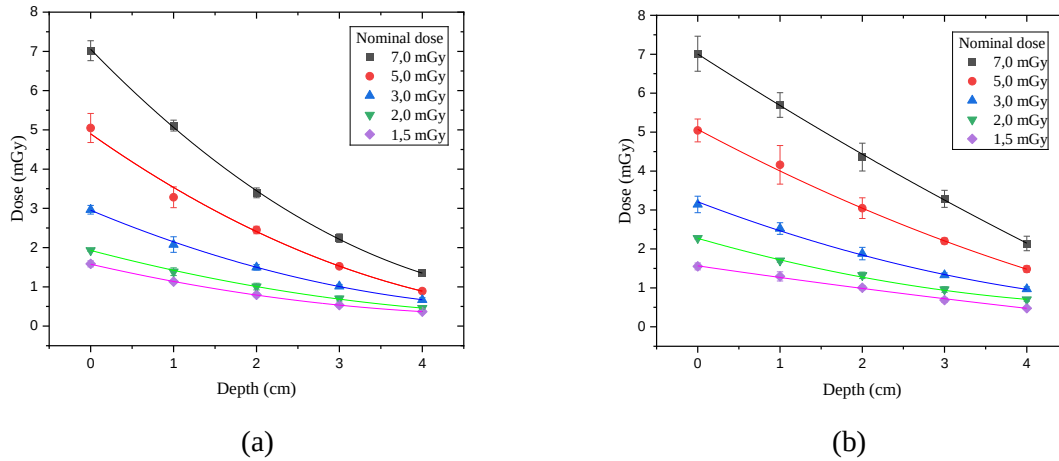


Figure 4: Depth dose distribution for ABS material at radiation qualities of a) RQA-M2 and b) RQA-M4, for different material thicknesses. A progressive dose reduction is observed with increasing thickness, demonstrating the radiation attenuation behavior as a function of the incident radiation quality.

4 Conclusions

The experimental characterization of 3D-printed ABS and PETG materials demonstrated their feasibility as homogeneous substitutes for breast tissues in mammographic applications. PETG exhibited attenuation behavior similar to that of glandular tissue, while ABS more closely resembled adipose tissue. The depth dose profiles showed a strong correlation between thickness and attenuation for both evaluated energies. The $\text{CaSO}_4:\text{Dy}$ dosimeters showed satisfactory performance in terms of reproducibility, uniformity, and dose response. These results indicate that the materials studied, in combination with the TLDs used, represent an effective, low-cost, and customizable alternative for the development of physical phantoms applied to mammography dosimetry.

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Global usability of mobile CT units in stroke diagnosis: A literature review

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Abstract

Stroke remains a leading cause of mortality and long-term disability worldwide, especially in countries with significant regional disparities in healthcare access, such as Brazil. Rapid diagnosis is critical to minimize neurological damage and improve outcomes. Computed tomography (CT) plays a pivotal role in differentiating ischemic from hemorrhagic stroke and guiding timely therapeutic interventions. However, limited access to CT scanners, particularly in remote or underserved areas, often results in delayed diagnosis and suboptimal care. This literature review explores the global use of mobile CT units in stroke diagnosis, focusing on their geographic distribution, impact on clinical response times, and potential to broaden diagnostic coverage in areas lacking conventional infrastructure.

A structured review was conducted covering publications from the past five years, using keywords including “mobile CT,” “stroke diagnosis,” and “access to imaging technology.” High-impact studies were selected based on relevance and methodological quality, emphasizing practical applications, system integration, and clinical outcomes. Findings demonstrate that mobile CT technology has been increasingly adopted to reduce time-to-diagnosis and expand coverage beyond traditional hospital settings. These mobile units are particularly valuable in pre-hospital care settings, such as ambulances and mobile stroke units, enabling early thrombolysis and other time-sensitive treatments that can significantly improve functional recovery.

Despite its promise, the widespread adoption of mobile CT faces several barriers, including high implementation and maintenance costs, complex operational logistics, and the challenge of integrating mobile imaging into existing healthcare systems. Furthermore, access disparities between urban centers and rural or remote areas remain a significant concern. Policy support and strategic partnerships with the private sector are essential to scale up deployment and ensure equitable access.

In summary, mobile CT scanners represent an innovative approach to addressing gaps in stroke care, with the potential to reduce diagnostic delays and improve patient outcomes in underserved regions. Their integration into emergency care protocols and healthcare delivery models could redefine stroke management globally. The insights from this review support further research, policy development, and investment strategies aimed at incorporating mobile imaging technologies into standard clinical practice.

Keywords: Stroke; Mobile CT; Computed Tomography; Radiology.

1. Introduction

Stroke is a leading cause of death and disability worldwide, with low- and middle-income countries bearing a disproportionate burden. In Brazil, stroke has consistently ranked among the top causes of mortality, accounting for over 99 000 deaths in 2020 alone (SBAVC, 2021). Despite advances in acute stroke care, timely diagnosis remains a major barrier, particularly in underserved regions where access to imaging is limited. Rapid neuroimaging is essential for distinguishing between ischemic and hemorrhagic stroke, guiding appropriate treatment, and improving outcomes (Adams et al., 2007; Powers et al., 2019).

Intravenous thrombolysis remains the cornerstone of ischemic stroke management but is time-sensitive, with maximal benefit occurring within 4-5 hours of symptom onset (Hacke et al., 2008). Delayed treatment significantly increases mortality risk and the likelihood of long-term disability. However, access to computed tomography (CT), the gold standard for initial stroke imaging, is uneven in Brazil; only 25% of public hospitals have 24-hour CT availability, and travel times to imaging facilities in remote areas frequently exceed therapeutic windows (SBAVC, 2021).

Mobile stroke units (MSUs), which integrate CT scanners into pre-hospital emergency vehicles, have emerged as a promising strategy to overcome diagnostic delays. MSUs enable real-time imaging, remote specialist consultation via telemedicine, and immediate initiation of treatment. Recent trials have shown that MSUs significantly reduce door-to-needle times, increase thrombolysis rates, and improve functional outcomes at 90 days (Ebinger et al., 2022; Grotta et al., 2023). In the BEST-MSU trial, for example, patients treated in MSUs were more likely to achieve functional independence and less likely to die within 3 months compared with those receiving standard care (Grotta et al., 2023).

Cost-effectiveness analyses have also supported the scalability of MSUs. In high-incidence settings, their implementation may reduce health system costs by decreasing the need for long-term rehabilitation and institutional care (Wang et al., 2024). Yet, operational challenges persist, including high upfront costs, workforce training, and system integration.

This review synthesizes the recent literature on the global application of mobile CT technology in stroke diagnosis, with a particular focus on its impact in low-resource settings. By examining patterns of adoption, clinical outcomes, and barriers to implementation, we aim to provide insights into how mobile imaging can help address longstanding inequities in acute stroke care.

2. Methodology

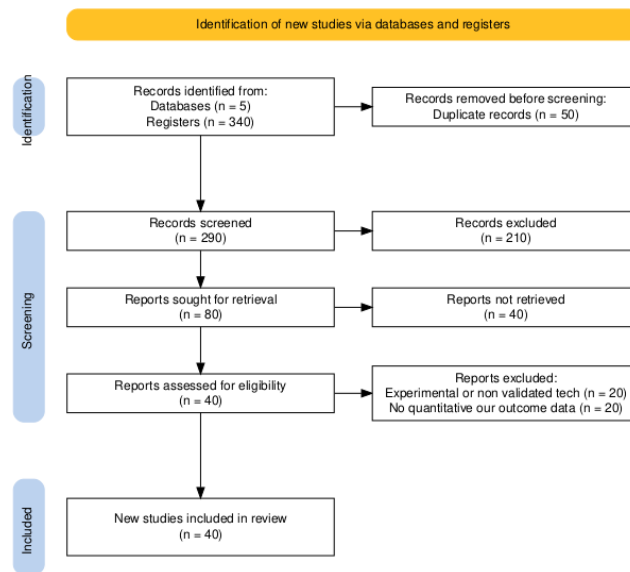
For this review, a range of peer-reviewed scientific articles were analyzed, focusing on stroke, the importance of computed tomography (CT) for diagnostic purposes, and the current global landscape regarding the use of Mobile Stroke Units (MSUs) and mobile CT scanners. The review explored the potential advantages and disadvantages of these technologies, highlighted cities where such systems have been deployed at scale and assessed real-world data on their clinical impact.

The literature search strategy included keywords such as “Portable computed tomography scanner”, “Portable CT”, “Mobile CT”, “Mobile computed tomography”, and “Mobile stroke unit”. The review focused on articles published within the last five years, from 2020 to 23 January 2025. No geographical restrictions were applied, as the objective was to evaluate the global impact of this technology.

Inclusion criteria consisted of publications in high-impact journals, studies assessing the implementation and clinical impact of mobile CT systems, and reports detailing the geographical distribution of such technologies. Exclusion criteria included experimental technologies without validated clinical application and articles lacking quantitative data or clinical outcome analysis.

Using this strategy, 40 articles were selected and categorized according to journal, country, publication date, and main findings. These data were subsequently used to construct graphs and tables, based on the quantitative analysis of the selected studies, which will be presented in the following sections. Below is

the PRISMA flowchart 01 illustrating the methodology of the review (Haddaway, Page, Pritchard, & McGuinness, 2022).



Flowchart 01 illustrating the methodology of the review (Haddaway, Page, Pritchard, & McGuinness, 2022).

3. Results and Discussions

The search yielded a selection of relevant articles, exemplified below (Table 01) by the five most recent studies (for the complete table, see Appendix 01: Article Selection and Classification). The inclusion of publications in high-impact journals such as the New England Journal of Medicine (NEJM) and Journal of the American Medical Association (JAMA) highlights the global scientific interest and robust evidence supporting the use of mobile computed tomography (CT) technology. Moreover, the presence of articles in specialist journals such as Die Radiologie, European Stroke Journal, and Neuroradiology underscores the growing engagement of medical imaging and neurology experts in evaluating this technology's role in stroke diagnosis.

Table 01: Five most recent studies from the selection

LINK	JOURNAL	COUNTRY	PUBLICATION DATE
Association of Mobile Stroke Unit Care and Spending, Utilization, and Death in New York City Journal of the American Heart Association	Journal of the American Heart Association	United States	Jan-25
Portable head CT motion artifact correction via diffusion-based generative model - ScienceDirect	Computerized Medical Imaging and Graphics	United Kingdom	Jan-25
Driving healthcare forward: The potential of mobile MRI and CT units in streamlining radiological services in Saudi Arabia - A narrative review - ScienceDirect	Journal of Medical Imaging and Radiation Sciences	Canada	Dec-24
Bringing imaging to the people: Enhancing access and equity in healthcare through mobile imaging - ScienceDirect	Journal of Medical Imaging and Radiation Sciences	Canada	Dec-24
Mobile computed tomography at the Munich Oktoberfest Die Radiologie	Die Radiologie	Germany	Oct-24

3.1 Geographic Distribution of Publications

Analysis of the geographic origin of the publications revealed a marked concentration of research activity in the United Kingdom and the United States, as illustrated in the accompanying graph (Figure 01). Notably, Latin America is underrepresented, with no articles originating from this region among the 40 most relevant and recent studies identified. This disparity suggests that mobile CT technology remains largely unexplored and underutilized in Latin American countries. Given the high incidence of stroke and limited access to CT imaging in rural and remote areas, especially in Brazil, there is an urgent need to foster research and adoption of mobile CT solutions in this region.

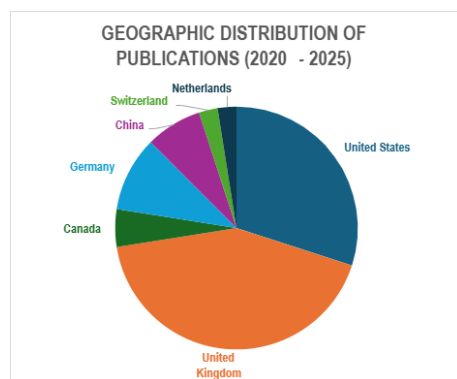


Figure 01: Geographic Distribution of Publications

3.2 Temporal Trends in Publications

The temporal analysis demonstrated a significant increase in the number of relevant publications over recent years, particularly in 2024 (Figure 02). This upward trend reflects growing scientific and clinical interest, likely driven by technological advancements and expanding implementations of mobile CT units worldwide.

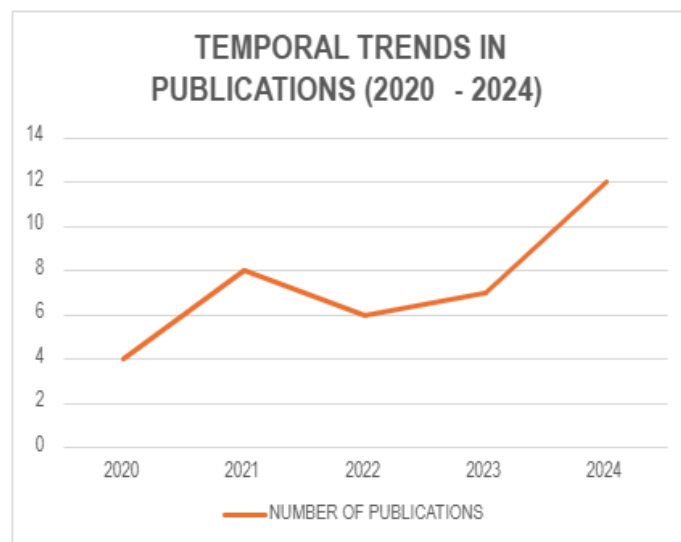


Figure 02: Temporal Trends in Publications

3.3 Hospitals and Regions Utilizing Mobile CT

From the dataset, a list of hospitals employing mobile CT technology was compiled to assess regional adoption and impact (Table 02). Again, a strong concentration was observed in Europe and North America, particularly in Germany, Denmark, and the United States, indicating broader implementation in high-income countries. This pattern suggests that adoption may be contingent upon financial resources and advanced healthcare infrastructure.

In Germany, prominent centers such as Charité Berlin, Heidelberg University Hospital, and institutions in Munich and Saarland have integrated mobile CT systems, supported by favorable public policies and funding for hospital innovation. Notably, deployment at large-scale events like the Munich Oktoberfest illustrates applications in emergency mass gatherings.

In the United States, leading institutions including the Cleveland Clinic and the University of Texas have adopted mobile CT, reinforcing the technology's role in reference hospitals.

Similarly, Scandinavian countries such as Denmark and Norway (including Aarhus, Stavanger, and Copenhagen University Hospital) showcase mobile CT adoption, likely reflecting health policy initiatives aimed at decentralizing care and improving diagnostic access in remote locations.

Importantly, the literature review did not identify any mobile CT units operating in Brazil, Argentina, or other developing countries. However, governmental sources indicate the existence of at least one operational unit in the Brasília region, Federal District, Brazil, highlighting a potential gap between published data and on-the-ground initiatives.

Table 02: Reference sites using mobile CT scanners

ARTICLE	JOURNAL	PUBLICATION DATE	MOBILE CT LOCATION	COUNTRY
Mobile computed tomography at the Munich Oktoberfest Die Radiologie	Die Radiologie	Oct-24	Munich Hospital / Oktoberfest MSU	Germany
Advances in pre-hospital care - operational experiences from the Melbourne mobile stroke unit	Vessel Plus	Feb-21	Melbourne Mobile Stroke Unit	Australia
Association Between Dispatch of Mobile Stroke Units and Functional Outcomes Among Patients With Acute Ischemic Stroke in Berlin - PubMed	The Journal of the American Medical Association	Feb-21	Charité - Universitätsmedizin Berlin	Germany
Mobile Stroke Unit Management in Patients With Acute Ischemic Stroke Eligible for Intravenous Thrombolysis Cerebrovascular Disease JAMA Neurology JAMA Network	The Journal of the American Medical Association	Oct-24	Cleveland Clinic Mobile Stroke Treatment Unit	United States
Prospective, Multicenter, Controlled Trial of Mobile Stroke Units - PubMed	New England Journal of Medicine	Sep-21	University of Texas Health Science Center at Houston	United States
Potential effects of a mobile stroke unit on time to treatment and outcome in patients treated with thrombectomy or thrombolysis: A Danish-German cross-border analysis - PubMed	European Journal of Neurology	Sep-24	Aarhus University Hospital	Denmark

Prehospital computed tomography in a rural district for rapid diagnosis and treatment of stroke - Jørgen Ibsen, Maren Ranhoff Hov, Gunn Eli Tokerud, Julia Fuglum, Marianne Linnerud Krogstad, Marie Stugaard, Hege Ihle-Hansen, Christian Georg Lund, Christian Hall, 2024	European Stroke Journal	Sep-24	Stavanger University Hospital	Norway
Initial experience with a state-of-the-art mobile head CT scanner for use in neurointensive care units Die Radiologie	Die Radiologie	Apr-24	Saarland University Hospital	Germany
State-of-the-art mobile head CT scanner delivers nearly the same image quality as a conventional stationary CT scanner Scientific Reports	Scientific Reports	Mar-24	Heidelberg University Hospital	Germany
Mobile stroke units based on rural emergency medical stations for pre-hospital intravenous thrombolysis of stroke in remote areas: Neurological Research: Vol 46 , No 12 - Get Access	Neurological Research	Oct-24	Klinikum der Universität München	Germany
Comparison of image quality between a novel mobile CT scanner and current generation stationary CT scanners Neuroradiology	Neuroradiology	Nov-22	Rigshospitalet - Copenhagen University Hospital	Denmark

The results highlight the pivotal role of mobile computed tomography (CT) units in reducing the time from symptom onset to treatment, which directly improves recovery and decreases post-stroke disability. Mobile Stroke Units (MSUs) enable pre-hospital diagnosis, expanding access to care in regions with limited hospital infrastructure and enhancing clinical outcomes. A recent Journal of the American Heart Association (JAHA, 2025) study reported that patients transported by MSUs were significantly more likely to receive intravenous tissue plasminogen activator (tPA) than those transported by conventional ambulances (49.9% vs. 9.4%). Another JAHA study (2021) found that patients managed via MSUs had lower median Modified Rankin Scale scores at three months, indicating less disability. Moreover, integration of CT scanners in MSUs facilitates direct transfer to interventional suites, reducing door-to-puncture times from 94.5 to 41 minutes (Stroke, 2020). Evidence from rural Germany (European Stroke Journal, 2024) and Canada (Scientific Reports, 2021) further supports mobile CT's role in decreasing treatment delays and limiting unnecessary transfers. Additionally, a New England Journal of Medicine study (2021) of over 1,400 patients demonstrated that 55% of MSU-treated patients achieved functional independence at 90 days, compared to 44.4% with standard care.

Despite their potential, mobile CT-equipped stroke units face significant implementation challenges. High acquisition and operational costs, complex logistics, and integration barriers with healthcare systems hinder widespread adoption, particularly in low- and middle-income countries where stroke incidence and access disparities are greatest. The scarcity of studies in these regions, especially the absence of robust data from Latin America, further restricts the generalizability of current evidence. Heterogeneity in implementation models, clinical protocols, and evaluation metrics also complicates direct comparisons across settings. While new systems from Siemens Healthineers or NeuroLogica/Samsung demonstrate image quality comparable to fixed CT units and enable both bedside and mobile applications, their adoption remains limited by requirements for adequate ambulance infrastructure, integration with clinical workflows and telemedicine platforms, and high implementation costs. These factors underscore that the successful expansion of mobile CT technology depends not only on technical capabilities but also on sustainable funding strategies, workforce training, and careful adaptation to local healthcare contexts.

4. Conclusions

This systematic review underscores the growing scientific interest in the use of mobile computed tomography (CT) for acute stroke care, particularly over the past decade. The synthesis of 40 selected studies reveals a consistent pattern: mobile CT integration, especially through Mobile Stroke Units (MSUs), significantly shortens time to treatment and improves clinical outcomes, including higher rates of thrombolysis, faster initiation of thrombectomy, and better functional independence at follow-up.

Importantly, the data highlight mobile CT's transformative potential in extending diagnostic reach to underserved and geographically remote regions, areas historically limited by delayed access to imaging and specialist care. However, widespread implementation remains constrained by substantial financial, logistical, and infrastructural barriers, particularly in low-resource settings.

To unlock its full potential, mobile CT deployment must be coupled with supportive public policy and structural investment. Innovative funding models, such as public-private partnerships, outcome-based subsidies, and integration into national emergency response systems, will be pivotal. Technological innovation must also continue to focus on miniaturization, cost reduction, and deployment flexibility, including airborne MSUs and solutions for mass-casualty events.

Among emerging private-sector innovations, the SOMATOM On Site CT scanner from Siemens Healthineers exemplifies how industry can accelerate bedside imaging with quality standards comparable to fixed units. Its successful application in ICUs and mobile platforms represents a compelling case for how portability and diagnostic performance can coexist, with measurable impacts on hospital flow and patient safety.

Ultimately, mobile CT is not merely a technological advancement, it is a paradigm shift. By decentralizing imaging and bringing diagnostics closer to the point of need, this innovation can reshape acute stroke pathways. To ensure equitable global access, stakeholders must converge in a shared commitment to overcome disparities and reimagine stroke care beyond the walls of traditional hospitals.

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Appendix 01

LINK	JOURNAL	COUNTRY	PUBLICATION DATE
Association of Mobile Stroke Unit Care and Spending, Utilization, and Death in New York City Journal of the American Heart Association	Journal of the American Heart Association	United States	Jan-25
Portable head CT motion artifact correction via diffusion-based generative model - ScienceDirect	Computerized Medical Imaging and Graphics	United Kingdom	Jan-25
Driving healthcare forward: The potential of mobile MRI and CT units in streamlining radiological services in Saudi Arabia - A narrative review - ScienceDirect	Journal of Medical Imaging and Radiation Sciences	Canada	Dec-24
Bringing imaging to the people: Enhancing access and equity in healthcare through mobile imaging - ScienceDirect	Journal of Medical Imaging and Radiation Sciences	Canada	Dec-24
Mobile computed tomography at the Munich Oktoberfest Die Radiologie	Die Radiologie	Germany	Oct-24
Mobile stroke units based on rural emergency medical stations for pre-hospital intravenous thrombolysis of stroke in remote areas: Neurological Research: Vol 46 , No 12 - Get Access	Neurological Research	United Kingdom	Oct-24
Mobile Stroke Unit Management in Patients With Acute Ischemic Stroke Eligible for Intravenous Thrombolysis Cerebrovascular Disease JAMA Neurology JAMA Network	The Journal of the American Medical Association	United States	Oct-24
Potential effects of a mobile stroke unit on time to treatment and outcome in patients treated with thrombectomy or thrombolysis: A Danish-German cross-border analysis - PubMed	European Journal of Neurology	United Kingdom	Sep-24
Prehospital computed tomography in a rural district for rapid diagnosis and treatment of stroke - Jørgen Ibsen, Maren Ranhoff Hov, Gunn Eli Tokerud, Julia Fuglum, Marianne Linnerud Krogstad, Marie Stugaard, Hege Ihle-Hansen, Christian Georg Lund, Christian Hall, 2024	European Stroke Journal	United Kingdom	Sep-24
Mobile Stroke Units in Acute Ischemic Stroke: A Comprehensive Systematic Review and Meta-Analysis of 5 “T Letter” Domains	Cardiology in Review	United States	Aug-24
Rescue equipment should include portable medical imaging systems - ScienceDirect	Science Bulletin	China	Jun-24
Initial experience with a state-of-the-art mobile head CT scanner for use in neurointensive care units Die Radiologie	Die Radiologie	Germany	Apr-24
Mobile imaging trailers: A scoping review of CT and MRI modalities - ScienceDirect	Radiography	United Kingdom	Mar-24
State-of-the-art mobile head CT scanner delivers nearly the same image quality as a conventional stationary CT scanner Scientific Reports	Scientific Reports	United Kingdom	Mar-24
Can We Miniaturize CT Technology for a Successful Mobile Stroke Unit Roll-Out? IEEE Conference Publication IEEE Xplore	IEEE	United States	Dec-23
Impact of Mobile Stroke Units on Patients With Large Vessel Occlusion Acute Ischemic Stroke: A Prespecified BEST-MSU Substudy Stroke: Vascular and Interventional Neurology	Stroke: Vascular and Interventional Neurology	United States	Dec-23
P124/272 Initial encounter with a cutting-edge mobile head CT scanner (Somatom On.site) in a neuro-intensive care unit Journal of NeuroInterventional Surgery	Journal of NeuroInterventional Surgery	United Kingdom	Aug-23
Prehospital Stroke Care Part 2: On-Scene Evaluation and Management by Emergency Medical Services Practitioners - PubMed	International Journal of Stroke	United Kingdom	May-23
Prehospital Stroke Care Part 1: Emergency Medical Services and the Stroke Systems of Care - PubMed	International Journal of Stroke	United Kingdom	Apr-23
The first mobile photon-counting detector CT: the human images and technical performance study - IOPscience	Physics in Medicine and Biology	United Kingdom	Apr-23

Mobile CT and cloud service makes CT inspection fast and affordable	e-Journal of Nondestructive Testing (eJNDT)	Germany	Feb-23
Comparison of image quality between a novel mobile CT scanner and current generation stationary CT scanners Neuroradiology	Neuroradiology	Germany	Nov-22
Top-Level Design and Simulated Performance of the First Portable CT-MR Scanner IEEE Journals & Magazine IEEE Xplore	IEEE	United States	Sep-22
Mobile Stroke Units: Evidence, Gaps, and Next Steps Stroke	International Journal of Stroke	United Kingdom	Mar-22
Comparison of Mobile Stroke Unit With Usual Care for Acute Ischemic Stroke Management: A Systematic Review and Meta-analysis - PubMed	The Journal of the American Medical Association	United States	Mar-22
Comparison of Mobile Stroke Unit With Usual Care for Acute Ischemic Stroke Management: A Systematic Review and Meta-analysis Cerebrovascular Disease JAMA Neurology JAMA Network	The Journal of the American Medical Association	United States	Feb-22
Bringing CT Scanners to the Skies: Design of a CT Scanner for an Air Mobile Stroke Unit	Applied Sciences	Switzerland	Jan-22
Feasibility study of portable multi-energy computed tomography with photon-counting detector for preclinical and clinical applications Scientific Reports	Scientific Reports	United Kingdom	Nov-21
Prospective, Multicenter, Controlled Trial of Mobile Stroke Units - PubMed	New England Journal of Medicine	United States	Sep-21
High-resolution CT with arch/neck/head CT angiography on a mobile stroke unit Journal of NeuroInterventional Surgery	Journal of NeuroInterventional Surgery	United Kingdom	Aug-21
Pre-Hospital Diagnosis in Mobile Stroke Unit - ScienceDirect	Journal of Stroke and Cerebrovascular Diseases	United States	Jul-21
Stroke care equity in rural and remote areas - novel strategies	Vessel Plus	China	May-21
Pre-hospital triage of suspected acute stroke patients in a mobile stroke unit in the rural Alberta Scientific Reports	Scientific Reports	United Kingdom	Mar-21
Association Between Dispatch of Mobile Stroke Units and Functional Outcomes Among Patients With Acute Ischemic Stroke in Berlin - PubMed	The Journal of the American Medical Association	United States	Feb-21
Advances in pre-hospital care - operational experiences from the Melbourne mobile stroke unit	Vessel Plus	China	Feb-21
Mobile stroke unit versus standard medical care in the management of patients with acute stroke: A systematic review and meta-analysis - Nida Fatima, Maher Saqqur, M Shazam Hussain, Ashfaq Shuaib, 2020	International Journal of Stroke	United Kingdom	Jun-20
Economic evaluation of the Melbourne Mobile Stroke Unit - Joosup Kim, Damien Easton, Henry Zhao, Skye Coote, Garveeta Sookram, Karen Smith, Michael Stephenson, Stephen Bernard, Mark W Parsons, Bernard Yan, Patricia M Desmond, Peter J Mitchell, Bruce CV Campbell, Geoffrey A Donnan, Stephen M Davis, Dominique A Cadilhac, 2021	International Journal of Stroke	United Kingdom	Jun-20
Mobile Stroke Unit Computed Tomography Angiography Substantially Shortens Door-to-Puncture Time Stroke	International Journal of Stroke	United Kingdom	Apr-20
Mobile Stroke Units: Bringing Treatment to the Patient Current Treatment Options in Neurology	Current Treatment Options in Neurology	United States	Feb-20

Impact of Body Mass Index on Radiation Doses in Interventional Radiology

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Abstract

Interventional radiology is a well-established minimally invasive alternative to conventional surgery, offering shorter recovery times and reduced hospital costs. However, fluoroscopy-guided procedures remain among the primary sources of ionizing radiation exposure, posing risks to both patients and healthcare professionals. This study aimed to establish institutional Typical Values (TVs) for radiation dose, considering variations in Body Mass Index (BMI), and to assess the impact of anthropometric differences on patient exposure in a high-volume hemodynamic unit. A retrospective analysis was conducted on 2,200 procedures performed between December 2022 and March 2024, including coronary and vascular angiographies and angioplasties.

Median air kerma at the reference point ($K_{a,r}$) and fluoroscopy time were higher in therapeutic procedures (1,112.0 mGy and 9.03 minutes for coronary angioplasty) compared to diagnostic ones (369.5 mGy and 2.97 minutes for coronary angiography). Stratification by BMI, based on WHO criteria, showed a progressive increase in radiation dose with higher BMI, ranging from 185.0 mGy in underweight patients to 796.0 mGy in those with obesity class III. Fluoroscopy time also tended to be longer in higher BMI categories, reaching up to 83.0 minutes in overweight patients. Comparison between overall TVs and BMI-adjusted TVs revealed notable differences in some procedures. The largest relative error was observed in vascular angiography (18.6%), while coronary procedures showed smaller variations, indicating that BMI adjustments can significantly influence dose reference levels, particularly in diagnostic interventions.

These findings underscore the importance of defining dose reference parameters that account for both procedure type and patient anthropometric profile to optimize radiological practice. The methodology adopted, focusing on a single fluoroscopy system and BMI-based segmentation, contributes to improved standardization, technical consistency, and data reproducibility.

Keywords: typical values, interventional radiology, radiation dose, BMI.

1 Introduction

Interventional radiology is one of the major innovations of modern medicine, enabling less invasive diagnostic and therapeutic procedures that reduce recovery time and hospital costs (Sothmann et al., 2021; Tristram et al., 2022). These procedures are performed using fluoroscopy, a real-time imaging technique that allows visualization of anatomical structures and guidance of medical devices during interventions. However, the frequent use of fluoroscopy in such procedures represents one of the main sources of ionizing radiation exposure for both patients and professionals (Tristram et al., 2022; Ashworth et al., 2021; Metaxas et al., 2025). In certain cases, the radiation dose received by a patient during a single procedure can reach alarming levels (Shah et al., 2015).

This exposure becomes even more critical in populations with overweight and obesity, whose prevalence has been increasing at alarming rates (Shah et al., 2015). Projections indicate that by 2044, about 75% of Brazilian adults will be overweight (Tristram et al., 2022; Shah et al., 2015). Body composition directly influences absorbed dose, as patients with higher body mass index (BMI) require greater current and exposure time to obtain adequate images (Tristram et al., 2022; Yanch et al., 2009; Ector et al., 2007). Studies have shown that obese patients may receive up to 1.8 times more radiation than those with normal weight. Therefore, it is crucial to understand how anthropometric profiles affect radiation exposure (Shah et al., 2015; Yanch et al., 2009; Ector et al., 2007).

Considering this reality, the ALARA principle (As Low As Reasonably Achievable) becomes even more relevant. Widely adopted in clinical practice, it states that radiation exposure should be kept as low as possible, provided that diagnostic or therapeutic objectives are still achieved (Ashworth et al., 2021; Metaxas et al., 2025). In critical contexts such as interventional procedures, clinical management remains the priority, even if it involves high doses (Meyer et al., 2017). Nevertheless, this does not exempt healthcare services from the responsibility to monitor, analyze, and optimize exposures (Metaxas et al., 2025; De Monte et al., 2020). For this purpose, the International Commission on Radiological Protection (ICRP), through Publication 135 (Vaño et al., 2017), recommends the use of Diagnostic Reference Levels (DRLs) and Local Diagnostic Reference Levels (LDRLs) as tools to identify practices that deviate from usual standards (Vaño et al., 2017; IAEA, 2011).

DRLs and LDRLs are usually defined as the third quartile of dose distributions collected from large patient groups (De Monte et al., 2020; IAEA, 2011). While effective for identifying practices that differ from typical patterns, these indicators may not accurately reflect the operational reality of specific services, especially where protocols or patient characteristics vary widely. In this context, Typical Values (TVs) stand out because they represent the median of dose distributions in a single institution or equipment, offering greater statistical stability and sensitivity for detailed analysis of local practice. All these indicators can be calculated from parameters such as the reference point air kerma ($K_{a,r}$), kerma-area product (PKA), total fluoroscopy time, and the number of acquired images (Vaño et al., 2017; Vano et al., 2011; Ruiz-Cruces et al., 2016).

This study chose to use TVs due to their ability to reflect the habitual clinical practice of a specific service, minimizing the interference of variables such as differences between equipment, operators, or institutional protocols (Costa et al., 2024; Skrk et al., 2021). Although DRLs, LDRLs, and TVs are all effective tools for dose assessment, they are often defined based on “standard” patients weighing between 60 and 70 kg. This limitation reduces their applicability in clinical settings with significant anthropometric variability, as is the case in most populations (Skrk et al., 2021; Marshall et al., 2000; Vieira et al., 2022). To overcome this limitation, different authors have suggested the use of BMI as a complementary variable in the analysis of radiation exposure, as it more accurately reflects body composition and X-ray beam attenuation, thus allowing assessments that are more compatible with the real anthropometric profile of adult patients undergoing interventional procedures (Shah et al., 2015; Vieira et al., 2022).

Shah et al. (2015) marked an important starting point by demonstrating the influence of BMI on patient dose in interventional procedures, highlighting its potential role in dose optimization strategies. This

finding opened the door for incorporating BMI into the discussion of radiation exposure, but it has not yet translated into a structured methodology for defining practical indicators, such as Typical Values (TVs). In this context, the present study advances the field by systematically integrating BMI into the determination of TVs across a large sample and multiple procedure types, thereby providing concrete support for optimization strategies that are more closely aligned with the real patient population.

2 Materials and Methods

2.1 Equipment

All examinations were performed using a fluoroscopy system, the GE Innova IGS 320 (GE Healthcare), located in the hemodynamics unit of a large hospital that serves as a referral center for interventional procedures. The equipment was properly assessed in accordance with the quality control protocols established by the Brazilian Health Regulatory Agency (ANVISA) through Collegiate Board Resolution (RDC) No. 611/2022 and Instruction Normative (IN) No. 91/2022 (ANVISA, 2022a; ANVISA, 2022b) as well as the international recommendations described in the IAEA TECDOC 1958 (IAEA, 2021). All parameters met the required acceptance criteria, ensuring the safe and proper operation of the system.

2.2 Selection of Evaluated Procedures

The choice of interventional procedures evaluated in this study was based on two criteria: average fluoroscopy time and frequency of performance in routine clinical practice. An analysis was conducted of the examinations performed in the hemodynamics unit, including both elective and emergency procedures. Based on these data, the modalities with the highest volume and the greatest potential radiation dose to patients and staff were selected: vascular angiography, vascular angioplasty, coronary angiography, and coronary angioplasty.

2.3 Data Acquisition

This retrospective study was conducted between December 2022 and March 2024, encompassing 2,200 examinations with complete records in the hospital system. Patients aged 18 years or older who underwent the previously selected procedures were included. Sociodemographic data (age, sex, weight, and height) were collected manually. Air kerma at the reference point ($K_{a,r}$, in mGy) and total fluoroscopy time (in minutes) were obtained from dose reports generated by the fluoroscopy system itself, in accordance with ICRP Publication 135, which recommends these parameters as the most suitable for monitoring radiation exposure (Vaño et al., 2017).

2.4 Determination of Typical Values

The determination of TVs was conducted in accordance with the recommendations of ICRP 135 (Vaño et al., 2017), which defines the TV as the median of dosimetric variables, representing a robust central value that is less susceptible to the influence of extreme values. Medians were calculated globally, by procedure type, by BMI categories, and by procedure type stratified by BMI, as detailed below.

The sample included patients with weights ranging from 40 to 186 kg and heights from 1.35 to 1.95 m, encompassing the full anthropometric diversity observed in the service. Although ICRP 135 (Vaño et al., 2017) recommends the use of “standard patients” (60–80 kg) for the calculation of Diagnostic Reference Levels (DRLs), this approach does not reflect the actual population served. By using BMI as a parameter for population representation—being a criterion widely recommended by the World Health Organization (WHO) (WHO, 2000) and internationally adopted in population studies, clinical protocols, and epidemiological research—it is possible to integrate weight and height into a single metric, more accurately characterize body profile, and classify the population into clinically relevant categories for medical practice. Unlike weight alone, which does not account for differences in stature, BMI provides a

more accurate depiction of anthropometric diversity, allowing this study to more precisely assess the impact of these variations on TVs and radiation exposure.

2.5 Global Determination of Typical Values

A global TV was estimated based on all measurements of Ka,r (mGy) available in the database, without any stratification by procedure type, age group, sex, or patients' anthropometric profiles. This analysis considered the entire set of 2,200 examinations, reflecting the average clinical practice of the service during the study period.

2.6 Determination of Typical Values by Procedure Type

TVs were also estimated stratified by type of interventional procedure, based on the individual analysis of Ka,r values recorded for each examination. To achieve this, data were grouped according to the performed modality (vascular angiography, vascular angioplasty, coronary angiography, and coronary angioplasty), allowing the construction of specific distributions for each group. From these distributions, the median Ka,r was calculated.

2.7 Determination of Typical Values by Body Mass Index (BMI)

The analysis of the influence of anthropometric profiles on exposure levels was carried out through stratification of patients based on BMI. BMI was individually calculated using recorded weight (kg) and height (m) data, applying the following formula:

$$BMI = \frac{weight(kg)}{height^2(m^2)} \quad (1)$$

With patients previously classified by BMI categories according to the criteria established by the WHO (WHO, 2000), the analysis of Ka,r values (mGy) was conducted within each category: underweight (BMI < 18.5 kg/m²), normal weight (18.5–24.9 kg/m²), overweight (25.0–29.9 kg/m²), obesity class I (30.0–34.9 kg/m²), obesity class II (35.0–39.9 kg/m²), and obesity class III (≥ 40.0 kg/m²). For each of these categories, the Ka,r TVs were determined.

2.8 Determination of Typical Values Adjusted by Procedure and BMI

An estimation of TVs was also carried out for each type of interventional procedure, considering the stratification of patients by the predefined BMI categories. For each procedure, the median Ka,r was calculated within each BMI category, and the adjusted TV was obtained through a weighted mean of these medians, using the proportion of patients in each BMI category as the weighting factor. The sample included patients ranging from 40 to 186 kg in weight and 1.35 to 1.95 m in height. While ICRP 135 (Vaño et al., 2017) recommends establishing DRLs using a standard patient population (60–80 kg), this criterion does not reflect the full anthropometric variability observed in clinical practice. Therefore, instead of applying the ICRP weighting scheme, which is designed for national or regional benchmarks, this study adopted a weighting based on the actual distribution of patients across BMI categories. This approach provides adjusted TVs that are more representative of the institution's patient population and minimizes biases related to body habitus variability.

Finally, the adjusted TVs were compared to the values obtained without BMI stratification, and the impact of the adjustment was assessed by calculating the relative percentage error between the general and adjusted TVs. This methodology enabled the evaluation of the influence of patients' body profiles on typical exposure levels for each interventional procedure.

3. Results

A total of 2,200 examinations were analyzed, distributed among four types of procedures, covering both clinical and demographic aspects. Table 1 presents the characterization of the sample, showing for each procedure the number of patients by sex, as well as the mean age, BMI, fluoroscopy time, and Ka,r TVs (Typical Values of the reference point air kerma). In the column “Ka,r TV (mGy)”, the Typical Values correspond to the medians of Ka,r for each procedure. The value of 413.0 mGy shown in the “Total” row is not the arithmetic mean of the other rows, but rather the median calculated from all 2,200 examinations combined, without stratification by procedure type, in accordance with ICRP 135 recommendations.

Table 1: Sample distribution by procedure and overall total, including sex distribution, mean age, BMI, fluoroscopy time, and Ka,r TV.

Procedure	Female	Male	N	Age (years)	IMC (kg/m ²)	Ka,r TV (mGy)	Mean Time (min)
Vascular Angiography	11	14	25	63,4	27,4	147,0	4,33
Vascular Angioplasty	22	46	68	57,2	27,1	132,0	8,32
Coronary Angiography	625	965	1590	62,0	27,8	369,5	2,97
Coronary Angioplasty	174	343	517	63,0	30,5	1112,0	9,03
Total	832	1368	2200	-	-	413,0	-

Figure 1 shows boxplots of the Ka,r distribution for each type of procedure. The displayed medians were used to define the TVs, as recommended by ICRP 135 (Vaño et al., 2017).

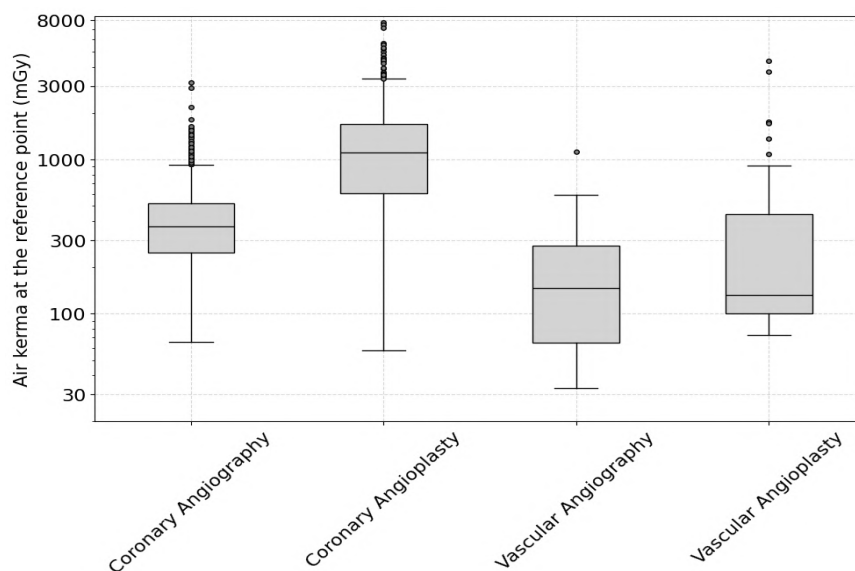


Figure 1: Distribution of Ka,r values by type of interventional procedure. *Note: The y-axis is presented in logarithmic scale to improve visualization of data spanning several orders of magnitude.*

Considering the influence of anthropometric profile on radiation exposure, the data were organized by BMI categories according to the WHO (WHO, 2000) classification. Table 2 presents the distribution of patients, the median Ka,r values (mGy), and the maximum fluoroscopy time recorded for each category.

Table 2: Distribution of radiation exposure indicators and fluoroscopy time by BMI categories.

BMI Category	N	Ka,r TV (mGy)	Maximum Time (min)
Underweight (<18)	43	185.0	13.9
Normal weight (18–24.9)	497	268.0	56.4
Overweight (25–29.9)	897	415.0	83.0
Obesity I (30–34.9)	511	534.0	52.4
Obesity II (35–39.9)	159	641.0	38.4
Obesity III (≥ 40)	93	796.0	27.0

Figure 2 complements this analysis by showing the distribution of Ka,r values by BMI category, highlighting variations in doses associated with different body profiles.

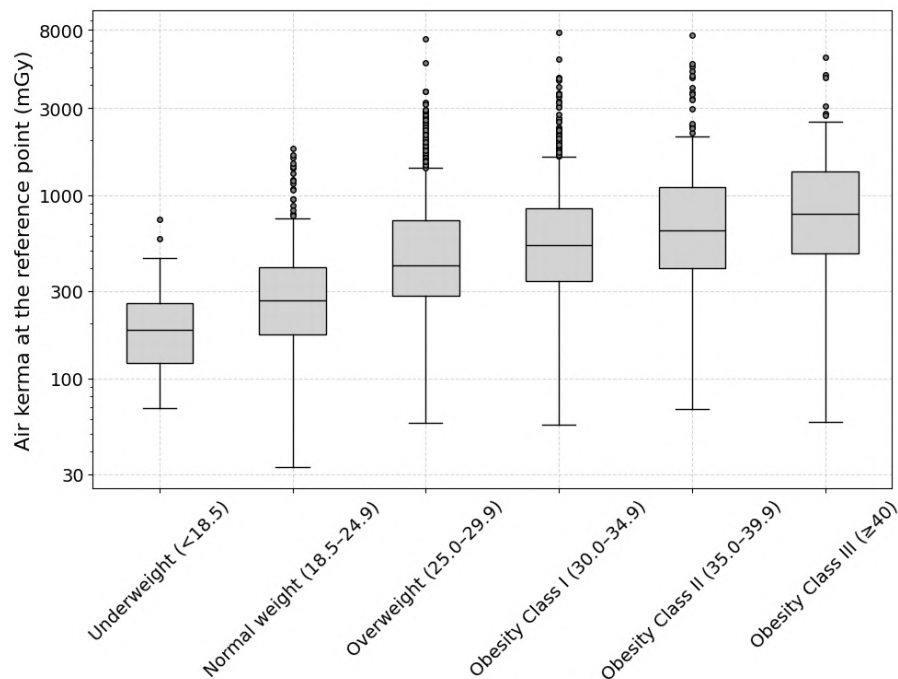


Figure 2: Distribution of Ka,r values (mGy) by BMI category.

Table 3 shows the Ka,r TVs for each type of interventional procedure, the TVs adjusted by BMI stratification, and the relative percentage error between them. *Note: The y-axis is presented in logarithmic scale to improve visualization of data spanning several orders of magnitude.*

Procedure	Ka,r TV (mGy)	Ka,r TV by BMI	Relative error (%)
Vascular Angiography	147,0	174,36	18,61
Vascular Angioplasty	132,0	143,22	8,50
Coronary Angiography	369,5	373,10	0,97
Coronary Angioplasty	1112,0	1113,40	0,12

4. Discussion

The analysis of 2,200 interventional procedures performed on a single device enabled the definition of TVs based on the median Ka,r and fluoroscopy time, in accordance with ICRP 135 guidelines (Vaño et al., 2017). The sample was predominantly composed of overweight patients, with mean BMIs ranging from 27.1 to 30.5 kg/m² across the different procedures, highlighting the importance of considering anthropometric profiles in defining reference parameters.

A trend of increasing dose with higher BMI was observed, as shown in Table 2 and Figure 2. Patients with Class III obesity had a median Ka,r of 796.0 mGy, while those with normal weight showed values around 268.0 mGy. Maximum fluoroscopy time was also longer in higher BMI categories, reaching 83 minutes for overweight patients compared to 13.9 minutes for underweight patients. These findings demonstrate that TVs defined without BMI consideration may underestimate or overestimate doses, compromising optimization strategies.

Comparison between general TVs and BMI-adjusted TVs showed significant variations across procedures. The highest relative error was found in vascular angiography (18.6%), followed by vascular angioplasty (8.5%). Coronary angiography and coronarioplasty showed smaller differences, with relative errors of 0.97% and 0.12%, respectively. Although absolute differences were small in some cases, the results indicate that BMI adjustment can have a meaningful impact on dose level definition, especially in diagnostic procedures such as vascular angiography.

Previous studies, such as (Tristram, 2022), highlighted TVs as essential tools for dose control but did not consider variations due to BMI or patient characteristics. This study, focusing on a single device and including BMI stratification, offers a more sensitive approach to the diversity of the patient population served, reducing operational variability, since all procedures were performed on the same fluoroscopy system, under consistent clinical and technical protocols established by the institution, and conducted by trained operators following a uniform local standard. This standardization minimizes the influence of differences related to hardware, acquisition settings, and individual operator techniques on the dose indicators, thereby enhancing the consistency and comparability of the typical values obtained.

The research conducted by (Vieira, 2022), performed in institutions in Minas Gerais, Brazil, marked a significant advance by proposing DRLs adjusted by BMI categories in interventional procedures. Their results confirmed that patients with higher BMI receive significantly higher doses, reinforcing the impact of anthropometric variation on radiation exposure. The present study not only corroborates the findings of Vieira et al. (2022) but also complements them by adopting TVs organized by procedure type and BMI category within a single institution. This approach enabled greater control over exposure factors and revealed differences that could be masked in global analyses. For vascular procedures, BMI-adjusted TVs showed a relevant impact, with variations of up to 18.6% in angiography and 8.5% in angioplasty. For coronary procedures, the differences were smaller (0.97% in coronary angiography and 0.12% in coronary angioplasty), but they should not be overlooked. From a radiological protection perspective, even small variations may become meaningful when accumulated across high-volume services or in patients undergoing multiple procedures, reinforcing the importance of reporting both substantial and subtle differences.

Some limitations of this study should be acknowledged. The complexity of procedures was not explicitly stratified, even though it is recognized as a factor that can influence patient dose. In addition, Ka,r values were obtained directly from the equipment, which inherently involves uncertainties associated with system calibration and manufacturer specifications, typically on the order of 10%. These aspects, however, do not compromise the robustness of the present findings, as all data were acquired under standardized protocols and within the same institution, ensuring consistency and comparability. Rather, they highlight opportunities for refinement in future multicenter studies, where greater heterogeneity in protocols and patient populations may require additional correction factors.

4. Conclusion

The results obtained allow for a more precise characterization of radiation exposure in interventional procedures, contributing to protocol optimization and safer clinical practices. The use of TVs organized by procedure type and anthropometric profile, combined with the analysis of technical variables, provides direct support for dose management at the institutional level. This approach, grounded in the recommendations of ICRP 135 (Vañó et al., 2017), proves applicable to routine clinical practice and can be replicated by other institutions with similar characteristics.

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Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)

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Abstract

Based on universal principles, the International Commission on Radiological Protection (ICRP) developed the concept of Diagnostic Reference Levels (DRLs). DRLs assist in optimizing practices and protocols, thereby promoting radiological safety. The Dose-Area Product (DAP) is a dosimetric parameter that quantifies the total radiation output from an X-ray tube during a procedure. The aim of this study was to investigate the DRLs for medical radiographs by estimating the dose-area product (DAP), to optimize radiological practice. The methodology was based on data collection from radiographs of the chest, abdomen, pelvis, lumbar spine, and thoracic spine performed in the radiology department of a public hospital. This study is part of the research project “Radiological Scenario in the University Hospital of the SUS” (CAAE: 71737417.9.0000.5149). Data were extracted from Digital Imaging and Communications in Medicine (DICOM) files and patient records. For analysis, the following were collected: DAP (mGy.cm²), peak kilovoltage (kVp), milliamper-second (mAs), projections, weight, height, and body mass index (BMI). The data were analyzed and presented in tables. The results showed the total number of patients for the Chest, Lumbar Spine, Thoracic Spine, Pelvis and Abdomen radiographs were 127, 20, 10, 14 and 5, respectively. The total number of projections for the Chest, Lumbar Spine, Thoracic Spine, Pelvis and Abdomen radiographs were 368, 38, 20, 19 and 15, respectively. The median DAP values for Chest radiographs were 4.6 mGy.cm² for the posteroanterior (PA) projection and 9.1 mGy.cm² for the lateral (L) projection. For the Lumbar Spine, the median DAP was 19.5 mGy.cm² in the anteroposterior (AP) projection and 27.7 mGy.cm² in the (L) projection. For the Thoracic Spine, the median DAPs were 10.8 mGy.cm² (AP) and 17.1 mGy.cm² (L), respectively. For the Pelvis (AP), the median DAP was 21.8 mGy.cm², and for the Abdomen (AP) it was 20.9 mGy.cm². These results indicate that the median DAP increases with the thickness to be radiographed, thus the (L) projection presented a greater DAP than the (AP/PA) projections. Median DAPs by weight group were also assessed. For the Chest (PA) and (L), and Lumbar Spine (AP and L), the median DAPs in the 50–69 kg weight group compared to the 70–90 kg weight group were in accordance with the exposure factors applied and patient weight. For the Thoracic Spine (AP and L), Pelvis (AP), and Abdomen (AP), the median DAPs for the 50–69 kg weight group were slightly higher than those for the 70–90 kg weight group, which is associated with the choice of technical parameters for these radiographs. Digital radiology has shown remarkable technological advances over the past decade. Although it has the potential to reduce patient doses, it is essential to apply appropriate exposure factors to achieve high-quality diagnostic images. This study highlights the importance of investigating DRLs for medical radiographic examinations to improve image quality and ensure radiological protection.

Keywords: DRL's; Dose-area product; Radiological protection; Medical radiography.

1 Introduction

The dose-area product (DAP) or Kerma area product (KAP) is a tool of monitoring the dose received by patients during radiological procedures. DAP values are generated based on the exposure factors used during image acquisition. These values are subsequently made available through the Digital Imaging and Communications in Medicine (DICOM) system, an international standard that standardizes information on radiographic examinations (Bushberg *et al.*, 2011).

As a result, optimization is a principle of radiation protection established by the NN-3.01 standard of the Brazil National Nuclear Energy Commission (CNEN) of 04/18/2024, which is based on the reasonableness between good diagnostic quality, dose reduction, and cost-benefit. Its principles are used to reduce DAP during medical exposures, ensuring the safety and radiation protection of individuals exposed to ionizing radiation during the disease diagnosis in the radiography departments (CNEN, 2024).

At the international level, since 1990, the diagnostic reference levels (DRLs) introduced by the International Commission on Radiological Protection (ICRP) aim to optimize radiological examinations. For medical radiography, its objectives cover data from five anatomical regions: thorax, abdomen, pelvis, lumbar spine and thoracic spine. The aim is to study the doses generated from the exposure parameters such as peak voltage (kVp), milliamperage per second (mAs) and current (mA) selected to obtain radiographic images. These parameters should be according to the anthropometric characteristics of the patients, such as weight, height, and body mass index (BMI). Thus, it is possible to review and improve techniques in future radiological procedures (ICRP, 2017).

Therefore, the objective of this study was to investigate the DRLs of medical radiographs in a radiology department, using the dose-area product (DAP) to optimize radiological practice.

2 Methodology

The research was carried out in a radiology sector of a public hospital and is part of the research project "Radiology Scenario at the SUS Assistance University Hospital" CAAE: 71737417.9.0000.5149 of the Federal University of Minas Gerais (UFMG), linked to the extension project "Radiology care: bringing together the experiences of the university, the public hospital and the community" (SIEX-UFMG 402801).

To investigate the DRLs of medical radiography, the DAP (mGy.cm²) of radiographs of the anatomical regions of the thorax, abdomen, pelvis, lumbar spine and thoracic spine were surveyed from two X-ray equipment, over a period of 30 days. The types of projections of each anatomical region and the exposure factors: kVp, mA and mAs were surveyed for these examinations. Patient demographics, such as age, sex, weight, height, and BMI index, were collected and correlated.

Radiographic image data were obtained using the Picture Archiving and Communication System (PACS), dosimetric parameters extracted from Digital Imaging and Communications in Medicine (DICOM) files, and patient records. The two X-ray equipment analyzed are from the company VMI Medical and have a Direct Radiography (DR) system to obtain the images. The equipment has a minimum kVp of 40 and a maximum of 150; mAs variation from 0.1 to 630 mAs and a total filtration thickness of 3.7 mm Al (VMI, 2017).

2 Results and Discussions

Over a period of 30 days, medical radiograph data were collected from 176 patients from the anatomical regions of the Thorax, Lumbar Spine, Thoracic Spine, Pelvis and Abdomen. These anatomical regions

are the ones that contribute most to the effective dose in the patient (IAEA, 2024). Table 1 shows the demographic data of the patients who underwent medical radiographs in the radiology department.

Table 1 – Demographic data of the patients for each type of examination: sex, age, weight, height, and BMI.

Exam (total)	Female Mean/Standard Deviation (SD) (Minimum-Maximum)				Male Mean/Standard Deviation (SD) (Minimum-Maximum)			
	Age (years)	Weight (kg)	Height (m)	BMI	Age (years)	Weigh (kg)	Height (m)	BMI
Chest (127)	62 (19-90)	67 ± 8.6 (51-80)	1.57 ± 0.1 (1.42-1.78)	27 ± 4.0 (19-38)	58 (20-84)	76 ± 9.2 (60-90)	1.72 ± 0.1 (1.47-1.93)	26 ± 3.5 (20-37)
Lumbar Spine (20)	64 (26-85)	65 ± 9.6 (50-79)	1.60 ± 0.0 (1.53-1.63)	26 ± 4.2 (22-34)	61 (51-75)	76 ± 0.0 (76-76)	1.79 ± 0.0 (1.79-1.79)	24 ± 0.0 (24-24)
Thoracic Spine (10)	71 (56-85)	62 ± 8.6 (53-79)	1.58 ± 0.0 (1.51-1.63)	25 ± 4.1 (21-34)	—	—	—	—
Pelvis (14)	71 (47-101)	78 ± 13.6 (54-88)	1.61 ± 0.0 (1.54-1.68)	30 ± 5.0 (23-35)	61 (47-76)	—	—	—
Abdomen (5)	54 (34-57)	69 ± 5.2 (64-74)	1.62 ± 0.0 (1,62-1.62)	28 ± 0.0 (28-28)	66 (66-66)	—	—	—

The Chest in relation to the number of radiological examinations of females and males by anatomical region had 74 females and 53 males. For the Lumbar Spine region, there were 17 female and 3 male exams. For the Thoracic Spine region, there were 10 exams, all females. For the pelvis region, there were 11 females and 3 males and for the abdomen region, 4 females and 1 male. According to the IAEA, for DRLs studies, a minimum of 20 patients is indicated for the medical radiography modality (IAEA, 2024). Although the Thoracic Spine, Pelvis and Abdomen regions did not reach this minimum value, they were included in the study, as the data will be part of a six-month survey.

Regarding the weight of patients for both sexes, they were in the range of 50 to 90 kg determined by the IAEA for DRLs studies (IAEA, 2024). In the region of the Lumbar Spine, the minimum and maximum weight of males was the same, due to the small number of exams collected. In the regions of the Thoracic Spine, Pelvis and Abdomen, weight data were not identified in the examinations surveyed of males.

The IAEA does not establish a standard height for the determination of DRLs, in this study the mean height of all anatomical regions of the female sex was obtained of 1.59 m and for the male sex mean of 1.75 m.

According to the World Health Organization (WHO), BMI between 25 and 29.9 indicates overweight for both sexes (WHO, 2025). The results showed for the regions of the Thorax, Lumbar Spine, Thoracic Spine and Abdomen, the BMI was overweight for females and for the Pelvis region, the BMI was obesity above 30. For males, the BMI for the Chest region indicated overweight and for the Lumbar Spine region indicated normal weight of 24 BMI, the other regions did not have information identified in the exams.

Table 2 shows the exposure factors collected from medical radiographs of the Chest (AP, PA, and L), Lumbar Spine (AP, OBL, and L), Thoracic Spine (AP and L), Pelvis (AP and “Frog-Leg”), and Abdomen (AP- supine position (S) and AP- orthostatic (ORT)) regions: exam and projection, number of projections, kVp, mAs, and mA. The standard deviation (SD) was calculated, and the minimum and maximum values were presented.

Table 2 - Technical parameters of medical radiographs: projections, number of projections, kVp, mAs, mA.

Exam and Projection	Number of Projections	kVp - mean/SD (min.-max.)	mAs - mean/SD (min.-max.)	mA
Chest - Anteroposterior (AP)	2	125 ± 0.0 (125-125)	11 ± 1.4 (11-11)	280
Chest - Posteroanterior (PA)	129	104 ± 11.4 (75-128)	9 ± 4.1 (3-40)	280
Chest - Lateral (L)	129	115 ± 11.6 (75-138)	15 ± 6.0 (3-40)	280
Lumbar Spine - AP	20	82 ± 5.9 (74-93)	61 ± 22.9 (25-112)	280
Lumbar Spine - L / Oblique (Obl)	18	83 ± 6.2 (75-100)	68 ± 25.8 (25-112)	280
Thoracic spine - AP	10	80 ± 3.6 (75-85)	38 ± 10.3 (25-56)	280
Thoracic spine - L	10	82 ± 4.9 (75-88)	42 ± 12.6 (25-62)	280
Pelvis - AP/ "Frog-Leg"	19	76 ± 6.5 (70-90)	49 ± 26.0 (8-110)	280
Abdomen - AP S/AP ORT	15	78 ± 11.5 (75-120)	37 ± 8.5 (10-46)	280

Table 2 shows that the mean kVp for Chest X-rays in (PA) and (L) projections was above 100, as provided for in the adult patient examination protocols. The mean mAs for L projection was higher than for PA, as expected, due to the greater thickness of the region in this radiographic position, according to the literature (Bontrager *et al.*, 2018). It should also be considered that the exposure factors of high kVp and low mAs are part of the examination protocol for obtaining a long gray scale (low contrast) for the visualization of all anatomical structures of the Chest. Chest examination is the one with the highest number of projections, as it is one of the most performed medical radiographs in imaging departments, with various purposes, such as monitoring probes in hospitalized patients. (Bontrager *et al.*, 2018).

For Lumbar Spine AP and L, Pelvis AP, and Abdomen AP the mean kVp and mean mAs are also within the range of the literature (Bontrager *et al.*, 2018). The Thoracic Spine also shows this result, except for the L projection, in which the mean should be around 65 mAs, a parameter responsible for the increase in dose, signaling an optimization of the protocol. However, by increasing the sample of exams in the research, these values may be closer to the literature. The tube current was standardized for these anatomical regions for the two X-ray equipment, according to the manufacturer's Table of Radiological Techniques.

To define DRLs of a single room or a small number of X-ray rooms of a medical radiography installation (less than 10 rooms), it is recommended to establish the Median Typical Value (IAEA, 2024). In this study, the median typical value was used.

Table 3 shows the statistical values of DAP (mGy.cm²) collected from medical radiographs of the Chest, Lumbar Spine, Thoracic Spine, Pelvis, and Abdomen of all 127 patients.

Tab 3- DAP values of medical radiographs: minimum, maximum, mean, standard deviation, and median.

Exam - Projection (Number of projections)	DAP minimum (mGy.cm²)	DAP maximum (mGy.cm²)	DAP mean (mGy.cm²)	Standard deviation	DAP Median (mGy.cm²)
Chest - AP (2)	16.0	18.0	17.0	1.4	17.0
Chest - PA (129)	1.0	25.0	6.0	4.9	4.7
Chest - L (129)	1.6	44.6	11.9	7.2	9.1
Lumbar Spine - AP (20)	9.0	97.0	24.8	19.3	19.5
Lumbar Spine - L / Obl (18)	9.0	97.0	30.8	17.6	27.7
Thoracic spine - AP (10)	8.3	22.0	13.0	5.1	10.8
Thoracic spine - L (10)	9.0	30.0	18.4	6.6	17.1
Pelvis - AP / "Frog-Leg" (19)	3.8	71.2	26.0	18.8	21.8
Abdomen - AP S / AP ORT (15)	6.0	23.2	20.0	4.0	20.9

It can be observed that the median DAP for the L Chest is 51,6% higher than for the PA Chest, due to the necessary exposure factors of kVp and mAs for thicker structures. The median DAP of the AP chest was higher than for the PA and L projections, but due to the small number of projections, it does not yet represent a mean as expected.

For the Lumbar Spine, the median DAP of the AP and L/Obl projections had a difference of 29.6% and for the Thoracic Spine in the AP and L projections, there was a difference of 36.8%. These results corroborate the exposure factors used for radiographs of these anatomical regions.

It can be observed that the median DAP of the Pelvis and Abdomen present similar results, but according to the *National Diagnostic Reference Levels guide* (GOV.UK, 2024), the median DAP of the Abdomen should be 12% higher than the median DAP of the Pelvis. Then, more tests should be collected for evaluation.

Another analysis is presented in Table 4, which shows the median DAP (mGy.cm²) and means with SD of kVp, mAs of the radiographs: Chest (AP, PA and L), Lumbar Spine (AP, Obl and L), Thoracic Spine (AP and L), Pelvis (AP and "Frog-Leg") and Abdomen (AP-S, AP-ORT). These medical radiographs were separated into two groups of patients according to weight: group between 50 and 69 kg and group between 70 and 90 kg.

Table 4 - Median DAP, kVp (mean/SD) and mAs (mean/SD) values of medical radiographs separated by patient weight group.

Exam and Projection	Group - Weight 50 to 69 kg			Group - Weight 70 to 90 kg		
	kVp mean/SD	mAs mean/SD	DAP median	kVp mean/SD	mAs mean/SD	DAP median
Chest - PA	101 / 6.9	8 / 3.4	4.0	107 / 11.0	9 / 3.6	5.0
Chest - L	111 / 11.3	14 / 4.1	8.0	115 / 11.5	16 / 6.8	9.0
Lumbar Spine - AP	80 / 0.0	56 / 33.7	16.0	83 / 2.9	61 / 6.6	22.0
Lumbar Spine - L	80 / 0.0	57 / 35.7	17.9	85 / 3.3	74 / 18.9	40.0
Thoracic Spine - AP	79 / 4.4	40 / 11.6	12.0	80 / 0.0	38 / 9.9	11.2
Thoracic Spine - L	81 / 5.4	43 / 15.4	18.0	85 / 0.0	42 / 4.4	17.0
Pelvis - AP / "Frog-Leg"	70 / 0.0	39 / 0.0	19.0	72 / 3.0	36 / 17.0	17.0
Abdomen - AP S/AP ORT	75 / 0.0	41 / 5.5	20.9	83 / 18.4	33 / 11.7	20.3

It can be observed that the median DAP of the Chest PA and L of the Group (50 to 69 kg) is lower than that of the Group (70 to 90 kg), corroborating the exposure factors used in the exams for groups of different biotypes. The median DAP for the Lumbar Spine also presents this result between the two groups.

The median DAP of the Thoracic Spine, Pelvic and Abdomen radiographs showed slightly higher values in the group (50 to 69 kg) than in the group (70 to 90 kg), which was not expected. The difference in median DAP between the Thoracic Spine AP Groups was 0.8 mGy.cm² and 1 mGy.cm² for L projection. The difference in median DAP between the Abdomen AP Groups was 2.0 mGy.cm² and 0.6 mGy.cm² in the Pelvis AP Groups. This result suggests a review of the exposure factors used and a greater number of exams for evaluation.

When evaluating the exposure factors of the radiographs used, it was noted that the mAs were slightly higher, ranging from 1 mAs to 8 mAs in the Groups (50 to 69 kg) in relation to the Groups (70 to 90 kg), which directly influences the dose received by the patient, i.e., the DAP. According to Bontrager *et al.* (2018), mAs is the main controlling factor of radiographic density, responsible for controlling the number of X-rays emitted by the X-ray tube and the duration of exposure. There is a relationship of proportionality between the current of the tube and the production of X-ray photons, because when the mAs is doubled, the amount or duration of the X-rays emitted is doubled, increasing the amount of radiation, increasing the probability of increasing the dose in the patient.

The DAP values presented in the study are below the values of international DRLs, but there is a need for a larger data collection, with the evaluation of the image quality related to the response of the image detector.

The literature shows us the importance of determining the DRLs for medical radiography in Brazil. Studies conducted by Freitas and Yoshimura (2003) showed the variation in dose in chest radiological examinations of PA and L in the state of São Paulo, in which the dose can be reduced by appropriate changes in exposure factors, without loss of image quality. In 2009, Freitas and Yoshimura conducted a survey of the DRLs of the most performed medical radiographs in Brazil, where they showed that exposure factors should always be reviewed for dose reduction in patients, especially pediatric patients.

In a study by Osibote and Azevedo (2007) carried out in the state of Rio de Janeiro, it was shown that the entrance skin dose (ESD) were below the international DRLs references. However, it indicated the need for training of Radiological Techniques professionals to optimize the doses in patients.

In the scenario where digital systems (DR) technology is increasingly widespread in the world, research by Alshamrani, Alkenawi, Alghamdi, *et al.* (2021), investigated DAP values of more common radiographic examinations compared with international DRLs. The results showed DAP values below the international DRLs for some medical radiographs and others above, in which it related an analysis of the response to the type of image detector.

In a study carried out by Zarghani, Masoud and Pandesh (2023) evaluated doses with a DAP meter for the most common medical radiographs, with the aim of implementing DRLs to optimize the patient's dose and image quality. The research affirms that the implementation of a quality control program is necessary to optimize the dose in patients.

4 Conclusions

The results of this study investigated the minimum, maximum, mean and median DAP of medical radiographs of the chest, abdomen, pelvis, lumbar spine, and thoracic spine of a total of 176 patients from a radiology department. The median DAPs are considered typical values for the determination of DRLs in a radiology department with few X-ray rooms. Thus, the calculated DAP values were below the international NRDs. However, some median DAPs of the same anatomical region are higher for projections in PA than L, evidencing the use of non-optimized exposure factors. The number of medical radiographs of the chest and lumbar spine met the minimum for the study and the other anatomical regions it is necessary to increase this number, but it was possible to start their analyses. The weight of the patients was between 50 and 90 kg, with BMI indicating overweight in the majority.

It can be concluded that for a better evaluation, a greater collection of medical radiographs, evaluation of image quality, training of Radiological Techniques professionals, as well as the establishment of national guidelines aimed at good practices for optimizing the doses received by patients are necessary.

Finally, based on this study, we intend to continue collecting data for an analysis that substantiates the proposal to optimize examination protocols for medical radiographs of the chest, abdomen, pelvis, lumbar spine, and thoracic spine. To improve the quality of the diagnostic image and contribute to the radiological protection of the radiology department studied.

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Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics

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Abstract

This study explores the application of supervised machine learning (ML) algorithms to predict outcomes of patient-specific quality assurance (PSQA) in intensity-modulated radiotherapy (IMRT) and volumetric-modulated arc therapy (VMAT). A cohort of 133 anonymized clinical radiotherapy plans covering treatment sites such as prostate, head and neck, pelvis, central nervous system (CNS), and stereotactic radiosurgery (SRS) was analyzed. From each DICOM-RT (Digital Imaging and Communications in Medicine – Radiation Therapy) plan, 60 quantitative features were extracted, encompassing multileaf collimator (MLC) dynamics, dose distribution characteristics, and complexity indices including Modulation Complexity Score (MCS), Average Leaf Speed (ALS), and Leaf Sequence Variability (LSV). To address class imbalance between QA pass/fail outcomes, a synthetic dataset of 66 perturbed plans simulating realistic delivery deviations was generated, complemented by the use of the Synthetic Minority Oversampling Technique (SMOTE) [8]. Feature importance analysis identified MCS, ALS, and dose irregularity metrics as key predictors of QA failure. The optimized Random Forest classifier achieved a weighted F1-score of 0.99 during cross-validation and 0.92 on the independent test set. The results corroborate findings in the literature that support the integration of ML models to enhance PSQA accuracy and sensitivity beyond conventional gamma-based analysis [1, 10, 7, 9]. By leveraging DICOM-RT metadata, this study highlights the feasibility of implementing predictive algorithms in radiotherapy QA pipelines, improving workflow efficiency and reducing dependence on manual dosimetric verification.

Keywords: radiotherapy, machine learning, DICOM-RT

1. Introduction

The integration of artificial intelligence (AI) techniques into quality assurance (QA) workflows in radiotherapy has garnered significant attention in recent years, with the primary objective

of enhancing treatment accuracy and optimizing clinical resources. A growing body of literature has demonstrated the potential of machine learning algorithms to identify treatment discrepancies and improve the reliability of patient-specific QA procedures [1, 2]. Predictive modeling approaches that leverage plan complexity metrics and historical QA data have shown considerable promise in anticipating QA outcomes, thereby minimizing the need for extensive physical measurements and contributing to increased operational efficiency [3, 4]. This study seeks to promote a deeper integration between academic investigation and clinical practice, fostering the development of intelligent computational tools to support radiotherapy QA processes. Concurrently, gamma analysis continues to represent a cornerstone in the verification of dose delivery accuracy in radiotherapy. Despite its widespread clinical adoption, conventional QA methodologies frequently fall short in detecting subtle deviations and long-term trends, which underscores the growing relevance of combining traditional techniques with data-driven approaches [5]. The transition toward AI-assisted QA paradigms reflects a broader shift in the field, emphasizing adaptability, automation, and evidence-based decision-making. The present work aims to investigate the application of machine learning in PSQA in radiotherapy, with a particular emphasis on the assessment of delivery complexity and the prediction of gamma passing rates. The proposed approach is grounded in the extraction of quantitative features from planning and delivery data, with the objective of training a predictive model capable of classifying treatment plans according to TG-218 gamma criteria [5].

2. Methodology

2.1 Dataset Construction and Feature Extraction

The implementation was carried out in Python, leveraging specialized libraries such as pydicom for parsing DICOM files, and pandas for organizing extracted data into relational structures. The pipeline was designed in a modular fashion, with dedicated scripts for handling each DICOM modality: RTPLAN, RTDOSE, and RTIMAGE.

The RTPLAN parser was responsible for extracting treatment beam configurations, delivery times, and prescribed doses. Iterative loops were employed to traverse patient datasets and store relevant metadata. For RTDOSE files, the parser converted raw voxel data into three-dimensional matrices, enabling the computation of aggregate dosimetric parameters such as mean and maximum dose within defined regions of interest. In turn, the RTIMAGE module focused on retrieving portal image metadata, including acquisition timestamps and pixel intensity values, ensuring their alignment with the corresponding treatment arcs.

A total of 133 radiotherapy treatment plans were evaluated, encompassing a variety of anatomical sites, including 70 prostate, 15 head and neck, 5 central nervous system (CNS), 3 stereotactic radiosurgery (SRS), 20 pelvis, and a general 20 as "other" category. For each plan, a structured dataset was built containing 60 distinct metrics extracted from DICOM-RT files. These metrics included parameters such as prescribed dose, maximum dose, number of fractions, and total monitor units (MU).

The data were organized using the pandas library, which enabled efficient data manipulation and supported exploratory data analysis. Visual analytics were developed using matplotlib and seaborn to aid in the identification of patterns and detection of potential outliers. For example, scatter plots were used to examine the relationship between monitor units and acquisition duration, while histograms helped visualize inter-patient variability in dose per fraction.

EPID-based measurements were incorporated for real-time delivery validation. A dedicated script segmented treatment plans by arc, guided by acquisition timestamps extracted from RTIMAGE metadata. Matching algorithms were employed to associate EPID frames with specific RTPLAN segments. Spatial filtering techniques were applied to reduce image noise, followed by dose profile extraction and comparison with RTDOSE data through metrics such as mean percentage difference and gamma index similarity.

The validation process encompassed both manual and automated cross-checks. Scripts compared extracted metadata against expected clinical protocol values to identify inconsistencies or outliers. For instance, plans where the total dose derived from RTDOSE deviated substantially from RTPLAN prescriptions were flagged and, in less than 1% of cases, excluded due to formatting anomalies. Temporal consistency of EPID acquisitions was also verified to ensure synchronization with documented treatment phases.

A validation module was integrated into the pipeline to assess the integrity of input files before processing. This routine detected and excluded corrupted or incomplete datasets. Performance optimization was achieved by enabling batch processing and parallel computation through Python's multiprocessing library, significantly reducing execution time across the dataset comprising thousands of records from 133 patients. Additionally, a logging routine was incorporated to capture progress, runtime errors, and diagnostic information, enhancing traceability and reproducibility.

These efforts culminated in the construction of a robust, well-structured database, which serves as the foundation for future developments in predictive modeling and automated QA optimization in radiotherapy workflows.

2.2 Balancing QA Datasets with Simulated Delivery Errors

To improve the predictive performance of the machine learning models developed in this study, a synthetic cohort of 66 patient treatment plans was generated to simulate QA failures. These synthetic plans were derived by applying controlled perturbations to the original dataset, such as altering multileaf collimator (MLC) trajectories, beam-on times, and dose distributions. The objective was to emulate realistic delivery deviations that could result in failing gamma index criteria, thereby increasing the representation of the minority class (QA failures) in the training dataset.

This strategy aligns with the general recommendations found in recent reports on the implementation of online treatment monitoring (OTM) systems, which emphasize the importance of evaluating system robustness using diverse clinical scenarios and simulated errors [6, 7]. The original dataset, composed predominantly of QA-approved cases, presented a marked class imbalance that could negatively impact model training. Unbalanced datasets tend to bias classifiers toward the majority class, reducing sensitivity to clinically relevant errors. To mitigate this issue, the Synthetic Minority Oversampling Technique (SMOTE) was used in this case [8]. SMOTE generates new samples of the minority class (QA failures) by interpolating between existing observations in the feature space, thereby enhancing the diversity and representativeness of this class during model training [8]. The use of SMOTE in this context is justified by the need for a more balanced training set, which improves model generalizability and reduces the risk of false negatives—cases in which failing plans are incorrectly classified as acceptable. Such errors could lead to undetected deviations in clinical practice, a risk highlighted in QA guidelines for IMRT and VMAT [9]. Additionally, the balanced dataset facilitates robust evaluation metrics such as sensitivity, specificity, and area under the receiver operating characteristic (ROC) curve, as emphasized in recent studies applying machine learning to radiotherapy QA [1, 10, 11]. Through this data augmentation and balancing process, the final model was trained on a dataset with equal representation of passed and failed QA cases. The trained Random Forest classifier achieved high performance metrics in the test phase, including an F1-score of 0.92, which reflects a strong balance between precision (low false positives) and recall (low false negatives). These results highlight how crucial it is to address class imbalance when designing supervised learning approaches for radiotherapy quality assurance.

3. Results

A comprehensive dataset encompassing 133 patients was analyzed, generating an extensive collection of records with 60 features related to treatment planning and delivery per arc. Following meticulous data preprocessing and validation, supervised machine learning models were developed to predict quality assurance (QA) outcomes, classified as either pass or fail based on gamma analysis using the 3%/3 mm criteria with a 10% low-dose threshold, as stipulated by the AAPM TG-218 guidelines [5].

A total of 184 treatment arcs were assessed, of which 128 (69.6%) satisfied the gamma passing criteria, while 56 (30.4%) were identified as failing. A pronounced discrepancy was observed between the original (Modified = 0) and perturbed (Modified = 1) datasets: 91% of the failing arcs were associated with modified plans, underscoring the model's sensitivity to variations in plan complexity and dose delivery precision.

To address the inherent class imbalance and enhance model robustness, a synthetic oversampling technique was employed, resulting in a balanced dataset with 128 instances for each class (pass and fail). A Random Forest classifier was subsequently trained and evaluated using five-fold cross-validation, achieving a mean F1-weighted score of 0.992 ± 0.009 , indicative of exceptional predictive performance across validation subsets.

On an independent test set comprising 77 instances, the classifier attained an overall accuracy of 92%. The confusion matrix, presented in Table 1, elucidated the model's classification performance, revealing only six misclassifications: three false positives (FP), where passing arcs were erroneously predicted as failing, and three false negatives (FN), where failing arcs were incorrectly predicted as passing. The true positives (TP, correctly predicted failing arcs) and true negatives (TN, correctly predicted passing arcs) dominated, affirming the model's capability to accurately differentiate between clinically acceptable and suboptimal treatment deliveries.

	Predicted Pass	Predicted Fail
Actual Pass	45 (TN)	3 (FP)
Actual Fail	3 (FN)	26 (TP)

Table 1. Confusion Matrix for the Random Forest Classifier on the Test Set

Note: TN = True Negative, FP = False Positive, FN = False Negative, TP = True Positive.

The model achieved robust performance across both QA outcome classes. For failing arcs, it yielded a precision of 0.97 and recall of 0.87, while passing arcs presented the inverse trend, with precision and recall of 0.88 and 0.97, respectively. These complementary patterns resulted in balanced F1-scores of 0.92 for both classes, indicating the model's ability to effectively distinguish between acceptable and unacceptable treatment deliveries without bias toward the majority class.

The feature importance analysis identified the Modulation Complexity Score (MCS) as the primary predictor of quality assurance (QA) outcomes, with an importance score of 0.292, followed by Average Leaf Speed (ALS) at 0.190 and the Irregularity Factor at 0.142. These results underscore the critical influence of dynamic multileaf collimator (MLC) parameters on QA failures. Other notable features include the number of Monitor Units (MU_1, 0.073), Maximum Dose (0.034), and geometric field parameters such as Field_Y_1 (0.018). These findings highlight that both dosimetric and geometric characteristics of the treatment plan play significant roles in determining QA performance, with MLC-related dynamics being particularly impactful, as shown in Figure 1.

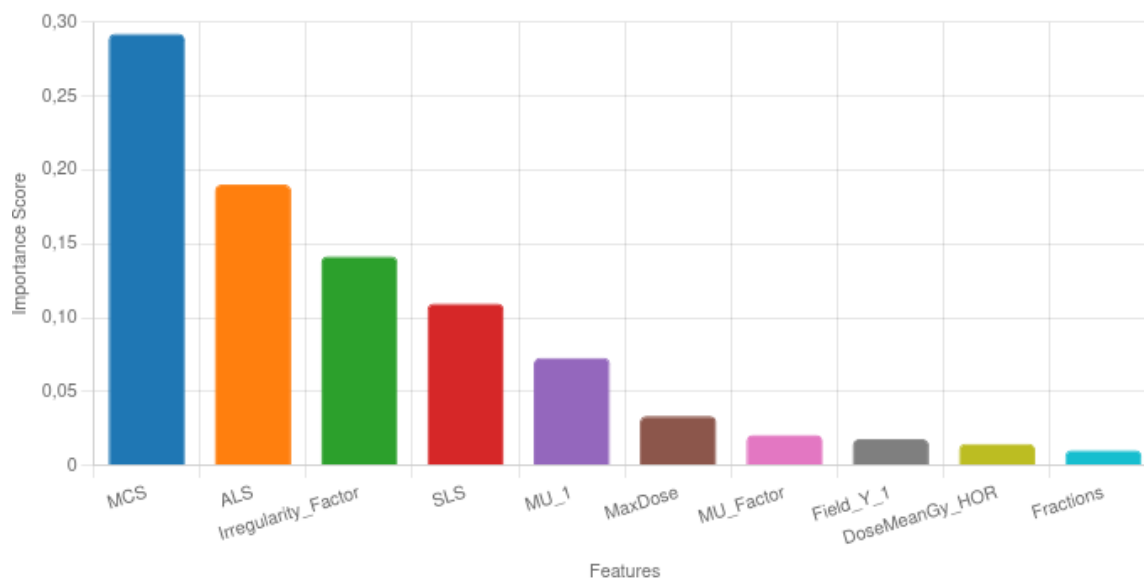


Figure 1. Feature Importance Scores for the Random Forest Classifier on the Test Set

The low incidence of false positives (3/77) minimizes the risk of unwarranted clinical interventions for acceptable plans, while the equally low rate of false negatives (3/77) ensures that most suboptimal plans are detected, thereby enhancing patient safety. These results validate the robustness of the Random Forest classifier and its potential for integration into clinical QA workflows for VMAT plans in radiotherapy.

4. Discussion

The high F1-score of 0.92 on the independent test set substantiates the robustness of the Random Forest classifier and its capacity for generalization to unseen data. Beyond overall accuracy, a detailed analysis of the confusion matrix reveals the critical implications for clinical implementation. The model's low misclassification rates—just three false positives and three false negatives on the test set—indicate its potential as a reliable screening tool to enhance the efficiency of patient-specific quality assurance (PSQA) workflows.

The minimal rate of false positives (FP) is particularly significant for clinical efficiency, as it suggests that the model would rarely flag an acceptable plan for unnecessary manual review. This minimizes the expenditure of clinical resources. Conversely, the rate of false negatives (FN) represents the most critical concern from a patient safety perspective, as these are clinically failing plans that the model incorrectly classified as acceptable. While this rate is low, it underscores that the model should function as a decision support tool rather than a complete replacement for established dosimetric verification procedures. Its primary role is to assist medical physicists in prioritizing which plans require the most rigorous review, thereby optimizing resource allocation without compromising safety.

The feature importance analysis offers valuable insight into the physical determinants of QA failure. The Modulation Complexity Score (MCS) and Average Leaf Speed (ALS) emerged as the most influential predictors, which aligns with the well-documented clinical observation that highly complex treatment plans requiring rapid multileaf collimator (MLC) movements are inherently more susceptible to delivery errors. This finding validates the model's physical foundation, as it effectively learns to identify the same risk factors that contribute to real-world delivery discrepancies. The successful classification of these complex cases was significantly enabled by the use of a synthetic dataset of realistic failure scenarios, a crucial step in mitigating the class imbalance inherent in clinical QA data.

Despite its promising performance, this study has several limitations. The dataset, while augmented, is derived from a single-institution cohort. Consequently, the model's generalizability across diverse clinical environments with varying equipment and protocols remains to be fully validated. Additionally, the simulated perturbations, although informed by clinical experience, may not encompass the full spectrum of potential delivery errors. Future research should focus on validating this predictive framework using multi-institutional data. Further improvements could be achieved by integrating more sophisticated machine learning architectures, such as deep learning, and by incorporating real-time data from treatment log files to enhance predictive accuracy and facilitate the development of an online monitoring system.

5. Conclusions

This study presents a comprehensive approach to predictive modeling of radiotherapy plan quality assurance (QA) through the integration of clinical data, complexity metrics, and machine learning techniques.

Descriptive and exploratory analyses revealed substantial variability across anatomical regions, particularly in parameters such as prescribed dose and maximum dose per fraction, emphasizing the clinical heterogeneity inherent to radiotherapy planning. Notably, pelvic treatments exhibited the highest variability in dose delivery, underscoring the need for site-specific QA strategies.

The inclusion of synthetically generated failure cases, SMOTE, led to a more balanced training distribution, allowing the model to better distinguish between passing and failing QA scenarios. This improved representation of both classes contributed directly to the robustness of the Random Forest classifier, which achieved a weighted F1-score of 0.99 during cross-validation and 0.92 when evaluated on an independent test set.

The analysis of feature importance highlighted the relevance of plan complexity metrics, with the Modulation Complexity Score (MCS), Average Leaf Speed (ALS), and Sequence Leaf Speed (SLS) emerging as the most influential predictors of gamma passing rate. These findings are consistent with the literature and reinforce the clinical interpretability of model outputs.

In summary, the proposed framework demonstrates that supervised machine learning models, informed by quantitative radiotherapy metrics, can effectively support QA decision-making. By reducing dependence on manual measurement procedures and enabling early identification of potentially suboptimal plans, this approach contributes to the development of more efficient, data-driven, and personalized QA processes in contemporary radiotherapy practice.

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Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: observational study

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Abstract

Pediatric chronic kidney disease-mineral and bone disorder represents a complex clinical spectrum, manifesting as skeletal alterations, metabolic disturbances, and extra-skeletal calcifications. Additionally, vascular calcifications and correlative cardiovascular disease (CVD) are substantial risk factors for mortality in this population, with CVD accounting for up to 30% of deaths in children undergoing dialysis. Computed tomography (CT) is a usually clinical examination method to analyze vascular calcification performed by calcium scoring CT scan (CTcs) protocol. Due to the rarity of such findings in pediatric patients, the CTcs is primarily restricted to specific clinical scenarios and research protocols, which formed the context of this study. Given the lack of reported volumetric CT dose index (CTDI_{vol}) values in the literature for pediatric patients undergoing CTcs, this study aimed to compare the CTDI_{vol} value determined by the CT scanner with results obtained from the exposure of thermoluminescent dosimeters (TLDs) positioned on the patient. To achieve this, six calibrated TLD100 dosimeters were affixed to the mid-axillary line, at the level of the fourth intercostal space. The CTcs were performed on a Philips, 64-channel Brilliance CT scanner, which undergoes rigorous quality control at the university hospital. The results show the mean CTDI_{vol}, indicated by the CT scanner, was 4.00±0.04 mGy, while measurements performed with the TLD-100 show a mean CTDI_{vol} of 4.87±1.35 mGy. It means an increase of 22% of the dose on the patient, when compared by the results of CTDI_{vol} coming from the scanner, demonstrating greater accuracy in dosimetry obtained with TLDs. Further studies are underway using modern CT scanner that can implement image quality for vascular calcifications associated with reduced radiation doses, thereby adhering to the ALARA (As Low As Reasonably Achievable), principle of radiation protection 4.

Keywords: chronic kidney disease; pediatric CT; thermoluminescent dosimeter, CTDI_{vol}

1 Introduction

More than fifty years have passed since the invention of computed tomography (CT) by Ambrose and Hounsfield in 1971 in the United Kingdom. During this time, this method has revolutionized imaging diagnostics, paving the way for volumetric imaging of the body for medical diagnoses and new clinical findings (McCollough, et al. 2023), (Smith-Bindman et al. 2025), (Zhang, L, et al. 2021), (Kardos et al. 2024), (Hsieh, J., Flohr, T., 2021).

Pediatric chronic kidney disease-mineral and bone disorder presents a complex clinical spectrum, manifesting as skeletal alterations, metabolic disturbances, and extra-skeletal calcifications. Furthermore, vascular calcifications and correlated cardiovascular disease (CVD) are significant risk factors for mortality in this population, with CVD accounting for up to 30% of deaths in children undergoing dialysis (Lalayiannis, A. D., et al. 2023). CT scans are commonly used as clinical examination method to analyze vascular calcification, performed by calcium scoring CT scan (CTcs) protocol (Zhang, L, et al. 2021), (Kardos et al. 2024), (Eddy, F. K., et al. 2021), (Mese, I., et al. 2024). However, due to the rarity of such findings in pediatric patients, CTcs is primarily restricted to specific clinical scenarios and research protocols, which forms the context of this study.

Due to the potential of biological effects of X-rays, the International Commission on Radiological Protection (ICRP) provides recommendations and guidance to mitigate the risks associated with ionizing radiation (ICRP 103, 2007). CT radiation can be linked with increased cancer incidence; therefore, it is important to minimize X-ray exposure, especially in pediatric patients who are more sensitive to the biological effects of the radiations. The International Commission on Radiation Units and Measurements (ICRU) published recommendations for dose assessment in CT, specifically regarding the CT Dose Index volume (CTDI_{vol}), (ICRP 102, 2007), (Smith-Bindman, et al. 2019), (Eddy, F. K., et al. 2021), (Mese, I., et al. 2024).

The CTDI_{vol} is a measure of the main dose in a CT examination, calculated from the CT Dose Index (CTDI) for a 100 mm length. It is frequently used to estimate the absorbed dose by the patient. These quantities should not be confused with the patient dose; this is only a referential dose index applied in a quality control CT program to evaluate the dose at a specific condition, expressed in *mGy*. The CT scanner consoles provide CTDI, CTDI_{vol}, and the dose length product (DLP) in a report after image acquisition; these are calculated by the CT scan's own computer software. (Guimarães, A. et al. 2024), (ICRP 102, 2007), (ICRU 74, 2005), (Bos, D., et al. 2020), (Smith-Bindman, R., et al. 2019).

The application of thermoluminescent dosimeters (TLDs) for measuring X-ray radiation doses and subsequently evaluating CTDI_{vol} is well-documented, owing to their accuracy, reproducibility, and ideal ability to measure a wide dose range. Among these, lithium magnesium fluoride (LiF:Mg,Ti), commercially referred to as TLD-100, is particularly well-suited for X-ray dosimetry in patients. Its efficacy in this context is attributed to several key properties, such as: TLD-100 exhibits good sensitivity and low energy dependence within the diagnostic X-ray energy spectrum, and its tissue-equivalent density minimizes image artifacts when positioned on a patient (Guimarães, A., et al. 2024), (da Silva, J. A., et al. 2023), (Rawat, N. S. et al. 2023), (Daros, K. A. C, et al. 2012), (Matsubara, K.,2017), (Dixon, R. L., 2003), (Kikuti, C., et al. 2015), (Horowitz, Y., et al. 2018).

Given the lack of reported CTDI_{vol} values in the literature for pediatric patients undergoing CTcs, this study aimed to compare the CTDI_{vol} value determined by the CT scanner with results obtained from the TLD-100 positioned on the patient.

2 Materials and Methods

This study's procedures were authorized in advance by the institutional Research Ethics Committee (protocol #2130/11).

Pediatric CTcs scans were performed at the university hospital using a Philips Brilliance 64-channel CT scanner, which is subject to stringent quality control. The CTcs protocol parameters verified in this study were the tube kilovoltage, current-time product, slice thickness, pitch, field of view (FOV), filter, and CTDI_{vol}. The average body mass index (BMI) of the patients was calculated by dividing weight by the square of height, in $kg.m^{-2}$, for each patient.

Dosimetric evaluations were conducted on 19 pediatric patients with clinical indications for CTcs. For these evaluations, a total of twenty Harshaw TLD-100s were utilized. Specifically, six TLD-100s, sealed in thin calibrated plastic, were affixed to the patient's skin along the mid-axillary line at the level of the fourth intercostal space.

Prior to patient irradiation, the TLD-100s underwent a thermal treatment protocol: one hour at 400°C in a Quimis muffle furnace, followed by two hours at 100°C in an Olidelf-cz sterilization oven. TLD readings were acquired using a Harshaw model 4000 reader with default parameters set for TLD-100 (Daros, K. A. C, et al. 2012), (Kikuti, C., et al. 2015), (Horowitz, Y., et al. 2018).

For each TLD-100, a conversion factor (f_{TL}), expressed in $mGy.nC^{-1}$, was calculated using equation 1.

$$f_{TL} = \frac{\bar{D}}{\bar{L}} \quad (1)$$

where, $\bar{D}$ is the dose in mGy and $\bar{L}$ is the thermoluminescent response signal, in nC .

$\bar{D}$ was established as the average of three X-ray exposures from a Philips Compact Plus 500 equipment at 120 kV and 20 mAs. The half-value layer (HVL) was 5.2 mmAl at 120 kV. These doses were measured on top of a polymethyl methacrylate phantom scatter, with an area of 106 cm² and a height of 4.5 cm, using a calibrated Radcal model 10X6-6 ion chamber.

$\bar{L}$ is the average of the thermoluminescent response signal for each TLD-100 exposed under the same conditions as the ion chamber (Guimarães, A. et al. 2024), (da Silva, J. A. et al. 2023), (Daros, K. A. C, et al. 2012), (Kikuti, C., et al. 2015), (Matsubara, K., 2017).

The comparison between the CTDI_{vol} value determined by the CT scanner and the results obtained from the TLD-100 dosimeters positioned on the patient was evaluated by these results.

3 Results and Discussion

Table 1 shows the CTcs protocol parameters, which remained consistent across all scans. The coefficient of variation (CV) for BMI was approximately 9%, indicating an adequate and relatively low level of dispersion within the dataset. This suggests that the measurements and observations for this study were consistent.

Table 1: Average BMI of the patients and CTcs protocol parameters.

Average BMI*	Parameters					
	Voltage	Current x time	Slice thickness	Pitch	FOV**	Filter
	(kV)	(mAs)	(mm)			
15.46±1.48	120	55	2.5	0.75	240	Standard

* body mass index; ** Field of view.

Figure 1 displays the relative $CTDI_{vol}$ for the patients, obtained through both TLD measurements and calculations from the CT scanner's software. The relative $CTDI_{vol}$ values in Figure 1 were normalized based on the average $CTDI_{vol}$ results. Specifically, the average $CTDI_{vol}$ reported by the CT scanner's computer was 4.00 ± 0.04 mGy, while the average $CTDI_{vol}$ measured by TLD-100 was 4.87 ± 1.35 mGy.

These results suggest that $CTDI_{vol}$ measurements taken with TLD-100 offer a more realistic assessment. This is because the $CTDI_{vol}$ computed by the CT scanner's software is generated under the assumption of ideal performance for a brand-new system (Dixon, R. L., 2003).

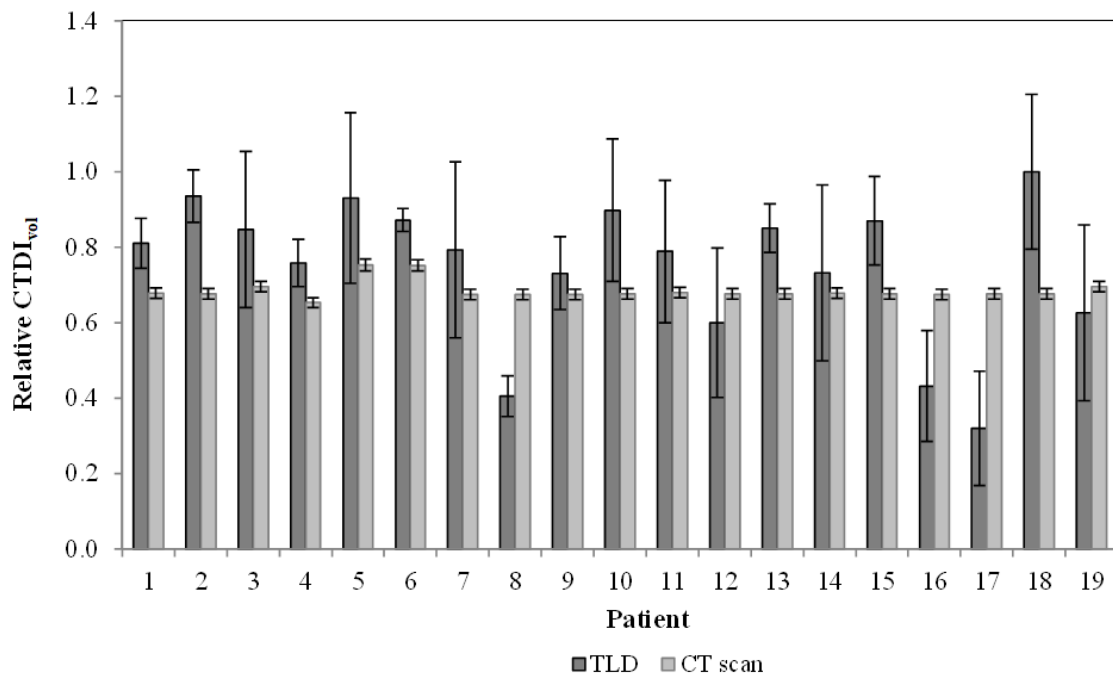


Figure 1: Relative $CTDI_{vol}$ for the patients, obtained through both TLD measurements and calculations from the CT scanner's software.

An effective dose estimate in CT is based on several parameters, including DLP (ICRP 102, 2007; ICRU 74, 2005). Despite their adequate BMI, the patients' varying heights resulted in different DLP values, which were calculated from the CT scanner's software. This, combined with the need to ensure high-quality images, presents challenges for CTce optimization in the patient group studied. These new results will be integrated into future studies.

4 Conclusions

In this observational study, we found that the radiation dose estimated by CTDI_{vol} from pediatric coronary artery calcium scoring CT scans, performed to evaluate chronic kidney disease-mineral and bone disorder was potentially 22% higher than the CTDI_{vol} results provided by the scanner. This highlights the greater accuracy of dosimetry obtained with TLD-100s positioned on the patients.

The use of TLDs, such as TLD-100, for measuring X-ray radiation doses and evaluating CTDI_{vol} was confirmed, aligning with findings from numerous previous studies.

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Radiotherapy Planning with VMAT in Halcyon 2.0: Impacts of Isocenter Configuration on Dosimetric Parameters and Treatment Efficiency

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Abstract

This study evaluates the performance of three radiotherapy planning strategies applied to a simulated case of distant bone metastases, using the Halcyon 2.0 linear accelerator. The techniques investigated included: a plan with a single extended isocenter, one with two isocenters in each lesion, and another with an isocenter centralized between the lesions. The prescribed dose was 30 Gy in five fractions, applied with the Volumetric Modulated Arc Therapy (VMAT) technique. Parameters analyzed included the isodose volume corresponding to 50% of the prescribed dose (V50%), the minimum dose in the target planning volume (PTV), the homogeneity index (IH), the maximum dose in the spinal cord, and the total number of monitor units (MU). Data were subjected to the Shapiro-Wilk test to assess normality and analysis of variance (ANOVA) to identify differences between plans. The results show that there were no statistically significant differences between the analyzed methods ($p > 0.05$), indicating equivalent performances. However, each technique presented unique characteristics: single isocenter planning demonstrated greater efficiency in beam time and MU, while plans with multiple isocenters were better-preserved organs at risk. This study suggests that the three techniques offer consistent and similar results, allowing flexibility in the choice of planning, depending on the anatomical and clinical characteristics of the patient. The analysis reaffirms the ability of Halcyon 2.0 to meet complex demands with high efficiency and dosimetric quality.

Keywords: Radiotherapy Planning; Bone Metastases; Halcyon 2.0.; Isocenter Strategies.

1 Introduction

Please use this template to create your final full-length paper, which should be between 6 and 10 pages in length. Metastases represent one of the main complications associated with cancer, characterized by the dissemination of tumor cells to sites distant from the primary tumor. Among the most affected sites are the lungs and bones, which frequently present multiple secondary lesions and impose significant therapeutic challenges. Bone metastases affect approximately 280,000 individuals annually in the United States. These lesions predominantly affect the spine, with a higher incidence in the thoracic region (60–70%), followed by the lumbosacral (20–25%) and cervical (10–15%). In addition, bone metastases in long bones occur more frequently in the lower extremities (76%) (Cilla et al., 2024).

In the context of radiotherapy, the presence of multiple metastases requires advanced approaches that ensure precise dose distribution, preservation of adjacent organs, and treatment efficiency. Strategies such as stereotactic body radiotherapy (SBRT) associated with the volumetrically modulated arc technique (VMAT) have been widely studied to maximize tumor coverage and reduce toxicity in normal tissues (1).

Given this high incidence, technological advances in radiotherapy are essential to provide more effective and personalized treatments, minimizing the impact of the disease on patients' quality of life (Garbar et al., 2024).

The development of linear accelerators has been essential for the evolution of radiotherapy, allowing the implementation of more sophisticated techniques adapted to clinical needs. Halcyon 2.0 (Varian Medical Systems, Palo Alto, CA, USA), shown in Figure 1 and commercially launched in July 2018, represents a significant advance in this area, bringing improvements in relation to the previous model. Among its main updates, we highlight the acquisition of images in kV, the extension of the maximum treatment length to 36 cm with the use of multiple isocenters, an effective resolution of the multileaf collimator (MLC) of 0.5 cm, and the introduction of dynamic beam flattening sequences, providing more homogeneous beam profiles, ideal for three-dimensional conformal radiotherapy (Varian Medical Systems, 2018).

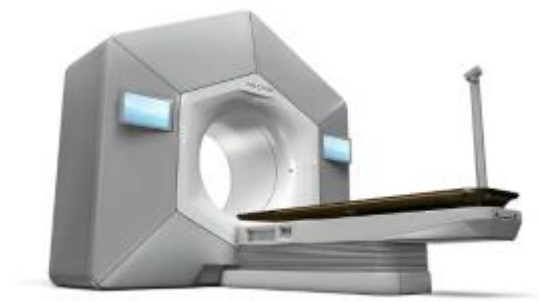


Figure 1. Layout of Varian's Halcyon linear accelerator. Available at: <https://www.varian.com/ptbr/products/radiotherapy/treatment-delivery/halcyon>.

Most of the Halcyon 2.0 hardware remains unchanged from its predecessor, including beam shaping and data characteristics, MLC dimensions, dosimetric gap, transmission, inter-blade leakage, and the integrated electronic portal imaging device (EPID) with portal dosimetry capabilities and treatment workflow optimization. These features ensure system reliability and enable the application of advanced treatment techniques with operational efficiency (Laugeman et al., 2020).

One of the distinguishing features of Halcyon 2.0 is the ability to create treatment plans with fields greater than 28 cm in length using a double isocenter (extended isocenter). This feature allows for the expansion of target volume coverage and is particularly useful for targets with large separation and in clinical scenarios that require multiple isocenters, such as in multifocal treatments. The use of this approach seeks to optimize the accuracy of dose distribution, ensuring better preservation of organs at risk (Varian Medical System, 2018).

According to Van Timmeren et al. (2022), for cases of multiple lung metastases, the mean lung dose was slightly higher (0.5 Gy) in the single isocenter technique compared to the conventional multiple isocenter technique. From a technical point of view, multiple lesions generally require an individual plan for each lesion, resulting in a prolonged treatment time. The use of a single isocenter to treat multiple lesions simultaneously is a fast and efficient strategy, reducing the risk of setup errors and intra-fraction shifts, which improves patient adherence to treatment (Van Timmeren et al., 2022).

However, the single-isocenter approach presents technical challenges as the number or distance between lesions increases. In such cases, careful evaluation of the technical requirements is necessary, as in the VMAT technique, to ensure that the dose distribution is adequate. The combination of SBRT with VMAT

for single-isocenter cases may require complex and heterogeneous dose distributions, with steep dose gradients outside the planned target volumes (PTVs), and extensive sparing of organs at risk (1).

The study by Jayadevan et al. (2024) demonstrated that the multiple isocenter technique provided comparable PTV coverage to the single isocenter technique, with a mean liver dose of 5.4% lower, sparing an additional 4.5% of normal liver volume. These results indicate that for patients with larger compromised liver volume, the multiple isocenter approach may be more advantageous, despite the 1.5-fold increase in total monitor units and beam-on time. Nevertheless, no significant differences were observed in the maximum dose to organs at risk or in gamma analysis between the two approaches (Jayadevan et al.,2024).

Given these considerations, the Halcyon 2.0 linear accelerator represents a significant advancement in image-guided radiotherapy, enabling greater efficiency in dose delivery and improving the precision of treatment of multiple lesions. The possibility of using a single isocenter reduces the total treatment time and minimizes setup errors, while the multiple isocenter approach can offer additional benefits, such as lower mean dose to adjacent organs and better tissue preservation. In the context of bone and lung metastases, advanced techniques such as these play a crucial role in improving clinical outcomes, enabling more effective and personalized treatments.

The present work aims to analyze the technical and clinical capabilities of the Halcyon 2.0 linear accelerator in the radiotherapy of multiple bone metastases, with emphasis on the comparison between single isocenter and multiple isocenter approaches.

2 Materials and Methods

A case simulation involving two bone lesions located at T11 and L1, separated by approximately 6 cm along the longitudinal axis, was performed. Treatment planning was conducted in the Eclipse system, version 16.1.2 (Varian), using Halcyon equipment and the volumetric modulation technique per arc (VMAT), with the application of two complete arcs and collimator orientations at 5° and 355°. The dose prescribed to the PTV was 600 cGy per fraction, totaling five fractions.

Three plans were made: the first, illustrated in Figure 2, with the extended isocenter available in Halcyon; the second, with 2 isocenters, each centered on one of the lesions, as shown in Figure 3; and, finally, the third, with a single isocenter positioned between the two lesions, as shown in Figure 4.

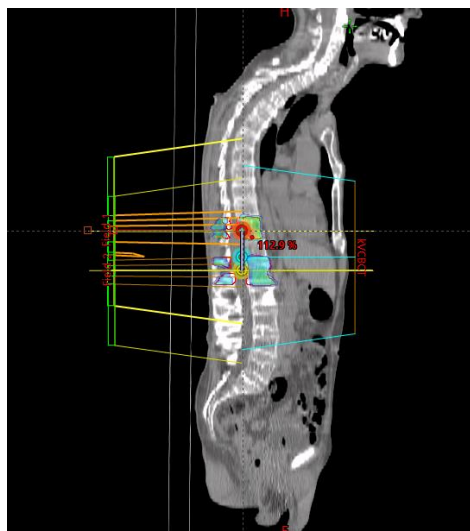


Figure 2. Radiotherapy plan with extended isocenter. Source: The Author (2024).

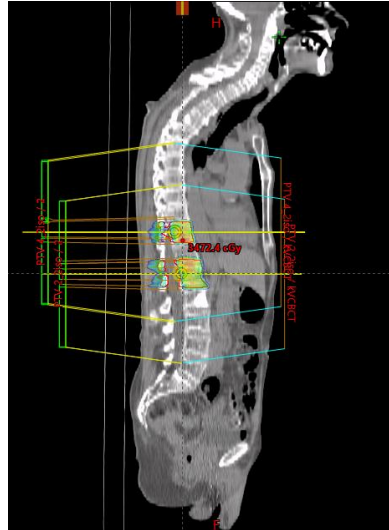


Figure 3. Radiotherapy plan with 2 isocenters each in the center of a lesion. Source: The Author (2024).

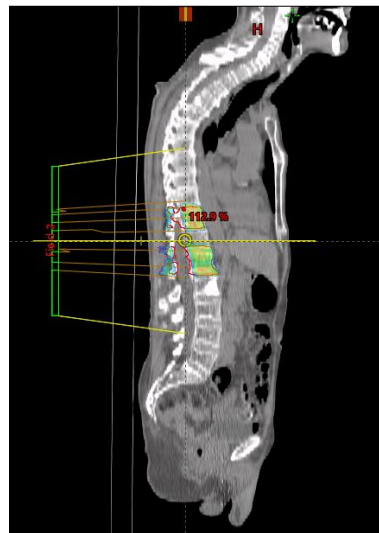


Figure 4. Radiotherapy plan with 1 isocenter between the two lesions. Source: The Author (2024)

During the planning process, several parameters were analyzed for comparison between the three methods, including:

- The volume of isodose corresponding to 50% of the prescribed dose (V50%),
- The minimum dose at PTV (Dmin PTV),
- The homogeneity index (HI),
- The maximum dose received by the spinal cord (Dmax spinal cord),
- The total number of monitor units (UM total).

The results of these analyses are summarized in Table 1. All plans were normalized to the 98% curve.

The homogeneity index (HI), as suggested by ICRU 83 (2011) and calculated according to Equation 1 below, was used to assess the uniformity of dose distribution in the PTV, where values close to zero indicate greater (ideal) homogeneity and higher values reflect greater dose variation within the target volume. The ideal HI is equal to zero, representing a homogeneous dose distribution in the PTV (Gregoire et al., 2011).

$$IH = 100. \frac{D_{2\%} - D_{98\%}}{D_{50\%}}$$

- D2%: Dose received by 2% of the PTV volume (reflects the maximum dose).
- D98%: Dose received by 98% of the PTV volume (reflects the minimum dose).
- D50%: 50% dose.

In addition, the variance will be analyzed using the ANOVA method, which tests the equality of three or more population means based on the verification of sample variances. Single-factor ANOVA is indicated for populations with normal distribution and homogeneous variances (or similar standard deviations). In this way, it will be verified whether the population means are statistically equal or not, evaluating the existence of any treatment effect between the different groups (<https://edisciplinas.usp.br/enrol/index.php?id=41810>).

Before applying ANOVA, it is essential to verify whether the normality assumption is met, as it is a prerequisite to ensure the validity of the results. To this end, the Shapiro-Wilk test will be used, one of the most robust and widely used methods to assess data normality. This test, originally developed for samples smaller than 50, has become preferred due to its high sensitivity to deviations from normality resulting from asymmetry, kurtosis, or both (Mendes, 2003). Its application allows us to confirm whether the data follows a normal distribution, validating the use of ANOVA to analyze variances between groups.

Thus, the Shapiro-Wilk test will be used as a preliminary step in the analysis, ensuring that the normality assumptions are met before proceeding with the application of ANOVA. If data presents significant deviations from normality, it will be necessary to consider alternatives, such as data transformations or nonparametric methods, to guarantee results.

3 Results

Table 1 presents the main parameters analyzed for the three different plans evaluated, involving two isocenter configurations: a plan with two isocenters and two plans with only one isocenter, one of which was positioned between the lesions and the other with an extended isocenter. The parameters evaluated include the isodose volume corresponding to 50% of the prescribed dose (V50%), the minimum dose in the PTV, the IH, the maximum dose received by the bone marrow, and the total MU. These data allow for a comparison of the performance of the plans in terms of dose coverage, preservation of adjacent tissues, and efficiency in treatment delivery.

Table 1. Parameters were analyzed for each of the three different plans.

	1 plane 2 isocenters	2 plans	1 isocenter
V50%	888.75	923.46	847.88
Minimum dose of PTV	22.49	23.09	22.57
IH	0.233	0.261	0.2606
Maximum bone marrow dose	27.78	27.53	25.94
UM total	2,184.1	4,504.7	2,632.5

Source: The author (2024).

Initially, the Shapiro-Wilk test was applied to evaluate the distribution of the data, with the objective of verifying the adequacy of the assumption of normality. The result obtained, with a value of $p \geq 0.05$, indicates that the null hypothesis of normality was not rejected, allowing the data to be considered as normally distributed and validating the subsequent use of parametric tests.

With the normality of the data confirmed, an analysis of variance (ANOVA) for repeated measures was performed, adopting a significance level of 0.05, to identify statistically significant differences between the means of the three designs. The results were obtained and presented in Table 2. The repeated measures ANOVA test is particularly suitable for this context, as it considers the correlation between observations

obtained from the same experimental group under different conditions, reducing the error variance and, thus, increasing the sensitivity to detect differences between the groups.

Table 2. Anova – single factor.

ANOVA			
Source of variation	Between groups	Within groups	Total
SQ	634,400.5707	23,911,951.62	24,546,352.19
GI	2	12	14
MQ	317,200.2854	1,992,662.635	
F	0.159184		
<i>p-value</i>	0.85461		
F-critical	3.885294		

Source: The author (2024).

4 Discussion

The results obtained in this study indicated that there are no statistically significant differences between the means of the evaluated plans, as evidenced by the $p\text{-value} = 0.854611$. This finding suggests that the three planning methods analyzed presented equivalent performances about the variable of interest. This conclusion contrasts with the results reported by Van Timmeren et al. (6), who observed a slight increase in the mean lung dose (0.5 Gy) in the single isocenter technique compared to the conventional multiple isocenter technique (6).

Furthermore, the findings of this study differ partially from the results presented by Jayadevan et al. (7), who reported a 5.4% lower mean liver dose in the multiple isocenter plans compared to the single isocenter. In our study, the absence of a statistical difference between the plans suggests that, under the conditions analyzed, the evaluated methods present similar performance, and it is not possible to point out a clear superiority of one over the others (7).

From a clinical perspective, the statistical equivalence between the plans reinforces the need to consider additional factors when choosing the ideal strategy for each patient, such as treatment time, ease of execution, and individual anatomical variations. While previous studies suggest specific advantages for each approach, our results indicate that, under the conditions tested, either technique can be used without compromising the quality of treatment. This highlights the importance of an individualized assessment, in which the decision for the most appropriate technique must consider not only the dosimetric parameters but also operational and clinical aspects that may impact treatment adherence and efficacy.

4 Conclusions

Based on the results of the statistical analysis, it is concluded that the three plans evaluated do not present significant differences in terms of performance, as indicated by the high $p\text{-value}$ (0.854611) in the repeated measures ANOVA test. The confirmation of the normality of the data by the Shapiro-Wilk test, followed by the absence of statistical significance in the comparisons between the group means, suggests that the plans produce consistent and equivalent results for the variable of interest.

This equivalence between the plans implies that, for the clinical context and the specific parameters analyzed, any of the three methods can be used without compromising the expected performance, since the differences observed are not statistically significant.

Thus, it can be considered that the three plans offer similar dose coverage and distribution profiles in PTV, providing flexibility in choosing the most convenient method for clinical practice, without compromising the quality of treatment.

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Three-dimensional dose mapping of gold-198 nanoparticles: A comparative study of 3D-printed matrices using Fricke xylenol gel and Monte Carlo simulations

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Abstract

Gold-198 nanoparticles (¹⁹⁸AuNPs) stand out as promising theranostic agents, whose clinical application requires precise dose distribution assessment, for which the Fricke xylenol gel (FXG) dosimeter emerges as an ideal three-dimensional mapping tool. The study performed a comparative evaluation of 3D-printed containers for ¹⁹⁸AuNPs in FXG dosimeters, examining how different materials influence dose deposition patterns while identifying the most suitable material for three-dimensional dosimetry of nanoscale radiation sources. The ¹⁹⁸AuNPs were synthesized following the method described by Kannan et al. (2006) and Katti et al. (2006), yielding the characteristic wine-red colloidal solution. Nanoparticle size distribution and elemental composition were characterized using Dynamic Light Scattering (DLS) and Transmission Electron Microscopy (TEM). The FXG dosimeters were prepared following the method described by Taveres et al. (2024) and stored in CT Vista compatible vials with cylindrical containers until gelation. The cylindrical containers were 3D printed (ABS [acrylonitrile-butadiene-styrene], ASA [acrylonitrile-styrene-acrylate], PETG [polyethylene terephthalate glycol], and PLA [polylactic acid]) using a Bambu Lab X1 Carbon printer, with standardized parameters. Dosimetric measurements were performed after 48-hour exposure to 1 ml of ¹⁹⁸AuNPs contained in 3D-printed containers with corresponding Monte Carlo simulations using MCNP 4C code to replicate experimental conditions. Dose uniformity (>4 mm) was observed across materials and dimensions, while significant differences (<2 mm) were detected due to ¹⁹⁸Au beta particles (simulation-confirmed), validating 3D-printed containers as viable dosimetric platforms for ¹⁹⁸AuNP studies.

Keywords: 3D dosimetry; Nanoparticle dosimetry; 3D printing.

1 Introduction

Technological advances in medicine have driven the development of cancer treatments and diagnostics that improve patient survival rates, though the major challenge remains eliminating tumors while sparing healthy tissues. In this context, radioactive gold nanoparticles (¹⁹⁸AuNPs) emerge as promising theranostic tools that integrate diagnosis and therapy into a single agent, offering innovative approaches to cancer treatment.

Gold nanoparticles, being high atomic number materials, show great potential for radiation therapy due to their dose enhancement effects from secondary electrons and targeted therapeutic capabilities (Hainfeld et al., 2008; Fortin et al., 2018). However, their clinical application requires specific dosimetric protocols. Conventional methods fail to accurately measure dose distribution around these liquid nanosources while posing contamination risks that compromise detector sensitivity. The Fricke Xylenol Gel (FXG) dosimeter, when combined with proper nanoparticle containers, provides an important solution for 3D dosimetry.

3D dosimetry employs volumetric detectors to fully characterize dose distribution in target tissues, optimizing radiation treatments (Darafsheh, 2021). For ^{198}Au sources, which produce steep dose gradients requiring high resolution (Avizonis et al., 1992; Syed, 2009), gel dosimetry, particularly FXG, stands out for its tissue equivalence and ability to generate complete 3D dose maps from a single irradiation (Ibbott, 2004; Valente et al., 2016).

The classical Fricke dosimeter operates through radiation-induced oxidation of Fe^{2+} to Fe^{3+} in aqueous medium (Fricke and Morse, 1927). Its gel version, stabilized with chelating agents like xylenol orange, enables spatial dose mapping through color variation, allowing direct correlation between attenuation and 3D dose distribution (Schreiner, 2015). When combined with 3D-printed tissue-equivalent matrices FXG enables customizable and anatomically relevant dosimetry, bridging the gap between idealized simulations and radioactive nanoparticles dosimetry.

Meanwhile, Monte Carlo simulations can model radiation interactions with ^{198}Au NPs in exquisite detail, serving as the gold standard for theoretical dose calculations (Al-Yasiri et al., 2019; Villate and Hahn, 2019; Seniwal et al., 2021). Direct comparison between experimental FXG measurements and Monte Carlo predictions is essential to validate techniques and facilitate clinical implementation of these nanoparticles.

Despite growing interest in ^{198}Au NPs, their 3D dose distributions have rarely been evaluated using combined experimental and computational methods. Previous work has focused primarily on isolated Monte Carlo simulations (Al-Yasiri et al., 2019; Villate and Hahn, 2019; Seniwal et al., 2021; Hahn and Villate, 2021; Villate et al., 2022; Sgrignoli et al., 2024; Koka et al., 2024; Angelocci et al., 2025). There is a critical need for integrated studies combining experimental 3D dosimetry with Monte Carlo validation to improve nanoparticle dosimetry accuracy and applicability.

To advance nanoparticle dosimetry, this study aims to evaluate ^{198}Au NP dose distributions in distinct 3D-printed matrices using both FXG dosimetry and Monte Carlo simulations. The results will establish a robust protocol for radioactive nanoparticle dosimetry. The methodology embraces nanoparticle synthesis and characterization, FXG preparation, 3D containers fabrication, experimental analysis, and computational validation, with detailed protocols presented in subsequent sections.

2 Methodology

This work begins with the production and characterization of ^{198}Au NPs, in which the synthesis method is carefully selected to control size and morphology, critical parameters for therapeutic applications. Subsequently, we detail the preparation and characterization of the FXG dosimeter, establishing experimental parameters to ensure precise and reproducible 3D dose mapping. The experimental setup incorporates 3D-printed containers, with technical justification for selected materials and geometries, while Monte Carlo simulations model the identical irradiation geometry by incorporating source spectrum, material properties, and particle transport for theoretical analysis of dose distribution.

2.1 Gold Nanoparticles (^{198}Au NPs)

This study adapted the Turkevich method (1951) with modifications from Kannan et al. (2006) and Katti et al. (2006) to synthesize gum arabic-functionalized radioactive nanoparticles. Irradiation of ^{197}Au ($2 \times$

$10^{13} \text{ n}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}$) yielded ^{198}Au , which was dissolved in $\text{HCl}:\text{HNO}_3$ (3:1), purified, and reacted with gum arabic (0.2 mM, 1:1 ratio) at 100°C , followed by reduction with sodium citrate (39 mM) and NaOH (13 mM) to form wine-red $^{198}\text{AuNPs}$. Characterization included DLS for hydrodynamic diameter and TEM for core morphology and size distribution.

2.2 Fabrication of Prototype Containers via 3D Printing

Prototype modeling was developed using Autodesk Fusion 360 software, creating hollow cylindrical geometries with diameters of 18 mm and 23 mm, featuring a central hole and walls with a uniform thickness of 1 mm. For additive manufacturing, a Bambu Lab X1 Carbon printer was used, configured in Bambu Studio software with a 0.4 mm diameter nozzle and a layer resolution of 0.08 mm in High Quality mode. The extrusion temperatures were optimized for each material: ABS at 270°C , ASA at 260°C , PETG at 250°C , and PLA at 220°C , ensuring dimensional quality and prototype reproducibility. The entire printing process was conducted in a controlled environment to minimize thermal variations that could affect the dimensional accuracy of the parts.

2.3 Fricke Xylenol Gel Dosimeter (FXG)

The study employed a modified version of the Cavinato and Campos (2010) protocol, as adapted by Tavares et al. (2024), using an optimized xylenol orange concentration to minimize imaging artifacts. The FXG dosimeter was prepared by dissolving 5% bovine gelatin (270 Bloom) in 75% of ultrapure water at 70°C , then incorporating the remaining components (50 mM H_2SO_4 , 1 mM NaCl, 1 mM ferrous ammonium sulfate, and 0.01 mM xylenol orange in 25% water) at 35°C . After homogenization, the gel was stored in custom flasks and refrigerated for 12 hours before OCT analysis using a Vista 16 scanner (590 nm) with VistaScan software, which measured radiation-induced optical density changes for 3D dose mapping (Oldham et al., 2014; Olding et al., 2010).

The Vista 16 OCT scanner analyzed the FXG before and after irradiation. Measurements were taken 48 hours post-exposure to 1 mL of $^{198}\text{AuNPs}$ (activity: 16.65 MBq or 0.45 mCi) housed in 3D-printed containers alongside the FXG dosimeter. Two containers each of ABS, ASA, PETG, and PLA were used, with cylindrical dimensions of either 18 mm (diameter) $\times$ 40 mm (height) $\times$ 1 mm (wall thickness) and 23 mm $\times$ 40 mm $\times$ 1 mm (Figure 1). These were refrigerated during exposure to $^{198}\text{AuNPs}$ to prevent natural oxidation that could compromise results through chemical reactions.



Figure 1: FXG flasks with prototypical holders filled with $^{198}\text{AuNPs}$. Left holder dimensions: 18 mm $\times$ 40 mm $\times$ 1 mm. Right holder dimensions: 23 mm $\times$ 40 mm $\times$ 1 mm.

2.5 Monte Carlo Model

Monte Carlo simulations were performed using MCNP 4C code (Werner, 2017) to model the complete experimental setup, including the phantom, FXG, and nanoparticle solution ($^{198}\text{AuNPs}$) with exact dimensions and compositions. The study evaluated absorbed doses (F6 tallies) across thirty 0.5 mm radius

× 0.1 mm thickness cylindrical volumes (total 3 mm depth), with the first 1 mm representing holder wall dose and subsequent 2 mm quantifying FXG dose.

Twenty simulations compared four holder materials (ASA, ABS, PETG and PLA) against a no-holder reference, testing two diameters (18 mm and 23 mm) per material. Each configuration simulated both β -decay electrons and γ /X-ray emissions. Figure 2 illustrates the geometric model, comparing holder sizes (Figure 2a), tally positioning (Figure 2b).

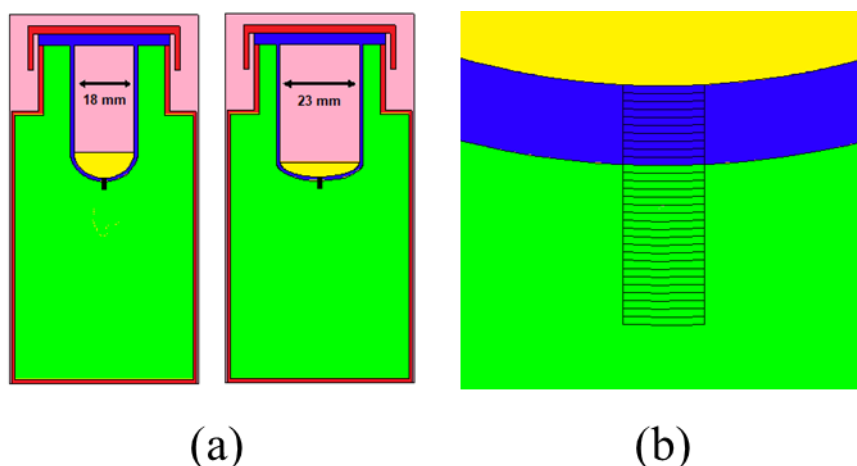


Figure 2: Geometrical model simulated using MCNP. (a) Sagittal plane view of the geometry for 18 mm (left) and 23 mm (right) holders. Color key: pink = air; yellow = $^{198}\text{AuNP}$ solution; green = FXG; red = phantom walls (PET); blue = holder (different material for each simulation). (b) Close-up of the tallied region for the holder.

3 Results and Discussion

TEM images revealed spherical nanoparticles with an average diameter of 8 nm, demonstrating high size and shape uniformity (Figure 3). DLS results showed an average hydrodynamic diameter of 40 nm (Figure 4) with a polydispersity index of 0.31, confirming a moderately narrow size distribution and good colloidal stability in solution.

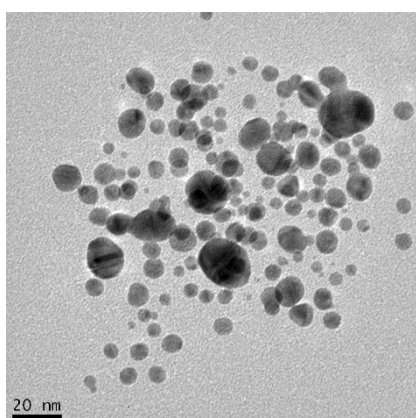


Figure 3: TEM image of $^{198}\text{AuNPs}$.

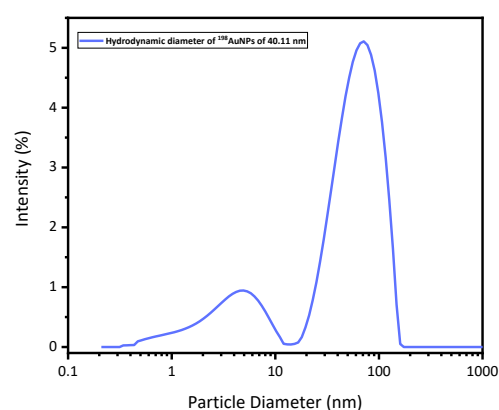


Figure 4: DLS analysis of particle size distribution.

The significant difference between TEM (8 nm) and DLS (40 nm) measurements demonstrates the formation of an efficient gum arabic coating around the nanoparticles. This feature is particularly relevant

for biomedical applications, as the polymeric coating not only provides colloidal stability but also enhances biocompatibility and therapeutic efficacy (Katti et al., 2006; Madhusudhan et al., 2019). The combined TEM-DLS analysis enabled comprehensive nanoparticle characterization, verifying both the metallic core morphology and the complete system's colloidal behavior.

The methodology developed to evaluate FXG dosimeters exposure in contact with $^{198}\text{AuNPs}$ -loaded holders revealed no macroscopic visual changes in the flasks after 48 hours of irradiation (Figure 5).

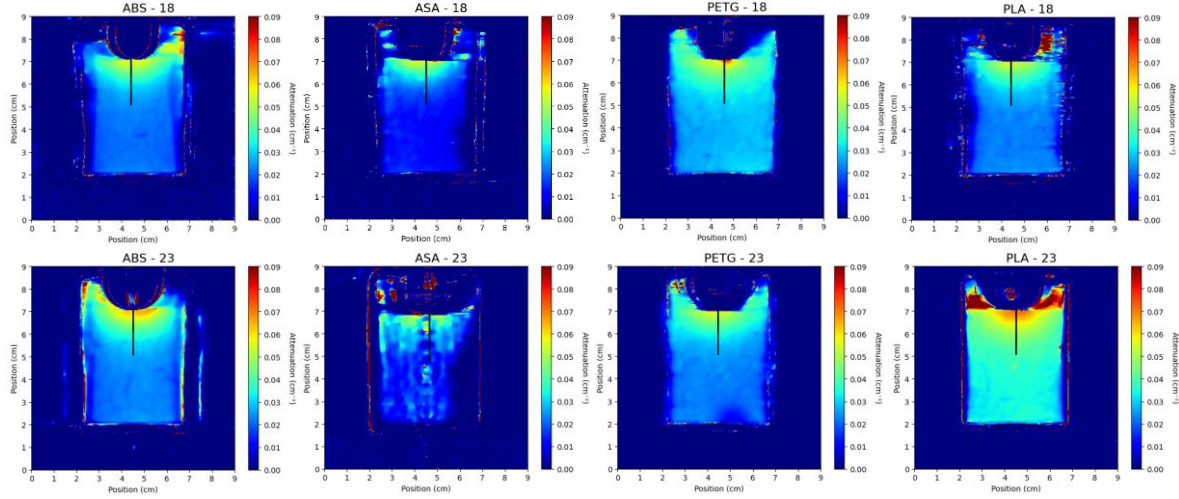


Figure 5: Dose distribution by optical attenuation in FXG dosimeters exposed to $^{198}\text{AuNPs}$ (48 h) using ABS, ASA, PETG, and PLA holders (18 mm and 23 mm diameters). The black lines indicate the regions where the points were selected to obtain the depth dose distribution curve.

These results demonstrate OCT's sensitivity in detecting spatial dose variations, even in the absence of visible changes, aligning with previous studies highlighting FXG's efficacy for small field dosimetry mapping (Oldham, 2014; Schreiner, 2015). Likewise, three-dimensional OCT analysis identified a characteristic dose deposition pattern along the flasks' X-axis, with higher intensity at the liquid-holder interface (Figure 6).

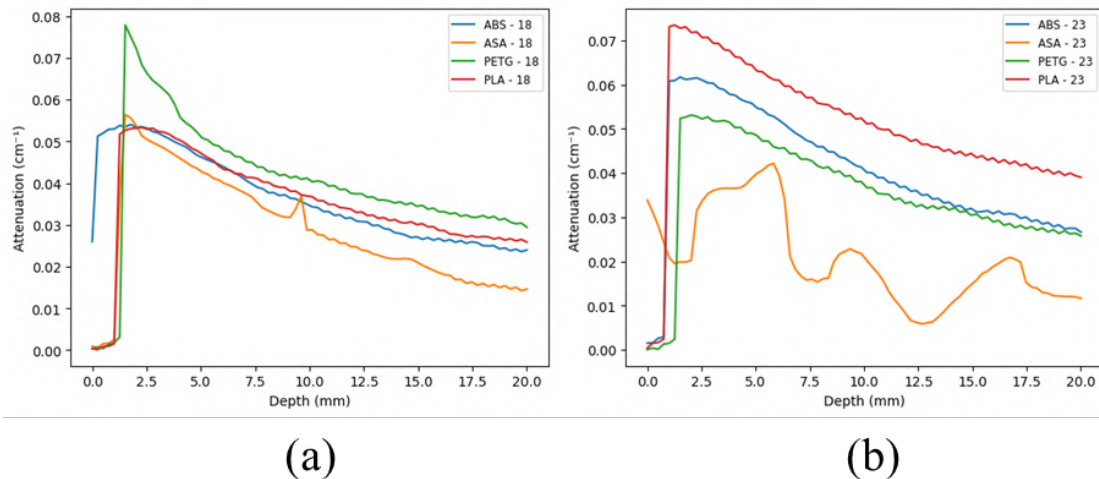


Figure 6: Attenuation curves as a function of depth for ABS, ASA, PETG, and PLA materials with diameters of 18 mm (a) and 23 mm (b).

For the 18 mm-diameter holders, PETG demonstrated superior attenuation with a sharp peak (~2 mm depth) and sustained higher values across all depths. ABS, ASA, and PLA showed comparable profiles,

with ABS slightly exceeding ASA, while PLA maintained stable intermediate performance. These findings confirm PETG's optimal radiation attenuation in the 18 mm configuration. Parallel analysis of 23 mm-diameter revealed the holders manufactured from ABS, PETG, and PLA exhibit relatively similar attenuation profiles, though with distinct intensity peaks, a characteristic attributed to differences in their elemental composition and material density. Among these, ASA demonstrated particularly irregular behavior, showing significant fluctuations in its attenuation curve. This instability appears directly related to imperfections generated during the demolding process of the holders, which compromise surface uniformity in contact with the dosimetric gel. Such irregularities result in more pronounced artifacts during optical tomography image reconstruction, adversely affecting the final resolution of the obtained images.

Simulated depth-dose profiles for $^{198}\text{AuNPs}$ in 18 mm and 23 mm diameter holders are presented in Figures 6-9. Comparative analysis reveals subtle variations in dose distribution as a function of holder diameter.

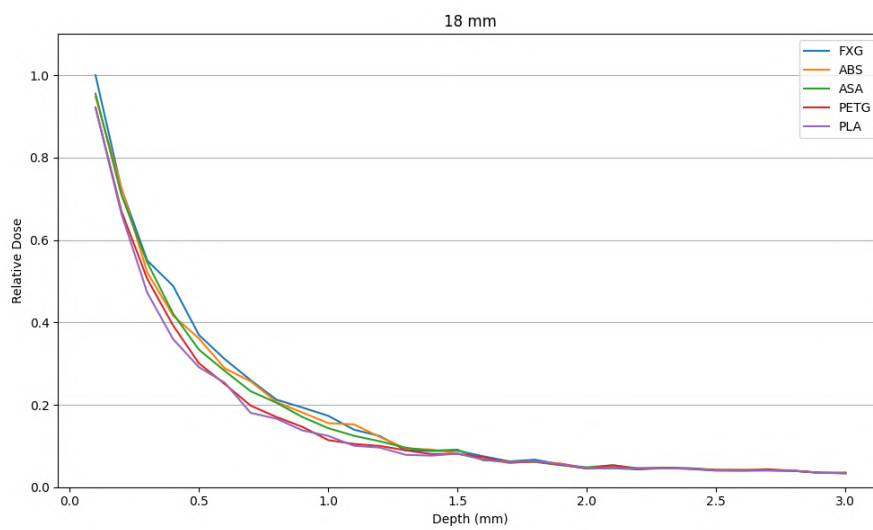


Figure 7: Relative dose-depth profiles for $^{198}\text{AuNPs}$ in ABS, ASA, PETG, and PLA holders with 18 mm diameter.

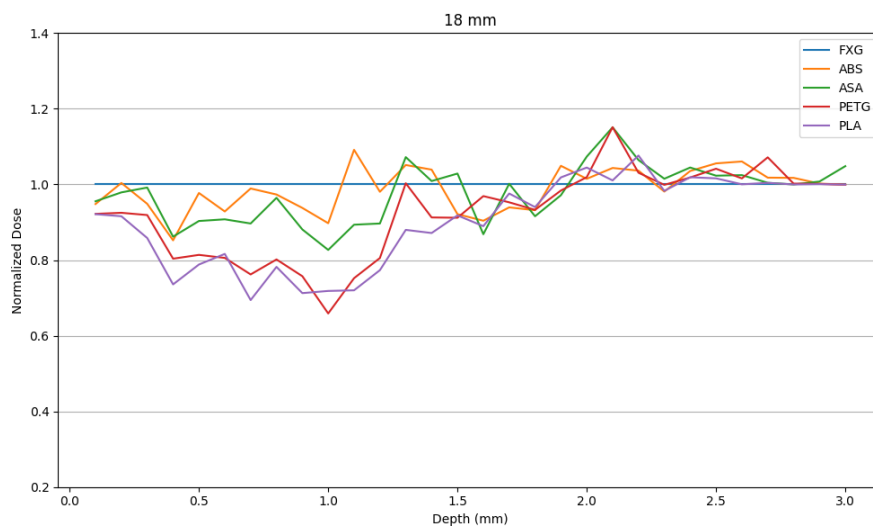


Figure 8: Normalized depth-dose curves from $^{198}\text{AuNPs}$ in 18 mm diameter holders (ABS, ASA, PETG, PLA).

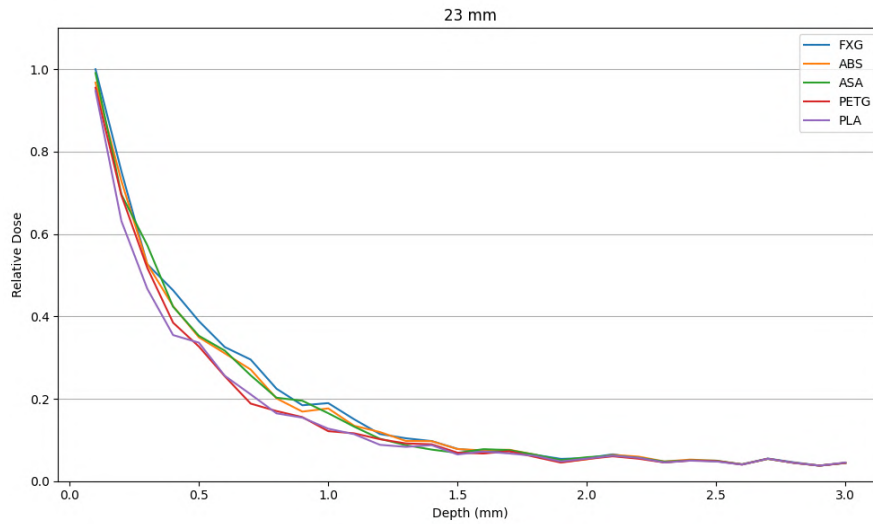


Figure 9: Relative dose-depth profiles for $^{198}\text{AuNPs}$ in ABS, ASA, PETG, and PLA holders with 23 mm diameter.

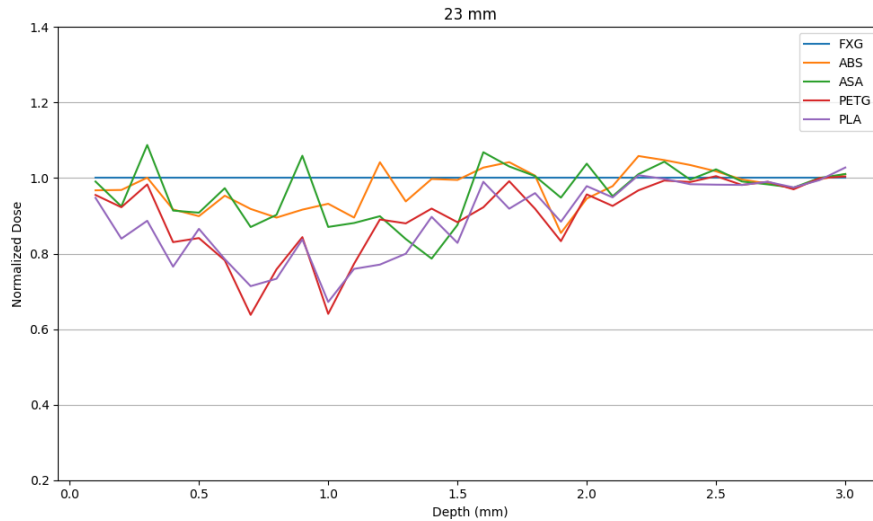


Figure 10: Normalized depth-dose curves from $^{198}\text{AuNPs}$ in 23 mm diameter holders (ABS, ASA, PETG, PLA).

The relative dose graphs (Figures 6 and 8) clearly demonstrate the distinct influence of β^- particles and photons on energy deposition at different depths. In the superficial region (0 - 1 mm), the interaction is dominated by β^- particles emitted from $^{198}\text{AuNPs}$, characterized by intense and localized dose deposition (Angelocci et al., 2025). This effect is particularly evident in the first 0.5 mm, where the high LET (Linear Energy Transfer) of electrons results in significantly elevated dose values, with FXG showing the highest relative doses in this region.

The materials ABS, ASA, PETG, and PLA exhibit a similar pattern (Figures 7 and 9), though with subtle variations in initial attenuation (0 - 0.5 mm) that reflect differences in their physical properties and elemental composition (Alssabbagh et al., 2017; Bustillo et al., 2024; Bento et al., 2024). Beyond 1 mm depth, corresponding to the holder wall thickness, a marked transition occurs in the dose profile. This transition is corroborated by the experimental results from optical attenuation maps shown in Figure 5, which demonstrate more uniform dose deposition contributing significantly at greater distances.

In this deeper region, the contribution of β^- particles becomes less relevant due to their short range, and dose deposition becomes dominated by secondary radiation generated from β^- particle interactions, as well as primary γ photons and X-rays from the source (Delacroix et al., 2002; Angelocci et al., 2025). The convergence of dose curves beyond 1 mm depth indicates that material composition has secondary influence on dose distribution in this region, making polymeric materials viable alternatives to FXG for applications involving depth measurements.

4 Conclusions

The dose profiles showed that all tested polymeric materials (ABS, ASA, PETG, and PLA) exhibited attenuation compatible with beta radiation, with higher dose deposition in the superficial layers, a behavior expected due to the high LET of electrons. Compared to the FXG dosimeter, ABS showed better agreement throughout the depth, while PETG and PLA underestimated the dose in the initial regions. Although ASA performed well in the simulation, difficulties in dismantling the mold compromised its experimental assessment. From about 2 mm depth onwards, all materials showed results similar to FXG, indicating dosimetric viability in deeper regions.

These results suggest that, although promising, the approach still requires some adjustments, as short-range sources, such as those used in this study, demand more precise and in-depth analysis. This underscores the relevance of optical computed tomography, which emerges as a promising technique for evaluating dose deposition more comprehensively, even in regions with limited range.

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Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLD and Radiochromic Film

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Abstract

External beam radiation therapy (EBRT) is widely used for uveal melanoma to preserve the eye and achieve effective tumor control. However, late side effects remain a concern, reinforcing the need for dose reduction to mitigate long-term toxicity. This study aims to validate a 3D-printed ocular tissue-equivalent anthropomorphic phantom, with enhanced anatomical accuracy based on the ATOM® phantom, through end-to-end dosimetric analysis using radiochromic film and thermoluminescent dosimeters (TLDs) for quality assurance (QA) in EBRT. Gafchromic EBT3 films and LiF:Mg,Ti microcube TLDs were used for dose assessment in target volumes and organs at risk (OAR). Calibration and irradiation were performed using a Varian TrueBeam linear accelerator with a 6 MV photon beam. Film dosimetry, with films positioned in an oblique plane along the optic nerve, was processed using the *Film2Dose* software and compared to the Varian Eclipse® treatment planning system (TPS) via gamma analysis. TLDs were also placed in this region for point dose verification. In the film analysis, dose uncertainties on the order of 3% were observed. Gamma analysis yielded a mean passing rate of 99.9% (3%/3 mm/10% threshold), 98.6% (TG-218 reference criteria: 3%/2 mm/10% threshold), and 98.3% (2%/2 mm/10% threshold). TLD results were within 3σ of TPS predictions, indicating statistical compatibility, with a maximum discrepancy of 16%, mainly attributed to the cone-beam computed tomography dose not accounted for in the TPS. Preliminary results demonstrated the phantom's reliable performance for dose verification in ocular radiotherapy. Ongoing improvements in dosimeter positioning, reproducibility, and dose assessment, along with additional dosimetric testing, aim to support dosimetry in other organs at risk, such as the lens and eye fundus, and to facilitate its implementation in routine clinical quality assurance practice.

Keywords: Tissue-equivalent anthropomorphic phantom; 3D printing; Dosimetry; Radiochromic films; Thermoluminescent dosimeters; Eye Radiotherapy.

1. Introduction

Ocular cancers pose a significant threat not only to vision but also to the overall health and survival of affected patients. Tumor development processes may affect the eye either primarily or through metastasis. The main primary tumors include conjunctival and uveal melanoma, retinoblastoma, conjunctival and intraocular lymphoma, and ocular surface squamous neoplasia. The most common metastatic processes originate from breast and lung cancers (Williams & Di Nicola, 2021).

Uveal melanoma affects the iris, ciliary body, and choroid, accounting for about 83% of ocular melanoma cases. It is the most common primary intraocular tumor in adults, especially between 50 and 70 years of age. In the United States, its incidence is estimated at 6 to 7 cases per million individuals per year. Despite effective treatment of the primary tumor, systemic metastases develop in up to 50% of

patients within 10 years, often leading to death (Huang & Guo, 2024; Jager et al., 2020; Williams & Di Nicola, 2021; Xu et al., 2025). In Brazil, 2,166 cases were reported between 2000 and 2016 (Lucena et al., 2020). Retinoblastoma is the most frequent ocular cancer in children, with approximately 95% of cases diagnosed before the age of five. It accounts for 11% of all cancers diagnosed during the first year of life. The estimated incidence is 1 in 16,000 live births, corresponding to around 300 new cases annually in the United States (Williams & Di Nicola, 2021; Xu et al., 2025).

Treatment options include observation, local resection, chemotherapy, and radiation therapy—either via ophthalmic plaque or external beam. Enucleation is reserved for cases where globe- and vision-sparing approaches are not feasible (Maheshwari and Finger, 2018). External Beam Radiation Therapy (EBRT) plays a key role, particularly for uveal melanoma. Modalities include Proton Beam Radiation Therapy (PBRT) and stereotactic radiosurgery/fractionated stereotactic radiosurgery (SRS/FSRS), delivered via linear accelerators (linacs) or Gamma Knife. Due to PBRT's high cost and limited availability, and the need for specialized teams for brachytherapy, SRS/FSRS has become the preferred alternative. Linac-based SRS/FSRS has demonstrated local control rates of up to 95.4% and eye preservation rates of 94.6% (Yilmaz et al., 2024).

However, EBRT is associated with well-known side effects. Radiation retinopathy occurs in 12.7–81% of cases, neovascular glaucoma in 8–42%, cataracts in 83% within five years, and optic neuropathy in 4–64% (Yilmaz et al., 2024). Such adverse effects have reinforced the need to optimize isodose distribution during treatment planning, particularly in regions close to organs at risk (OAR), to minimize radiation-induced complications (Dunavoelgyi et al., 2013).

To support treatment quality and safety, Medical Physics employs tissue-equivalent anthropomorphic phantoms that mimic human biological tissues and enable dosimetric studies involving ionizing radiation exposure (DeWerd and Kissick, 2014). These phantoms are essential for ensuring accurate dose delivery, treatment effectiveness, and patient radioprotection. However, it's known that commercial phantoms often lack detailed anatomical representation of ocular structures, limiting research in this region. Previous studies have developed 3D-printed anthropomorphic eye phantoms, based on the ATOM® phantom, to assess the dose to OAR—such as the lens, eye fundus, and optic nerve—during ocular EBRT, using thermoluminescent dosimeters (TLD) for point dosimetry (Pereira et al., 2022). Later adaptations enabled the use of radiochromic film as well for planar dosimetry (Fidelis, 2024; Fidelis et al., 2024). These phantoms were designed to replace slices 3 and 4 of the ATOM phantom's head.

This study aims to validate these 3D-printed ocular tissue-equivalent anthropomorphic phantoms, with enhanced anatomical accuracy based on the ATOM® phantom, through end-to-end dosimetric analysis using radiochromic film and TLD for quality assurance (QA) in EBRT.

2. Materials and Methods

2.1. Phantoms Description

The phantom for radiochromic film dosimetry (Fig. 1a) was designed to allow film placement in two distinct planes. The first is an oblique plane aligned along the anatomical trajectory of the optic nerve (Fig. 1b), oriented at a 21° angle to the left relative to the sagittal plane, extending from the posterior segment of the ocular globe to the optic canal within the cranial cavity. This configuration enables dose assessment along the full extent of the optic nerve, as well as part of the ocular globe. This setup allows for dose evaluation of midline ocular structures, including the lens, retina, and optic nerve.

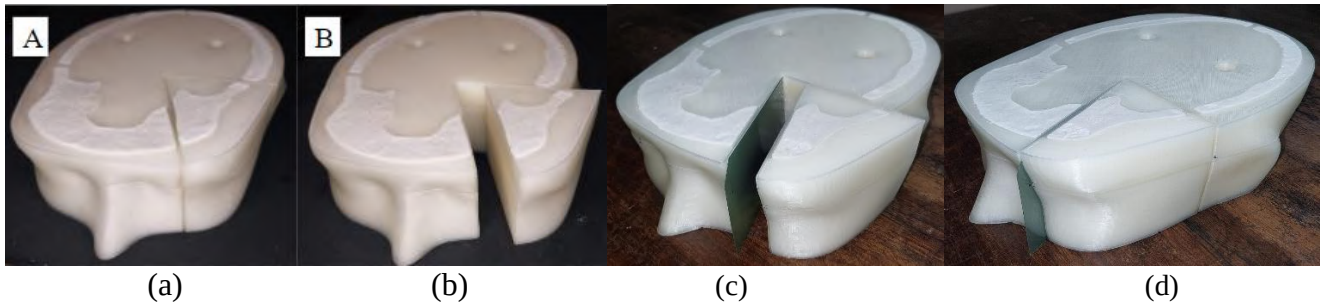


Figure 1: Anthropomorphic phantom for radiochromic films dosimetry. (a) Complete phantom. (b) Oblique slice. The positioning of the radiochromic film in the oblique-plane section (c) and its fixation by clamping between the phantom components (d) are shown.

The TLD dosimetry phantom (Fig. 2) was developed in a similar manner to the previous one. However, the left eye, also fabricated using 3D printing, is a separate component from the rest of the head. This phantom contains small holes in left eye and optical nerve region to allow the insertion of TLD (LiF:Mg,Ti) microcubes for point dose measurements in the regions of interest (i.e., lens, eye fundus, optic nerve, and planning target volume—PTV).

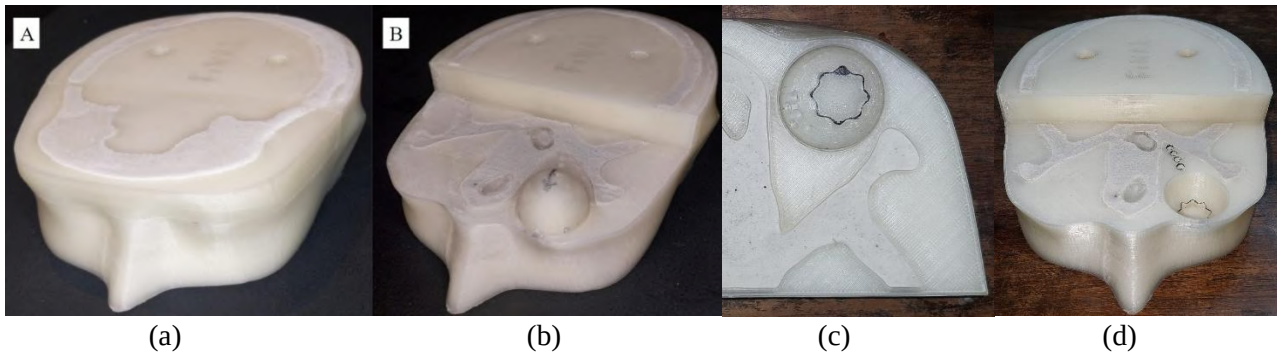


Figure 2: Phantom for TLD dosimetry. (a) Complete phantom. (b) With the front part removed, the detachable eye is visible. (c) Bottom view of the eye with the star-shaped male fixation. (d) Front view without the eye, showing TLD holes in the optic nerve and the female fitting in the base, securing the eye with ≤ 0.2 mm movement.

2.2. Treatment Planning for the Phantoms

The end-to-end dosimetric analysis involves treating the developed phantom as a patient, that is, subjecting it to all the steps required for dose delivery in EBRT. Initially, both phantoms underwent computed tomography (CT) scanning. The acquisition was performed using a Philips Brilliance Big Bore CT scanner, installed at Hospital Quinta D'Or in Rio de Janeiro, Brazil. The scanner has a resolution of 2.56 pixels per millimeter and a voxel size of $0.3906 \times 0.3906 \times 1$ mm³. The institution's standard protocol for head and neck CT includes 120 kVp, a tube current-time product of 300 mAs and 1 mm slice thickness. After the CT acquisition, the irradiation planning was carried out using the Varian Eclipse treatment planning system (TPS), version 16.1. Within the software, it was possible to delineate the structures of interest, such as the eye, lens, and optic nerve. A PTV was also defined within the inner region of the eye globe to simulate choroidal melanoma to be treated.

Three dynamic arcs were planned with multileaf collimators (MLC) shaped to the PTVs with a 3 mm margin. In this configuration, both phantoms can be irradiated using the same treatment plan, with different dose levels achieved by simply rescaling the monitor units (MU) for comparative analysis. It is important to note that the prescribed dose to the PTV in the radiochromic film phantom was 4 Gy, while in the TLD phantom it was 1 Gy. These planning choices are crucial because, above 1 Gy, LiF:Mg,Ti no longer exhibit a linear dose–response curve, transitioning into supralinearity, which should be avoided (Horowitz, 2001). On the other hand, the uncertainties associated with EBT3 film dosimetry increase significantly at doses below 1 Gy, which we also sought to avoid (Akdeniz, 2024).

2.3. Radiochromic Film Dosimetry

2.3.1. Film Calibration

Accurate dose measurement using radiochromic films requires a proper calibration protocol. In this study, Gafchromic EBT3 films were employed. Using a precision cutting tool, the films were sectioned into $4 \times 4 \text{ cm}^2$ squares. Irradiation was performed with a Varian Clinac IX linac, delivering a 6 MV photon beam. During the irradiation setup, the films were placed within a $10 \times 10 \text{ cm}^2$ field in surface, at a source-to-surface distance (SSD) of 100 cm, positioned at a depth of 10 cm of solid water and above 5 cm of solid water for considering backscatter radiation.

Based on this configuration, the dose was calculated using the linear accelerator's commissioning and routine quality assurance data. The planned dose range for calibration was from 0.5 to 6.5 Gy and two films for background measurements. After irradiation, and following a 24-hour interval, the films were scanned using an Epson Expression 11000XL scanner. To avoid curvature and scanning inconsistencies across the film surface, a 2 mm thick acrylic plate was fabricated and placed over the films during scanning. The scanned films were then imported into the *Film2Dose* software (Alves, 2017), described with more details by (Alves, 2014), for calibration curve calculation based on optical density and the planned dose for each film.

2.3.2. Phantom Irradiation and Film Analysis

The phantom was irradiated using a clinical Varian TrueBeam linear accelerator with a 6 MV photon beam, properly commissioned for the delivery of treatments with small fields (i.e., radiosurgery) in accordance with IAEA TRS-483 (International Atomic Energy Agency, 2017). So far, dosimetric analyses have been performed only on the oblique plane of the phantom. After scanning, the irradiated film had its optical density data converted into dose using the *Film2Dose* software. During treatment planning, the dose distribution on the plane where the film would be positioned was exported from the TPS in DICOM (Digital Imaging and Communications in Medicine) format. This allowed the *Film2Dose* software to compare the dose distributions from the TPS and the radiochromic film using the Gamma index (Das et al., 2022).

This index has significant clinical relevance, as it is recommended by Task Group No. 218 (TG-218) of the American Association of Physicists in Medicine (AAPM) as the universal tolerance and action criteria for intensity modulated radiation therapy (IMRT) verification quality assurance (Miften et al., 2018). The objective of this index is to compare the delivered treatment dose distribution predicted by the TPS with that obtained experimentally using radiation detectors, as this comparison represents a key criterion for the approval of patient treatment plans. The gamma index evaluation in this work was performed using the criteria 3%/3 mm/10% threshold, 3%/2 mm/10% threshold (as recommended by TG-218), and 2%/2 mm/10% threshold. These criteria were selected due to their widespread adoption in clinical practice.

2.4. TLD Dosimetry

2.4.1. TLD Reproducibility and Calibration

To evaluate the uncertainty associated with TLD dosimetry, a reproducibility test was performed. In this study, 49 TLD (LiF:Mg,Ti) microcubes — 1 mm edge-length cubes — were characterized. These dosimeters were selected for their small dimensions, which are comparable to or smaller than the OARs under investigation, thus ensuring the required spatial resolution.

In this process, the TLDs underwent appropriate thermal treatment (400°C for 1 hour and 100°C for 2 hours) for annealing in a PTW-TLDO oven. Afterward, they were irradiated using a Varian Clinac IX linear accelerator with a 6 MV photon beam. The irradiation setup consisted of a $20 \times 20 \text{ cm}^2$ field at the surface, with a SSD of 100 cm, and the detectors positioned at a depth of 5.5 cm in solid water, with an additional 5 cm of solid water for backscatter radiation (Fig. 3). The dosimeters were irradiated with 10 MU, which under the specified conditions correspond to a dose of 8.91 cGy.

Following irradiation, the TLDs were subjected to a pre-readout thermal treatment at 100°C for 15 minutes. The readout was performed using a Harshaw 3500 TLD reader. This procedure was repeated five times on different days. Based on the TLD responses, the coefficient of variation was evaluated individually for each dosimeter. Dosimeters with coefficients of variation above 4% were excluded from the study.



Figure 3: TLD reproducibility and calibration setup.

The calibration process followed the same irradiation setup, thermal treatment protocols, and readout procedures as previously described. To construct the calibration curve, the dosimeters were irradiated with 10 to 150 MU, corresponding to 8.91 to 133.59 cGy, respectively.

The five most reproducible TLDs were defined as control dosimeters and were consistently irradiated with the 8.91 cGy dose from the characterization study. This procedure allowed for the definition of an individual correction factor for each calibration and phantom dosimetry, based on the prior reproducibility analysis. Moreover, this step enabled the verification of potential systematic errors throughout the dosimetry process. The correction factor for each dosimetry process, $\bar{f}$, presented in Equation 1, corresponds to the mean value of the individual correction factors, f_n , calculated for each of the five control dosimeters, as shown in Equation 2. The dose values obtained in each dosimetry round were then corrected using $\bar{f}$.

$$\bar{f} = \frac{\sum_{n=1}^5 f_n}{5}. \quad (1)$$

$$f_n = \frac{M_{dosimetry}}{\bar{M}_{reproducibility}}, \quad (2)$$

where $M_{dosimetry}$ is the measured value in a specific dosimetry process, and $\bar{M}_{reproducibility}$ is the mean of the readout values obtained during the reproducibility study for each dosimeter. Thus, the corrected readout values (Eq. 3) are used for dose conversion through the calibration curve.

$$M_{corrected}[nC] = M_{dosimetry}[nC] \times \bar{f}. \quad (3)$$

2.4.2. Phantom Irradiation and TLD Analysis

The irradiation of the TLD phantom was also carried out using a clinical Varian TrueBeam linear accelerator with a 6 MV photon beam (Fig. 4). Up to this stage, point doses have been evaluated at four distinct locations along the optic nerve. At this stage of the study, measurements at locations other than the optic nerve (Fig. 5) within the phantom have not yet been conducted due to experimental challenges related to dosimeter positioning. Based on the doses measured by the TLDs, it will be possible to compare them with the corresponding dose values predicted by the TPS at these locations (Fig. 5). The prescription dose to the PTV in this phase was 1 Gy, using the same delivery parameters as those applied in the film phantom irradiation, with MU values rescaled accordingly. The TLD dosimetry followed a protocol similar to the calibration procedure.



Figure 4: TLD phantom irradiation setup.

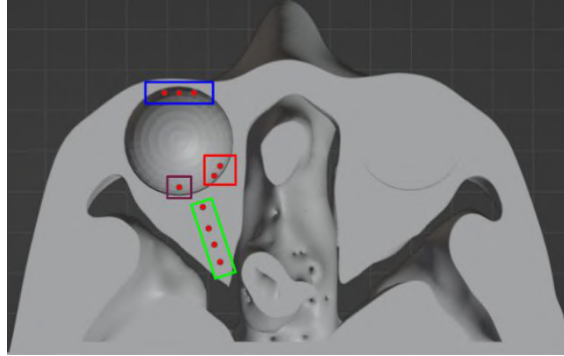


Figure 5: 3D model of the TLD phantom. The dosimetry points are highlighted in color: optic nerve (green), lens (blue), eye fundus (purple), and PTV (red).

The measured readouts will allow for direct comparison with the planned doses at the respective TLD positions in the treatment planning system. In this study, the comparison was performed by calculating the relative discrepancy between the experimental and TPS values, as well as through a statistical compatibility analysis.

The uncertainty of the TLDs irradiated with the phantom was determined from the combined standard uncertainty, considering both the individual uncertainty of each dosimeter (coefficient of variation) and the maximum acceptable uncertainty in dose delivery for radiotherapy treatments, $\sigma_{radiotherapy} = 5\%$, as defined by the International Commission on Radiation Units and Measurements (International Atomic Energy Agency, 2024) (Eq. 4).

$$\sigma_{readout} = \sqrt{\sigma_{TLD}^2 + \sigma_{radiotherapy}^2} \quad (4)$$

3. Results and Discussions

3.1. Radiochromic Film Dosimetry

The calibration films were properly scanned, and upon being imported into the *Film2Dose* software, the calibration curve was established (Fig. 6). The curve also shows a good fit to the experimental data, with a coefficient of determination (R^2) of 0.9997. An associated mean dose uncertainty of 6.1 cGy was also observed.

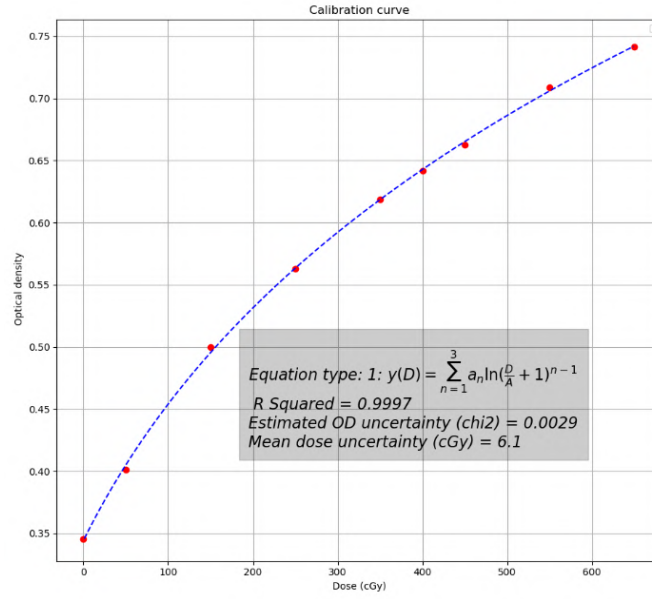


Figure 6: Film calibration curve.

After calibration, the phantom was irradiated with different films on three separate occasions, following the setup shown in Figure 4. The irradiated and scanned films (Fig. 7a), once imported into the *Film2Dose* software, had their optical density converted into dose using the calibration curve (Fig. 7b). The dose distribution from the TPS was also analyzed (Fig. 7c).

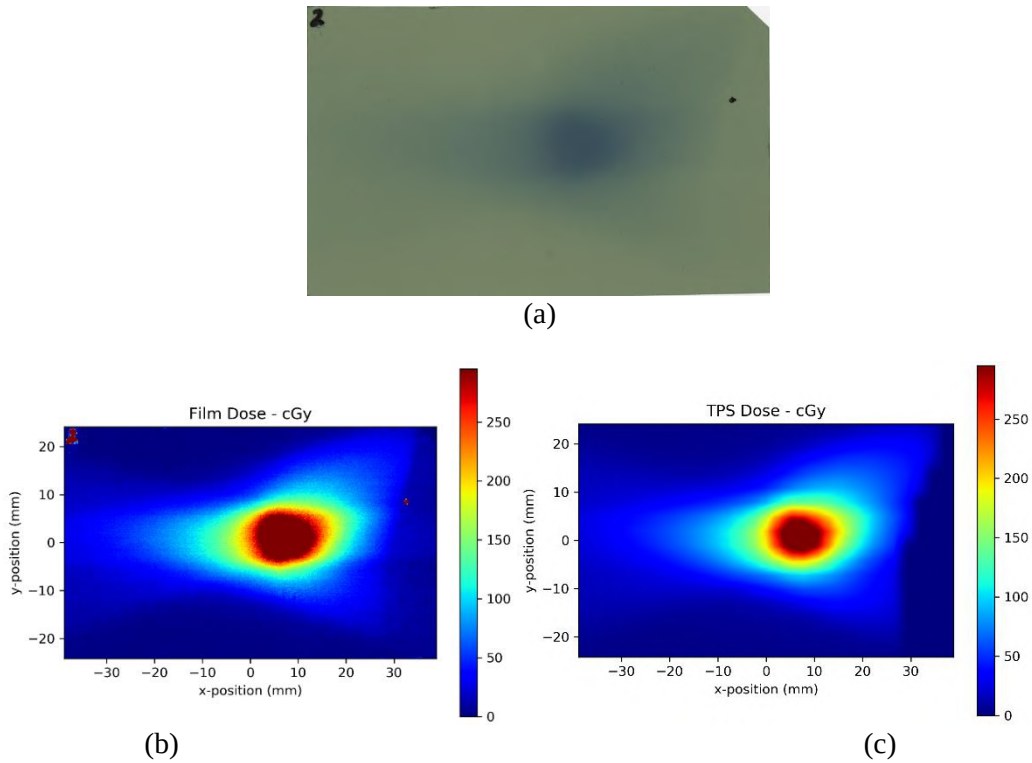


Figure 7: (a) Scanned irradiated film. (b) Dose distribution on the film. (c) Dose distribution in the TPS.

After the conversion, gamma index analysis with the TPS was also performed. The histogram (Fig. 8a) shows the percentage of passing points and provides an overview of the point-by-point analysis. If the gamma index value for an experimental point relative to the corresponding TPS point is less than or equal to 1, the point is considered equivalent and approved. The spatial distribution of these values is shown in Figure 8b.

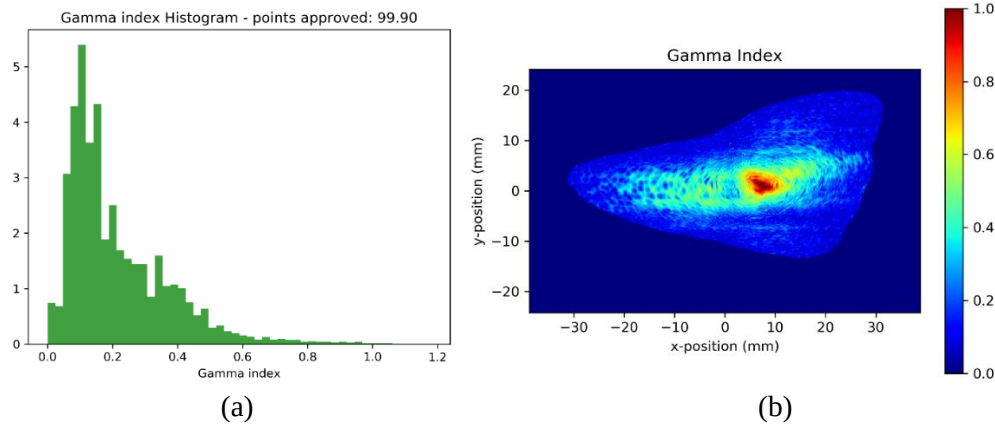


Figure 8: (a) Gamma index histogram showing the percentage of passing points. (b) Spatial distribution of the gamma index. These results correspond to the 3%/3 mm/10% threshold criterion for the second film analyzed.

Gamma analysis yielded a mean passing rate for the three films of 99.9% (3%/3 mm/10% threshold), 98.6% (TG-218 reference criteria: 3%/2 mm/10% threshold), and 98.3% (2%/2 mm/10% threshold). As recommended by TG-218, the universal tolerance limit for treatment plan acceptance is a gamma passing rate above 95%. Therefore, the results obtained in this study can be considered clinically acceptable, demonstrating that the developed phantom and the film analysis methodology could produce reliable and reproducible quality assurance results.

3.2. TLD Dosimetry

Reproducibility was evaluated for the 49 TLDs available. Table 1 shows the reproducibility results for five of the forty-nine TLDs evaluated in this study. Dosimeters A1 to A4 were used to determine the dose at specific points in phantom's optic nerve.

Table 1: Reproducibility results of five TLDs.

TLD ID	Mean (nC)	Standard Deviation (nC)	Coefficient of Variation (%)
A1	39.03	0.83	2.13
A2	39.84	1.20	3.02
A3	36.13	1.30	3.59
A4	37.04	0.63	1.69

Calibration curves were determined individually for all 49 TLDs. The corrected readout values ($M_{TLD} \times \bar{f}$) were used to generate the calibration curves. The calibration curve for dosimeter A4 is presented in Equation 5, exhibiting an R^2 value of 0.9987. All TLDs exhibited R^2 values greater than 0.99, confirming a strong linear correlation and reliable calibration accuracy.

$$D [cGy] = 0,2222 \left[\frac{cGy}{nC} \right] \times M_{corrected}[nC] + 1,6861[cGy]. \quad (5)$$

Dosimeters A1 to A4 underwent all thermal treatments and were irradiated as described in Section 2.4.2. The corrected readout measured doses in TLDs, combined uncertainty, the values extracted from the TPS, and the discrepancy analysis are presented in Table 2.

Table 2: Dosimetry data of the TLDs at the optic nerve.

TLD ID	Corrected readout (nC)	Measured Dose (cGy)	TPS Dose (cGy)	Relative discrepancy
A1	100.8	24.8 ± 1.3	22.4	11%
A2	69.7	17.8 ± 1.0	15.7	14%
A3	53.0	14.6 ± 0.9	12.6	16%
A4	45.2	11.8 ± 0.6	11.1	6%

The results presented in Table 1 show significant discrepancies between the measured and planned doses. The measured values are consistently higher than those predicted, indicating the presence of a systematic error in the dosimetry process. This additional dose is likely attributable to the exposure from Cone-Beam CT (CBCT) imaging, not included in the TPS dose calculation, performed on the linear accelerator for phantom positioning prior to treatment. Although CBCT employs a lower-energy beam technique (100 kV/150 mAs) compared to the therapeutic beam (6 MV), its contribution to the total dose is not negligible. In a study by Alvarenga (2022), CBCT doses delivered by the Varian TrueBeam were evaluated using TLDs for the same imaging protocol, revealing dose values of 0.59 ± 0.03 cGy near the left optic nerve region and up to 1.10 ± 0.06 cGy in the left eye region. In the present work, the positions corresponding to TLDs A4 and A1 are spatially close to these anatomical locations, supporting the hypothesis that the observed overdose is due to CBCT exposure.

No correction factors were applied for small-field dosimetry since, as reported by Azangwe et al. (2014), TLD micro-cubes exhibit correction factors close to unity across field sizes from 0.6×0.6 to 10×10 cm², with deviations below 1% relative to the alanine water reference dose (maximum 0.9% at 2.4×2.4 cm²).

Despite the observed discrepancies, the experimental measurements remain within 3σ compatibility with the TPS-calculated values, yielding compatibility test results of 1.8, 2.1, 2.2, and 1.1 for dosimeters A1, A2, A3, and A4, respectively.

4. Conclusions

This study aimed to validate 3D-printed, tissue-equivalent anthropomorphic eye phantoms for dosimetric quality assurance analyses in EBRT using radiochromic films and TLDs.

The preliminary results indicate that the phantoms demonstrated promising performance in their first measurements.

The radiochromic films irradiated in the oblique section of the phantom and compared with the TPS using the gamma index achieved clinical approval across all evaluated criteria.

The TLD analysis revealed a systematic error associated with dose contribution from CBCT imaging prior to treatment, which is not accounted for by the TPS.

Based on these findings, further developments are underway, including axial plane dose evaluation using radiochromic films; CBCT dose quantification at each TLD measurement point; and refinement of the 3D-printed phantom design to address fixation issues of TLDs at additional positions (lens, eye fundus, and PTV), aiming to complete the validation process of the proposed eye phantoms.

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Validation of an alanine-based postal dosimetry system for quality control in radiotherapy

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Abstract

External dosimetry audits play an important role in enhancing the quality and safety of radiotherapy procedures by identifying systematic errors and fostering consistency in clinical practices. However, despite their proven importance, access to such auditing services remains limited for many radiotherapy centers, particularly in resource-constrained settings. In this context, the present study aimed to validate a postal quality assurance (QA) system for photon radiotherapy, which is currently under development at the Department of Nuclear Energy of the Federal University of Pernambuco (DEN/UFPE), using alanine-EPR dosimeters. The proposed system consists of a setup designed to position and irradiate six dosimeter holders simultaneously, thereby allowing the execution of beam audits under both reference and non-reference conditions. To assess the system's performance, measurements obtained using the DEN/UFPE setup were compared with those from the TLD-based dosimetry audit program of the Brazilian National Cancer Institute (PQRT/INCA), as well as with a PTW 30013 ionization chamber, used as a reference under identical irradiation conditions. The postal QA system evaluates central-axis and off-axis dose, beam quality, flatness, and symmetry. The results of the intercomparison demonstrated strong concordance among the different systems, with dose deviations relative to the ionization chamber not exceeding 1.2% for the DEN/UFPE system and 0.9% for the PQRT/INCA system. These findings confirm the reliability, accuracy, and effectiveness of the proposed alanine-based postal dosimetry audit system for clinical implementation.

Keywords: Dosimetry audit; Quality control; Alanine-EPR dosimetry.

1 Introduction

Dosimetry audit is a key component of quality assurance programs in radiotherapy, since radiation oncology treatments require a high degree of accuracy both in terms of the level of absorbed dose in tumor tissues and the geometric precision of radiation delivery (Dimitriadis et al., 2017). In addition to assisting identify and resolve problems, external dosimetry audits also reduce uncertainty and increase the consistency of dosimetry protocols among participating institutions (Dimitriadis et al. 2023; Grochowska et al., 2013). Over time, this type of service has improved dosimetric accuracy in many

radiotherapy centers and helped establish and maintain appropriate medical physics practices (Lye et al., 2019; Misson et al., 2024).

The International Atomic Energy Agency (IAEA), in collaboration with the World Health Organization (WHO), was the pioneer in implementing postal dosimetry assessment systems for quality control in radiotherapy. Over five decades of operation, the IAEA/WHO dosimetry audit program has conducted 13,756 audits in 136 countries across six geographic regions (Izewska et al., 2020). During these audits, the dose measured using dosimeters sent by mail is compared to the dose reported by the participant. Agreement within $\pm 5\%$ is considered acceptable; deviations outside this limit require the adoption of follow-up measures. In an analysis of the quality of reference dosimetry in radiotherapy centers participating in this program over the course of its 50 years, Izewska et al. (2020) report a significant increase in the overall percentage of acceptable results, from about 50% in its early years to 99% currently, although there is some variability in results among participating countries.

Regarding the types of dosimeters typically adopted by available postal audit systems, there is a predominance of passive dosimeters, such as thermoluminescent (TLD) and optically stimulated luminescence (OSLD) dosimeters, in addition to alanine, radiophotoluminescence (RPLD) dosimeters, and radiographic films. For assessments conducted under non-reference conditions, most audit networks use ionization chambers and films, followed by other types of dosimeters such as TLD, alanine, diodes, diamond, and dosimeter arrays. Typical uncertainties associated with measurements using passive dosimeters generally range from 1% to 3% ($k = 1$) (Izewska et al., 2018).

Despite the existence of a number of national networks dedicated to dosimetry audit, access to this service by radiotherapy centers on a global scale is still insufficient to meet current needs. In a survey to assess the global availability of dosimetry audits, based on the IAEA audit network database, Izewska et al. (2018) reported that each year, an increasing number of radiotherapy facilities are being implemented, and the demand for quality control assessments exceeds the operational capacity of dosimetry audit networks. Therefore, expanding the availability of these audits becomes essential for strengthening clinical dosimetry in radiotherapy and ensuring adequate protection for cancer patients undergoing radiation treatment.

In this context, the objective of this study was to develop and validate a postal quality control system for radiotherapy, based on alanine/EPR dosimeters. This system allows the evaluation of parameters such as central-axis dose, off-axis dose, depth dose, as well as the quality, flatness, and symmetry of the photon beam, through a single irradiation. Alanine dosimeters were chosen for this purpose because they offer significant advantages in several metrological aspects when compared to other types of dosimeters, such as TLDs. Among these advantages are excellent post-irradiation stability, a wide linear dose response range, high reproducibility and accuracy, as well as a response independent of radiation energy above 100 keV and low dose rate dependence (Regulla and Deffner 1982; Borgonove et al. 2007; Waldeland and Malinen 2011; Baffa and Kinoshita, 2014), which is particularly relevant in audits covering different clinical settings.

2 Material and methods

2.1 Alanine EPR-dosimetry

The alanine-EPR dosimetry is based on the measurement of the concentration of free radicals in crystalline α -alanine, generated by the interaction with ionizing radiation. These radicals are detected by an Electron Paramagnetic Resonance (EPR) spectrometer, which records microwave energy absorption over a specific magnetic field range. The EPR spectrum of irradiated alanine (Figure 1) corresponds to the first derivative of the absorption line, and the central line amplitude (h) is proportional

to the concentration of free radicals and, therefore, to the absorbed dose (ISO and ASTM, 2013; Baffa and Kinoshita, 2014; Goodman et al., 2016).

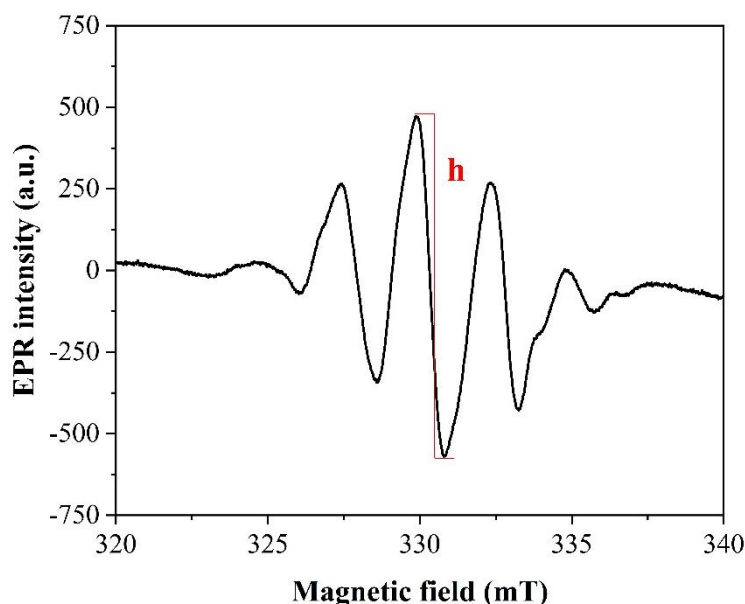


Figure 1: A typical alanine-EPR spectrum, where 'h' represents the amplitude of the central resonance line.

Alanine dosimeters used in this study were manufactured by A  rial (Illkirch, France) with a diameter of 4 mm and thickness of 2.5 mm. Each dosimeter had a mass of 37.40 ± 0.21 mg and was composed of 93% alanine and 7% binder. These dosimeters do not require any pre-irradiation or pre-readout processing, unlike TLDs, which typically demand such procedures. Dosimeter measurements were performed using a Magnettech ESR 5000 EPR spectrometer (Bruker, Bremen, Germany), operating in the X-band frequency range. The microwave power and modulation amplitude applied were 10 mW and 0.7 mT, respectively. The modulation frequency was 100 kHz.

To account for potential stability fluctuations in spectrometer performance, the alanine EPR signal intensity was normalized by the signal intensity of the ruby reference sample. This synthetic ruby crystal was located within the resonant cavity and was measured simultaneously with each alanine dosimeter.

The alanine-EPR response was quantified by measuring the peak-to-peak amplitude of the most intense EPR peak, which corresponds to the amplitude of the center line in each dosimeter's spectrum, as mentioned before. In a similar manner, the ruby EPR response was evaluated by determining the amplitude between the maximum and minimum points of its corresponding spectrum. In addition to normalization by the ruby EPR response, the alanine EPR response was normalized by the dosimeters' mass.

It should also be added that the alanine-EPR response was previously characterized for photon beams used in radiotherapy by evaluating its reproducibility, linearity, and dependence of its response on both dose rate and beam energy, aiming to ensure its reliability under typical clinical conditions (Albino et al., 2022).

2.1.1 Calibration curve

For the calibration curve, a set of alanine-EPR dosimeters was irradiated in a True Beam STX linear accelerator (Varian, California, USA) with doses ranging from 2 Gy to 30 Gy. In addition, the irradiation setup included: 6 MV photon beam, dose rate of 600 MU/min and radiation field of 10×10 cm².

The EPR measurements obtained were used to construct the calibration curve, which represents the dosimeter normalized response – by both the dosimeter mass and the amplitude of the ruby-EPR signal – as a function of absorbed dose. Calibration factor was subsequently derived from the slope of the linear fit to the resulting curve.

2.2 DEN/UFPE postal audit system

The quality control system (Figure 2) which is under development at the Department of Nuclear Energy of the Federal University of Pernambuco (DEN/UFPE) basically consists of a support containing two acrylic (polymethyl methacrylate-PMMA) rods with a diameter of 2.5 cm each, whose centers are 10 cm apart. These rods support six dosimeter holders in total and these holders were manufactured via 3D printing on a Form2 Desktop 3D printer (Formlabs, Somerville, Massachusetts) using two types of resins: Black V4 and Flexible (Formlabs, Somerville, Massachusetts), both with a relative density of 1.09 g/cm³, according to the manufacturer. Each dosimeter holder carries two alanine pellets.

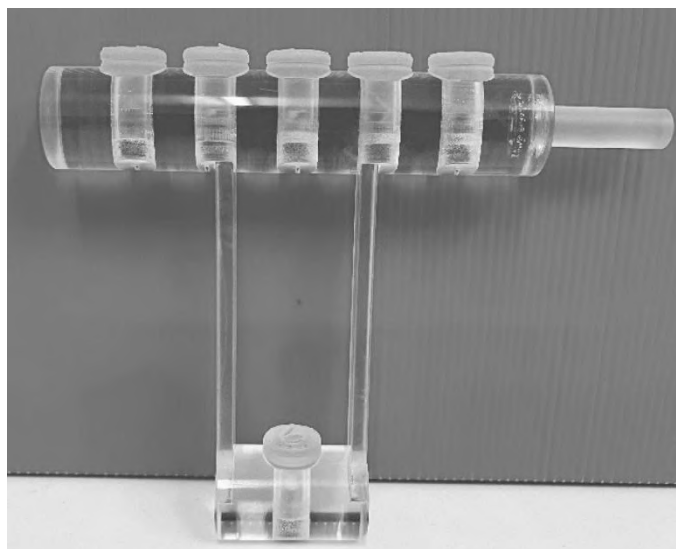


Figure 2: DEN/UFPE postal audit system containing acrylic rods and alanine dosimeter holders.

The complete setup allows the six dosimeter holders to be irradiated simultaneously, enabling the evaluation of central axis dose at 10 cm and 20 cm depths, off-axis dose at 2 cm and 4 cm, in addition to the quality (TPR_{20,10}), flatness and symmetry of the radiation beam. These parameters are assessed according to TRS-398 (IAEA, 2000) and TEC DOC 1151 (INCA, 2000).

2.3 Intercomparison with the TLD postal audit service of PQRT/INCA

To validate the DEN/UFPE system, an intercomparison was performed with the postal audit service of Quality Program in Radiotherapy (PQRT) of the Brazilian National Cancer Institute (INCA), which uses TLD-100 powder dosimeters as the detector element.

The two systems were individually positioned in a 1D Scanner water phantom (Sun Nuclear, Florida, USA) and irradiated in a True Beam STX linear accelerator (Varian, California, USA) with 6 MV energy, dose rate of 600 MU/min, and doses of 10 Gy for the DEN/UFPE system and 2 Gy for the PQRT/INCA system. The dosimetric measurements obtained with these systems were compared with those of a Farmer 30013 ionization chamber (PTW, Freiburg, Germany), irradiated under the same conditions. The setup used for irradiation was: SSD equal to 100 cm, 10 x 10 cm² radiation field, and depth of 10 cm.

Figure 3 shows the irradiation setup used for both the DEN/UFPE system (Figure 3a) and the PQRT/INCA system (Figure 3b).

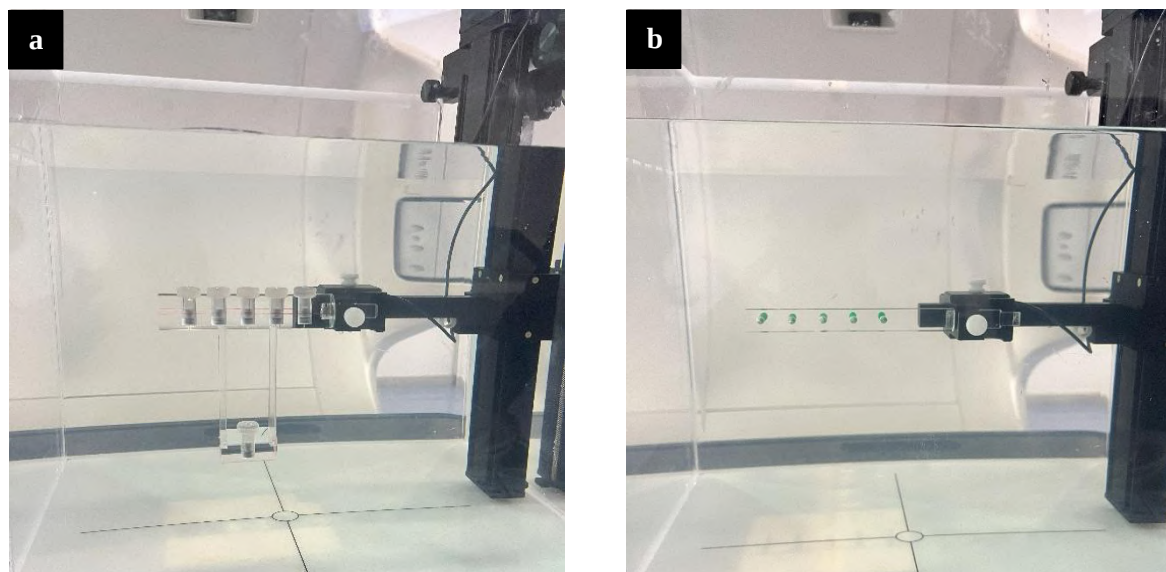


Figure 3: Irradiation setup to (a) DEN/UFPE system and (b) PQRT/INCA system.

3 Results and discussion

3.1 Calibration curve

Figure 4 shows the calibration curve of alanine dosimeters for 6 MV radiation energy. A linear regression was applied to the experimental data points, demonstrating that the alanine-EPR response exhibits a linear behavior within the evaluated dose range, with a determination coefficient (R^2) of 0.9999. This finding is consistent with previous studies reported in the literature (Helt-Hansen et al., 2009; Baffa and Kinoshita, 2014). As described by Regulla and Deffner (1982), the dose-response curve for alanine typically maintains linearity from a few grays up to around 100 kGy.

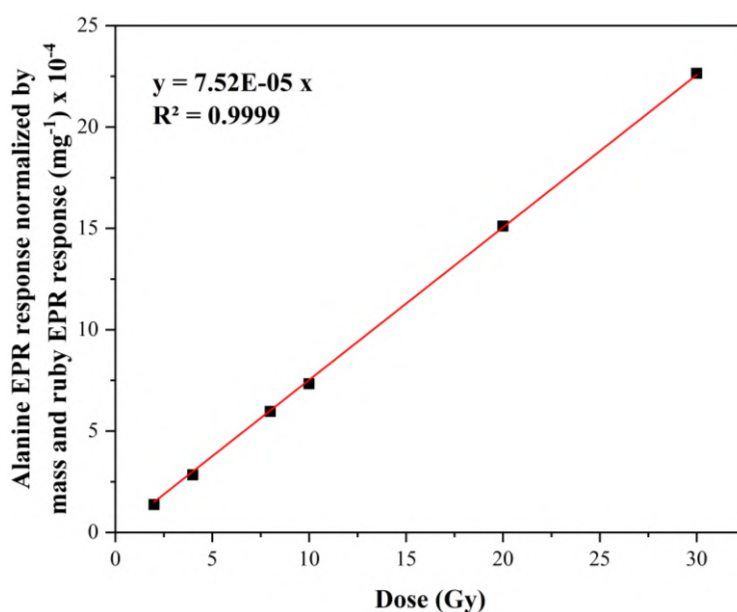


Figure 4: Alanine calibration curve for 6 MV radiation energy.

3.2 Validation of DEN/UFPE postal audit system

The validation of the postal QA system proposed in this work was conducted through an intercomparison with the Brazilian dosimetry audit service of the PQRT/INCA, which is a national reference in the field. This program has been in operation for 25 years and evaluates the physical parameters of photon beams under both reference and non-reference conditions. It is estimated that from 2003 to 2018, the PQRT/INCA evaluated 708 linear accelerator beams, performing 5,445 tests (Araújo et al., 2019).

Both systems were evaluated in terms of central-axis dose, off-axis dose, flatness, and beam symmetry. Furthermore, measurements taken with the PTW 30013 ionization chamber were used as a reference to determine the percentage deviations from the values provided by each system. Table 1 presents the results of this intercomparison. In determining off-axis doses, the ‘+’ symbol represents the measurement points on the right side of the central axis, while the ‘-’ symbol corresponds to those on the left side, taking as a reference a front view of the gantry.

Table 1: Results of the intercomparison between PQRT/INCA and DEN/UFPE postal audit systems. The deviation values correspond to the percentage differences between the measurements of alanine or TLD dosimeters and ionization chamber.

Parameter	Value measured		Deviation %	Value measured		Deviation %
	PQRT/INCA	Ionization chamber		DEN/UFPE	Ionization chamber	
Dose in the central axis in reference conditions	2.03 Gy	2.02 Gy	0.43%	9.99 Gy	9.97 Gy	0.22%
Off-axis dose at -2 cm	2.01 Gy	2.05 Gy	-0.77%	9.95 Gy	9.98 Gy	-0.39%
Off-axis dose at +2 cm	2.03 Gy	2.02 Gy	0.66%	9.84 Gy	9.96 Gy	-1.15%
Off-axis dose at -4 cm	1.98 Gy	1.97 Gy	0.90%	9.68 Gy	9.70 Gy	-0.17%
Off-axis dose at +4 cm	1.98 Gy	1.96 Gy	0.87%	9.77 Gy	9.67 Gy	1.08%
Beam flatness	1.37%	1.60%	-0.23%	1.57%	1.60%	-0.03%
Beam symmetry	0.31%	0.28%	0.03%	-0.92%	0.28%	-1.20%

Based on the data presented, it can be seen that the differences between the values measured with the DEN/UFPE system and the ionization chamber did not exceed 1.2%. For the PQRT/INCA system, the maximum percentage deviation was 0.9%. This agreement between the results obtained with both systems indicates the reliability and accuracy of the system proposed, which is capable of verifying important photon beam parameters with precision compatible with the standards of a well-established program such as INCA.

4 Conclusions

The system developed by DEN/UPPE demonstrated satisfactory performance and was technically consistent for dosimetry audit purposes. The agreement with the results of an already established national program highlights the robustness of the proposed system, confirming its suitability for independent assessments of dosimetry quality in radiotherapy treatment units. Therefore, the next steps involve distributing this system to different radiotherapy services across the country, with the aim of strengthening quality control and ensuring the safety of treatments offered to patients.

Acknowledgements

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Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the Triga IPR-R1 nuclear reactor

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Abstract

This study investigates the mechanical performance of the thermoplastic polymer Ingeo™ 3D850, a high-heat resistant variant of polylactic acid polymer (PLA), in the context of its potential use for components involved in the visual inspection of nuclear reactor fuel elements. Specifically, the study considers the conditions at the Triga IPR-R1 reactor operated by the Nuclear Technology Development Center (CDTN) in Brazil, where the inspection of fuel rods is part of an aging management plan enforced by regulatory authorities. Samples were printed using Ingeo™ 3D850 according to ASTM D638 Type V specifications and subjected to gamma irradiation doses ranging from 7 kGy to 175 kGy at the CDTN Gamma Irradiation Laboratory (LIG). The tensile strength of the samples was subsequently measured, revealing a marked reduction in mechanical performance with increasing radiation dose. Notably, resistance to tensile stress decreased progressively, with high-dose exposure resulting in significant material embrittlement. Comparisons with previously published data on standard PLA (Ingeo™ 4043D) indicated that PLA-HT outperforms traditional PLA beyond the 70 kGy threshold, suggesting better long-term viability in high-radiation applications. A detailed assessment of the radiation dose gradient within the reactor pool was performed using historical and theoretical models. These findings offer practical insights into material performance and allow for predictions regarding the operational lifespan of 3D-printed components used in the Triga IPR-R1 reactor.

Keywords: Triga IPR-R1 reactor; Gamma-rays; Tensile strength;

1 Introduction

Periodic visual inspections of nuclear reactors are one of the main strategies for mitigating risks associated with the aging of its components, especially regarding the structural integrity of fuel elements. In Brazil, the Triga IPR-R1 nuclear reactor a well-type reactor, immersed in an aluminum-lined cylindrical tank, 1.92 m in diameter and 6.62 m deep, located at the Nuclear Technology Development Center (CDTN) in

Belo Horizonte, has been in operation since the 1960s. The permanence of several fuel elements in the core since their first criticality poses technical challenges regarding corrosion and the maintenance of the mechanical properties of metal claddings, requiring a rigorous approach to periodic inspections, in accordance with guidelines from the International Atomic Energy Agency (IAEA) and regulations from the National Nuclear Energy Commission (CNEN).[8][10][2]

One of the proposals under development to improve the visual inspection process for these fuel elements involves the design of a specific support, partially manufactured through additive manufacturing (3D printing), that allows for controlled rotation of the element, being located approximately two meters under the water, facilitating image capture with underwater cameras. Selecting suitable materials for the construction of this device requires careful evaluation of their thermal and mechanical properties, considering the presence of ionizing radiation. In this context, the use of technical polymers, such as Ingeo™ Biopolymer 3D850 (commercially known as PLA-HT), has attracted interest due to its high dimensional stability, superior thermal resistance to conventional PLA (4043D), and feasibility for 3D printing.[9]

However, the application of these materials in environments subject to prolonged exposure to gamma radiation requires analyzing their response to the effects of ionizing radiation. It is widely documented that polymers can undergo oxidative degradation or crosslinking depending on the absorbed dose, significantly altering critical properties such as tensile strength and thermal behavior.[6][7]

Therefore, this work aims to investigate the mechanical behavior of PLA-HT after exposure to different doses of gamma radiation, simulating the actual conditions to which it would be subjected inside the reactor vessel during visual inspections. Based on the experimental data obtained, we intend to evaluate the feasibility of using this material in applications involving the presence of ionizing radiation, contributing with technical subsidies for future implementations of plastic parts in nuclear environments.

2 Base properties of Ingeo™ 4043D and 3D850

The material Ingeo™ 4043D exhibits a high degree of multifunctional extrusion performance, resulting in a 3D printing filament with excellent printability characteristics, such as high detail resolution, strong adhesion to the printing platform (without the need for a heated bed), reduced warping, and minimal odor emission (i.e., absence of strong, greasy, or oily smells during printing).[9]

In contrast, Ingeo™ 3D850 is described as having a faster crystallization rate and improved thermal resistance in 3D printed parts. This material provides enhanced performance in formulations tailored to increase impact and heat resistance. Monofilaments produced with Ingeo™ 3D850 exhibit superior 3D printing features, including precise detail fidelity, reliable build-plate adhesion, low distortion, and negligible odor generation. Table 1 outlines some key differences between these two materials.[9]

Table 1: Properties between the materials.

Properties	Ingeo™ 4043D (PLA)	Ingeo™ 3D850 (PLA-HT)
Peak Melting Temperature (°C)	145-160	165-180
Ultimate Tensile Strength (MPa)	60	51
Glass Transition Temperature (°C)	55-60	55-60
Heat Distortion Temperature (°C)	55	80-90

It is widely recognized that polymeric materials undergo structural transformations when exposed to ionizing radiation. Lee (2015) reported that radiation exposure typically leads to the generation of free radicals and ions in polymers, alongside the initiation, alteration, or disruption of molecular bonds. The two primary radiation-induced phenomena are crosslinking and oxidative degradation.[6][7]

The dominance of one of these competing processes depends on several factors, including the molecular structure of the polymer and the specific irradiation conditions. These modifications have direct implications for the macroscopic properties of the irradiated materials.[7]

3 Gamma radiation measuring inside the Triga IPR-R1 nuclear reactor

Dose rates near the vicinity of a nuclear fuel element were measured, simulating the conditions under which printed components would be subjected to gamma radiation. Measures of the dose rate inside the central tube of the Triga IPR-RI were also taken.

According to internal documentation from CDTN (2025), measurements were carried out using a Farmer 2502/3 dosimeter, using a 0,6 cm³ sensor. The reactor was in criticality 3 days prior to the measuring. The resulting data is presented in Tables 2 and 3.

Table 2: Dose rate inside the reactor's central tube

Time (h)	Exposure (R)	Exposure Rate	Dose Rate (Gy/h)
0,035	727	20937,6	183,6
0,034	704	20773,8	182,1
0,034	702,5	20729,5	181,8
0,035	719	20542,9	180,1

Table 3: Dose rate in the fuel element vicinity

	Time (h)	Exposure (R)	Exposure rate	Dose rate (Gy/h)
Fuel element	0,018	60,5	3403,1	29,8
Background	0,049	10	203,4	1,8
Liquid dose rate			3200	28

3.1 Surface gamma radiation measuring

According to internal documentation from CDTN, measurements were carried out to determine gamma dose rates at the surface of the reactor well with one of the fuel elements at various depths. Fuel element was vertically repositioned in increments, ranging from a depth of 2,7 meters up to 50 cm. The detector was placed 1,0 meter above the water surface. It is worth noting that these measurements were originally performed in 2002. The resulting data is presented in Table 4.

Table 4: Dose rate at 1m outside the reactor well

Distance of the fuel from the surface (cm)	Dose rate (Gy/h)
250	0,0005
200	0,00119
150	0,0048
140	0,00879
130	0,0145
120	0,027
110	0,046
100	0,08
90	0,135
80	0,261
70	0,506
60	0,96
50	1,658

4 Methodology

A total of 50 samples were fabricated using PLA-HT filament. All samples were printed in accordance with the dimensional specifications defined for “Type V” samples, as outlined on pages 3 and 4 of ASTM D638 – 14. This standard defines each specimen as having an overall length of 63,5 mm and a thickness of 3,2 mm. Of these samples, 45 were subjected to irradiation, grouped in sets of five, forming nine irradiated groups and one control group.[3]

The tensile strength of 3D-printed components can be maximized through optimal print geometry, specifically, by utilizing a 45° infill angle. This technique was adopted in the production of all test specimens in order to maximize their strength. [5]

The irradiated groups were exposed to gamma radiation doses of 7 kGy, 14 kGy, 21 kGy, 28 kGy, 35 kGy, 50 kGy, 93 kGy, 130 kGy, and 175 kGy.

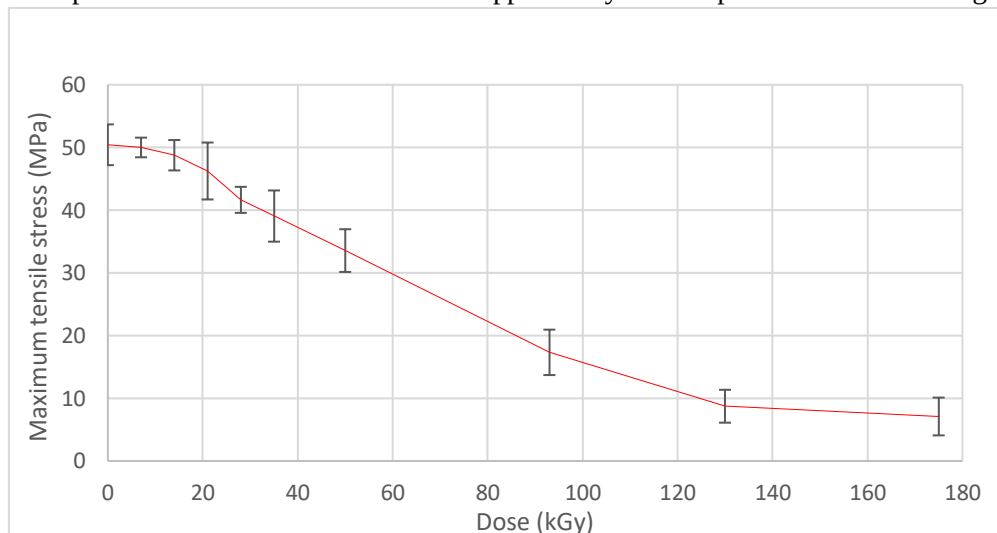
The Gamma Irradiation Laboratory (LIG), located within the CDTN facilities, operates a Category II panoramic multipurpose irradiator. The unit was manufactured by MDS Nordion (Canada), model/serial number IR-214, type GB-127, and is equipped with a dry Co-60 (Cobalt-60) source with a maximum activity of 2,200 TBq (60,000 Ci). The irradiation of the specimens was carried out in accordance with the pre-established doses.

Following irradiation, tensile tests were conducted at the Mechanical Testing Laboratory of CDTN, which is certified to perform standardized mechanical characterizations in compliance with technical norms. The laboratory evaluates the mechanical behavior of materials under various types of mechanical stress conditions. The samples were tested for

5 Results analysis

A graph was generated based on the collected data of the tensile stress evaluations and is presented in Figure 1, illustrating the relationship between maximum tensile stress sustained by the samples and the corresponding radiation dose.

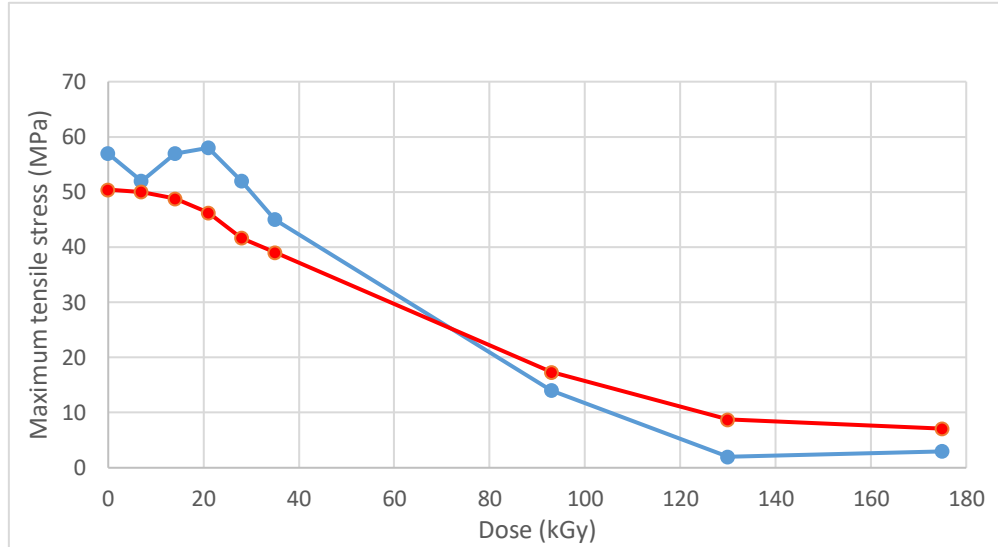
Figure 1: Graph of the maximum tensile stress supported by the samples in relation to the gamma dose.



From the analysis of this data, a pronounced embrittlement effect is evident in the material when exposed to gamma radiation, particularly at higher dose levels.

A comparative analysis was also performed using data reported by Alsabbagh (2020), represented in Figure 2. These results indicate that the Ingeo™ 4043D material exhibits superior tensile strength performance up to approximately 70 kGy. Beyond this dose range, Ingeo™ 3D850 demonstrates slightly enhanced mechanical resistance under tensile loading.[1]

Figure 2: A comparative analysis was conducted between PLA (blue) and PLA-HT (red) samples.



This may be attributed to the degradation of polymer chains, primarily through chain scission and oxidative degradation. The latter mechanism is likely the predominant one, given that the irradiation was conducted in an oxygen-rich air environment at LIG. The observed increase in material brittleness can also be explained by oxidative degradation processes, which are known to prevail over crosslinking mechanisms in thin polymeric specimens.[1]

These findings provide a basis for estimating the operational lifespan of components fabricated from PLA or PLA-HT, taking into account both their proximity to radioactive fuel elements and the mechanical stress conditions to which they are subjected.

Based on the previously presented data, a comparative assessment between the two materials can be made, taking into account the onset threshold at which each material begins to exhibit a significant reduction in tensile strength, as well as their suitability for applications within the operational environment of the Triga IPR-R1 reactor. These considerations are summarized in Table 5.

Table 5: Time estimates for PLA and PLA-HT under the Triga working conditions

	Significant embrittlement threshold (kGy)	Estimated Time on fuel element vicinity (h)	Estimated time on reactor central tube (h)	Estimated time 50 cm outside the reactor with fuel at 50 cm underwater (h)
PLA	28	1000	153	16887
PLA-HT	21	750	115	12665

Experimental results indicate that PLA-HT undergoes a steep decline in tensile strength beginning at approximately 21 kGy. Using this threshold and the gamma dose rate measurements obtained previously, it can be inferred that components fabricated from PLA-HT could maintain structural integrity for approximately 750 hours when positioned close to the fuel element, without experiencing significant mechanical degradation.

In the reactor's central tube, where gamma radiation levels are substantially higher, the estimated operational lifetime of PLA-HT components is reduced. Based on the experimentally determined average dose rate, it is estimated that the material could perform for approximately 115 hours under such conditions before exhibiting noticeable loss in tensile strength.

An additional observation pertains to operations conducted near the reactor surface, such as fuel handling or hoisting procedures. In these scenarios, PLA-HT components are considered relatively safe for extended periods. Assuming the fuel element is elevated to a height of 50 cm above the waterline and the PLA-HT component is positioned 1 meter above the surface, the expected radiation exposure is

sufficiently low to allow the material to retain its tensile properties for several years without significant degradation.

When analyzing the behavior of PLA, it becomes evident that this material exhibits greater resistance to gamma radiation compared to PLA-HT. However, for applications outside the reactor pool, PLA-HT remains preferable due to its advantageous properties such as improved dimensional stability and ductility.

4 Conclusions

The results of these tests provide an insight into the mechanical behavior of the Ingeo™ 3D850 polymer (PLA-HT) when subjected to ionizing gamma radiation, specifically within the operational context of the Triga IPR-R1 nuclear reactor. By systematically exposing 3D-printed specimens to a range of gamma doses and subsequently measuring their tensile strength, it was possible to establish a clear correlation between radiation dose and mechanical performance. A significant decline in tensile resistance was observed beginning at approximately 21 kGy, which marks a threshold for the onset of structural embrittlement in PLA-HT under these conditions.

When compared to standard PLA (Ingeo™ 4043D), PLA-HT demonstrated inferior resistance to gamma-induced degradation at lower doses but slightly outperformed PLA in structural applications requiring dimensional stability and ductility beyond the 70 kGy mark.

Using dose rate data collected from various zones within the reactor lifetime estimates for printed components were determined. These projections suggest that PLA-HT can remain mechanically functional for approximately 750 hours when positioned near a fuel element, and up to 115 hours in the more intense gamma radiation field of the central reactor tube. In low-radiation surface environments, such as during fuel-handling operations at shallow depths, the material's resistance to degradation permits a much longer operational lifetime, potentially extending to several years without significant loss in structural integrity.

These findings support the conditional use of PLA-HT for components involved in visual inspection systems in nuclear reactor environments, provided that exposure conditions like duration and radiation intensity are adequately managed. Furthermore, this study reinforces the necessity of integrating material-specific radiation response data into the engineering design process, especially in critical applications where mechanical reliability over time is essential.

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Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining

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Abstract

Acid mine drainage (AMD) is a significant environmental concern in mining regions due to the high concentrations of dissolved metals in water, often exceeding regulatory limits. This study investigates the potential of natural adsorbent materials—specifically, Japanese-origin zeolites RT22 and HR17—for the removal of metals from acidic waters collected from two drainage basins (BIA and BNF) at a decommissioned uranium mine in Poços de Caldas, Brazil. These waters are characterized by low pH values (3.45–3.67) and elevated concentrations of aluminum, manganese, and uranium, underscoring the need for effective remediation strategies. Adsorption experiments were conducted by contacting 0.1 g of each material with 100 mL of AMD solution for 24 hours under constant agitation. The pH of the solutions was monitored at 0, 1, and 24 hours, showing a significant increase—approaching the lower legal limit for effluent discharge (pH 5)—indicating partial neutralization of the acidic waters. Elemental analysis of the post-adsorption solutions using inductively coupled plasma optical emission spectrometry (ICP-OES) revealed complete removal of iron and a notable reduction in aluminum concentrations, particularly with HR17. Although manganese concentrations in solution remained relatively stable, analysis of the solids after adsorption indicated retention of manganese and uranium, especially in HR17, suggesting its superior adsorption capacity under acidic conditions. The results demonstrate that both RT22 and HR17 are effective in removing key contaminants from AMD, with HR17 showing greater potential. The increase in pH and the retention of uranium and manganese support the use of these natural zeolites as cost-effective and environmentally friendly materials for AMD treatment and environmental remediation.

Keywords: Acid drainage; Uranium; Mining; Adsorbents; Metal removal.

1 Introduction

Acid mine drainage (AMD) is recognized as a major environmental challenge in mining areas worldwide and is addressed within the framework of the United Nations Sustainable Development Goals (UNDP and UN Environment, 2018). This phenomenon arises when sulfide-bearing minerals undergo oxidation upon

exposure to air and water, often accelerated by microbial activity, resulting in acidic waters rich in dissolved metals. The environmental risks associated with AMD have prompted extensive research into both passive and active treatment strategies (Gaikwad & Gupta, 2008; Tong *et al.*, 2021; Ighalo *et al.*, 2022; Daraz *et al.*, 2023; Jiao *et al.*, 2023). The toxicity of these waters can lead to severe ecological consequences, including bioaccumulation in food chains and the decline of sensitive species.

Given the variability in geological and operational conditions across mining sites, the composition of AMD can differ significantly, necessitating site-specific treatment approaches and the development of materials capable of effectively removing target metal ions under local conditions (Sicupira *et al.*, 2015; Mokgehle & Tavengwa, 2021; Nguyen *et al.*, 2022). Among the most prevalent contaminants are iron, aluminum, and manganese (Mohammed & Yaacob, 2016; Dlamini *et al.*, 2019; Lobo-Recio *et al.*, 2021). Additionally, the recovery of rare earth elements (REEs) from AMD has gained interest due to their growing industrial demand and relatively high concentrations in these effluents (Lozano *et al.*, 2020; Moraes & Ladeira, 2021; Cicek *et al.*, 2023).

Adsorption stands out as a widely used and cost-effective method for removing metal ions from contaminated waters (Goher *et al.*, 2015; Tran *et al.*, 2018), especially when combined with pH adjustment techniques (Markovic *et al.*, 2020). Natural zeolites, in particular, have shown high efficiency in adsorbing heavy metals due to their structural and chemical properties, availability, and low cost (Wulandari *et al.*, 2020; Morante-Carballo *et al.*, 2021; Velarde *et al.*, 2023). Their performance depends on factors such as Si/Al ratio and the presence of exchangeable cations (Sobus *et al.*, 2020). These materials have also demonstrated potential in the removal of radioactive elements like cesium and uranium (Battsetseg *et al.*, 2024; Johan *et al.*, 2015; Munawar *et al.*, 2023).

In uranium mining, the presence of radionuclides such as uranium, thorium, radium, and radon adds complexity to environmental remediation efforts (Ladeira & Gonçalves, 2008; Dinis & Fiúza, 2021; Ma *et al.*, 2021; Petrounias *et al.*, 2023). Uranium, in particular, can be oxidized to its hexavalent form (U(VI)) in the presence of Fe^{3+} , forming soluble uranyl complexes with carbonate and sulfate ions (Bhatti & Tuovinen, 2023). Its removal by adsorption is influenced by factors such as pH, contact time, and initial concentration.

The Poços de Caldas alkaline complex in southeastern Brazil, a Mesozoic geological formation rich in U, Th, and REEs (Schorscher & Shea, 1992), hosts a former uranium mine known as the Ore Treatment Unit (UTM). Operating from 1982 for 13 years, the site generated large volumes of waste capable of producing AMD at a rate of $1.1 \times 10^5 \text{ m}^3/\text{month}$. Although a treatment system based on uranium co-precipitation with calcium hydroxide is in place, it results in the generation of calcium diuranate sludge (Pereira *et al.*, 2018). This study aims to evaluate the performance of two natural adsorbent materials of volcanic origin from Japan—RT22 and HR17—in removing aluminum, iron, manganese, and uranium from AMD collected at the decommissioned uranium mine in Poços de Caldas, contributing to the development of more effective and sustainable treatment alternatives.

2 Experimental Procedure

The water samples analyzed in this study, referred to as BIA and BNF, were collected from two acid drainage basins associated with uranium mining activities. These basins are located within the Caldas Decommissioning Unit (UDC), in the municipality of Caldas, southern Minas Gerais, Brazil. The acidic nature of the waters is attributed to their interaction with waste rock piles rich in sulfide minerals, which promote the generation of AMD.

To assess the acidity of the samples, pH measurements were performed using a Digimed DM-22 pH meter, with a resolution of 0.01, at a controlled temperature of 20°C. For elemental analysis, the concentrations of metals in the BIA and BNF solutions were determined using Inductively Coupled Plasma Tandem Mass Spectrometry (ICP-MS/MS), with an Agilent 8800 instrument. The system operated at a radiofrequency power of 1550 W and a nebulizer gas flow rate of 1.09 L/min, equipped with

nickel sampler and skimmer cones. Two gas modes were employed: a no-gas mode for elements without interference (sample depth 7 mm), and a helium mode for elements affected by interferences (sample depth 10 mm). Nitric acid was used for both acidification and sample digestion, and calibration was performed using a multielement standard solution (2A, SPEX, Horiba Jobin Yvon).

2.1 Adsorption Tests

The adsorbent materials RT22 and HR17 were tested for their ability to remove metals from the BIA and BNF solutions. Each experiment involved adding 0.1 g of adsorbent to 100 mL of solution in Erlenmeyer flasks, followed by continuous agitation for 24 hours using a bench shaker. All tests were conducted in duplicate, resulting in a total of four experiments. During the adsorption process, the pH of the solutions was monitored at three intervals: immediately after contact (0 h), after 1 hour, and after 24 hours, using the same equipment and conditions described previously. After the contact period, the mixtures were vacuum filtered through membrane filters. The concentrations of iron, manganese, and aluminum in the filtrates were then quantified using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES), with a Varian LIBERTY RL Sequential instrument.

Figure 1 presents a schematic representation of the experimental procedure used in this study to evaluate the adsorption efficiency of the natural materials RT22 and HR17 in removing metals from acid mine drainage (AMD). The workflow begins with the collection of water samples from two drainage basins, BIA and BNF, located at a decommissioned uranium mining site. The samples were first analyzed for pH and metal concentrations using ICP-MS/MS. Subsequently, adsorption experiments were conducted by placing 0.1 g of each adsorbent in contact with 100 mL of the AMD solution under continuous agitation for 24 hours. During this period, pH measurements were taken at three time points: 0, 1, and 24 hours. After the contact time, the solutions were vacuum filtered, and the concentrations of iron, manganese, and aluminum in the filtrates were determined using ICP-OES. This diagram provides a clear and concise overview of the experimental methodology applied in the study.

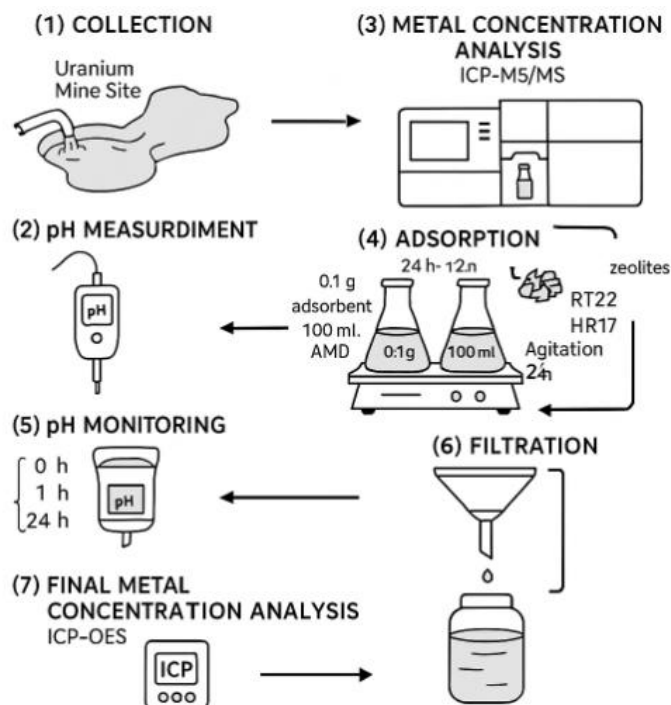


Figure 1: Schematic representation of the experimental procedure for evaluating metal adsorption from acid mine drainage using natural zeolites (RT22 and HR17).

3 Results

The following section presents the results obtained from the characterization of acid mine drainage (AMD) solutions and the subsequent adsorption experiments using natural zeolites RT22 and HR17. Initially, the physical and chemical properties of the water samples collected from the Bacia Ivan Antunes (BIA) and Bacia Nestor Figueiredo (BNF) are described, including visual observations and elemental composition. These data provide the baseline conditions for evaluating the performance of the adsorbent materials. The results of pH variation and metal concentration before and after adsorption are then discussed, highlighting the effectiveness of each material in removing key contaminants such as iron, aluminum, manganese, and uranium.

3.1 Characterization of Acid Drainage Basins and Initial Water Composition

The acid mine drainage solutions referred to as BIA and BNF were obtained from two distinct drainage basins, each named accordingly. These basins are located within the Caldas Decommissioning Unit (UDC), in the municipality of Poços de Caldas, Minas Gerais, Brazil (coordinates: 21°58'S, 46°30'W). Their physical characteristics and locations are illustrated in Figure 2, which presents photographic views of both basins.



Figure 2: Photographic views of the BIA (left) and BNF (right) acid drainage basins located at the Caldas Decommissioning Unit in Poços de Caldas, Brazil.

The acid mine drainage solutions used in this study were collected from two distinct basins: Bacia Ivan Antunes (BIA) and Bacia Nestor Figueiredo (BNF), both located at the Caldas Decommissioning Unit (UDC) in Poços de Caldas, Minas Gerais, Brazil (21°58'S, 46°30'W). These basins are visually represented in Figure 2, which highlights their contrasting physical characteristics. The water in BIA appears more yellowish, a coloration attributed to the presence of precipitated iron compounds, while BNF exhibits a deeper reddish-brown hue, indicating differences in composition and possibly in the oxidation states of dissolved metals. These visual distinctions reflect the unique geochemical conditions of each basin, which are relevant to the adsorption behavior observed in subsequent experiments. The characterization of the composition and acidity of the solution can be seen in Table 1, with the measured element concentration and pH values.

Table 1 presents the elemental concentrations in the BIA and BNF acid mine drainage solutions, as determined by ICP-MS/MS. Both samples exhibit notably high levels of aluminum, exceeding 100 mg·L⁻¹, followed by manganese concentrations above 55 mg·L⁻¹. Calcium was also present in significant amounts, with approximately 29 mg·L⁻¹ in BIA and 20 mg·L⁻¹ in BNF. Other elements detected at elevated levels include zinc, potassium, magnesium, uranium, sodium, and strontium.

Table 1: Concentration of elements present in AMD solutions, pH values and legislative parameters

	BIA		BNF		LIMITS BY LEGISLATION ¹
pH	3.45 ± 0.05		3.67 ± 0.05		5 - 9
	Concentration [μg L ⁻¹]	RSD [%]	Concentration [μg L ⁻¹]	RSD [%]	Concentration [μg L ⁻¹]
Li	16.8	0.2	12.5	2.4	-
Be	34.0	2.7	34.9	2.2	-
Na	3599.2	3.6	1980.7	1.7	-
Mg	7028.4	1.0	5776.4	5.3	-
Al	109991.0	0.7	102498.8	0.9	-
K	9363.7	1.0	7913.2	3.7	-
Ca	28582.6	2.5	19986.3	3.1	-
Mn	67201.0	1.8	58816.1	1.3	1000
Fe	138.5	1.1	242.9	1.4	15000
Co	68.4	1.3	49.8	0.7	-
Ni	44.3	1.2	73.7	0.8	2000
Cu	6.0	4.1	11.7	4.3	1000
Zn	9474.0	1.7	8497.4	0.8	5000
Ga	238.2	2.3	164.7	1.1	-
As	40.0	3.4	28.6	1.0	500
Se	94.3	1.8	72.1	3.9	300
Rb	39.3	0.1	21.1	0.5	-
Sr	2267.4	1.6	1396.2	0.7	-
Cd	24.5	0.8	24.8	1.4	200
Ba	753.1	3.1	580.2	0.5	5000
Pb	69.3	0.3	31.7	1.3	500
Th	140.6	1.2	103.2	3.1	-
U	5001.5	0.9	4515.2	1.5	30

Component concentrations below the detection limit (V, Cr, Ag, Cs, Tl) were suppressed. ¹ The values were obtained from CONAMA (2011), with the exception of the limit for uranium, which is a provisional recommendation updated in 2012 by the WHO.

When compared to the limits established by environmental regulations, the concentrations of manganese, zinc, and uranium are particularly concerning. Manganese levels are approximately 60 times higher than the legal threshold, zinc about twice the limit, and uranium exceeds the recommended value by nearly 170 times. These findings underscore the urgent need for effective treatment strategies, not only due to the high metal content but also because of the extreme acidity of the waters—both BIA and BNF exhibit pH values below 4, well below the acceptable range of 5 to 9 for effluent discharge.

Interestingly, despite the visual evidence of iron precipitation in the BIA basin (as seen in Figure 2), the measured concentration of dissolved iron in both samples was relatively low, below 0.250 mg·L⁻¹. This suggests that a significant portion of the iron may already be present in particulate or precipitated form. Additionally, thorium was detected in trace amounts, with concentrations below 0.150 mg·L⁻¹, indicating the presence of other radionuclides of potential environmental concern.

3.2 Adsorption Efficiency and pH Variation in AMD Solutions

The graph in Figure 3 illustrates the variation in pH over a 24-hour period for AMD solutions from the BIA and BNF basins treated with two different adsorbent materials (RT22 and HR17), as well as control tests using distilled water. The pH of the BIA samples (RT BIA and HR BIA) starts below 4.0 and shows a gradual increase over time, with HR17 promoting a slightly higher pH elevation compared to RT22. For the BNF samples, both RT BNF and HR BNF begin with higher initial pH values—around 4.5—and also exhibit a modest increase, with HR17 again showing a slightly better performance.

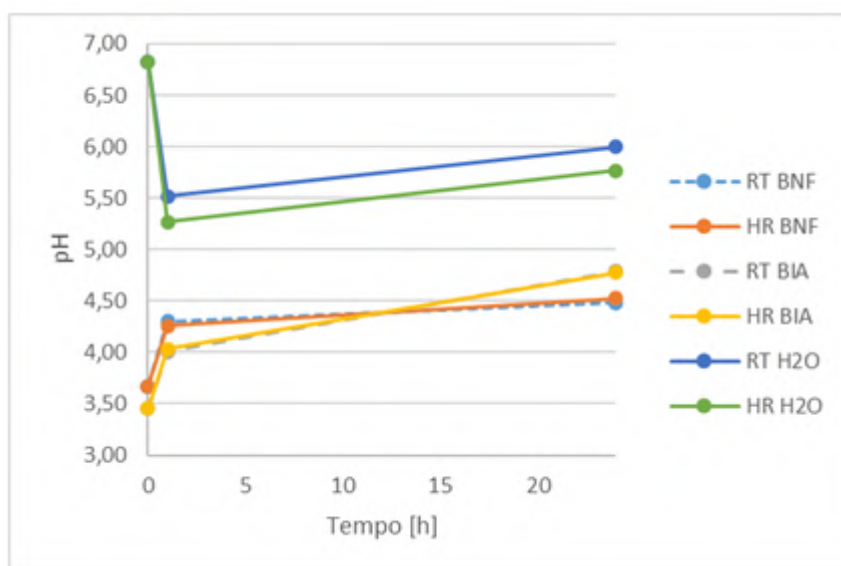


Figure 3: pH variation of BIA and BNF solutions over 24 hours during adsorption experiments with RT22 and HR17, including control tests with distilled water.

In the control samples (RT H₂O and HR H₂O), the pH starts near neutral (~6.5), with RT H₂O showing a slight initial drop followed by recovery, while HR H₂O remains relatively stable. These results suggest that both adsorbents contribute to a partial neutralization of the acidic solutions, with HR17 demonstrating a slightly greater buffering effect, particularly in the BIA sample. This pH increase is relevant, as it may enhance the conditions for metal precipitation or adsorption in subsequent treatment steps.

After the adsorption experiments, the remaining solutions were vacuum filtered and analyzed by ICP-OES to determine the concentrations of iron (Fe), manganese (Mn), and aluminum (Al). The results, summarized in Table 2, reflect the effectiveness of the adsorbent materials RT22 and HR17 in removing these metals from the AMD solutions. Iron was effectively removed in all cases where adsorbents were applied, with concentrations falling below the detection limit of 0.1 mg·L⁻¹. This indicates a high affinity of both RT22 and HR17 for iron under the tested conditions. In contrast, manganese concentrations remained relatively unchanged, suggesting limited adsorption capacity for this element in the short contact time used.

Aluminum showed a more variable response. In the BIA solution, RT22 reduced aluminum by approximately 18%, while HR17 achieved a reduction of about 30%. In the BNF solution, the reductions were around 14% for RT22 and a more significant 38% for HR17. These results suggest that HR17 is more effective than RT22 in adsorbing aluminum, particularly in the BNF sample. Interestingly, small amounts of aluminum were detected in the control samples with distilled water (RT H₂O and HR H₂O), likely due to leaching from the adsorbent materials themselves. This may explain the slight decrease in pH observed in these controls, especially in HR H₂O, which showed the highest aluminum release among them.

Table 2: ICP-OES results from adsorption tests

	Fe		Mn		Al	
	<i>Concentration</i>	<i>RSD</i>	<i>Concentration</i>	<i>RSD</i>	<i>Concentration</i>	<i>RSD</i>
	[mg L ⁻¹]	[%]	[mg L ⁻¹]	[%]	[mg L ⁻¹]	[%]
BIA	2.1	0.8	83.0	0.6	141.8	1.3
RT BIA	< 0.1	-	87.7	3.0	116.3	26.9
HR BIA	< 0.1	-	82.7	0.3	100.0	0.1
BNF	0.2	2.8	73.0	0.7	123.4	0.3
RT BNF	< 0.1	-	73.2	0.2	106.0	0.4
HR BNF	<0.1	-	71.7	1.5	76.1	1.2
H2O	< 0.1	-	<0.1	-	<0.1	-
RT H2O	< 0.1	-	< 0.1	-	0.5	12.9
HR H2O	< 0.1	-	< 0.1	-	0.3	23.8

Overall, the data confirm that both materials are effective in removing iron and moderately effective for aluminum, with HR17 showing superior performance. Manganese, however, appears more resistant to removal under the tested conditions, indicating the need for complementary treatment strategies or longer contact times.

4 Discussion

The results obtained from the adsorption experiments demonstrate the potential of natural zeolites RT22 and HR17 in the treatment of acid mine drainage (AMD), particularly in reducing the concentrations of iron and aluminum, and in partially neutralizing the acidity of the solutions. The complete removal of iron from both BIA and BNF solutions, as indicated by concentrations below the detection limit (<0.1 mg·L⁻¹), confirms the high affinity of both materials for this element under acidic conditions. This behavior is consistent with previous studies that highlight the efficiency of natural zeolites in adsorbing transition metals from aqueous media (Goher *et al.*, 2015; Wulandari *et al.*, 2020).

Aluminum removal was also observed, although to a lesser extent. HR17 showed superior performance compared to RT22, reducing aluminum concentrations by approximately 30% in BIA and 40% in BNF. This difference may be attributed to the higher aluminum content and possibly greater surface reactivity of HR17, as suggested by its composition and structure. The partial removal of aluminum is relevant, as this element is typically present in high concentrations in AMD and contributes significantly to its toxicity (Lobo-Recio *et al.*, 2021).

In contrast, manganese concentrations remained relatively stable in solution after treatment, indicating limited adsorption under the tested conditions. This result aligns with findings from Munawar *et al.* (2023), who reported that manganese removal by zeolites is more effective after prior removal of iron and aluminum, suggesting that competitive adsorption or pH-dependent speciation may limit Mn uptake in early treatment stages.

The pH variation observed during the experiments further supports the effectiveness of the adsorbents. Both materials promoted a gradual increase in pH, with values rising from below 3.7 to nearly 4.8 after 24 hours. Although still below the regulatory threshold for effluent discharge (pH 5–9, CONAMA, 2011), this shift indicates a partial neutralization of the acidic environment, which is essential for improving the conditions for subsequent treatment steps or natural attenuation processes (Markovic *et al.*, 2020).

Overall, the results confirm the potential of RT22 and especially HR17 as low-cost, effective materials for the initial treatment of AMD. Their ability to remove key contaminants and increase pH supports their application in integrated remediation strategies, particularly in regions with similar geochemical profiles to the Poços de Caldas uranium mine.

5 Conclusions

Conclusion. The present study reinforces the viability of using natural zeolitic materials, specifically RT22 and HR17, as adsorbents for the treatment of acid mine drainage (AMD) from uranium mining areas. The materials demonstrated the ability to reduce the concentration of key contaminants and to promote a partial neutralization of the acidic environment, which is essential for improving water quality and enabling further remediation steps. These findings are particularly relevant in the context of decommissioned mining sites, where long-term environmental impacts persist even after the cessation of extractive activities.

Among the two materials tested, HR17 consistently showed superior performance in terms of pH stabilization and aluminum removal, indicating that its physicochemical properties may be more favorable for adsorption under acidic conditions. Although manganese removal was limited under the conditions tested, the results suggest that further optimization of the process could enhance its effectiveness. The complete removal of iron from solution is a particularly promising outcome, as this element is often a major contributor to AMD toxicity and staining.

Overall, the study contributes to the growing body of research supporting the use of low-cost, naturally occurring materials for environmental remediation. The application of RT22 and HR17 aligns with sustainable development goals by offering an accessible and environmentally friendly alternative for AMD treatment. These materials hold potential not only for improving water quality in mining-impacted areas but also for integration into broader treatment systems, especially in regions with similar geological and environmental conditions.

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Development of a Distributed Monitoring System for Active Radiological Protection

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Abstract

To provide a complementary layer to conventional area monitoring in radiative facilities, a distributed monitoring system was developed as an active barrier within the defense-in-depth strategy (CNEN, 2024). This system integrates Internet of Things (IoT) and digital twin (DT) technologies. IoT has several applications in the nuclear field, such as the environmental radiation monitoring network implemented at CERN (Manzano *et al.*, 2021); proposed network for monitoring research laboratories where radioactive materials are handled (Rocha *et al.*, 2020); and systems designed to detect radioactive sources outside regulatory control (Tran-Quang; Dao-Viet, 2022). The state of the art in DT applications within the context of ionizing radiation has recently been reviewed in an article that synthesizes the main findings across several academic databases (Macedo *et al.*, 2024). These include the use of digital twins as virtual representations of equipment (Sumini *et al.*, 2024), as well as their implementation in nuclear power plants for energy production monitoring and operational simulation (Kochunas *et al.*, 2021). The system developed in this study, named the Intelligent System for Dosimetry Data Monitoring and Collection (in Portuguese *Sistema Inteligente de Monitoração e Coleta de Dados de Dosimetria – SIMCDD*), is primarily based on open-source and low-cost components, with an emphasis on adaptability and customizability. This enables the system to be configured according to the specific monitoring requirements of each facility. While most studies focus on applying digital twins to simulate equipment or processes in nuclear power plants, SIMCDD proposes a novel integration of low-cost IoT networks with digital twins accessible via local web interface, specifically aimed at active radiological protection in monitored environments. This approach represents a practical and innovative advancement by enabling real-time interaction between a distributed sensor network and an accessible three-dimensional (3D) digital model. It adds an additional layer of active radiological protection using open and affordable technologies, thereby increasing the feasibility of implementing defense-in-depth strategies in smaller-scale institutions. The network of devices measures ambient dose equivalent rates using Geiger-Müller (GM) detectors managed by Arduino microcontrollers. Data is transmitted to a gateway, which stores it in a server-side database. Users can access monitoring data through a web interface hosted on the local network. The same server also hosts a customizable digital twin of the monitored facilities. A partial digital twin of CDTN is currently under development. In the future, real-time data collected by the IoT detectors will be associated with a 3D model of specific locations within CDTN, where detectors may be deployed.

Keywords: Internet of Things; Digital Twins; Radiation Dosimetry; Monitoring; Open Source.

1 Introduction

The CNEN-NN-3.01 Standard (2024) establishes the basic requirements for radiation protection, defining the responsibilities and actions that must be adopted to ensure the safety of individuals and the environment in scenarios involving exposure to ionizing radiation, where the standard is applicable. It

also aims to prevent undesired effects caused by such exposures. For the purpose of controlling or assessing exposure to ionizing radiation, the measurement of specific related quantities is performed. This process is known as monitoring (CNEN, 2024).

Radiation dosimetry aims to determine the amounts of energy transferred by ionizing radiation sources. It is used in monitoring to determine quantities known as dosimetric quantities, which involve radiation interaction factors with matter as well as radiation field factors (ICRU, 2011; LNMRI, 2011; CNEN, 2024). In facilities where ionizing radiation is used, monitoring methods are adopted according to the activities performed. One such method is area monitoring, which consists of designating a workplace area to be monitored through measurements taken at different points. The selection of these points and the number of instruments must ensure adequate representation of the activities carried out (IAEA, 2014, 2018).

The Internet of Things (IoT) can be defined as a network of identifiable devices, embedded with intelligence and equipped with sensing and actuation capabilities, enabling the connection of objects and people through the internet (Rayes; Salam, 2019). Emerging technologies such as IoT have seen significant growth in applications within the nuclear field, such as the IoT-based environmental radiation monitoring network at CERN (Manzano *et al.*, 2021); an IoT network proposed for monitoring research laboratories handling radioactive materials (Rocha *et al.*, 2020); and systems for detecting orphan radioactive sources (Tran-Quang; Dao-Viet, 2022). This technology enables the design of a distributed monitoring system associated with a digital twin, allowing monitored data to be linked to a three-dimensional model of the facilities.

The digital twin (DT) is a concept defined as a mapping of the real world into the digital world, enabling real-time interaction between the two (Yao *et al.*, 2023). A recent article discussed the state of the art of DT applications in the field of ionizing radiation, compiling the main findings from various databases (Macedo *et al.*, 2024), such as the use of DTs as virtual representations of equipment (Sumini *et al.*, 2024), or their application in nuclear power plants, focusing on energy production and system simulations (Kochunas *et al.*, 2021).

Given this context, the present work initiated the development of a distributed monitoring system as an active barrier within the defense-in-depth strategy. This system combines Internet of Things and digital twin technologies to provide an additional layer of active radiation protection using open and low-cost technologies, thereby expanding the applicability of defense-in-depth to smaller institutions. This active layer complements monitoring in occupational environments by enabling faster identification of variations in radiation fields in monitored areas—variations that cannot be promptly detected by passive systems. This system, named the Intelligent System for Dosimetry Data Monitoring and Collection (in Portuguese *Sistema Inteligente de Monitoração e Coleta de Dados de Dosimetria* – SIMCDD), is composed of wireless networks to enable real-time monitoring, allowing for continuous and accurate analysis of radiation exposure. The use of low-cost devices facilitates the deployment of active detectors alongside passive ones, and the implementation of a digital twin developed with open technologies and accessible via the local network. The direct, real-time association between distributed sensors and an accessible three-dimensional model represents a practical and unprecedented advancement in this field.

2 Materials and methods

The version of the system presented in this work was developed to enable test-mode operation and to perform proof of concept. For this purpose, a minimal system architecture was established, consisting of: a detection device; active communication with a gateway; an operational server; data writing to a database; and a preliminary data visualization webpage. This setup allows for evaluation of the system's operation, enabling early identification of points for improvement and correction. Figure 1 shows a schematic representation of this minimal architecture.

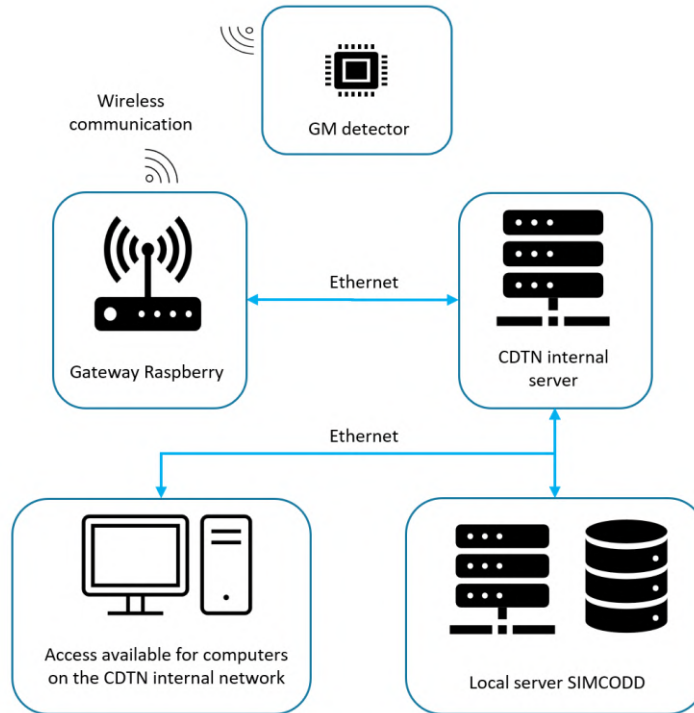


Figure 1: Minimal architecture used for SIMCodd.

The smart detector developed for on-site monitoring consists of a Geiger-Müller type detection module, controlled by an Arduino Nano, a widely used open-source platform for developments of this nature. This module also includes a 16x2 LCD display for local reading of the data and an audible alarm for alerts. The GM tube used is the Geiger J305, which operates at 380 V. The module’s circuit board contains the necessary components to support the detector’s proper functioning and power supply. The board’s interface also allows adaptations to include additional components.

Through this interface, while keeping the Arduino Nano as the controller, the NRF24L01 module was added to enable wireless communication via radio frequency in the 2.4 GHz ISM band (Industrial, Scientific, and Medical). This component is a transceiver (capable of both sending and receiving data) that supports communication over 126 physical channels and can interact with up to six other modules per physical channel through logical channels. It allows for data packets of up to 32 bytes to be transmitted, with ranges varying from 100 to 1,000 meters when using a suitable external antenna (Nordic, 2007; Eletrogate, 2024). This module can consume up to 115 mA during peak transmission, which exceeds the current supply limits of the Arduino. Therefore, a 18650 battery module was added to provide adequate power to the device. This feature is also crucial to enable the device’s mobility, expanding its usability in locations where power outlets are not available.

Initially, the device assembly was performed using a temporary setup with a breadboard and auxiliary wiring, allowing for testing and easy modifications. This approach simplifies adjustments in case changes to components or layout are required. Once the final hardware configuration is defined, a custom printed circuit board (PCB) will be designed to integrate the GM module and other components, providing a definitive, production-ready assembly. The current assembly of smart device components can be seen in Figure 2.

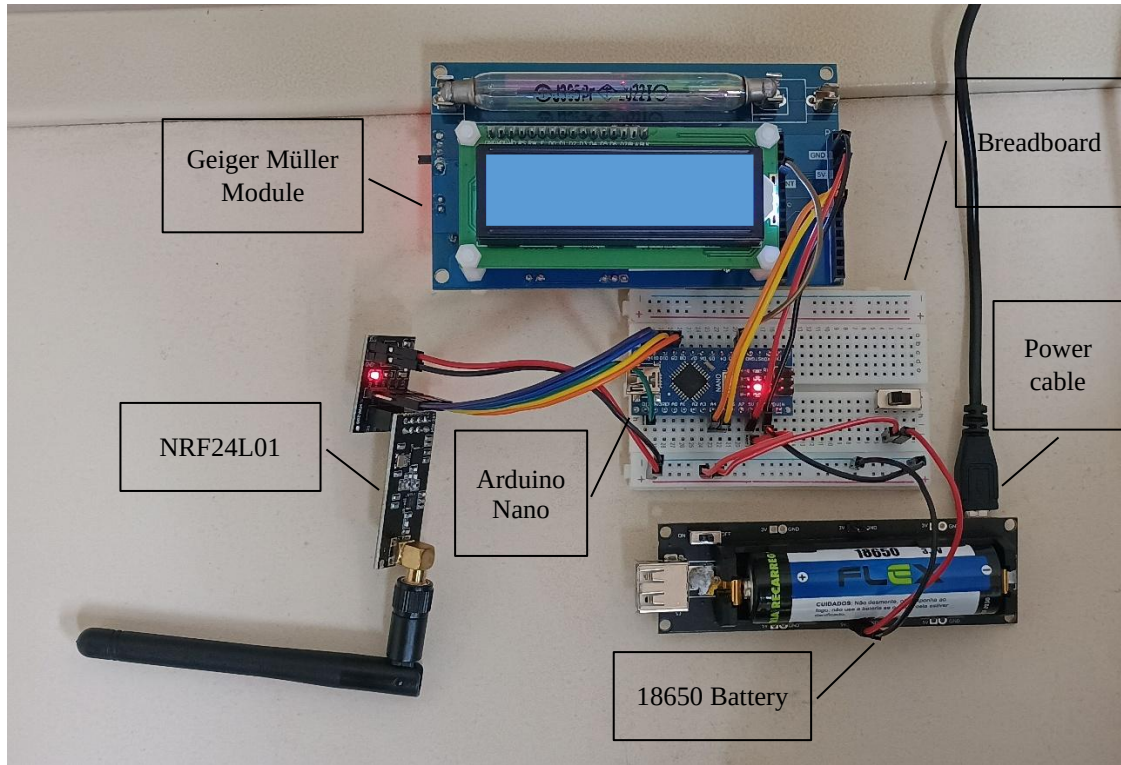


Figure 2: Assembly of the smart detection device.

This detector was calibrated at the Dosimeter Calibration Laboratory (LCD/CDTN), in the Gamma Calibration Room, using an STS-brand irradiator, model OB85/3, with a Cs-137 source. The calibration scope adopted was ambient dose equivalent rate. Readings were taken from 2.5 $\mu\text{Sv/h}$ up to 80 $\mu\text{Sv/h}$, with source emission traceability starting at 4 $\mu\text{Sv/h}$. Even so, these parameters and limitations were considered acceptable for this study and for the proposed application.

The reference dose rate used for detector configuration is based on the whole-body effective dose limit for occupationally exposed individuals (OEIs), as established by CNEN-NN-3.01 (2024), Subsection III – Dose Limitation, which sets the annual limit at 20 mSv per administrative year (January to December). Accordingly, the dose rate value used was 2.28 $\mu\text{Sv/h}$, as shown in Equation (1).

$$\dot{D} = \frac{dD}{dt} = \frac{20,000}{8760} = 2.2831 \left[\frac{\mu\text{Sv}}{h} \right] \quad (1)$$

The gateway plays a fundamental role in the device network, as it typically requires greater processing power, more elaborate routines to manage devices and received data, and interfaces with the server and database. Due to these requirements, the board chosen for developing the gateway is a Raspberry Pi 3B, a compact computer with 40 General Purpose Input/Output (GPIO) pins, allowing for connection to external modules, sensors, and other boards. This board features 1 GB of RAM and native interfaces for Wi-Fi, Ethernet, HDMI and USB.

It supports several Linux distributions, including an official version optimized for the hardware, based on Debian Linux, called Raspberry Pi OS. This is the operating system used for the gateway, offering development capabilities in C/C++ and Python for the creation of custom routines for specific tasks, in addition to the native functionalities of the board and OS. Therefore, the gateway's operational routine was developed in C/C++, using libraries for GPIO access, NRF24L01 module communication, and database connection. The gateway assembly can be seen in Figure 3.

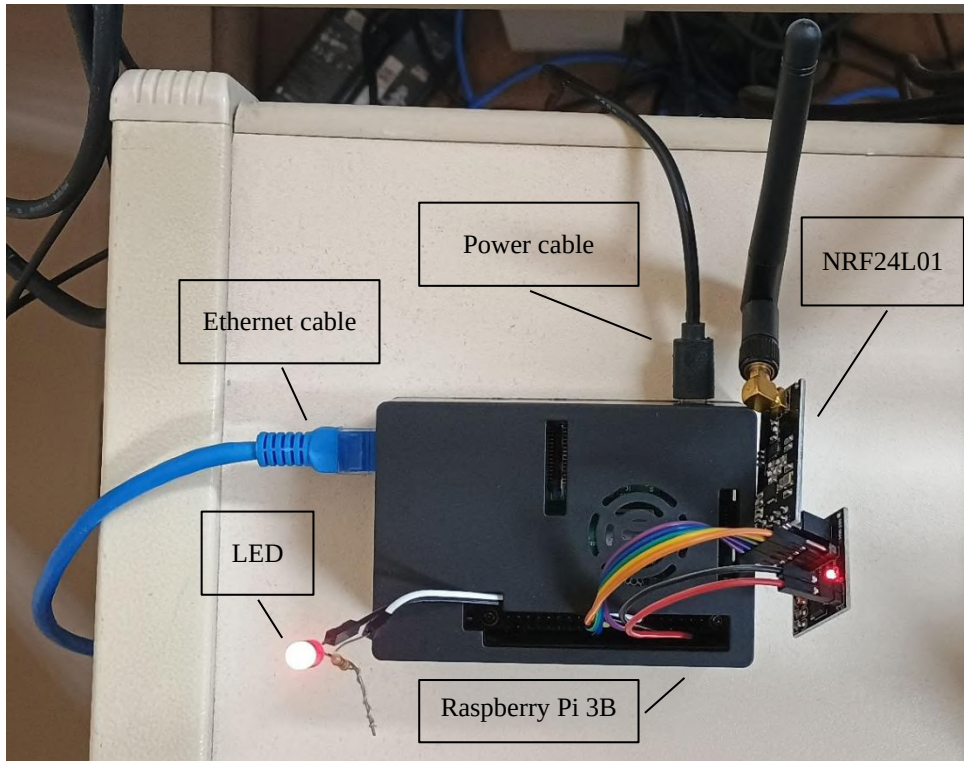


Figure 3: Gateway assembly.

The data collected by detection devices are centralized at the gateway but are not stored locally, neither by the gateway nor the detectors, only displayed temporarily on the LCD. The data is saved at the time the gateway writes them to the local server's database. This server runs Ubuntu 22.04 (Linux) and functions both as a web server and a database server. The server hardware is a SilverStone CS01-HS desktop with an AMD FX-9830P processor, 16 GB of RAM, a 480 GB SSD, and a 4 TB HDD. This computer is intended for local data storage and edge computing. The applications developed for querying, visualizing, and utilizing the stored data are made available through webpages fed by the system's database.

The web server used to host the application pages is Apache, an open-source HTTP/HTTPS server that requires minimal system resources to operate. Apache is a secure and widely adopted solution by companies and institutions of all sizes and has remained the most popular web server on the internet since 1996 (ASF, 2024). The database server employed is MariaDB, also open-source software, which enables structured data storage. It is a relational database, where data can be accessed through the Structured Query Language (SQL) interface, and is one of the most widely used database systems worldwide (MariaDB, 2024). This local server is connected to CDTN's internal network and can be accessed via the institutional domain, making the applications available internally to any computer connected to the wired network. Limiting access to the local network is also a cybersecurity measure, ensuring the entire system adheres to the same information security policies and requirements already in place at the center.

Initially, the applications were developed to allow real-time visualization of monitoring data through a chart and complementary information. This first version offers a minimal interface to support the start of test-mode operation and proof of concept. In parallel, development of the digital twin (DT) will be carried out, enabling data to be visualized in association with a 3D model of CDTN's facilities, which will be monitored using smart detectors.

For the development of CDTN's DT, 3D modeling will be performed using FreeCAD, an open-source software that generates models in the IFC format (Industry Foundation Classes). FreeCAD is a parametric 3D CAD modeler that enables easy modeling of geometries. It is compatible with multiple operating systems (cross-platform), highly versatile, and supports various open file formats such as STEP, STL,

DXF, and IFC. These features facilitate their integration into collaborative development workflows and interoperability with other tools (FreeCAD, 2025). IFC is a global open standard for digital descriptions of built assets, developed by buildingSMART International, alongside other standards that address the industry's interoperability challenges (BSI, 2025). IFC files can be visualized through standard web browsers using open-source JavaScript libraries specifically designed for this type of application, adding interactivity to browser-based web applications. The 3D model files will be stored on the Apache server, which will serve them to browsers accessing the application. Through the JavaScript libraries, these files can be loaded and rendered directly in the user's browser, without requiring specialized or high-performance software to access CDTN's digital twin. Figure 4, illustrates the representation of the integrated system and access to the digital twin.

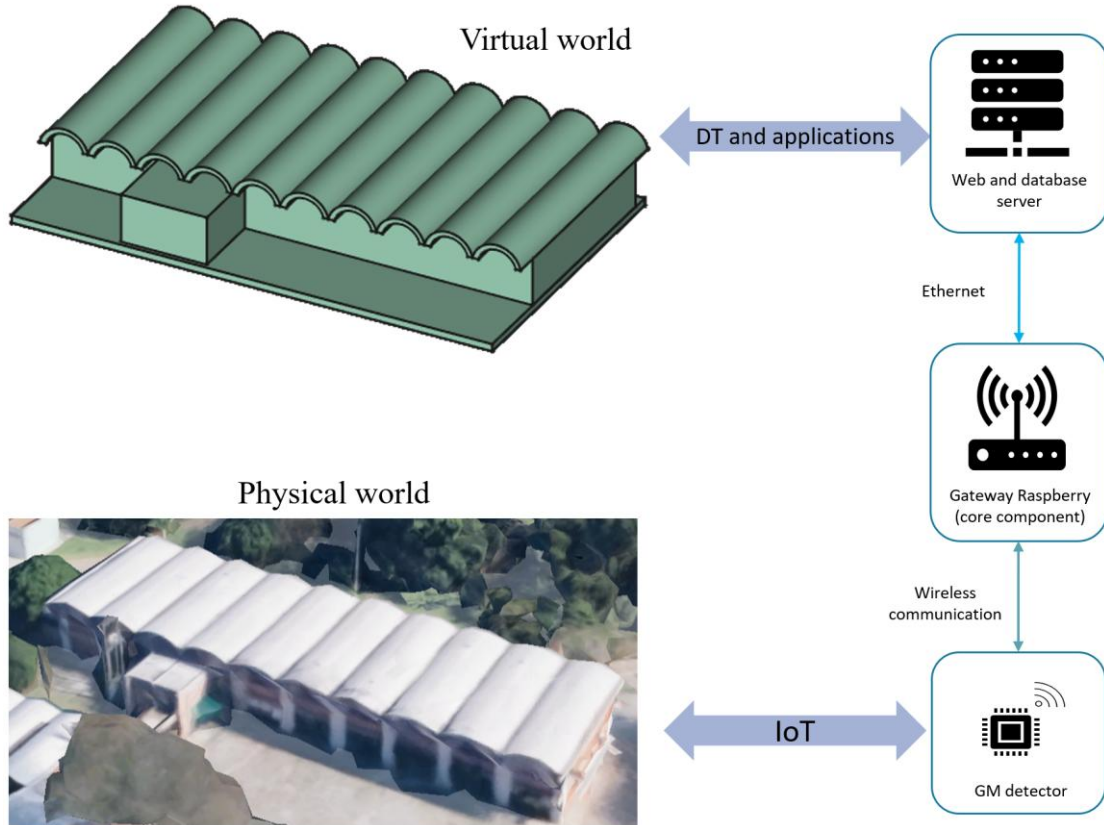


Figure 4: Integrated system representation.

In this context, the developments and testing aim to achieve a distributed monitoring system, integrated via a wireless network, which enables measurement data to be recorded in a database and provides access to the 3D facility model through the local network. The main tests carried out in this phase were: evaluation of the functionality of the code developed for each device; assessment of data transmission time; database write stability; and data availability on the visualization pages.

3 Results and discussion

Smart devices provide feedback messages after attempting to transmit data to the gateway, allowing for monitoring of both transmission status and communication stability. These messages are displayed on the LCD screen along with the radiation readings. Figure 5 shows examples of successful and failed data transmission feedback

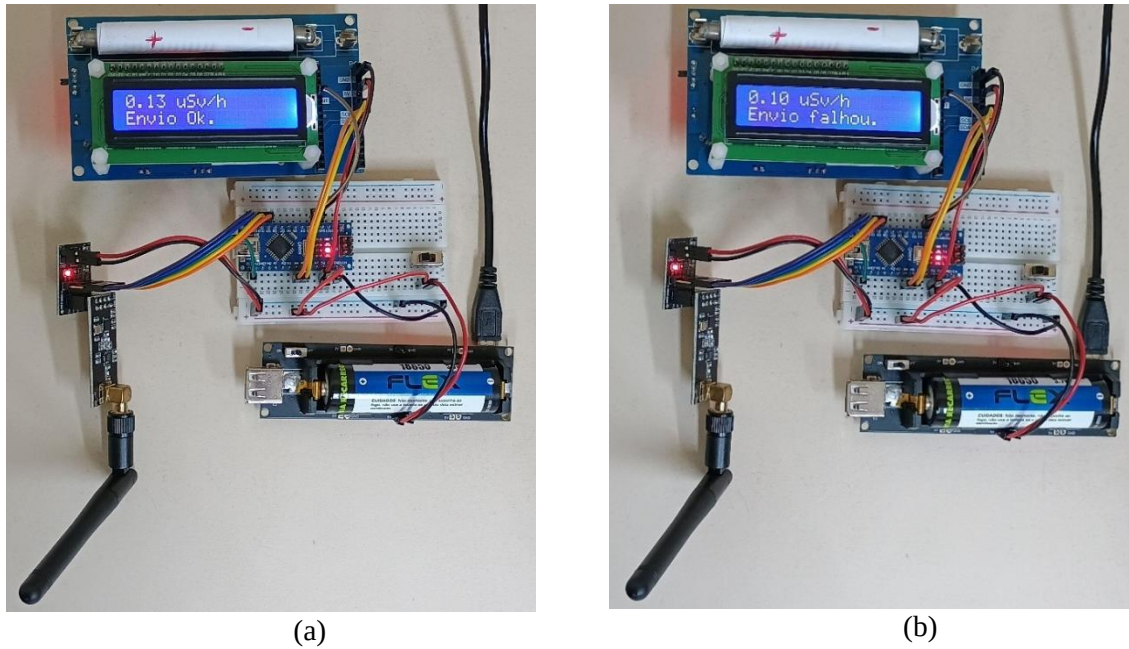


Figure 5: Device reading and data transmission feedback. (a) Successful transmission. (b) Failed transmission.

During detector calibration, measurement errors ranging from 10% to 30% were observed, with all results falling below the expected values. The error increased in correlation with the dose rates, being more pronounced at higher dose rates, particularly above 10 $\mu\text{Sv/h}$. These levels of deviation from expected values exceed the accuracy parameters targeted for smart devices. For this reason, a new calibration will be performed in order to obtain a correction factor or model to be implemented in the detector. This evaluation process has already been initiated and remains ongoing.

To monitor the operation of the gateway, information on the received data packets and their recording in the database is displayed in the terminal of its operating system. In addition, to assess communication stability, packet loss is also monitored. A progressive improvement in packet reception and system uptime was observed, as shown in Table 1. However, the system still experienced communication failures after a certain period of continuous operation. The use of a module operating in the 2.4 GHz band is subject to significant interference and noise, as it shares this frequency range with technologies such as Wi-Fi and Bluetooth. This communication instability is not considered acceptable within the performance requirements set for the system.

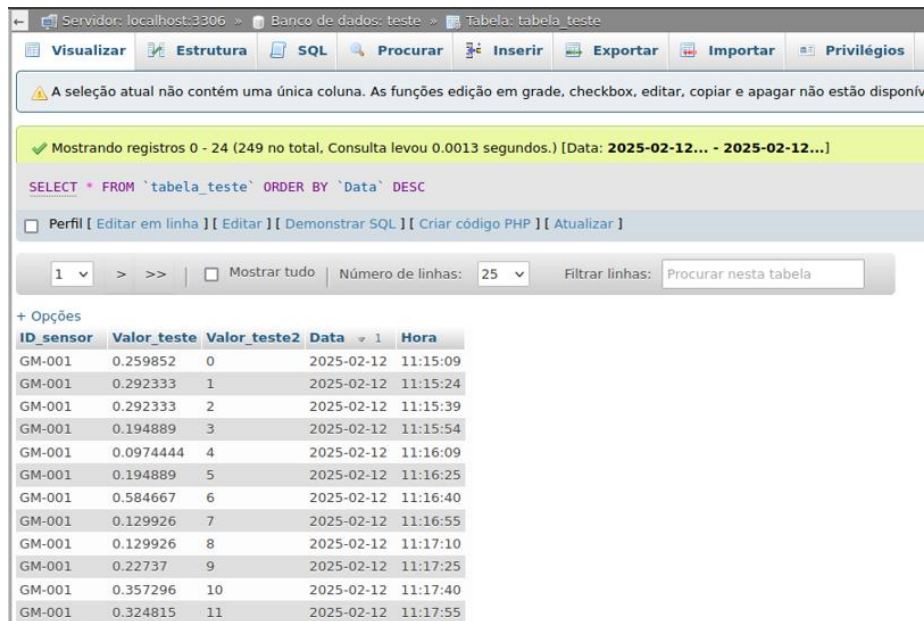
Table 1: System performance testing data.

Channel (MHz)	Transmission power mode	Uptime (h)	Packet loss (%)
2500	Low	19.3	28.15
2402	High	94.8	1.917
2402	High	29.4	None
2402	High	140.5	None

These results were obtained using the system's minimal configuration, with a single smart device in operation. When additional devices are integrated into the network, the routines will be updated, and these evaluations will be repeated to reflect the new system conditions.

The web server was configured and connected to CDTN's internal network. This configuration included setting the hostname within the network, enabling security layers, and structuring the server's directories

and files. The pages created can be accessed within the user domain network at the address simcodd.usuarios.cdn.br. At this stage, a test database was created to store temporary data collected during the proof-of-concept tests, as shown in Figure 6.



ID_sensor	Valor_teste	Valor_teste2	Data	Hora
GM-001	0.259852	0	2025-02-12	11:15:09
GM-001	0.292333	1	2025-02-12	11:15:24
GM-001	0.292333	2	2025-02-12	11:15:39
GM-001	0.194889	3	2025-02-12	11:15:54
GM-001	0.0974444	4	2025-02-12	11:16:09
GM-001	0.194889	5	2025-02-12	11:16:25
GM-001	0.584667	6	2025-02-12	11:16:40
GM-001	0.129926	7	2025-02-12	11:16:55
GM-001	0.129926	8	2025-02-12	11:17:10
GM-001	0.22737	9	2025-02-12	11:17:25
GM-001	0.357296	10	2025-02-12	11:17:40
GM-001	0.324815	11	2025-02-12	11:17:55

Figure 6: Data stored in the test database as written by the gateway.

These data were used to populate the interface created for visualizing the readings and accessing the information collected during the tests. Figure 7 shows the graph generated from the dose rate data recorded over time, including the corresponding date and time of each reading. This screen currently displays all data stored in the database. As system development progresses, functionalities will be added to allow filtering by time periods and other improvements to the data query interface for system users.

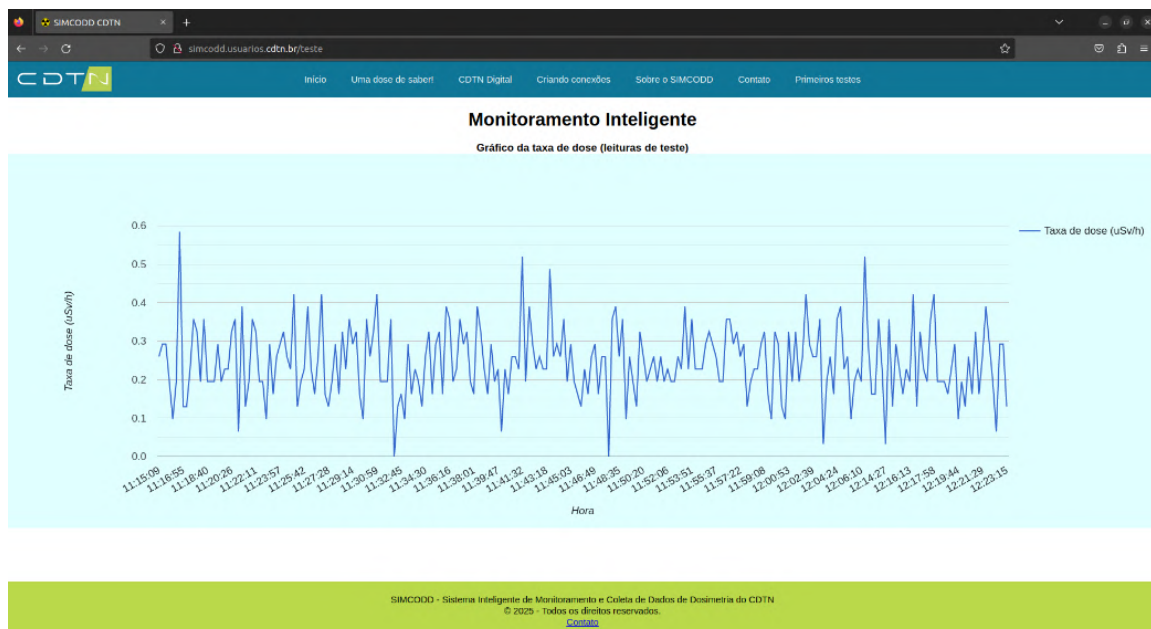


Figure 7: Graph of test readings.

For the development of the digital twin, work has begun on building the 3D model visualization environment, in which information from the devices and monitoring points will be linked. It has not yet been possible to load geometries of CDTN facilities. However, it is possible to visualize an example 3D model available in one of the libraries used for developing the DT environment, as shown in Figure 8.

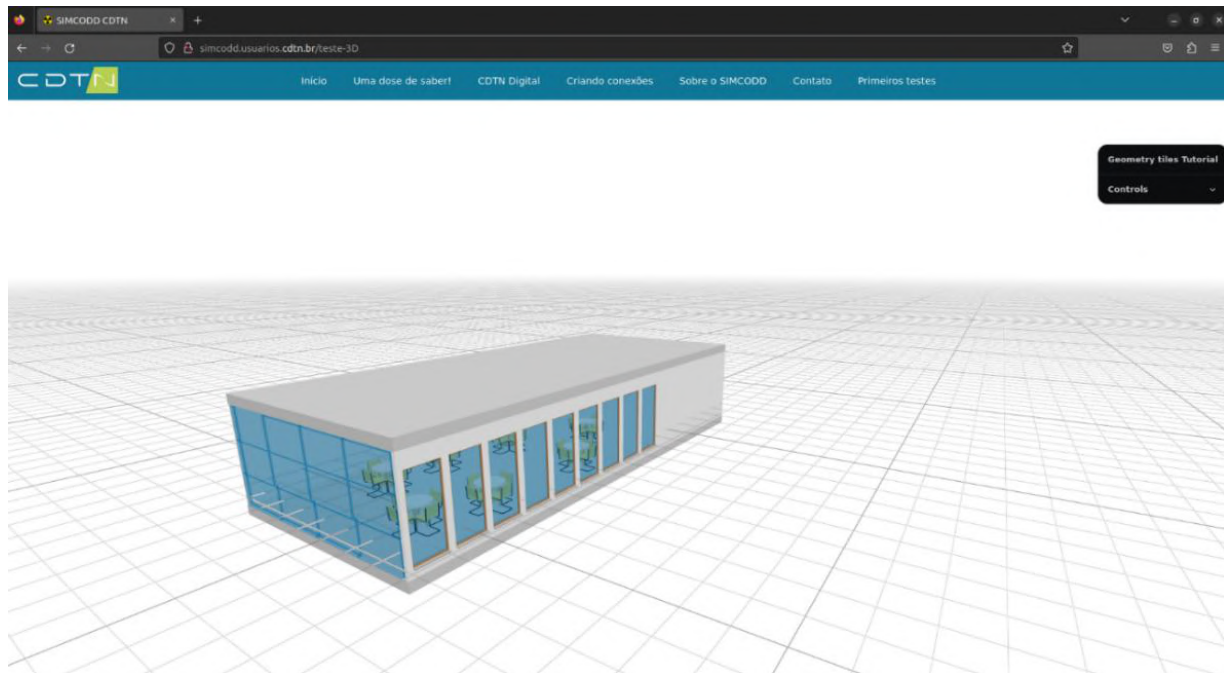


Figure 8: Example visualization of a 3D model.

4 Conclusions

Defining a minimal system configuration enabled the proof of concept to be carried out, demonstrating the system's ability to perform and transmit measurements, record them in a database, and make this data accessible within the local network. This proof of concept validates the system as a viable tool to be used as a complementary layer to area monitoring in radiative facilities, serving as an active barrier within the defense-in-depth strategy.

The direct association, in real-time, between distributed sensors and an accessible three-dimensional model represents a practical and unprecedented advancement in this field. The implementation of a digital twin that allows real-time monitoring of an occupational area can serve as a protective measure for occupationally exposed individuals (OEIs), enabling the use of active detectors alongside passive ones and thereby reducing part of the occupational exposure risks.

The adoption of open technologies and low-cost components expands the applicability of defense-in-depth strategies to smaller institutions, allowing them to integrate an active monitoring layer in addition to current practices. Considering the use of five detectors and a standard desktop computer as a server, a system like the one presented here can be implemented at a cost of less than 10,000 BRL, a value significantly lower than that of commercial systems currently available on the market.

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Dose Evaluate and Suggestion of Typical Value for Head CT Headache Protocols: a Preliminary Study

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Abstract

Headache is one of the most prevalent disorders in the world population^[1] and, in some cases, computed tomography (CT) scans are an essential tool for evaluating this condition. Therefore, knowledge about the doses in CT head scans, to which these individuals are subjected, is important to evaluate the possibility of optimizing protocols. Objective of this study is to evaluate the volume computed tomography dose index (CTDI_{vol}) and dose length product (DLP) performed in adult CT head protocols for clinical indications of headache in order to suggest a “typical value”. Preliminary data were collected in the first quarter of 2025 in two different CT equipment of the university hospital. Typical value was estimated according to ICRP 135^[2] and the effective dose was calculated using dose coefficients obtained by Monte Carlo^[3]. The results demonstrated that the typical value for adult head CT protocols with clinical indication of headache is 35.2 mGy and 825.1 mGy.cm to CTDI_{vol} and DLP, respectively. Effective doses (E) estimated by the dose coefficient range from 1.1 to 2.6 mSv. The mean effective dose was 1.9 ± 0.6 mSv. The mean age of the patients was 54 years. No significant relationship was found between the dose values and the physical characteristics of the patient based on the body mass index because these are head CT protocols. Typical values are an important tool for monitoring CT doses in hospitals and clinics. Through continuous monitoring and evaluation of doses for specific clinical indications is possible to identify atypical practices and evaluate the possibility of “optimizing” protocols under the “as low as reasonably achievable” (ALARA) radiation protection principle.

Keywords: Computed Tomography, typical value, dosimetry, headache

1 Introduction

Headache is one of the most prevalent disorders in the world population (Stovner et al, 2022) and, in some cases, computed tomography (CT) head scans are an essential tool for evaluating this condition. Head CT scans are among the most commonly performed examinations (Modlinska et al, 2024) (Dieckmeyer et al, 2023) (Tan W S et al, 2023). Thus, evaluate radiation dose from this scans is a way to optimizing protocols. ICRP 135 (2017) suggests diagnostic referenve level (DRLs) and typical values as a way to evaluate and optmize doses from diagnostic exams that use ionizing radiation.

According to ICRP, DRLs are fundamentally divided into national DRLs, when data samples are collected at a national level (country); regional DRLs, when values of already established national DRLs are gathered from different countries or a continent; and local DRLs, when the collected data originates from a neighborhood or city, for example, where the number of devices is small (10 to 20) or a national DRLs has not yet been established. Data collected for DRLs suggestions within each institution is evaluated using median values, and suggested DRLs are estimated by the value of the third quartile (3rd) or the 75th percentile of this data distribution.

However, “typical values” are estimated when the number of participating institutions or equipment is too small to determine local diagnostic reference levels. Typical values can be determined for a single institution by promoting a comparative method for dose evaluation and potential protocol optimization (ICRP, 2017). Due to being median values and involving few pieces of equipment in the analysis, “typical values” are more restrictive than the DRLs. Such reference values are not dose limits and should not be compared or obtained for evaluating individual patient examinations. It is important to emphasize that achieving good image quality for clinical diagnosis is a priority, taking into account the principles of justification and optimization of procedures (ICRP, 2007^a).

In CT scans, reference values are obtained through the Volume Computed Tomography Dose Index (CTDI_{vol} mGy) and Dose Length Product (DLP – mGy.cm). DRLs and “typical values” have been established based on anatomical regions, weight, and age, without considering clinical indications (Tan W S et al, 2023) and the variety of specific protocols existing in different institutions. The objective of this study is to evaluate CTDI_{vol} and DLP performed in adult CT head protocols for clinical indications of headache in order to suggest a “typical value.” Effective dose estimates for this procedure were also obtained.

2 Materials and Methods

Preliminary data were collected from 40 patients that underwent head CT scans for headache clinical indication in the first quarter of 2025. Two Philips 64 slice CT equipments were used for the study. Consent forms were signed and data from exams was obtained retrospectly from the local PACs. Image iterative reconstruction was used. Table 1 presents the aquisition parameters from head protocols.

Table 1: Aquisition parameters from CT equipments.	
Parameters	Head
Tube corrente (mAs)	200-250
Tube voltage (kV)	120
Slice thickness (mm)	0.5
Detector coverage (mm)	40mm
Pitch	0.4
Gantry rotation time (s)	0.5

Typical value was estimated according to ICRP 135 (2017) and the mean and standard deviation, effective dose was calculated using dose coefficients obtained by Monte Carlo (2007). Patients mean

age and standard deviation were calculated. Statistical analysis was conducted to evaluate the BMI influence on the CTDI and DLP values.

3 Results

The results demonstrated that the “typical value” (median) for adult head CT protocols with clinical indication of headache is 35.2 mGy and 825.1 mGy.cm to CTDIvol and DLP, respectively. Effective doses (E) estimated by the dose coefficient range from 1.1 to 2.6 mSv. The mean effective dose was 1.9 ± 0.6 mSv. Table 2. presents statistical analysis using Spearman’s rank correlation.

Table 2. Spearman’s rank correlation		
	BMI	
	ρ	p-value
CTDI	0.184	0.25
DLP	0.097	0.55

Statistical analysis did not presented significative correlation $p > \alpha$ to 0.05 level significance. Table 3 presents typical values (median), 3rd quartile, mean and variation coefficient values from CTDIvol, DLP and effective dose. The mean age of the patients was 54 ± 15 years.

Table 3. Typical values, 3 rd quartile, mean and variation coefficient values			
	CTDIvol (mGy)	DLP (mGy.cm)	Effective dose (mSv)
Typical values (median)	35.2	825.1	1.73
3rd quartile	35.2	873.8	1.83
Mean	36.1 ± 4.2	898.9 ± 285.2	1.9 ± 0.6
Variation Coefficient (%)	11.6	32.0	32.0

Results showed a high standard deviation (285.2) and a greater coefficient of variation (32.0) derived from the DLP and effective dose data. These data may suggest positioning errors or poor delineation of the image acquisition limits during the scan in head protocols.

Meanwhile, the preliminary typical values of CTDIvol and DLP showed percentage values lower than those obtained in the study by Eray Atli et al. (2021), 52 mGy (-32.3%) and 988 mGy.cm (-16.5%) for CTDIvol and DLP, respectively.

4 Conclusions

This work demonstrated that head protocols for headaches clinical indication can be optimized due to the results related to DLP values. It is important to emphasize that there are few studies related to obtaining "typical values" (2nd - median) available in the literature. For this reason, it is necessary to disseminate this data aimed at improving diagnostic services in CT and reducing doses to patients, taking into account the clinical indication and the quality of the obtained image.

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Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via Gamma-ray Spectrometry and Algorithmic Data Processing

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Abstract

The accuracy in determining radiopharmaceuticals half-life constitutes a fundamental foundation for metrological reliability in various applications, particularly in nuclear medicine. Parameters such as administered activity, patient effective dose, and imaging time windows depend directly on this nuclear quantity. Errors in its quantification compromise diagnoses, generate waste of high-cost supplies, and expose patients to unnecessary risks. The intrinsic complexity of these measurements demands robust methodologies, such as instrumental fluctuations, radioactive decay during data acquisition and spectral interferences introduce critical uncertainties. The method using high-purity germanium gamma-ray spectrometry emerges as a technical solution by employing counting systems to correct spectrometer variations, establishing a reliable benchmark throughout the measurement process. Furthermore, the integration of computational techniques via Python revolutionizes spectrometric data processing. Signal processing algorithms can filter electronic noise, while curve-fitting routines (e.g., exponential decay models) automate complex calculations. Automated and reusable libraries enable real-time decay factor corrections, reducing human errors and analytical inconsistencies. Prior computational simulations optimize acquisition parameters (counting time, detector settings), while automation of repetitive calculations enhances reproducibility. Comparative studies demonstrate up to 40% reduction in expanded uncertainties when these techniques are combined, particularly for ultra-short half-life radionuclides such as ¹⁸F, ²⁰²Tl and ¹²³I. The impacts extend beyond the laboratory, affecting the entire clinical chain: precise calibration of dose calibrators, validation of quality control protocols, and calculation of expiration dates for radiopharmaceuticals kits directly derive from these measurements. For therapeutic radiopharmaceuticals, where minimal error margins define therapeutic efficacy versus toxicity, half-life accuracy becomes an ethical imperative. We present a hybrid methodology designed for reliability and metrological enhancement in determining nuclear decay parameters —such as half-lives—, particularly for radiopharmaceuticals applications using high-resolution gamma-ray spectrometry. This approach integrates advanced experimental techniques, computational tools and data science. It not only comply with regulatory requirements (CNEN NE 3.05, ISO 16117) but ensures that innovations in radiopharmacy can deliver effective clinical safety from bench to bedside.

Keywords: Half-life; Radiopharmaceutical; Gamma-ray spectrometry; Data algorithmic processing.

1 Introduction

Accurately determining the half-life of radiopharmaceuticals is an essential metrological parameter for quality assurance in diagnostic and nuclear therapy procedures. This quantity directly influences the dose administered to the patient, the optimal intervals for image acquisition, and the validity of radiopharmaceuticals. Small uncertainties in its quantification can result in undue exposure, erroneous clinical interpretations, and significant financial losses. For example, inaccuracies in the half-life values of radiopharmaceuticals, such as ^{18}F (109.74 min), ^{123}I (13.2 h), or ^{202}Tl (12.2 days), can directly impact the dose administered to the patient. This happens because radioactive activity decays exponentially and is directly proportional to the exposure or dose to be delivered. To illustrate, if the actual half-life of ^{18}F is 5% longer than the reference value (e.g., 115.3 min instead of 109.74 min), the dose calculation will underestimate the residual activity at the time of administration. As a result, the patient will receive a higher effective dose than planned, violating the principle of dose optimization. Conversely, if the actual half-life is shorter than the reference value, the calculation will overestimate the dose, leading to underadministration.

Traditionally, techniques based on gamma spectrometry with hyperpure Germanium (HPGe) detectors have been used for this purpose, most notably the Reference Sample method, which allows systematic corrections through comparison with traceable standards [1,2]. However, limitations associated with manual data processing—such as interpolation errors, flawed decay corrections, and limitations in statistical analysis—still represent critical bottlenecks in the reliability of results. In this scenario, the application of computational tools and big data techniques, combined with data integrity verification frameworks, interactive visualization, and automated mathematical modeling, significantly increases the robustness and reproducibility of analyses. The code developed in this study allows, for example, automated reading of large volumes of spectrometric data, application of logarithmic transformations, generation of custom graphs, and implementation of experimental acquisition validation routines, all in an integrated manner [7].

Automating these steps reduces systematic errors, eliminates subjectivity in manual analysis, and significantly reduces expanded uncertainty, especially relevant for short-half-life radionuclides such as ^{18}F , ^{202}Tl and ^{123}I [1, 3]. Furthermore, by incorporating big data principles, such as the simultaneous processing of multiple files and processing scalability, the proposed system aligns with the modern demands of large-scale monitoring and quality control. This work, therefore, proposes a hybrid methodology that integrates high-resolution gamma spectrometry, established metrological techniques, and advanced computational solutions to improve the determination of nuclear parameters, such as half-life. The proposal strictly meets current regulatory requirements (CNEN NE 3.05, ISO 20042:2019), while incorporating the practical and ethical advantages that automation and data science can bring to contemporary clinical radiopharmacy.

2 Materials and Methods

The proposed system used the next methods.

2.1 The importance of precision in the half-life of radiopharmaceuticals

^{123}I , ^{18}F , and ^{202}Tl were the radionuclides chosen for measurement due to their application in nuclear medicine for diagnosis and therapy. ^{123}I is a radionuclide widely used in nuclear medicine for thyroid imaging, brain neurotransmission studies (e.g., ^{123}I -FP-CIT for Parkinson's) and labeling of pharmacological compounds. Its decay occurs predominantly by electron capture (99.99%), with the emission of gamma radiation with a principal energy of 159 keV and a characteristic half-life of 13.22 h, and the production of ^{123}Te (Tellurium-123) as a stable daughter [10], according figure 1.

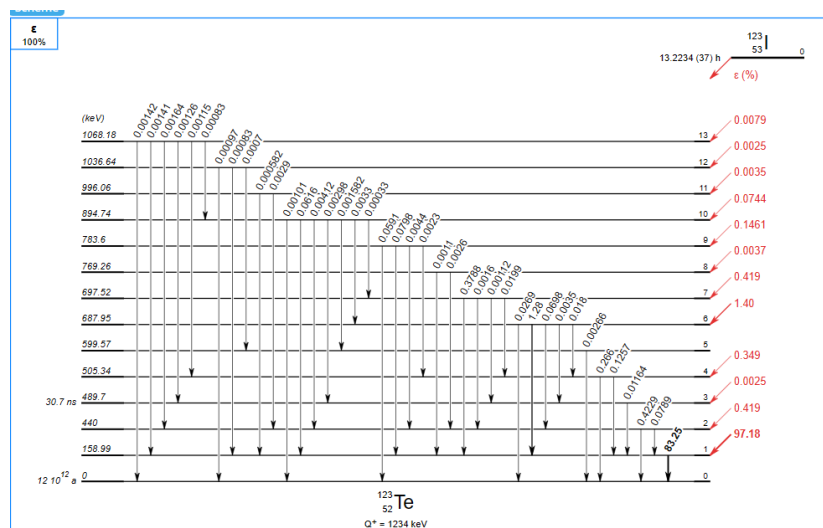


Figure 1: Decay scheme of ^{123}I

^{18}F is one of the most important radionuclides in Nuclear Medicine, especially for PET (Positron Emission Tomography). Its decay occurs predominantly by β^+ (positron) emission (96.7%), resulting in stable ^{18}O (Oxygen-18), with the emission of two 511 keV gamma photons (positron-electron annihilation), with a half-life of 1.82 h, as can be seen in figure 2.

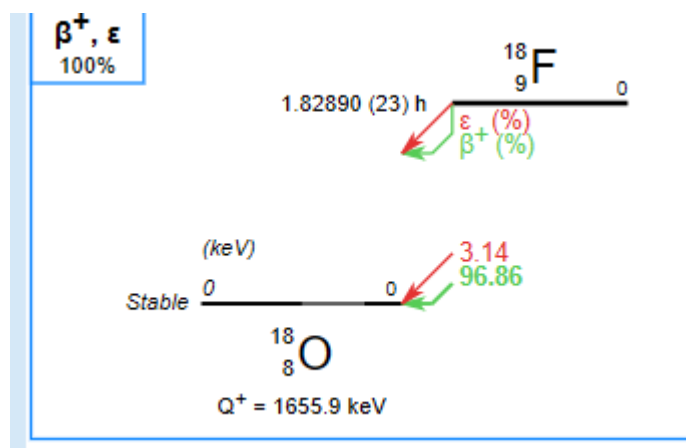


Figure 2: Decay scheme of ^{18}F

^{202}Tl is a radionuclide used in nuclear medicine, mainly in myocardial perfusion studies (alternative to $^{99\text{m}}\text{Tc}$ in patients with restrictions). Its decay occurs by electron capture (EC) (100%), transforming into stable ^{202}Hg (Mercury-202), with gamma ray emission at the main energy of 439 keV and a half-life of 12.23 days characteristic X-rays [10].

2.2 Advantages of gamma spectrometry in determining half-life

It is one of the simplest, well-known and widely used relative or indirect methods of measuring activity.

It only requires comparing the sample to be measured to a reference or, depending on the measurement conditions, prior knowledge of the detector's efficiency, which can be obtained experimentally through the use of radioactive standards. Therefore, when it comes to calibration, the method must be standardized, exhibit short- and long-term stability, be accurate within the application range, have repeatable procedures

and measurements, and be traceable to both the national and international metrological network. In addition to detecting the presence of radiation with energies ranging from 10 keV to 3 MeV, it allows the identification of the radionuclide and its calibration regardless of the presence of gamma-emitting radiochemical impurities [4]. After measuring the radionuclide sources over a sufficient time interval to achieve good counting statistics, the respective channels where the count in each photopeak is maximum are determined. The graphical representation of the photopeak energy as a function of the number of channels is called the system energy calibration curve and is generally represented by a straight line. See figure 3.

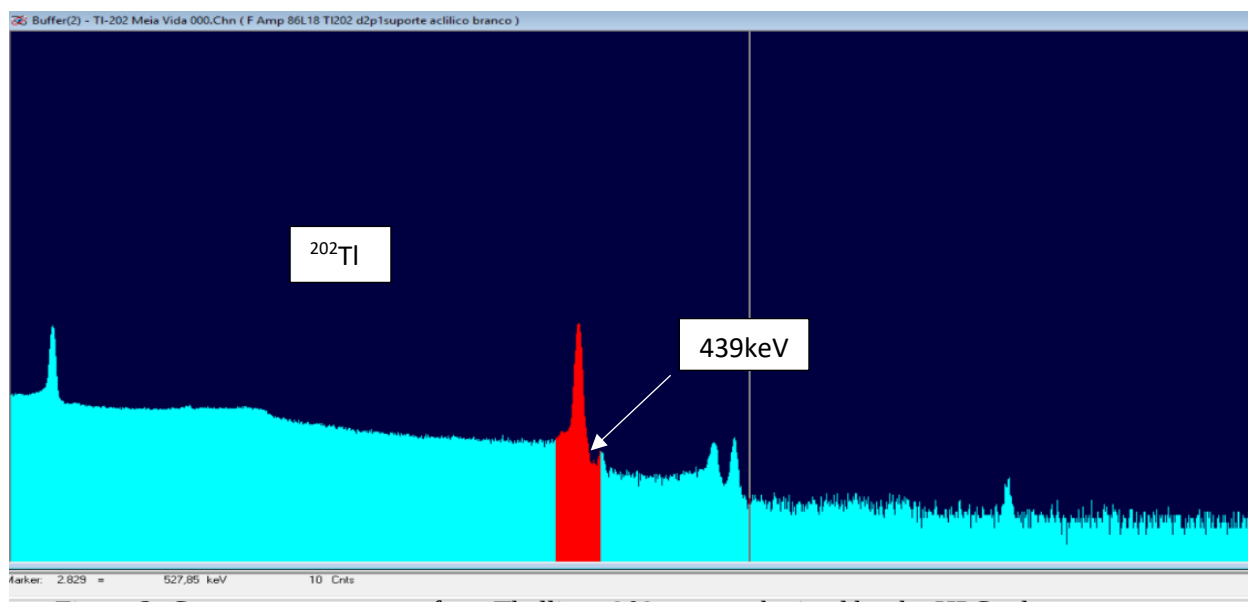


Figure 3: Gamma-ray spectrum for a Thallium-202 source obtained by the HPGe detector.

The methods used were gamma spectrometry, with the HPGe (brand: ORTEC; model: Gmx 70P495) available at the Ionizing Radiation Metrology Laboratory (IRD / CNEN) and traceable to primary measurement methods. For measurements performed with the HPGe detector, it was necessary to calibrate the spectrometer using the channel versus energy relationship. Data were obtained using a commercial system for acquiring and recording count data, Maestro Software [5]. The least squares method was used to fit the experimental data series to a mathematical expression. For measurements performed with the ionization chamber, after obtaining the data, a spreadsheet containing sample information in the form of input data such as counts, date, and time was used to obtain the half-life value. A statistical calculation procedure was introduced into the activity calculation spreadsheet, based on Chauvenet's data rejection criterion [8, 9].

2.3 Computational Tools Used in Data Analysis

Within the scope of the computational tools employed, the use of scientific programming routines enabled the automation of spectrometric data processing, including steps such as import, graphical visualization, statistical adjustment, and half-life calculation. The inclusion of the technical appendix documents the implemented algorithm in a traceable manner, ensuring analytical reproducibility, minimizing manual intervention, and reinforcing the metrological reliability of the results [7].

3 Results

Considering the exponential fit, where ^{18}F transforms into a stable isotope, with the number of atoms decaying per unit of time, we have:

$$\frac{dN(t)}{dt} = -\lambda N(t)$$

Integrating, we have:

$$N(t) = N_0 e^{-\lambda t}$$

The exponential function indicates that for equal time intervals, there is decay of equal fractions of radioactive atoms [8, 9].

Calling:

$$\ln N(t) = y$$

$$\ln N(0) = a$$

$$-\lambda = b$$

$$t = x$$

We have the reduced equation of the line, where the two variables have a dependent relationship:

$$y = a + bx$$

With the sequential acquisition of a certain number of spectra n , using the least squares method, and are determined, and consequently the half-life and associated uncertainty, as follows:

$$a = \frac{\sum y - b \sum x}{n}$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$(\lambda = -b) \quad e \quad (c_0 = e^a)$$

$$\lambda = \frac{\ln 2}{T_{1/2}}$$

and: $T_{1/2} = \ln 2 / \lambda$

The uncertainty was estimated as [8]:

$$\delta_{T_{1/2}} = \frac{0,693}{b^2} \delta_b^2$$

Then, here the results for a ^{18}F source with 300 measurements in a live time of 300 s obtained an estimated half-life of (110.40 ± 0.12) min.

Table 1 presents the half-life of the radionuclide ^{18}F determined in this study through a combination of gamma spectrometry and exponential fitting techniques, enhanced by data processing tools. The value obtained is compared with various results previously published by other laboratories.

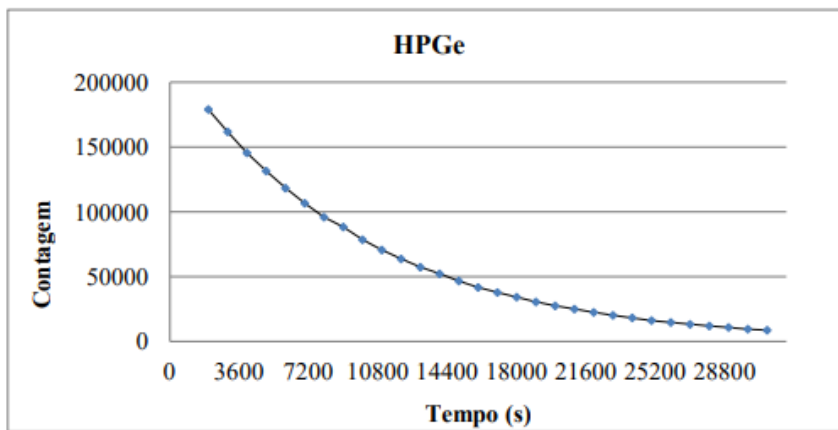


Figure 4: Decay graph for ^{18}F in successive intervals by the HPGe detector

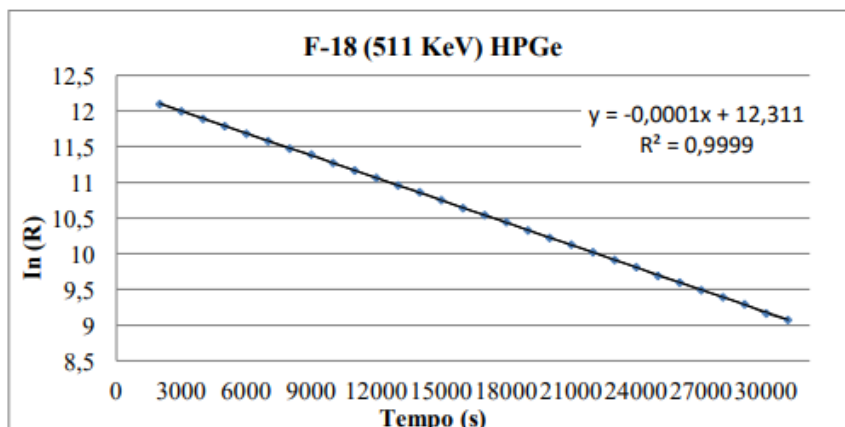


Figure 5: Graph for logarithm of counts as a function of time by the HPGe detector

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import os
4 caminho_arquivo = r'C:\Users\MAIEF\OneDrive\Documentos.xlsx'
5 if not os.path.exists(caminho_arquivo):
6     print(f"Erro: O arquivo não foi encontrado no caminho especificado: {caminho_arquivo}")
7 else:
8     df = pd.read_excel(caminho_arquivo)
9     print(df.head())
10    plt.figure(figsize=(10, 6))
11    plt.plot(df['Tempo(s)'], df['Log'], marker='o', linestyle='-', color='b')
12    plt.xlabel('Tempo(s)', fontsize=12)
13    plt.ylabel('Log da Corrente', fontsize=12)
14    plt.title('Gráfico de Log da Corrente vs Tempo(s)', fontsize=14)
15    plt.grid(True)
16    plt.show()
17
18 import numpy as np
19 from scipy.stats import linregress
20 # Extração dos dados
21 tempo = df['Tempo(s)'].values
22 log_corrente = df['Log'].values
23 # Ajuste da reta: log(C) = -At + b
24 slope, intercept, r_value, p_value, std_err = linregress(tempo, log_corrente)
25 # Cálculo da constante de decaimento (lambda) e meia-vida
26 lambda_ = -slope
27 meia_vida = np.log(2) / lambda_
28 # Propagação da incerteza (erro padrão da regressão)
29 incerteza_lambda = std_err
30 incerteza_meia_vida = (np.log(2) / lambda_**2) * incerteza_lambda
31 # Geração do gráfico com linha de ajuste
32 plt.figure(figsize=(10, 6))
33 plt.plot(tempo, log_corrente, 'bo', label='Dados experimentais')
34
35 plt.xlabel('Tempo (s)')
36 plt.ylabel('Log da Corrente')
37 plt.title('Ajuste Linear do Decaimento')
38 plt.legend()
39 plt.grid(True)
40 plt.show()
41
42 # Relatório parcial dos resultados
43 print(f"Constante de decaimento (λ): {lambda_: .6f} ± {incerteza_lambda: .6f} s-1")
44 print(f"Meia-vida (T1/2): {meia_vida: .2f} ± {incerteza_meia_vida: .2f} segundos")
45 print(f"Coefficiente de determinação (R2): {r_value**2: .4f}")

```

Figure 6: Python script for determining half-lives using the least squares method, including estimation of associated uncertainties and generation of illustrative graphs.

Table 1: Comparison of ¹⁸F half-life determined in this study with published values from other laboratories [10].

Laboratories	Half-life(min)	Year
DAMRI/CEA	109.74 ± 0,03	1991
LNHB/CEA	109.73 ± 0,03	2004
LNMRI	109.71 ± 0,02	2018
This study	110.41 ± 0,12	2025

4 Conclusions

With the developments achieved thus far, the proposed methodology reaches a high level of scientific automation, significantly reducing the subjectivity of the analysis and enhancing both data traceability and metrological consistency. The integration of high-resolution gamma-ray spectrometry, statistical modeling, and algorithmic data processing constitutes an innovative approach aligned with regulatory requirements and the operational needs of modern clinical radiopharmacy.

Up to this stage of the study, 300 experimental measurements have been performed using an HPGe detector, distributed among three clinically relevant radionuclides: 100 measurements of ¹²³I, 100 of ¹⁸F, and 100 of ²⁰²Tl. The acquired data have been processed automatically through custom-developed computational routines, enabling not only precise half-life calculations but also controlled estimation of associated uncertainties.

The set of adopted techniques establishes a solid foundation for the continuation of the study, aiming at the consolidation of a reliable, scalable metrological model applicable to different radionuclides, with direct impact on the quality and safety of nuclear medicine clinical applications.

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